Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency

B. Müller Mario Poneder Alexander Osika Kai Xue Peter Nguyen
Maarten Antonie Visser David Matscheko Brieuc de La Fournière

October 13, 2026

Paper release: `r2049` **Released:** October 13, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks.

Abstract

Observer-Patch Holography describes finite patches that record shared boundaries and repair disagreement. On a declared twelve-port branch, normalized repair response gives a rank-three metric quotient. Signed records and conservative seam currents are dense in its continuous Euclidean completion. Conservative Fibonacci record populations, with complete reads inside a shrinking metric radius and a declared model clock, have causal orders and normalized counts converging to flat $1 + 3$ spacetime. Their ordering fraction in interior causal diamonds tends to $1/10$. Segment-covering estimates control the finite causal and volume errors. For nonzero closed convex pointed displacement cones, operational boost covariance selects Lorentz shape; accessible interval counts reconstruct proper-time ratios. A separate positive scalar action on source-metric populations has controlled continuum detector readings. Modular covariance and a round celestial limit give Lorentz-frame kinematics. Positive null translations and finite tomography reconstruct stress tensors under locality, covariance and Ward hypotheses. With same-family physical stress identification, fixed-cap entropy stationarity, controlled small-ball and curvature limits, universal coupling, a vacuum reference and an independent scale, the Einstein equation follows up to a cosmological constant. Finite MaxEnt identities also determine the fixed-screen capacity-transfer law and de Sitter shock sign.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence,

geometric domain, action, representation, or clock remains a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Because nothing lies outside that structure, recovered structure needs no a-priori justification, nothing unrecovered may be assumed, and every change of description is implemented by the structure's own operations. Agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is the candidate that has to run and reproduce what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with its endogeneity clause; and least-informative state selection. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks remain declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

Derivation map

Normalized repair response gives one positive rank-three source Gram quotient E. Signed cumulative records and conservative seam words embed densely in its Euclidean completion. Its unit directions form S^2; adjoining a scalar line gives $\mathbb{R}\oplus E$ with quadratic form $t^2-\|x\|^2$. This connects the source metric to a four-dimensional Lorentz carrier without selecting a physical event clock or interpreting every algebraic direction as a measured signal.

The conditional implications on this common carrier are

$$\begin{aligned}
\text{nonzero closed convex pointed cone}\\ \text{finite irreducible operational rotations}\\ \text{one continuous boost direction}
&\Longrightarrow \text{Lorentz displacement cone}\\ &\quad\text{up to time orientation},\\
\text{conservative Fibonacci populations}\\ \text{complete local reads and } \Delta{=}a/c
&\Longrightarrow \text{flat causal-order limit}\\ &\quad\text{raw count–volume convergence},\\
\text{controlled order/count limit}\\ \text{accessible interval records}\\ \text{one reference interval}
&\Longrightarrow \text{proper-time ratios from counts},\\
\text{positive source-metric scalar action}\\ \text{controlled sampling and smooth data}
&\Longrightarrow \text{continuum detector bounds}\\ &\quad\text{and finite oscillator quantization},\\
\text{positive translations and finite null tomography}
&\Longrightarrow \text{conditional conserved } \langle T_{ab}\rangle,\\
\text{physical stress and universal coupling}\\ \text{entropy, small-ball and tensor-curvature limits}
&\Longrightarrow G_{ab}+\Lambda g_{ab}=8\pi G\langle T_{ab}\rangle.
\end{aligned}$$

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/main/docs/BASIS.md

These implications have distinct hypotheses. In particular, operational boost covariance acts on admissible histories and interventions; an algebraic action on the celestial sphere does not establish it. The finite proper carrier rotations and a continuous boost along one axis suffice to select the closed convex pointed causal cone by Theorem B.2. This theorem selects cone shape under its covariance premise, not a one-step read stencil or field equation.

The constructive event family uses actual conservative source records. Put $\varphi = (1+\sqrt{5})/2$ and $L = 2/\sqrt{\varphi+2}$. For consecutive Fibonacci numbers $q = F_n$ and $p = F_{n+1}$, with $n \geq 4$, its q^3 spatial records have a source-metric assignment to equal-volume cells with error $H_q = 2\sqrt{3}L/q$. Each event reads every preceding-layer record inside radius $a_q = L/\sqrt{q}$, including its own preceding record, and the model layer duration is $\Delta_q = a_q/c$, with $c > 0$. Segment subdivision bounds the reachable radius between $c(1-2h_q/a_q)T$ and cT. Since $h_q \leq H_q$ and $H_q/a_q \to 0$, these cones agree in the limit away from their null boundary. Proposition A.21 proves normalized raw counts converging to $dt\, d^3x$ volume. On fixed interior timelike intervals the ordering fraction tends to $1/10$. The finite path construction, the layered order, the flat reference constants, the count-clock enclosure, the generated-interval and strict-pair count limits and the $1/10$ limit for diamonds inside the sampling window are formalized in Lean; the general moving-tip finite volume-error estimate is an analytic proof. The finite readings of the family and of its carrier realization are recorded in Section B.11.

The population, complete-neighbour read rule, model tick and choice of one counted event per site and layer are supplied. Independent exact replay checks conservative currents, Gram-metric neighbours, actual writer versions, schedule changes and transmitted interventions at finite cutoffs. It does not turn those four controls into an asymptotic experiment or identify the tick with a matter clock. Under the same count limit, the observer readout $(|I|/|J|)^{1/4}$ converges to the proper-time ratio of two interior timelike diamonds. It reads accessible ancestry and counts, with a reference interval fixing units; its decoder needs no coordinate labels or oscillator. Theorem B.3 and Proposition B.4 state its limit and finite-error conditions.

Exact order embeddings form a separate finite interface. For a source spatial readback and positive height scale, a parent-edge speed bound maps authenticated ancestry into the future cone. Equal-height spatial separation and strict spacelike separation of increasing-height nonancestors give the converse, hence an injective order embedding. The source height is the attained longest authenticated-parent-chain length. These finite statements do not identify height with proper time, or element number with volume. A causal-set faithful embedding also requires the appropriate population density[1, 2]. The controlled source-record construction instead proves vanishing cone and count errors; it need not reflect null comparisons exactly at each finite resolution. Neither route selects curved topology or the physical interpretation of its readouts. Under the relevant continuum causality hypotheses, causal order and volume determine conformal geometry and scale[14, 15].

The minimal source-direction Einstein-shape theorem places supplied symmetric geometry and stress fields on the same finite event type. It asks for balance on nine fixed algebraic source directions, four discrete step maps, Ward/Bianchi identities, and step connectedness. Exact tomography derives all-null balance, the pointwise metric ambiguity, its constancy, and

$$\mathsf{geometry}_{ab} = \kappa\, \mathsf{stress}_{ab} + \Lambda \eta_{ab}.$$

The all-null relation is therefore a conclusion rather than a premise. The minimal package contains no normalization, universal-source, vacuum-calibration, or Newton-constant field. The nine-set is algebraic, not selected by provenance or invariant as a set under Lorentz transformations. No provenance edge selects a direction, tensor field, step, or balance. The ambient carrier dimension and signature are constructed independently by the source-causal carrier theorem; the poset dimension, fields, and their physical interpretation remain separate.

Identifying geometry with smooth curvature and stress with physical stress-energy requires the source-causal continuum, count–volume, stress, small-ball, coupling, scale, and remainder receipts on one family, together with either a tensor-curvature reconstruction converging to the Einstein tensor or the independent continuum small-ball/null-balance identification. Scalar-curvature convergence alone is diagnostic. The full `composedEinsteinBranch` remains available for that conditional physical composition; it does not construct its source-derived tower.

Stress reconstruction consumes same-source null-stress matching, locality/covariance, and the weak Ward balance; finite null tomography does not supply them. The celestial S^2 and the frame hyperboloid do not by themselves populate a spacetime. The source-causal continuum and Einstein conclusions are conditional theorems on their named receipts. Of the local-stress inputs, one finite precursor is in hand at one energy component: a face-local discrete Poynting law on the committed twelve-port carrier, exact at every step of the scaled Maxwell evolution, whose face sum is the committed global energy balance and whose density–flux pair is one conserved current in the source-free case (`Lean/Screen/LocalEnergyBalance.lean`). It supplies no momentum or stress components, covariance, or continuum limit, and the identification of the screen field energy with the modular energy the composition consumes is read against one declared clause: the clock-and-energy calibration row of the four-law surface, whose reference is the Gibbs state of a declared energy at one declared inverse temperature β. That row is inhabited by the screen field energy, and against it the modular energy ledger of any state is exactly β times its field-energy expectation plus $\log Z$, the cap first law takes the Clausius form $S(p) - S(\tau) = \beta(\langle H\rangle_p - \langle H\rangle_\tau) - D(p\|\tau)$ with the offset cancelled, the slope and the reference are determined by the ledger, the energy is free exactly up to an additive constant, and β is determined by the reference once the family and the step are declared; a two-point reference outside the Gibbs form has equal energy expectations with unequal ledgers, so the affine reading is specific to that clause. Nothing derives that the record reference is Gibbs and nothing selects β; the identification stays conditional on the declared row (`Lean/Thermodynamics/GibbsReferenceEnergyIdentification.lean`). The collar modular surface does not supply that inverse temperature. Its one thermal object, the reference on the realized branch of the declared maximum-entropy constraint, with modular Hamiltonian $K = \lambda \cdot S$ on the four joint sectors, has spectral weights equal to the Gibbs state of K at $\beta = 1$ in modular units, and for every nonzero β they equal the Gibbs state of K/β at β, so every temperature declaration inhabits the calibration row for the collar reference and two declarations with different β both do; a declared energy with two distinct levels pins β, a declared β pins the energy up to a constant, and the collar reference is Gibbs for an energy H at β exactly when $K = \beta H + \text{const}$. The collar functional is KMS at parameter one for the inner flow of K and at parameter β for the flow of K/β, so the KMS parameter and the inverse temperature trade against the energy scale. The identification of the collar weights with the record reference of the cap first law stays declared (`Lean/Thermodynamics/CollarTemperatureReading.lean`).

The sourced field equations on the committed complex are consistent in the following exact sense. The hopping load and hopping current of a declared port path satisfy the committed continuity equation at every port and every step, and a finite family of hopping charges satisfies it with total load equal to the sum of the charges. Along the scaled Ampère evolution with any current, the Gauss constraint propagates from one step to the next exactly when the sources satisfy the continuity equation, so a field history that solves the Ampère evolution with a family of hopping charges and starts from Gauss-consistent data stays Gauss-consistent at every step. The complete criterion is an equivalence: for declared sources, a field history solving the Ampère evolution and the Gauss constraint at every step exists if and only if the initial load is neutral and the sources are conserved; the existence direction is an explicit second-order recursion in the temporal gauge started from the canonical Coulomb field of the initial load. Because the port boundary conserves total load

on the closed carrier, one hopping charge alone has a Gauss-consistent history only when its charge is zero; a join of a worldline to the field sector therefore carries a neutral partner or a neutralising background. A neutral pair of crossing charges, with integer Green potential checked against the committed Green and incidence tables by a kernel check, inhabits the scaled Maxwell bundle with a nonzero current, and the unit crossing of every seam satisfies the continuity equation by a kernel check that fixes the sign convention between the crossing current and the incidence operator (`Lean/Screen/SeamChargeContinuity.lean`). The neutral-partner route is also realized in one finite action rather than by appending an independently prescribed background after variation. For $h \neq 0$, the source-free Maxwell window plus the two opposite minimal couplings and two clock actions is invariant under endpoint-vanishing gauge functions. With the seam-potential endpoints fixed, its field variations are equivalent to the interior sourced Ampère equations at $m < N$ and Gauss equations at $n < N + 1$. Replacing either path over a closed two-step interval while fixing the fields and other path gives an exact off-shell action difference, not by itself a path equation of motion. At $q = 1$, $h = 1/2$, the explicit Coulomb-started packet is stationary in the field slots, but for every $N \geq 1$ delaying either its positive seam-0 crossing or its negative seam-29 crossing, while fixing the fields and the other path, changes the same action by $5/12$ for every pair of clock-unit values within the fixed quadratic clock-action form. The charge-conjugate results therefore fail the declared closed-two-step exchange-stationarity condition in both path slots; applying both canonical exchanges simultaneously changes the action by $5/6$. These are only the displayed discrete exchanges, not continuum first-variation claims or a no-go for other paths, fields, or action shapes. The counterexample is specific to that Coulomb start. On the window $N = 1$, a second exact rational temporal-gauge history with the same crossings solves the sourced equations and leaves the same action unchanged under every in-window closed two-step replacement of either path. The exhaustive route check has two zero-difference endpoint routes and two triangular detours whose clock difference -4 is cancelled by interaction difference $+4$. The sources are nonzero and opposite. At the separately declared carrier unit $\tau = 3$, both crossings are timelike; identifying $\tau = h$ would instead make a crossing spacelike (`Lean/Screen/NeutralPairCoupledAction.lean`, `Lean/Screen/NeutralPairJointStationaryWitness.lean`). The step, charges, paths, rational history, clock unit, and action shapes are declared; the sign of the committed load relative to a physical charge, source selection, boundedness of the sourced history, a minimum or stability theorem, laboratory calibration, and any continuum limit are outside this statement.

The three-dimensional source metric has an exact local construction. For the declared twelve-port repair mean, the centered equal-port response kernel has an exact spectral decomposition. Dividing by its slow response before taking the infinite-step limit selects a rank-three projector. Its range in the twelve-port counting space, equipped with the induced Gram form, is an intrinsic Euclidean carrier. On the conditional signed cumulative port-record/load module, the limiting form has a three-dimensional radical on the real six-control extension. The integer module has no nonzero metric kernel, embeds densely in the resulting quotient, and completes to the same abstract continuous carrier, up to the unique isometry that preserves the port labels. The source load map is exactly onto $\mathbb{Z}^6$, with the antipodally equal loads as its kernel. Conservative seam boundaries have image

$$D_6 = \{z \in \mathbb{Z}^6 : \textstyle\sum_i z_i \equiv 0 \pmod 2\},$$

with Smith invariants $(1, 1, 1, 1, 1, 2)$. The pullback response metric makes this submodule dense in the same completion; the ordinary six-dimensional lattice metric on D_6 is a different metric and is not used. Cumulative record addition acts simply transitively on the exact record carrier and by isometries on its completion. The sixty directed seam labels form one proper carrier-rotation orbit and give the corresponding homogeneous internal convolution under the stated naturality

and unique-minimizer premises. The thirteen finite covariance/isometry table identities in the proper-action subpacket are checked with Lean's `native_decide`; they consequently trust the native compiler and runtime as well as the kernel and are not kernel-only receipts.

Every finite-step kernel has rank eleven on the centered space and rank six on the signed sector. The limiting response form supplies the rank-three metric and the completion topology. The rotation-covariant seam average selects a spatial operator under the stated feasibility and minimization hypotheses. A physical propagation law additionally requires the population, accepted dynamical moves and clock attachment. The source metric alone does not choose a preferred Cartesian frame or identify record addition with physical translation.

The Lorentz arrow is a composition with two independent inputs on one refinement tower: the finite cap-normal Bisognano–Wichmann certificate, whose clauses are cap-normal density and non-degeneracy, BW framing, support-order faithfulness, geometric support-flow group and continuity control, held-out oriented cross-ratio convergence, independently normalized geometric 2π-KMS comparison, and wrong-normalization separation, and an independently produced complete multiresolution algebra-state comparison package on the same tower. Geometry does not manufacture the modular state, and bare finite consensus implies neither input. The Einstein arrow consumes one common scaling family, universal coupling, a vacuum reference, and independent scale readouts.

The finite boundary-circle diagnostic uses half-filled ground states of anti-periodic nearest-neighbour hopping with Fermi velocity one, on rings of $16, 32, 64, 128$ sites. Summing every fitting odd hopping range of the half-ring modular Hamiltonian gives median relative profile errors $0.004808, 0.001152, 0.0002850, 0.00007106$ against the conformal profile. A fixed range-nine extraction instead increases from 0.0001124 at 64 sites to 0.0003381 at 128. Both extractions use the same Hamiltonian and state. The omitted contribution at each bond is bounded algebraically by $2\sum_{r>9} r|h_{i,i+r}|$ over fitting odd ranges, with $i = j + (1-r)/2$ at bond j; its evaluation here is numerical. These profile and transport cross-ratio comparisons establish finite diagnostics, not a KMS identity on a common cap algebra or a Cauchy-limit certificate (`code/geometry/modular_clock_instrumentation.py`).

The formal soldering layer separates event readback from frame choice. Once a coincidence quotient and invariant raw readback are supplied, the readback descends uniquely. A time-oriented affine Lorentz overlap cocycle and a compatible chart-coordinate family make Lorentz intervals chart independent, future-null directions covariant, and observer rest readbacks isometric under transport. The coordinate family must be identified with the quotient-descended readback; the source Gram quotient supplies its candidate spatial metric rather than a physical localization map.

Clock comparisons have a corresponding conditional interface. On a unit-timelike affine history, positive clock increments equal dimensionless proper time. Two ordered common events fix the positive-affine relation between two declared clock scales; a third gives a cross-multiplication check with no new fit parameter. These identities do not construct the clock histories or attach their rates to the source dynamics. The count clock provides a distinct, retrospective duration readout on the declared source-record family, with explicit resolution requirements when durations are accumulated along a moving worldline.

A kinematic mass-shell packet is machine-checked on the declared Hermitian Lorentz module (`Lean/Geometry/MassShellKinematics.lean`). For one declared mass parameter m and any future unit-timelike frame u, the scaled vector $p = m\,u$ has Lorentz square exactly m^2, and its scalar coordinate obeys $p_0^2 = m^2 + |\mathbf{p}|^2$. For positive m the scalar coordinate dominates m, with equality exactly when the spatial coordinates vanish, and at the standard frame it equals m. Every oriented Lorentz map carries the construction to the transported frame with the same Lorentz square, so the shell identity holds in every oriented chart, and a future null vector satisfies the zero-mass identity $v_0^2 = |\mathbf{v}|^2$. The mass parameter is a declared hypothesis on the declared module, stated in

its dimensionless convention with unit invariant speed. The packet identifies no coordinate with a physical energy readout and makes no physical rest-energy claim: the clock and energy calibration anchors and the observer-to-physical-spacetime attachment are not supplied.

A separate finite-graph precursor makes the discrete inverse-square route premise-complete. For a supplied integer layering, steady source law, shell equidistribution, and shell count cn^2 through a finite depth, exact divergence bookkeeping gives outward flux per shell edge $Q/(cn^2)$, with n^2 times that flux independent of the shell. An explicit three-vertex chain inhabits the premises. The same formalization also proves why the depth cutoff cannot be hidden: on a closed finite carrier, a globally steady drained source has zero total load, while an n^2 shell count imposed at every positive layer forces $c = 0$. These are finite conditional identities, not an identification of layer with physical radius, flux with gravity, or source with mass density.

Contents

1 Recovered Core: Foundations and Structural Branches **9**
- 1.1 Overview . . . 9
- 1.2 Model and Premises . . . 10
- 1.3 Information-Theoretic Tools . . . 19

2 Finite Screen Spectrum Theorem Package **31**
- 2.1 Overlap Consistency and Gluing . . . 62
- 2.2 Modular Flow and Lorentz Kinematics . . . 67
- 2.3 Gravity from Fixed-Cap Generalized-Entropy Stationarity . . . 79
- 2.4 Einstein-chain theorem and falsification boundary . . . 112

A Technical Details for the Specialist Derivations **114**
- A.1 Port-record metric completion . . . 114
- A.2 Lorentz and Einstein Bridge . . . 115
- A.3 Cosmological-Constant / Screen-Capacity Closure . . . 116
- A.4 Causal order on a source-metric population . . . 121
- A.5 Protected spatial records and mutable repair states . . . 126
- A.6 Source-population quadrature and detector errors . . . 127
- A.7 Readback through destructive seam averaging . . . 130
- A.8 Reusable classical readout with local feedback . . . 132
- A.9 Conditional full-family routing on the wired support . . . 133

B Recovery from finite federation logs **137**
- B.1 Native scalar records and the selection boundary . . . 138
- B.2 Conditional stored computation by supported pair means . . . 141
- B.3 Temporal acceptance, native continuation and preservation guards . . . 144
- B.4 Finite checkpoint selection and the shortest native two-record order . . . 146
- B.5 Eventual publication and the source-selection problem . . . 148
- B.6 From registered source operations to numerical reads . . . 154
- B.7 Repair instruments, recurrent influence, and erased records . . . 155
- B.8 Source selection and a sufficient record-transport law . . . 157
- B.9 Operational selection of the causal cone . . . 161
- B.10 A clock read from source interval counts . . . 163
- B.11 Observations of the causal manifold . . . 165

B.12 Spatial populations and a common scalar continuum action . . . 168
B.13 Action measure, record production and geometric density . . . 171
B.14 A common source-population scalar and quantum detector packet . . . 173
B.15 Time refinement of the same compact scalar detector . . . 177
B.16 An authenticated execution of the golden scalar action . . . 180
B.17 Volume work and finite field equations of state . . . 182
B.18 A clock reconstructed from the scalar records . . . 185
B.19 Calibrated clocks and an undetermined light-deflection coefficient . . . 187
B.20 A common scalar history with readout and restoration . . . 188
B.21 A bounded quantum probability of the same finite action . . . 190
B.22 The same quantum detector with reconstructed action-time uncertainty . . . 191
B.23 Repeated local quantum readout with controlled disturbance . . . 192
B.24 Finite-duration quantum readout on the same scalar action . . . 194
B.25 Finite local preparation of the scalar intervention . . . 197
B.26 Regional canonical reconstruction from two field layers . . . 198

C Local clocks, record gluing and information capacity 200

D M1 necessity: sparse reads and observable dependence 204

E Variational crossing area and a uniform read action 209

F Global vacuum fidelity of local scalar read programs 213
F.1 Exact mode error and a constructive higher-order step . . . 213
F.2 Sharp full-population thresholds . . . 215
F.3 The sparse ultraviolet scale and its repair . . . 216
F.4 Read cost, clock scope and an exact-preservation obstruction . . . 216

G Native time and coherent quantum transport 218

H Massive operational clocks and resolved local reads 223

I Native quantum dynamics with live noisy records 225

J Filled source vacuum and positive excitation energy 228

1 Recovered Core: Foundations and Structural Branches

Observer-Patch Holography is a reconstruction program for fundamental physics. On the declared screen-first global-support branch, physical data are charted by a horizon screen S^2, and the effective world is encoded by the mutual consistency of overlapping patch descriptions. This support screen is distinct from a local carrier boundary and a federation overlap nerve. On the producer branch, S^2 enters only after the carrier-to-support map satisfies the spherical incidence, mesh, cross-ratio, normalization, and refinement receipts. The legacy finite reference fixture does not supply that map because it has no composable seam triangle and its higher-overlap condition is vacuous. The source-derived producer instead supplies a twelve-chart, thirty-seam, twenty-face incidence nerve, nonvacuous seam-triangle cocycles, confluent phase repair, and a controlled oriented-S^2 support limit on one declared tower. Event, modular, scale, and laboratory attachments require separate premises. This recovered-core section develops fixed-cutoff overlap repair, the separated cofinal refinement-limit consensus bridge, collar recovery, and the geometry-readout route to Lorentz frame kinematics and conditional Einstein dynamics.

1.1 Overview

Overlap consistency first turns local patch states into a constrained gluing problem. The fixed-cutoff branch gives collar decompositions, edge centers, and finite repair dynamics on the declared physical quotient. Certified read-from provenance generates a finite informational poset and its exact longest authenticated-parent-chain height. Independently, the exact rank-three source Gram quotient and a real axis define a four-dimensional ambient target carrier with Lorentz inertia $(1, 3)$; source height enters only through the temporal coordinate of a scaled event placement. This is not an intrinsic dimension estimate for the finite poset, and the source unit directions are exactly the celestial S^2 and future-null-ray boundary. This removes a freely declared rank-four chart and fitted signature from the finite precursor. Modular flow on caps independently supplies Lorentz-frame kinematics and the three-dimensional observer-frame hyperboloid. Conservative Fibonacci-record populations, complete reads inside a shrinking metric radius, and a declared model clock give a separate flat $1 + 3$ causal and count-volume limit. This supplied law does not identify a physical source. The exact-embedding route to physical spacetime requires source-selected placements with two-way cone/order agreement; the controlled route permits uniformly vanishing causal discrepancy. Both require compatible physical clocks, count calibration and continuum geometry. Smooth Einstein promotion further requires same-family tensor-curvature convergence or the independent continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. Positive null translations supply directional charges, and finite null tomography reconstructs a symmetric tensor when its dependent-family linearity relations hold. Its interpretation as a local, covariant, conserved stress tensor additionally requires the declared same-source null-stress, locality/covariance, and weak-Ward supplies. Fixed-cap generalized-entropy stationarity, a uniform small-diamond limit, universal coupling, a vacuum reference, and independent scale readouts then imply the Einstein equation. Bare finite consensus stops at quotient normal forms, and the cap-normal H^3 chart stops at observer frames. Neither statement populates a spacetime.

The local pixel ratio $P \equiv a_{\mathrm{cell}}/\ell_\star^2$, the cosmic record capacity $N_\star = \log D_\star$, and the selected no-G scale certificate enter only on their declared quantitative branches. The source-causal continuum and Einstein theorems do not depend on the Standard Model matter or sector-completion branches.

1.2 Model and Premises

1.2.1 Observers and access model

The support-local algebra-state-record reduct of an observer patch is $O_{\mathrm{red}} = (P_O, \mathcal{A}(P_O), \rho_O, R_O)$, where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to P_O.
- ρ_O is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.
- R_O is a finite record algebra generated by stable internal record projectors within P_O.

An operational observer additionally carries overlap interface algebras and restriction maps, allowed update and repair instruments, and checkpoint data used for continuation. Observers are internal self-reading patterns in the global state. A patch and its compatible marginal supply a geometric or algebra-state chart. An operational observer requires the interface, record, repair, and checkpoint conditions.

1.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen S^2. Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state ω is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, ω restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

1.2.3 The premises A1–A3

This paper states the premises A1–A3 because both specialist derivations share their observer-overlap foundation. The flagship synthesis [25] gives the account of the theory, and the executable observer paper [26] records which of these premises the construction supplies. Branch-local hypotheses are written at the theorem that consumes them. The support-visible BW scaling theorem handles the scaling step. The source-causal continuum and Einstein theorems use the additional collar-recovery, edge-normalization, generalized-entropy, common-domain, stress, scaling, universal-coupling, vacuum-reference, and scale-readout interfaces stated in their own hypotheses; none of those interfaces belongs to A1–A3.

The screen premise A1 below defines the screen-first support branch. The producer branch reaches the same premise only after its carrier-to-support receipts establish the global support screen.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators

generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered

source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

The closure hypothesis. The premises A1–A3 describe the observer screen. The closure hypothesis states that the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the description and the described system are one object. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The closed object is the complete inhabited history, whose internal events and responses to interventions are its proposed intrinsic computation. This reading requires no external computer or playback time. It is distinct from fixedness of a repair state, a public record, or a probability law, and does not itself construct the history or prove a final cosmic endpoint.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point per declared map on its analytic domain. Interpreting that root as the laboratory

coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4}, compared with experimental standard uncertainty 2.1×10^{-8}; this is not agreement at measured precision. The gauge width supplies 99.2 percent of the gap between the bare root and the measured value; the residual 3.3904×10^{-4} is addressed to same-scheme hadronic transport to the Thomson endpoint.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the A1–A3 source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\!\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\mathrm{pres}} = N_0 \left(1 - \frac{P}{24}\right), \qquad \ln \frac{N_{\mathrm{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\mathrm{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

Collar recoverability, generalized-entropy structure, and quantum focusing lie outside A1–A3: they are classified gravity interfaces with their own theorems, countermodels, and required inputs, consumed field by field where the Einstein composition needs them. Sector content is not selected

by A1–A3: the conditional rank-45 tensor candidate does not establish physical families or a local Spin carrier, while scalar multiplicity and extra-sector completeness lack physical attachments. Each theorem below lists the declared completions it consumes.

1.2.4 Closure values and theorem-local technical data

The quantitative branches discussed here use one incomplete declared-map coordinate, one proposed capacity coordinate, and one selected no-G scale certificate:

$$P \equiv a_{\mathrm{cell}}/\ell_\star^2, \qquad N_{\mathrm{CRC}} \equiv \log\dim \mathcal{H}_{\mathrm{tot}}, \qquad \gamma_\star \equiv \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star \equiv \frac{3\pi}{\ell_\star^2}.$$

Here $P = P_\star$ denotes the unique mathematical root of the selected declared outer/inner map. No measured coupling enters that solve. Its physical Thomson-limit interpretation requires the same-scheme hadronic spectral transport, target-independent selection of that map, and a typed bridge proving that the two coordinates read the same physical quantity. A positive global coordinate would be defined by a source-only correctable public-record packet and the stable whole-fiber saturation equation $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_\star = \log D_\star$. The independent weak/Higgs common-load bridge gives $N_{\mathrm{bridge}} = \pi \exp[6\pi/(P\alpha_U(P))]$. On the source-forward branch $P = P_{\mathrm{fwd}}$, and that bridge is $N_0 = 3.532131543418936\ldots \times 10^{122}$, while the full weighted Planck base-ΛCDM comparison coordinate is $N_\Lambda = 3.31292709806038\ldots \times 10^{122}$, a mismatch of about 6.6 percent. On the finite collar branch, declared total reserve expectation $P_{\mathrm{fwd}}/4$ and six-class equidistribution give presence probability $P_{\mathrm{fwd}}/24$ for each class. If one class is physically selected, its scalar-weighted receipt is discharged, and its normalized survival factor acts on global capacity, it gives $N_{\mathrm{pres}} = N_0(1 - P_{\mathrm{fwd}}/24) = 3.292097877326465\ldots \times 10^{122}$. The alternative $N_0 e^{-P_{\mathrm{fwd}}/24} = 3.300072225377652\ldots \times 10^{122}$ requires a separate mean-count or continuum carrier. Neither global action is selected: exact positive compositional completions share the local survival datum and disagree globally. The named-law branch is not evaluable on this source class, and its horizon branch has no capacity object. Both comparisons are retrospective. Capacity-level neutrino side estimates tied to that branch value are comparison-only bookkeeping terms separate from the rejected weighted-cycle construction. The selected scale certificate supplies $\gamma_\star$, with $B_\star$ as the SI curvature display of the same branch. Once supplied, $\ell_\star^2 = 3\pi/B_\star$ becomes the usual Planck-area display after $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$. These are quantitative-branch quantities and add no premise to A1–A3.

Observation enters as an external test of the source-defined branches. In the OPH reverse-engineering reading, the universe is a closed mathematical fixed structure, and each readback object is specified independently of its measured target. For N, this object is set-valued before whole-fiber scalarization and its uniqueness theorem is the exact finite-size slack law, not a Banach contraction. Measurements may define comparison coordinates; they construct neither closure. The local map has a certified unique root. Global uniqueness requires the regulator-stable condition $s(D_\star) = 0 < s(D)$ elsewhere. A bounded target-clean counterfamily does not entail that condition under shared base agreement, positivity, the carrier bound, and executable finite controls. The exact verdict applies to this bounded completion class. The complete A1–A3 source contract has not been shown to contain the countermodels at every rung, and no executable-to-Lean bridge establishes that membership. Direct N is not evaluable on the incomplete source antecedent, and the stronger source-class verdict does not follow. A positive result needs a complete source antecedent, one physical zero, and a universe-level carrier attachment.

The spacetime and Einstein derivation uses the same convention as a compression estimate. A quantitative row enters that estimate only when the declared source map has no dependency path from the measured target or a calibrated proxy. If p_i upper-bounds the conditional accidental hit

probability of row i conditional on the other accepted rows, then

$$P_{\mathrm{acc}} \le \prod_i p_i.$$

No discriminating target-independent prediction enters this compression estimate. The electroweak checks use known targets and are comparison-only. Other positive results are theorem-level, conditional, comparison-only, or non-discriminating.

If both missing physical identifications are supplied, the two dimensionless coordinates determine dimensionless geometry, not an SI scale by themselves. On the de Sitter branch,

$$N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2}, \qquad P_\star = \frac{a_{\mathrm{cell}}}{\ell_\star^2},$$

so

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\mathrm{cell}} = \frac{3\pi P_\star}{N_\star}.$$

Equivalently, for the scale product $\mathcal{B}_\ell := \Lambda_\star N_\star$,

$$\mathcal{B}_\ell \ell_\star^2 = 3\pi, \qquad \mathcal{B}_\ell a_{\mathrm{cell}} = 3\pi P_\star.$$

These invariants survive the rescaling $\ell_\star^2 \mapsto \lambda^2 \ell_\star^2$, $a_{\mathrm{cell}} \mapsto \lambda^2 a_{\mathrm{cell}}$, and $\Lambda_\star \mapsto \lambda^{-2}\Lambda_\star$, while $\mathcal{B}_\ell \mapsto \lambda^{-2}\mathcal{B}_\ell$. Thus the SI value of $\mathcal{B}_\ell$, the SI area $\ell_\star^2$, and G_{SI} require the selected scale certificate; they are not derived from $P_\star$ and $N_\star$ alone. If the selected no-G scale certificate supplies $\gamma_\star$, equivalently $B_\star$, then

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star = \mathcal{B}_\ell = \Lambda_\star N_\star, \qquad \ell_\star^2 = \frac{3\pi}{B_\star}.$$

The branch certificate used for the SI gravity row is

$$B_\star = 3.60787391468\ldots \times 10^{70}\,\mathrm{m}^{-2},$$

hence

$$\ell_\star^2 = 2.61228030237\ldots \times 10^{-70}\,\mathrm{m}^2.$$

Rounded capacity displays such as $N_\Lambda \simeq 3.313 \times 10^{122}$ are useful cosmological labels, but they are not high-precision inputs for the Newton row; with the Planck-2018 curvature benchmark, using only the rounded capacity would shift the Newton row at the 8.4×10^{-4} level.

The capacity normalization uses the de Sitter static-patch entropy [52]. If

$$N_{\mathrm{patch}} = \left(\frac{r_{\mathrm{dS}}}{\ell_P}\right)^2$$

denotes the bare horizon area ratio, then

$$N_{\mathrm{scr}} = S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4\ell_P^2} = \pi N_{\mathrm{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

Using the Planck-2018 late-time de Sitter scale [54] gives $N_{\mathrm{patch}} \simeq 1.05 \times 10^{122}$, $N_{\mathrm{scr}} \simeq 3.313 \times 10^{122}$, and $\Lambda \ell_P^2 \simeq 2.845 \times 10^{-122}$.

The controlled scaling step for the Lorentz, null-modular, and Einstein statements is the support-visible BW scaling theorem. Fixed-cutoff regulator bookkeeping, quasi-local propagation, endpoint-Lipschitz control, and collar decomposition are cited from their preceding theorems.

Local information projection at the regulator scale. At the regulator scale ℓ_{UV}, consider the Axiom-3 subbranch carrying a global-extension interface and identity-proportional local references for which the state-determining weighted local objective is represented by one finite density matrix ω. The selected state maximizes the corresponding weighted local entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operator densities, each supported on a ball of radius $\leq r_0 = O(\ell_{\mathrm{UV}})$.
2. Constraint equations $\langle \sum_x O_a(x) \rangle = C_a$, one homogeneous global sum per density label a, where x runs over the cells of the UV lattice. The constraints are the N_{con} global sums, not one independent constraint per cell, so the number of independent constraints and of Lagrange multipliers is N_{con} (plus the optional global constraints below) on every lattice, independent of the cell count.
3. Optionally, a finite number of global constraints (total energy, charge).

This is the finite-matrix subbranch needed to derive the local Gibbs form of Lemma 2.6 from Axiom 3 with a cutoff-independent multiplier count. The global extension and the objective representation are explicit branch premises.

Clarification (conditional maximum randomness and thermal equilibrium). Axiom 3 is local state selection relative to an exact reference and a complete visible constraint set. It does not state that the universe is in thermal equilibrium. Non-equilibrium physics is modeled in the enlarged *local-multiplier* family obtained by replacing the homogeneous multipliers λ_a with slowly varying fields $\lambda_a(x)$, equivalently by imposing per-cell constraints $\langle O_a(x) \rangle = c_a(x)$. That enlarged family has dimension proportional to the number of UV cells, so it is a function-space approximation regime with gradient error terms. The refinement-stable branch and the induced maps $R_{\ell \to L}$ below concern the homogeneous N_{con}-parameter branch under the separate optimizer-pushforward interface. Controlled violations of exact Markov additivity appear under the explicit mixing hypotheses stated later in Section 2.3. Equilibrium is an approximation regime with explicit error terms.

Rotationally invariant constraint branch. Constraint sets are SO(3)-invariant on S^2.

Gauge as overlap redundancy at fixed cutoff. Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

Central-defect subbranch. Assume the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. Choose unitary overlap implementers with $U_{ji} = U_{ij}^{-1}$. On triple overlaps, when the failure of strict coherence is central, define

$$\Omega_{ijk} := U_{ij} U_{jk} U_{ki} = z_{ijk}, \qquad z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij} U_{jk} = z_{ijk} U_{ik}$. The defect is data of the implementer lift; writing only $\mathrm{Ad}(z_{ijk})$ would erase it because conjugation by a central element is the identity.

Collar double-scaling hypothesis. There is one refinement family with $\delta \to 0$, $\ell_{\mathrm{UV}} \to 0$, and uniform finite-range and strong conditional Gibbs mixing constants such that

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \leq c |\partial C|_{\mathrm{UV}} e^{-\delta/\xi}, \qquad \frac{\delta}{\xi} - \log |\partial C|_{\mathrm{UV}} \longrightarrow +\infty.$$

The second condition controls the complete boundary-prefactored envelope. The weaker ratio $\delta/\ell_{\mathrm{UV}} \to \infty$ does not imply it. Section 2.3 gives the finite-range theorem, explicit constants, a sufficient logarithmic schedule, and the finite-receipt fields. On the exact central-interface branch the CMI is zero without this mixing premise.

Geometric-subnet BW lift. The continuum Lorentz statement is asked only on the support-visible extracted geometric subnet for caps, together with the controlled collar/refinement limit that carries the Markov and recovery remainders explicitly. The finite cap regulators are type-I algebras, but the scaling-limit observer algebra may leave that class, so the theorem is automorphism-level

on the emitted geometric cap pair and the special type-I generator form is $K_C = 2\pi B_C + Z_C$ with Z_C central. The spacetime and Einstein paper proves the support-visible BW cap automorphism only when one cofinal branch carries the finite cap-normal certificate and the complete multiresolution algebra-state comparison package. It also proves the cap-normal H^3 chart: $q(\Omega) = (1, \Omega)$, $n_C = (\cot\alpha, \csc\alpha\, \mathbf{c})$, $n_{gC} = \Lambda_g n_C$, and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$, with cap half-spaces but no event-position consequence. Bare confluence underdetermines topology, dimension, and modular normalization by explicit countermodels. Separately, authenticated provenance generates the finite informational order and canonical source height. The exact rank-three source Gram quotient and an independent real axis define the ambient 1+3 Lorentz target carrier with an S^2 null boundary. After multiplication by a supplied positive `timeScale`, source height supplies only the temporal coordinate of an event placement. It is not an execution rank, intrinsic poset dimension, physical clock, or physical event time. The null-net-standardness packet supplies derived positive translations, while the source-causal continuum certificate keeps physical order, spatial placement, history-family custody, count–volume calibration, independent dimension, manifoldlikeness, topology, uniqueness, and curvature convergence explicit. The closure packets prove finite null tomography and finite entropy/MaxEnt identities; local stress, Ward conservation, generalized-entropy stationarity, smooth curvature, and the composed Einstein reading remain conditional on their named analytic and physical supplies. No source-derived family carrying the complete continuum packet is constructed (Theorems 4.3c–4.3h).

Derived quasi-local propagation and modular-locality control. At scale ℓ_{UV}, the automorphism group generated by the MaxEnt generator of Lemma 2.6, or more generally by any branch generator lying in the same bounded-support algebraic closure, has a finite Lieb–Robinson velocity v_{LR}: for local operators A, B supported on regions separated by distance d,

$$\|[A(t), B]\| \le c\|A\|\|B\| \min(|R_A|, |R_B|) e^{-(d - v_{\mathrm{LR}}|t|)/\xi}$$

for $d > v_{\mathrm{LR}}|t|$, where $\xi = O(\ell_{\mathrm{UV}})$.

This is the branch-internal control statement that turns the quasi-local structure of the local-Gibbs branch into explicit support control for time-evolved operators. It is part of the third-axiom branch stated at regulator scale.

> **Approximate modular covariance on the fixed-cutoff realized branch.** Under the fixed-cutoff realized presentation, Lemma 2.6 (local Gibbs), the derived quasi-local propagation bound above, and the collar double-scaling plus mixing hypotheses of Section 2.3, the modular flow $\sigma_t^{\omega,C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:
>
> $$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\mathrm{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\mathrm{mod}}|t|),$$
>
> where R^{+s} denotes the s-neighborhood thickening, v_{mod} is a "modular propagation velocity" controlled by v_{LR} and local norm bounds, and d is the distance from R to ∂C.
>
> **Heuristic motivation.** On the reduced-modular-locality branch of Lemma 4.1a-b, modular additivity localizes the nonadditive defect of $K_C = -\log\rho_C$ to the collar. The associated propagation receipt then bounds support spreading under $e^{iK_C t}$. In the rate-controlled collar limit of Section 2.3, the thickening vanishes in macroscopic units. This remains heuristic support for the tangent-limit package; a Lieb–Robinson bound for the global Gibbs generator alone does not prove locality of the reduced modular Hamiltonian.
>
> **Continuum-limit heuristic.** Define the induced region flow

$$f_t^C(R) := \lim_{\ell_{\mathrm{UV}}\to 0} R^{+v_{\mathrm{mod}}|t|}.$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$ becomes exact in the continuum limit, with error controlled by

$$\eta(\delta) \lesssim 2\sqrt{c|\partial C|_{\mathrm{UV}}}\, e^{-\delta/(2\xi)}$$

when CMI is measured in nats, with the full rate condition imposed on the prefactor.

The discussion above is heuristic support for the tangent-limit package used in Theorem 4.2.

Refinement-stable information-projection branch. Choose a family of coarse-graining channels $\Phi_{\ell\to L}$ between UV scale ℓ and IR scale L satisfying the separately declared optimizer-pushforward and refinement-tail interface. Then on the realized branch one may write

$$\Phi_{\ell\to L}\big(\omega_\ell(\lambda)\big) = \omega_L\big(R_{\ell\to L}(\lambda)\big),$$

for an induced map $R_{\ell\to L}$ on the common finite-dimensional homogeneous multiplier family. The existence of $R_{\ell\to L}$ as a self-map of that same family is the load-bearing cross-scale premise. Reusing the density labels a does not establish it. Coarse-graining a finite-range Gibbs family generically generates interactions outside the retained density list. The closure defect, the moment-matching I-projection realizing $R_{\ell\to L}$, and its trace-norm residual bound are constructed in Definition 2.6a and Lemma 2.6b. Exact vanishing of the defect along the realized branch remains part of the declared optimizer-pushforward interface. Under that interface, the regulator states lie in a shared multiplier family, and the realized branch is a persistent trajectory or invariant subset under the induced refinement maps. This state-side persistence statement does not upgrade fixed-cutoff edge labels to a transportable refinement-persistent sector colimit.

Fixed-cutoff realized presentation. At a UV scale ℓ_{UV}, local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluing is realized as a boundary action on a lifted finite-dimensional presentation. Section 2.3 proves the resulting fixed-cutoff collar package directly from overlap consistency on this realized presentation, while distinguishing the lifted fixed-point algebra from the sector-preserving collar algebra used in the edge-center (EC) theorem.

External mathematical inputs include SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015). The Einstein route also uses the fixed-volume area-variation identity for small geodesic balls and the local quadratic-polarization argument in the tensor upgrade. The null modular bridge and small-ball kernel are carried internally on the compact surface rather than imported from a separate EFT small-ball law.

Recovered-core theorem boundary. The Lorentz, null-modular, and Einstein recovered core runs through the support-visible automorphism theorem. Fixed-cutoff statements remain regulator statements. Full citations appear in the References.

1.2.5 Notation

- ρ_C: reduced state on cap C.
- $K_C := -\log \rho_C^\omega$: modular Hamiltonian of the reference state.
- B_C: geometric generator of the cap-preserving conformal dilation.
- $S_{\mathrm{gen}}(C)$: generalized entropy on a cap.
- ℓ_{UV}: UV length scale of the refined screen net.
- δ: collar width around a cap boundary.

1.3 Information-Theoretic Tools

1.3.1 Strong subadditivity and Markov states

For any tripartite state ρ_{ABC},

$$I(A:C \mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A:C \mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\mathrm{id}_A \otimes \mathcal{R}_{B\to BC})(\rho_{AB}).$$

1.3.2 Approximate recovery

If $I(A:C \mid B) \leq \varepsilon$ in nats, there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

1.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width δ define

$$B_\delta := \{x \in S^2 : \mathrm{dist}(x, \partial C) \leq \delta\}, \qquad A_\delta := C \setminus B_\delta, \qquad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with A_δ and D_δ interacting only through B_δ. The collar double-scaling hypothesis is the requirement that $I(A_\delta : D_\delta \mid B_\delta) \to 0$ in the refinement limit. At fixed regulator scale, the same finite-dimensional realized presentation supplies the patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

Finite quotient ensemble theorem. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

At least one action value must be finite, so that $Z_r > 0$. For a law on settled configurations only, restrict this construction to N_r, or set $S_r = +\infty$ outside N_r. A separately specified prior ν_r on Q_r instead induces the settled law $\mu_r^{\mathrm{nf}} = (n_r)_\# \nu_r$. These constructions agree only if

$$\sum_{x:\, n_r(x) = q} \nu_r(x) = \mu_r(q) \qquad (q \in N_r)$$

and μ_r has no mass outside N_r. For example, a retraction $n(0) = 0, n(1) = n(2) = 1$ sends the uniform law on three states to weights $(1/3, 2/3)$, whereas uniform counting on its two normal forms gives $(1/2, 1/2)$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to Q_{nf} is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#}\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r, self-adjoint constraints and a feasible set containing a faithful state,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

If the feasible states lie in a proper support projection P, first restrict to their joint support. The same multiplier argument uses $P(\log \sigma_r)P$ and $PF_{r,a}P$, exponentiated and normalized on $P\mathcal{H}$; replacing the reference by $P\sigma_r P$ before taking its logarithm is not valid without a commutation condition. An empty feasible set has no selected state. The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse maps, with $c_{tr} = c_{sr} \circ c_{ts}$ for $t \succeq s \succeq r$. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_{\#}\mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \left\| (c_{k+1,k})_{\#}\mu_{k+1} - \mu_k \right\|_{\mathrm{TV}},$$

then

$$\left\| (c_{nr})_{\#}\mu_n - \mu_r \right\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} \left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r \right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#}\mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \qquad \sum_{x : \pi_r(x) = q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y : \pi(y) = q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q) e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q') R(q', q)}{w(q) R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q,q')a(q,q') = \mu(q')R(q',q)a(q',q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant on the orthogonal complement of the common fixed space $\mathcal{F}_r = \bigcap_C \operatorname{Ran} E_C$:

$$\kappa_r = \inf_{0 \neq f \perp \mathcal{F}_r} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If every $c_C \geq \gamma_* > 0$, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0),$$

where P_0 projects onto $\mathcal{F}_r$. In finite dimension, $\kappa_r > 0$ whenever $\mathcal{F}_r^{\perp} \neq \{0\}$, since zero Dirichlet energy would put f in every fixed range. If all states are fixed, the displayed operator inequality is vacuous. Replacing $f \perp \mathcal{F}_r$ by $f \perp 1$ requires the additional ergodicity condition $\mathcal{F}_r = \operatorname{span}\{1\}$; a conserved nonconstant record otherwise makes that infimum zero. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt. Multiplying all rates by $s > 0$ rescales the generator and gap by s, while leaving the fixed space, stationarity and detailed balance unchanged. Numerical recognition therefore tests identities after removing a common rate scale and restores that scale to reported gaps. An unresolved positive gap is a precision limitation, not evidence that changing clock units changed the equilibrium.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \leq \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \leq 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Physical-vacuum requirement. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q,q') = J_r(q',q) \geq 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q,q')(f(q) - f(q')) + V_r(q)f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance.

Interpretation classes and required receipts. Every ensemble-facing run records its ensemble id, interpretation class, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are

seed, proposal, or repair noise,
conventional reference ensemble,
OPH-native quotient ensemble,
OPH physical vacuum,
OPH primordial field,
observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

Derived regulator realization. Choose a finite UV cellulation at scale ℓ_{UV} and let R be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary

gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The lifted boundary-invariant endomorphism algebra is

$$A_{\mathrm{inv}}(R) = \widetilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the fixed-cutoff realized presentation used throughout the collar analysis. The EC theorem works on the invariant-state realization and uses the sector-preserving induced collar algebra there; it does not apply to the entire fixed-point algebra on the unreduced tensor product. When the consensus paper speaks of quotient repair, the physical repair law is defined on this fixed-point / overlap-invariant data and any representative-level map is only a lift of that quotient update. Descent to the gauge quotient is therefore built into the physical-algebra formulation as quotient-level repair: the local map is $\mathrm{locRep}_\lambda : Q \to Q$, and the global map is the finite quotient normal-form operator

$$\mathrm{Rep}_\lambda = \overline{\mathrm{nf}}_\lambda : Q \to Q.$$

On the declared fixed-cutoff collar branch, the local repair step itself is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery channel. The declared fixed-cutoff branch adds repair completeness, together with control on the declared Petz domain where that branch is used. The consensus paper proves those clauses on a nontrivial rooted-tree packet-net export: finite packet labels live on a rooted tree, hidden labels are quotient-only, weighted parent-copy repair strictly lowers Φ, normal forms are exactly consistent packet assignments, and the classical full-support Petz channel is CPTP and trace-norm contractive on its positive-support-gap domain. Gauge-invariant observables factor through the quotient normal form on that verified branch. The declared touched-overlap acceptance contract makes Φ the finite-patch Lyapunov functional for primitive repair proposals on the scalar branch, but a proposal becomes a physical step only through the transactional acceptance layer: snapshot-current read sets, unchanged-outside-write validation, preserved boundary/sector/holonomy data, and exact well-founded descent. The read set is validation-complete: it contains every support of every measure term that can change under the write. Overlapping primitive proposals are aggregated by connected conflict component; the restriction-compatible union-collar glue supplies the canonical aggregate payload, while primitive members of that component do not commit independently. On the same fixed-cutoff quantum lift, the terminal expectation functional on each declared physical observable algebra is therefore unique even when microscopic representative lifts differ by gauge labels globally or by sector labels inside one quotient-local glued state. The unqualified coding statement at this stage is only a finite constraint-code statement: the overlap-net codewords are the globally consistent states $C = \Phi^{-1}(0)$. A bare graph does not determine code distance: the same graph can carry constant readouts with distance 1 or repetition constraints with distance $|V|$. Topological-code distance/min-cut, Knill–Laflamme resilience, spectral-gap convergence, BFT wall-clock liveness, and hardware search-work reduction therefore require their own certificates and are not imported by the core overlap graph. Exact descent gives termination. Confluence follows from an independently checked conflict-component diamond on the physical quotient, protected-support and protected-conflict completeness, and repair completeness. The theorem therefore gives one schedule-independent terminal quotient normal form from a fixed initial quotient state. A stronger same-boundary conclusion requires a preserved boundary/sector map and at most one consistent quotient extension in that boundary fiber. The layered functional carrier in the consensus paper proves the finite multi-edge, multi-step witness for $H_B \wedge H_{\mathrm{fib}}$; the functional selected-fiber theorem gives the rooted obstruction-check form by constructing the single candidate from the boundary/root data and then checking all non-tree, sector, and holonomy obstructions. On that branch, multiple same-boundary candidate interiors can be present, inconsistent candidates are eliminated, and all surviving candidates share one quotient normal form.

Normal-form hashes are public equality receipts after quotienting, not selectors among physically distinct minimizing endpoints. If accepted schedules terminate at different observer-facing quotient normal forms without a declared holonomy or higher-gauge obstruction, that repair law is outside the OPH consensus theorem. The neutral result of Ref. [16] packages this quantifier split exactly: under observation preservation and completeness, agreement modulo a silent equivalence of normal endpoints reached from all same-boundary sources is equivalent to injectivity of the induced boundary map on the consistent quotient. This cross-source criterion, same-source confluence, weak normalization, and all-schedule liveness are separate obligations. The named layered and functional carriers discharge the cross-source premise only on their stated domains. For separated cofinal refinement systems, the consensus paper adds an inverse-limit bridge: if finite-stage restriction maps commute with the quotient normal-form maps and holonomy maps, and if compatible families are visibly separated on cofinal finite stages, then finite normal forms and holonomy obstructions assemble into unique refinement-limit classes with finite-stage witnesses for nonzero holonomy. It also adds the RG-facing comparison: when a chosen coarse-graining channel shadows the finite-stage normal forms and obstruction maps with declared defects ε^n and ε^h, reconciling first and then coarse-graining gives the same macroscopic law readout as coarse-graining first and then reconciling, up to $\max\{\varepsilon^n, \varepsilon^h\}$; exact naturality is the zero-defect case. Exact normal-form naturality is discharged by a repair-morphism witness, and holonomy naturality by the corresponding cochain map. For distributed implementations, the same paper proves that a worker run presents one finite universe only when it starts from one global carrier and each event projects to a legal monolithic repair path, a physical stutter, or a certified rollback to an earlier committed projection. The required certificate names the global graph, initial state, partition, cut interfaces, observer registry, run/config/code hashes, linearized commits, restart roots, final monolithic normal form, and recomputed readout. Executor order is separate from observer history. Observer-clock naturality additionally requires a history-augmented quotient with semantic event keys independent of worker ID, repair iteration, queue position, timestamp, and retry metadata; a global observer registry descended from local registry groupoids with disjoint patch/cap/future namespaces and explicit lineage arrows; a state-preserving observer-algebra extraction functor from terminal augmented normal forms; a support-visible cap chart map natural with that extraction; and an operational clock instrument whose readout is calibrated up to the declared affine reparameterization and residual bound. The listed objects or their proof-producing finite certificates are prerequisites. In their absence, modular-order/readback naturality is a named premise rather than a consequence of implementation bookkeeping. The same quotient boundary applies to neutral-bulk readout. For each shard s, raw record rows Σ_s must first be quotient by the declared presentation groupoid Γ_s and then sent through the terminal normal-form map n_s, producing

$$X_s = n_s(\Sigma_s/\Gamma_s).$$

Interface maps $\tau_{ts} : U_{st} \to U_{ts}$ between these terminal charts must satisfy the identity, inverse, cocycle, and zero-cycle-holonomy checks before rows from different shards are compared. A global neutral readout is the quotient

$$Q_{\mathrm{vis}} = \left(\bigsqcup_s X_s \right) / \sim_\tau,$$

not the concatenation of shard-local tables. Channel features $F_{s,c}$ descend to Q_{vis} only when their domains and values are transported by the same atlas. For a complete declared channel set $\mathcal{C}_\star$, the neutral distance is

$$d_{\mathrm{neu}}(x, y) = \left(\sum_{c \in \mathcal{C}_\star} w_c \, d_c(F_c(x), F_c(y))^p \right)^{1/p}.$$

Here $p \geq 1$, and $\mathcal{C}_\star$ is finite. An infinite declared channel set is allowed only when the displayed weighted p-sum is finite for every compared pair. Under that finite-distance premise this is stated first as a quotient-visible pseudometric. It becomes a metric only after quotienting zero-distance feature-collision classes or proving that the channels jointly separate points. Pairwise available-channel comparison is explicitly excluded: missingness is handled only by complete cases, a fixed quotient-visible missing symbol, or a train-only imputation map labelled as an imputed-representation metric. The certificate records a finite channel list or a pairwise weighted-ℓ^p summability witness. Presentation changes (gauge representatives, port relabelings, observer order, schedules, and shard partitions) preserve the metric only when they induce a bijection of Q_{vis} together with channel isometries; refinement claims additionally require a cofinally vanishing distance tail modulus. Finite Euclidean claims require the double-centered Gram certificate

$$B = -\frac{1}{2} H(D^{\circ 2}) H \succeq 0$$

and noisy runs must report negative spectral mass, rank/effective-rank data, held-out stress, and planted/shuffled controls. Statistical certificates split independent generative shard batches before preprocessing; chart alignment, scaling, imputation, weights, graph construction, dimension choice, and thresholds are train/validation objects, and a test batch, seed, boundary condition, trajectory family, duplicate, or descendant appearing in more than one split blocks the claim. These theorems establish at most a common finite quotient metric or pseudometric. Identifying it with a physical Riemannian bulk or with Lorentzian/Einstein spacetime (via the source-causal continuum certificate of Theorem 4.3e, under its physical-order, refinement, count–volume, independent-dimension, manifoldlikeness, topology, and curvature receipts) requires the separate modular/geometric branch below. The same consensus surface is classified explicitly as a finite-state exact theorem package plus this controlled inverse-limit bridge: decidable normal-form computation with the Lyapunov step bound, automatic approximate stability only through the collar-local splice and record controls, a conditional fair-block contraction branch for long-run noisy approximate consensus at block indices, with deterministic block endpoints additionally required for the stated all-time expectation bound, and a verified rooted-tree packet-net subdomain. Computational universality for growing patch-net families belongs to the consensus paper's expressive-power boundary and is not a dependency for the recovered-core branches.

Fixed-cutoff packet closure map and invariant simplex. On any declared fixed-cutoff packet quotient Q whose repair relation is terminating and confluent, the normal-form map

$$N : Q \to Q_{\mathrm{nf}}$$

is well-defined and schedule-independent. It induces an affine OPH closure map on the probability simplex over packet states,

$$\mathcal{C}_Q : \Delta(Q) \to \Delta(Q), \qquad \mathcal{C}_Q\left(\sum_{q \in Q} p_q \delta_q\right) = \sum_{q \in Q} p_q \delta_{N(q)}.$$

Since Q is finite, $\Delta(Q)$ is a nonempty compact convex set and $\mathcal{C}_Q$ is a continuous affine self-map. Its image $\Delta(Q_{\mathrm{nf}})$ is invariant, and $\mathcal{C}_Q^2 = \mathcal{C}_Q$. This is an actual closure map on the fixed-cutoff packet-closed quotient branch. It is not the full Appendix/habitat closure-map theorem for arbitrary OPH state-and-law data; the first operation not internalized at that level is the general habitat lift from this finite packet quotient to the full compact-convex observer-supporting sector.

Finite quotient ensemble theorem. The closure map above is an idempotent normal-form projector. It sends any input law to terminal quotient normal forms, and every law supported on

those normal forms is a fixed point. Hence the physical probability law at regulator r must be supplied as an additional quotient-intrinsic object. Let

$$Q_r = \Sigma_r/\Gamma_r$$

be the finite physical quotient, or let $N_r = n_r(Q_r)$ be the quotient-normal-form subset when only settled configurations are intended to carry probability. The declared OPH quotient ensemble is specified by an intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Require $Z_r > 0$, equivalently at least one finite action value. If only normal forms carry probability, restrict the state space to N_r, or put infinite action outside it. A separately supplied prior ν_r induces $\mu_r^{\mathrm{nf}} = (n_r)_{\#}\nu_r$. It equals the displayed Gibbs law only when that law is supported on N_r and the prior's mass in each normal-form fiber equals its target weight. A uniform three-state prior with fibers of sizes one and two yields $(1/3, 2/3)$, not the uniform two-state law. The choice of m_r, S_r, or ν_r is load-bearing: uniform quotient counting, uniform representative counting pushed to the quotient, finite-groupoid weights, and trace weights of a declared quotient algebra are generally different.

Finite MaxEnt form. For a finite Q, positive m, and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

over the affine constraint surface

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has a unique full-support maximizer of the form below when the feasible set contains a full-support point (equivalently, the prescribed moment vector is in the relative interior of the attainable moment set):

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

This is the finite-dimensional Lagrange-multiplier argument with strict concavity of $\mathcal{H}_m$. If no full-support feasible point exists, the unique optimizer lies on a proper face; the same finite-multiplier formula holds only after restricting the state space and observables to the optimizer's support (or globally as an extended exponential family with infinite penalties). On a finite noncommutative quotient algebra, when the feasible set contains a faithful state, the corresponding density operator with faithful reference state σ_r is

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

Refinement and projective compatibility. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse-graining map. Exact compatibility of the weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q; necessarily $\alpha_{sr} = Z_s/Z_r$. The proof is just the equality

$$(c_{sr})_{\#}\mu_s(q) = Z_s^{-1} \sum_{q' \colon c_{sr}(q')=q} m_s(q')e^{-S_s(q')}$$

compared with $Z_r^{-1} m_r(q)e^{-S_r(q)}$. For the selected-prior route, if

$$c_{sr} \circ n_s = n_r \circ c_{sr}, \qquad (c_{sr})_{\#}\nu_s = \nu_r,$$

then functoriality gives $(c_{sr})_{\#}(n_s)_{\#}\nu_s = (n_r)_{\#}\nu_r$. Along a chain, finite-stage compatibility defines a unique projective-limit probability measure on the cylinder σ-algebra. If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#}\mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then total-variation contraction gives

$$\|(c_{nr})_{\#}\mu_n - \mu_r\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

Implementation invariance and representative lifting. If two implementations A, B carry quotient bijections $h_r : Q_r^A \to Q_r^B$ with

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then $(h_r)_{\#}\mu_r^A = \mu_r^B$. If an implementation stores representatives $\pi_r : \Sigma_r \to Q_r$, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling weights quotient states by orbit size and is physical only if representative counting is the declared base measure. A representative Markov kernel must also be quotient-lumpable: the transition probability to an orbit cannot depend on which representative of the source orbit is used.

Sampler correctness. A production sampler is a separate interface from a deterministic settler:

$$\texttt{settle}(q) \to n_r(q), \qquad \texttt{sample}(\mu_r) \to q.$$

For $w(q) = m(q)e^{-S(q)}$ and a proposal kernel $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q),$$

and hence $\mu P = \mu$. Irreducibility and aperiodicity give uniqueness and convergence on the finite state space. If a repair-informed proposal is $q' = R_{\mathrm{repair}}(q) + \eta$ with density g, the Hastings ratio must include

$$\frac{g(q - R_{\mathrm{repair}}(q'))}{g(q' - R_{\mathrm{repair}}(q))};$$

dropping this term generally changes the stationary law. For a lazy reversible finite chain with spectral gap γ,

$$\|P^n(x,\cdot)-\mu\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\mu(x)^{-1}-1}\,(1-\gamma)^n,$$

and the usual autocorrelation bound follows from the nontrivial spectral radius. Exact small regulators should therefore enumerate Q_r, w_r, P_r, detailed-balance error, stationarity error, refinement residuals, lumpability/orbit-size checks, and the exact spectral gap.

Gaussian screen continuation. For a real reduced screen field space V_r after background and dipole modes are removed, let K_r be positive definite and impose fixed expected quadratic release energy,

$$\mathbb{E}[q]=0, \qquad \mathbb{E}[q^{\mathsf{T}}K_r q]=d_r A_r, \qquad d_r=\dim V_r.$$

The unique maximum-entropy density relative to the declared Euclidean volume is

$$p_r(q)=\frac{\sqrt{\det K_r}}{(2\pi A_r)^{d_r/2}}\exp\left[-\frac{1}{2A_r}q^{\mathsf{T}}K_r q\right], \qquad \mathrm{Cov}(q)=A_r K_r^{-1}.$$

This is the relative-entropy proof against the displayed Gaussian. If instead every sample obeys $q^{\mathsf{T}}K_r q=d_r A_r$, the MaxEnt law is uniform on the ellipsoid, not Gaussian. Under a linear coarse field map $C_{sr}: V_s \to V_r$, exact Gaussian refinement is the covariance criterion

$$A_r K_r^{-1} = C_{sr}(A_s K_s^{-1})C_{sr}^{\mathsf{T}}.$$

For harmonic truncation, choose real modes $Y_{\ell m}$, $2 \le \ell \le L$, set

$$K_L Y_{\ell m}=\kappa_\ell Y_{\ell m}, \qquad \kappa_\ell=[\ell(\ell+1)]^{1+\theta/2},$$

and sample independent coefficients

$$a_{\ell m}\sim \mathcal{N}(0, A/\kappa_\ell).$$

Projection from L' to L then discards high modes and is exactly refinement-compatible. For irregular meshes, covariance should be primary under coarse-graining:

$$\Sigma_r = C_{sr}\Sigma_s C_{sr}^{\mathsf{T}}, \qquad K_r = A_r\Sigma_r^{-1}.$$

Equivalently, when eliminating fine variables from a precision matrix

$$K_s=\begin{pmatrix} K_{cc} & K_{cf} \\ K_{fc} & K_{ff}\end{pmatrix},$$

the coarse precision is the Schur complement

$$K_r = K_{cc}-K_{cf}K_{ff}^{-1}K_{fc},$$

not the principal block K_{cc}.

Physical vacuum and primordial premises. A stationary sampler is not a physical vacuum: the identity transition kernel leaves every measure stationary. A finite vacuum theorem needs a physical transfer operator. If T_r is real symmetric, positive definite, and strictly positive, Perron–Frobenius gives a simple largest eigenvalue $\lambda_{0,r}$ with positive eigenvector Ω_r. Then

$$H_r=-a_t^{-1}\log(T_r/\lambda_{0,r})$$

is positive semidefinite, Ω_r is the unique zero-energy state, and the Doob kernel

$$P_r(x,y) = \frac{T_r(x,y)\Omega_r(y)}{\lambda_{0,r}\Omega_r(x)}$$

is stochastic and detailed-balanced with

$$\mu_r(x) = \frac{\Omega_r(x)^2}{\sum_z \Omega_r(z)^2}.$$

Continuum identification further requires reflection positivity or an equivalent reconstruction certificate plus refinement compatibility of the transfer family. A screen covariance alone does not determine an unrestricted primordial curvature spectrum. Even the complete noiseless one-shell sequence for the thin-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

determines the three-dimensional correlation only on $0 \le s \le 2\chi_\star$ and has an infinite-dimensional kernel that persists under positivity. Source-derived uniqueness requires a physical dilation-intertwiner theorem or complete radial cross-covariance tomography. A finite prior gives a conditional continuation. Conventional free-scalar and lattice-gauge baselines may therefore be recorded. They instantiate neither an OPH-native physical vacuum nor a primordial source without these premises.

2 Finite Screen Spectrum Theorem Package

The screen-level spectrum theorem supplies the finite source object used by the cosmology branch. It does not supply temperature-temperature, temperature-polarization, polarization-polarization, lensing, likelihoods, or a physical cosmic microwave background interpretation. Those require the Boltzmann and data-contract theorems.

Definition 2.1 (finite screen regulator)**.** *For regulator r, a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

where $\mathcal{T}_r$ is a finite cellulation of the screen, $M_r \succ 0$ is the area/quadrature mass matrix, Γ_r is the hidden-presentation groupoid, $\mathcal{Q}_r = \Sigma_r/\Gamma_r$ is the physical quotient, and C_{sr}, J_{rs} are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require

$$f^{\mathsf{T}} M_r g \to \int_{S^2} f g \, d\Omega$$

with an explicit band-limited error bound

$$\left| f^{\mathsf{T}} M_r g - \int f g \, d\Omega \right| \le \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

Patch count alone is therefore not an angular-resolution certificate.

Definition 2.2 (geometric screen scalar)**.** *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

and let $\bar{J}_{X,r}$ be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define

$$q_{0,r} = \frac{1}{3}\log\frac{J_{X,r}}{\bar{J}_{X,r}}.$$

On a certified spherical chart, with $B_r = [1, n_x, n_y, n_z]$,

$$\Pi_{<2,r} = B_r(B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r, \qquad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

On an irregular screen, B_r is replaced by the certified generalized eigenprojector onto the finite $\ell = 0, 1$ subspace. Hard-coded feature z-scores or observer-summary weights are diagnostic proxies, not q_r.

Proposition 2.3 (quotient invariance)**.** *If a recharting U preserves the screen inner product and geometric readout,*

$$U^{\mathsf{T}} M_r' U = M_r, \qquad q_{0,r}' = U q_{0,r}, \qquad B_r' = U B_r,$$

then $\Pi_{\geq 2,r}' U = U \Pi_{\geq 2,r}$, and $q_r' = U q_r$.

Proof. Substitution gives

$$\begin{aligned}\Pi_{<2,r}' U &= U B_r (B_r^{\mathsf{T}} U^{\mathsf{T}} M_r' U B_r)^{-1} B_r^{\mathsf{T}} U^{\mathsf{T}} M_r' U \\ &= U B_r (B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r = U \Pi_{<2,r}.\end{aligned}$$

Subtracting from U gives the high-mode identity. □

Definition 2.4 (physical scalar precision and action)**.** *The scalar precision K_r must have a physical origin on $V_r = \operatorname{im} \Pi_{\geq 2,r}$. Allowed branches are:*

$$K_r^{\mathrm{Hess}} = \nabla^2 \Phi_r \Big|_{q=0},$$

for a quotient-visible mismatch or release free energy; a repair Dirichlet form

$$K_r^{\mathrm{diss}} = H_r - T_r^{\mathsf{T}} H_r T_r, \qquad T_r^{\mathsf{T}} H_r T_r \preceq H_r;$$

or the reversible generator form

$$K_r^{\mathrm{Dir}} = -\frac{1}{2}(L_r + L_r^{\dagger_H}).$$

A raw nonsymmetric repair matrix or caller-supplied κ is not a precision operator. The absolute normalization must be fixed independently, for example by

$$K_0 Y_{\ell m} = \ell(\ell+1) Y_{\ell m}$$

on the $\theta = 0$ branch. Once normalized,

$$S_{\mathrm{scr},r}[q] = \frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r = \frac{1}{2A_{q,r}} q^{\mathsf{T}} M_r K_r q.$$

Theorem 2.5 (finite MaxEnt screen covariance)**.** *Let $K_r e_i = \kappa_i e_i$ with an M_r-orthonormal basis of V_r, $\kappa_i > 0$, and $d_r = \dim V_r$. Among continuous densities in coefficients $q = \sum_i a_i e_i$ with $\mathbb{E}[a_i] = 0$ and*

$$\frac{1}{2}\mathbb{E}\langle q, K_r q \rangle_r = E_{q,r}^{\mathrm{src}},$$

the unique entropy maximizer is

$$d\mu_r(q) = Z_r^{-1} \exp\left[-\frac{1}{2A_{q,r}}\langle q, K_r q\rangle_r\right] dq, \qquad A_{q,r} = \frac{2E^{\mathrm{src}}_{q,r}}{d_r},$$

and

$$\mathbb{E}[a_i a_j] = \delta_{ij}\frac{A_{q,r}}{\kappa_i}.$$

Proof. The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature β_r. The expected energy is

$$E^{\mathrm{src}}_{q,r} = \frac{1}{2}\sum_i \kappa_i \frac{1}{\beta_r \kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus $A_{q,r} = \beta_r^{-1} = 2E^{\mathrm{src}}_{q,r}/d_r$. Strict concavity of entropy gives uniqueness. □

Remark 2.6 (discrete quotient branch)**.** *For a discrete finite quotient with intrinsic base weight $m_r(q)$, the exact Gibbs law is*

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.

Definition 2.7 (source release energy)**.** *Let m_r^0 be the tracially pointed base weight on the finite physical collar quotient. Let $F_{r,a}$ be a finite ledger of primitive collar observables: scalar occupancy, protected-center presence, released volume, conserved source charges, and the release clock. The ledger excludes screen energy, sky spectra, CMB residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. For source-emitted constraint values $c^{\mathrm{col}}_{r,a}$, the selected release law is*

$$\nu_r^{\mathrm{rel}}(x) = \frac{m_r^0(x)\exp[-\sum_a \lambda_{r,a}F_{r,a}(x)]}{Z_r}, \qquad \mathbb{E}_{\nu_r^{\mathrm{rel}}} F_{r,a} = c^{\mathrm{col}}_{r,a}.$$

Strict concavity of relative entropy gives uniqueness on full support. A boundary optimum uses the same form on its certified support. The constraint ledger, base weight, source DAG, multipliers, support, and refinement maps belong to the receipt.

For this independently selected quotient ensemble on released collar normal forms, define

$$E^{\mathrm{src}}_{q,r} = \frac{1}{2}\int \langle q_r(x), K_r q_r(x)\rangle_r \, d\nu_r^{\mathrm{rel}}(x).$$

The receipt must expose ν_r^{rel}, the base measure, K_r, d_r, $E^{\mathrm{src}}_{q,r}$, $A_{q,r}$, no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.

Proposition 2.8 (microscopic-law necessity)**.** *For fixed positive K_r, the Gaussian family $Z(A)^{-1}\exp[-\langle q, K_r q\rangle/(2A)]$ exists for every $A > 0$. The quadratic MaxEnt form therefore leaves one positive scale free. The source release law in Definition 2.7, or an equivalent source-only vacuum law, supplies the energy that fixes $A_{q,r} = 2E^{\mathrm{src}}_{q,r}/d_r$.*

Theorem 2.9 (rotational screen spectrum)**.** *If the continuum precision K_θ commutes with the $SO(3)$ action on scalar functions, then by Schur's lemma each $\mathcal{H}_\ell$ is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \qquad \ell \geq 2.$$

Then $\Lambda_\ell(0) = \ell(\ell+1)$, and the MaxEnt covariance gives

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

Consequently $\mathcal{D}_\ell^q = \ell(\ell+1)C_\ell^q/(2\pi) \sim (A_q/2\pi)\ell^{-\theta}$.

Definition 2.10 (source-derived conformal precision)**.** *Let $L_r \succeq 0$ be the normalized detailed-balanced scalar collar Dirichlet operator on V_r, with continuum limit $-\Delta_{S^2}$. Set*

$$B_r = (L_r + \tfrac{1}{4}I)^{1/2}, \qquad K_{\theta,r} = \frac{\Gamma(B_r + \frac{3}{2} + \frac{\theta}{2})}{\Gamma(B_r - \frac{1}{2} - \frac{\theta}{2})}.$$

For $-2 < \theta < 4$ on the retained $\ell \geq 2$ branch, this operator is positive. The gamma recurrence gives $K_{0,r} = L_r$ exactly. The finite receipt carries detailed balance, positivity, the $\ell(\ell+1)$ normalization, anisotropy splitting, and refinement residuals.

Theorem 2.11 (finite refinement error)**.** *Suppose that for $2 \leq \ell \leq L$*

$$\left|\frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1\right| \leq \varepsilon_{K,r}(L), \qquad \left|\frac{A_{q,r}}{A_q} - 1\right| \leq \varepsilon_{A,r}, \qquad \varepsilon_{K,r} < 1.$$

Then $C^q_{\ell m,r} = A_{q,r}/\lambda_{\ell m,r}$ satisfies

$$\left|\frac{C^q_{\ell m,r}}{A_q/\Lambda_\ell(\theta)} - 1\right| \leq \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

Proof. Write $A_{q,r} = A_q(1 + a_r)$ and $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$. Then

$$\frac{C^q_{\ell m,r}}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from $|a_r| \leq \varepsilon_A$, $|b_{\ell m,r}| \leq \varepsilon_K$. □

Theorem 2.12 (refinement-semigroup tilt)**.** *Let R_b be the scalar refinement map for scale ratio $b > 1$. If one isolated scalar covariance mode has positive eigenvalue $\lambda(b)$, $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$, and λ is continuous, then there is a unique real θ with*

$$\lambda(b) = b^{-\theta}, \qquad \theta = -\frac{\log \lambda(b)}{\log b}.$$

Proof. Set $g(t) = -\log \lambda(e^t)$. The semigroup law gives $g(t+s) = g(t) + g(s)$. Continuity makes $g(t) = \theta t$, hence the result. □

Definition 2.13 (edge-center reserve generator receipt)**.** *Write $s = \log b$ for logarithmic refinement thickness. Let $u_{\mathrm{full}}(s)$ be the scalar-conditioned covariance-survival cocycle across a full oriented collar, and let $u_q(s)$ be its source-facing half-collar restriction. The receipt requires*

$$u(s+t) = u(s)u(t), \qquad u(0) = 1,$$

strong continuity at zero, the source-derived full-collar density

$$-u'_{\mathrm{full}}(0) = \frac{P_\star}{24},$$

and the orientation-reversal coarea identity

$$-u_q'(0) = \frac{1}{2}[-u_{\mathrm{full}}'(0)] = \frac{P_\star}{48}.$$

These are infinitesimal generator statements. A one-step survival probability has exponent $-\log u_q(\log b)/\log b$.

Theorem 2.14 (edge-center tilt and repair-clock reconciliation)**.** *Under Definition 2.13,*

$$u_q(s) = e^{-\theta s}, \qquad \theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}.$$

In the coordinate $\theta = \kappa_{\mathrm{rep}}(P_\star - \varphi)$*, the same branch has*

$$\kappa_{\mathrm{rep}}^{\mathrm{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

The value e *is a separate diagnostic hypothesis unless an additional identity equates it with this source-derived coordinate.*

Proof. For $g(s) = -\log u_q(s)$, the cocycle law gives the continuous Cauchy equation. Hence $g(s) = \theta s$. Differentiation at zero and the half-collar identity give $\theta = P_\star/48$. The formula for $\kappa_{\mathrm{rep}}^{\mathrm{edge}}$ is algebraic. □

Theorem 2.15 (homogeneous radial source family)**.** *Assume source-stress closure, a single clock, freezeout, multicenter consistency, and translation/rotation covariance. Let* $C_\zeta = M_{\Delta_\zeta^2}$ *on the common physical* $d\ln k$ *mode basis. If a scale-natural source embedding transports refinement to physical dilation and its finite operator residual converges to*

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta, \qquad (D_s f)(k, \hat{k}) = f(e^{-s}k, \hat{k}),$$

then every positive measurable representative satisfies

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}$$

almost everywhere. Continuity gives the pointwise identity. A single shell has an infinite-dimensional radial kernel, including positive ambiguities; complete radial cross-covariances give the independent spherical-Hankel tomography route.

Proof. Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s}\Delta_\zeta^2(k)$ almost everywhere. In logarithmic coordinates, rational-translation invariance on a common full-measure set forces $e^{\theta t}\Delta_\zeta^2(e^t)$ to be constant almost everywhere. The one-shell and tomography statements follow from the correlation-restriction and spherical-Hankel theorems in the primordial bridge packet. □

Theorem 2.16 (thin-shell power-law lift)**.** *Assume* $q(\hat{n}) = Z_\star \Pi_{\ell \geq 2} \zeta_\star(R_\star \hat{n})$ *and*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}.$$

Then

$$C_\ell^q = 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) j_\ell^2(kR_\star),$$

and, for $-2 < \theta < 4$*,*

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta (k_\star R_\star)^\theta \frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)}.$$

Thus

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.$$

Proposition 2.17 (finite-window bound)**.** *Let* $\Psi_\ell(k) = \int dr\, W(r) j_\ell(kr)$, $W \geq 0$, $\int W dr = 1$, *with mean* $R_\star$ *and variance* σ_R^2. *If*

$$\delta_\ell(k) = \frac{k^2 \sigma_R^2}{2} \sup_{r \in \operatorname{supp} W} |j_\ell''(kr)|,$$

then

$$|C_{\ell,W}^q - C_{\ell,\mathrm{shell}}^q| \leq 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) \left[2\delta_\ell(k) + \delta_\ell(k)^2\right].$$

Proposition 2.18 (radial non-identifiability)**.** *For a finite radial basis,* $C = Ap$. *If* $A \in \mathbb{R}^{N_\ell \times N_k}$ *has rank* r, *then* $\dim \ker A = N_k - r$, *and every* $p + v$ *with* $v \in \ker A$ *gives the same screen spectrum. A physical radial reconstruction therefore requires either a source theorem restricting* p, *or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

Theorem 2.19 (promotion contract for the thin-shell power-law pipeline)**.** *Under the declared thin-shell lift and power-law source family above, one hash-locked source DAG may emit*

$$(q_r, S_{\mathrm{scr},r}, \theta, A_{q,r}, C_{\ell,r}^q, A_{\zeta,r}, \Delta_{\zeta,r}^2)$$

as a conditional primordial packet when the geometric-scalar, collar-law, release-energy, reserve-generator, conformal-precision, refinement, source-stress, single-clock, freezeout, adiabaticity, isocurvature, phase-coherence, thin-shell, radial-null, finite-window, and forward- residual receipts pass on that DAG. Any measurement, likelihood, fitted parameter, or measurement-calibrated ancestor fails the source packet. Physical temperature and polarization spectra require the independent Boltzmann, recombination, nuisance, covariance, and likelihood gates. This contract is specific to the displayed pipeline. A different source-only radial reconstruction or forward projection may use another fully typed contract, provided it certifies its own injectivity or null space, window and convergence control, forward residual, and no-target-leak provenance.

Target 2.20 (requirements for a source-derived screen spectrum)**.** *A concrete source-derived spectrum requires the following receipt families before primordial interpretation:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*
2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *the infinitesimal reserve generator, scalar refinement tilt, and angular screen spectrum with a finite error budget;*
4. *source-stress, clock, freezeout, radial-window, radial-null, forward-residual, and transfer firewall receipts.*

The theorem and algorithm packet is complete for the declared thin-shell power-law pipeline. No finite source DAG that passes this receipt set is supplied, and the list is not a no-go for a differently typed radial pipeline.

2.0.1 Source-only primordial bridge theorem

The finite screen covariance theorem is a screen theorem. To read it as a source-only primordial curvature spectrum one must add the following bridge objects and receipts. The dependency chain is

$$\boxed{\begin{aligned}&\mathrm{nf}(\mathcal{Q}_r) \to (\chi_r, u_r, h_r) \to \mathsf{BackgroundCurvatureStatus}_r \to (g, T_I) \to q_\zeta = \zeta_\rho \\ &\qquad \to \text{single clock} \to \text{freeze-out} \to \text{coherent growing mode} \to \Delta_\zeta^2(k).\end{aligned}}$$

No successful fit of a screen C_ℓ or a TT spectrum may replace one of these hypotheses.

Definition (background curvature branch label). The primordial bridge carries a finite record

$$\mathsf{BackgroundCurvatureStatus}_r = (\mathsf{BranchType}, \kappa, I_K, I_{\Omega_K}, \mathsf{Basis}, \mathsf{Top})$$

where

$$\mathsf{BranchType} \in \{\textsc{FlatExact}, \textsc{FlatAssumed}, \textsc{OpenCurved}, \textsc{ClosedCurved}, \textsc{Unresolved}\}.$$

FlatExact is allowed only after a direct theorem or conditional CMH receipt. FlatAssumed may run flat kernels but its outputs remain assumption-conditioned. Unresolved blocks physical interpretation. The H^3 observer-facing atlas is not this record; this record comes from the clocked FLRW slice and its spatial Levi–Civita holonomy or declared branch input.

Definition (source stress readout). On a reconstructed branch with relational metric g_{ab}, let

$$S_{\mathrm{src}} = \sum_I S_I[g, \Psi_I], \qquad T^I_{ab} = -\frac{2}{\sqrt{-g}} \frac{\delta S_I}{\delta g^{ab}}, \qquad T^{\mathrm{tot}}_{ab} = \sum_I T^I_{ab},$$

where I ranges over every active sector, including ordinary matter, radiation, neutrinos, OPH anomaly or repair sectors, and any auxiliary source kept in the run. At finite regulator this is a quotient-visible map

$$\mathsf{StressRead}_r : \mathrm{nf}(\mathcal{Q}_r) \longrightarrow \{T^I_{r,ab}\}_I.$$

All source stress components must be read from the same global normal form as the collar scalar.

Source-stress closure theorem. Assume each S_I is relationally diffeomorphism invariant, the source equations hold, sector exchanges satisfy

$$\nabla_a T^{ab}_I = Q^b_I, \qquad \sum_I Q^b_I = 0,$$

and $\mathsf{StressRead}_r$ descends to the physical quotient and converges. Then

$$\nabla_a T^{ab}_{\mathrm{tot}} = 0.$$

On the recovered Einstein branch,

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, T^{\mathrm{tot}}_{ab}.$$

The evidence bundle must also report the non-circular comparison

$$E^{ab}_T = T^{ab}_{\mathrm{src}} - \frac{G^{ab} + \Lambda g^{ab}}{8\pi G};$$

using the geometric right-hand side alone is a geometry diagnostic, not a source-only primordial receipt.

Total-energy frame and curvature covector. When $T^a{}_b$ has a unique future timelike unit eigenvector u^a, define

$$T^a{}_b u^b = -\rho u^a, \qquad h_{ab} = g_{ab} + u_a u_b,$$

and decompose

$$T^{ab} = \rho u^a u^b + p h^{ab} + \pi^{ab}, \qquad u_a \pi^{ab} = 0, \qquad \pi^a{}_a = 0.$$

Let $\Theta = \nabla_a u^a$ and let σ_{ab} be the shear. Total energy conservation gives

$$\dot\rho + \Theta(\rho + p) + \sigma_{ab}\pi^{ab} = 0, \qquad \dot f := u^a \nabla_a f.$$

On an expanding branch, define

$$p_{\mathrm{eff}} = p + \frac{\sigma_{ab}\pi^{ab}}{\Theta};$$

then $\dot{\rho} + \Theta(\rho + p_{\mathrm{eff}}) = 0$.

Theorem (exact total-curvature evolution). Work on an open flow domain on which $\Theta \neq 0$, $\dot{\rho} \neq 0$, and $\rho + p_{\mathrm{eff}} \neq 0$, so every quotient below is defined. Let

$$\dot{\alpha} = \Theta/3, \qquad \zeta_a = \nabla_a \alpha - \frac{\dot{\alpha}}{\dot{\rho}} \nabla_a \rho,$$

and

$$\Gamma_a^{\mathrm{eff}} = \nabla_a p_{\mathrm{eff}} - \frac{\dot{p}_{\mathrm{eff}}}{\dot{\rho}} \nabla_a \rho.$$

Then, exactly and without a long-wavelength approximation,

$$\mathcal{L}_u \zeta_a = -\frac{\Theta}{3(\rho + p_{\mathrm{eff}})} \Gamma_a^{\mathrm{eff}}.$$

Hence a barotropic effective source, $p_{\mathrm{eff}} = p_{\mathrm{eff}}(\rho)$, conserves ζ_a.

Collar scalar corollary. Assume that the quotient record emits a connected uniform-total-density cut and a collar-adapted spatial metric

$$h = \lambda^2 dr^2 + \sigma_{AB}(dx^A + \beta^A dr)(dx^B + \beta^B dr), \qquad \lambda > 0, \qquad \sigma_{AB} \succ 0.$$

The block-determinant identity gives

$$\sqrt{\det h} = \lambda \sqrt{\det \sigma_{AB}}.$$

The native readout premise is that J_X and its frozen reference are these spatial volume densities, with the same reference-density convention. The formula $J_X = \lambda\sqrt{\det \sigma_{AB}}$ without this metric attachment does not establish that premise. For relational rods Lie-dragged by the total-energy flow, the attached volume Jacobian satisfies

$$\frac{d}{d\tau} \frac{1}{3} \log J_X = \frac{\Theta}{3}.$$

Relative to a frozen background,

$$q_\zeta = \frac{1}{3} \log \frac{J_X}{\bar{J}_X}.$$

On the uniform-total-density cut,

$$q_\zeta = \zeta_\rho + \text{constant}.$$

After removing the monopole and dipole,

$$q = \Pi_{\ell \geq 2} q_\zeta = \Pi_{\ell \geq 2} \zeta_\rho, \qquad Z_q = 1.$$

This is the source-only quotient scalar on the OPH screen; by itself it has type ScreenCurvature, not PrimordialCurvature.

Definition (rank-one single-clock branch). Let

$$\mathcal{F}(x) = (\rho, p_{\mathrm{eff}}, \{\rho_I, p_I, Q_I\}_{I=1}^N)$$

be the scalar source map on the source moduli space. A branch is rank-one single-clock when $\operatorname{rank} d\mathcal{F} = 1$, $d\rho \neq 0$, every level set of ρ is connected, all sector velocities agree with u^a, and orthogonal momentum transfer vanishes. An approximate single-clock branch may instead carry an explicit interval bound on that transfer, but the bound must then be propagated into the transfer and curvature-evolution residuals. At finite cutoff this requires a quotient-visible clock χ_r and a factorization

$$\mathcal{F}_r = \widehat{\mathcal{F}}_r \circ \chi_r.$$

Theorem (single-clock adiabaticity). On an exact rank-one single-clock branch, and for the dynamically nondegenerate sectors I, J under discussion with $\dot{\rho}_I \neq 0$ and $\dot{\rho}_J \neq 0$,

$$p_{\text{eff}} = p_{\text{eff}}(\rho), \qquad S_a^{IJ} = 3(\zeta_a^I - \zeta_a^J) = 0,$$

and the intrinsic nonadiabatic pressure and transfer covectors obey

$$\Gamma_a^I = 0, \qquad \Xi_a^I = 0.$$

On the nonvanishing domain of the total-curvature theorem, therefore $\mathcal{L}_u \zeta_a = 0$. The proof is the chain rule through the single clock: every source scalar is a function of the same χ, and connected ρ-fibers remove residual branch labels.

Theorem (entropy repair gap). Let the linearized scalar repair/evolution block be

$$\delta Y_{n+1} = L_r \delta Y_n + \eta_n,$$

and let $P_\perp$ project away gauge directions, constrained zero modes, and the adiabatic growing direction. If there is a positive metric H_r such that

$$\|P_\perp L_r P_\perp\|_{H_r} \leq e^{-\gamma_r} < 1,$$

and the projected sector is not sourced by the complementary block,

$$P_\perp L_r = P_\perp L_r P_\perp,$$

then

$$\|P_\perp \delta Y_n\|_{H_r} \leq e^{-n\gamma_r} \|P_\perp \delta Y_0\|_{H_r} + \sum_{j=0}^{n-1} e^{-(n-1-j)\gamma_r} \|P_\perp \eta_j\|_{H_r}.$$

In the unforced case the orthogonal entropy, decaying, and repair scalar modes decay exponentially and converge to zero; they need not vanish after a finite number of steps. If the block-invariance identity is absent, the term $P_\perp L_r (1 - P_\perp) \delta Y_n$ must instead be included in the forcing bound. Eigenvalues alone do not certify this theorem for nonnormal repair matrices; a norm bound or a discrete Lyapunov inequality

$$L_\perp^\mathsf{T} H L_\perp \preceq e^{-2\gamma} H$$

is required.

Theorem (exact and approximate growing-mode coherence). Let $X(k)$ collect the scalar initial-condition variables for curvature, radiation, baryons, CDM/anomaly, neutrinos, and repair-sector scalars, and let $v_+(k)$ be the normalized adiabatic growing vector. Assume the single-clock and repair-gap theorems hold through readout and the growing/orthogonal splitting is invariant. If the initial orthogonal component is zero and no forcing or independent stochastic source has an orthogonal component, then

$$X(k) = v_+(k)\zeta(k), \qquad \mathcal{P}_X(k) = \mathcal{P}_\zeta(k) v_+(k) v_+(k)^\dagger.$$

Thus rank $\mathcal{P}_X(k) = 1$, source-side isocurvature fractions vanish, and all species share a deterministic growing-mode phase. Without those exact zero-component hypotheses, the repair-gap estimate gives only

$$X_n(k) = v_+(k)\zeta_n(k) + \varepsilon_{\perp,n}(k),$$

with $\varepsilon_{\perp,n}$ bounded by the preceding geometric-series inequality; strict contraction alone does not make it zero at finite readout. Finite runs must report λ_2/λ_1, $\beta_{\mathrm{iso}}(k)$, decaying-mode fraction, and phase-convention defects as source diagnostics.

Theorem (screen-to-primordial bridge). Assume the geometric record and causal-area metric receipts, the collar identity $q_\zeta = \zeta_\rho$, source-stress closure, rank-one normal form, positive repair gap, freeze-out, adiabatic growing-mode, isocurvature, and phase-coherence receipts all pass. Also assume the physical scale-bridge theorem: a source embedding with `source_embedding_hash`, a physical mode basis with `physical_mode_basis_id`, a calibrated comoving coordinate χ, mode-normalization and density-of-states receipts, and a no-post-hoc-calibration receipt, and a $\mathsf{BackgroundCurvatureStatus}_r$ that is not UNRESOLVED. Then the low-mode-removed OPH screen scalar is the source-only primordial curvature field restricted to the declared shell or radial window:

$$q(\hat{n}) = \Pi_{\ell_{\mathrm{src}}\geq 2} \int d\chi\, W(\chi)\zeta_\star(\chi\hat{n}).$$

For a thin shell, $q(\hat{n}) = \Pi_{\ell_{\mathrm{src}}\geq 2}\zeta_\star(R_\star\hat{n})$. The forward spectrum is exactly

$$C_\ell^q = 4\pi \int_0^\infty d\ln k\, \Delta_\zeta^2(k) |\Psi_\ell^{\mathsf{BranchType}}(k)|^2,$$

$$\Psi_\ell^{\mathsf{BranchType}}(k) = \begin{cases} \int dr\, W(r) j_\ell(kr), & \mathsf{BranchType} = \text{FLATEXACT or FLATASSUMED}, \\ \int dr\, W(r)\Phi_{\ell k}^{(-)}(r), & \mathsf{BranchType} = \text{OPENCURVED}, \\ \int dr\, W(r)\Phi_{\ell k}^{(+)}(r), & \mathsf{BranchType} = \text{CLOSEDCURVED}, \end{cases}$$

and for a thin shell

$$C_\ell^q = 4\pi \int_0^\infty d\ln k\, \Delta_\zeta^2(k) \left|\Psi_\ell^{\mathsf{BranchType}}(k; R_\star)\right|^2.$$

At finite regulator the continuum integral is represented by the finite spectral measure

$$d\nu_r(k) = \sum_j w_{rj}\delta(k - k_{rj}),$$

so the actual finite evidence receipt is

$$C_{\ell_{\mathrm{src}},r}^q = 4\pi \sum_j w_{rj}^{\log k}\, \Delta_{\zeta,r}^2(k_{rj}) |\Psi_{\ell_{\mathrm{src}},r}^{\mathsf{BranchType}}(k_{rj})|^2.$$

Every radial node, window, and mode bin is bound to the geometry hash, source-embedding hash, scale-certificate hash, boundary-condition hash, and mode-basis identifier, and the evidence bundle must emit radial kernel geometry compatibility receipt. The observed multipole L_{CMB} is not ℓ_{src}; it is the output index of the line-of-sight transfer solver. Cap eigenmodes, inverse cap-opening angles, and relabeled unit strings remain diagnostic by the dimensional, angular, cap-mode, and unit-string no-go propositions.

Physical scale dependency. The k_{rj} grid in the preceding equation is not a free producer field. It is consumed only after

physical spatial k receipt

recomputes the comoving k-intervals from the physical geometry, scale certificate, scale factor, finite eigenvalues, normalization, and refinement residuals. The radial kernel and source-screen projection additionally require

screen to physical k association receipt.

The source covariance is a finite positive operator on the common physical mode projectors:

$$C_{\zeta,r} \succeq 0.$$

Homogeneity and isotropy on the safe band require

$$P_I C_{\zeta,r} P_I = \mathcal{P}_\zeta(k_I) P_I + E_I$$

with a declared residual bound, together with an off-diagonal leakage report

$$\sum_{I \neq J} |P_I C_{\zeta,r} P_J|^2.$$

The same physical projector family is used by primordial covariance and anomaly response:

common primordial anomaly mode basis receipt.

Mode freezeout and common initial surface are separate gates:

physical mode freezeout map receipt, *physical freezeout surface receipt.*

The latter contains initial data and normal derivatives on one common spacelike surface.

Radial prior, null space, and residual receipt. The map above is a forward projection, not a free inversion. With finite bins p_j for Δ^2_ζ,

$$A_{\ell j} = 4\pi \int_{\text{bin } j} d\ln k\, b_j(k) |\Psi_\ell(k)|^2, \qquad C^q = Ap.$$

The evidence bundle must declare the basis, support, prior p_0, Q, positivity conditions, singular spectrum of $AQ^{-1/2}$, effective rank, null basis, resolution kernels, prior sensitivity, and every forward residual

$$r_\ell = C^q_\ell - \widehat{C}_\ell.$$

A finite prior, regularizer, positivity constraint, or square discretization selects one representative from a nonunique shell equivalence class. Its typed output is ConditionalRadialContinuation. A source-derived primordial packet using either radial route certified below requires the common source, scale, mode, null-space, positivity, ancestry, and non-fitting forward-residual receipts together with that route's exact uniqueness certificate. Another source-restricted injective construction may be used if it supplies its own orbit-separation theorem and the same physical and no-target-leak interfaces. If the geometry packet is imported, the result is *conditional physical*; a *source-native physical* result requires the native $\mathsf{CosmoGeomRead}_r$ theorem.

Complete radial-lift theorem packet. The screen-to-radial map is a forward restriction with a nonunique unrestricted inverse. For a flat source window W and geometric normalization Z_q,

$$C^q_{\ell,W} = 4\pi Z_q^2 \int_0^\infty d\ln k\, \Delta^2_\zeta(k) |\Psi_{\ell,W}(k)|^2, \qquad \Psi_{\ell,W}(k) = \int W(dr) j_\ell(kr).$$

The packet below certifies sufficient uniqueness routes through radial dilation, radial tomography and common-bulk consistency on unbounded radii. They are not asserted to be exhaustive or mutually exclusive. A source-derived parametric or dynamical law, an injective source-restricted basis, analytic or PDE constraints, or another fully typed orbit-separating construction remains admissible. A finite prior continuation has the type ConditionalRadialContinuation.

Theorem (one-shell correlation restriction and exact no-go). For a thin shell of radius R, the complete angular covariance satisfies

$$K_R(\mu) = Z_q^2 \xi_\zeta\Big(R\sqrt{2-2\mu}\Big) = \sum_{\ell\geq 0} \frac{2\ell+1}{4\pi} C_{\ell,R}^q P_\ell(\mu).$$

Hence even the noiseless sequence $\{C_{\ell,R}^q\}_{\ell\geq 0}$ determines the three-dimensional correlation only on $0 \leq s \leq 2R$. The full one-shell operator has an infinite-dimensional kernel, including positivity-preserving ambiguities. Explicitly, for any nonzero $g \in C_c^\infty((2R,\infty))$,

$$h_g(k) = \frac{2k^3}{\pi} \int_0^\infty r^2 g(r) j_0(kr)\, dr$$

has correlation perturbation $\xi_{h_g} = g$ and therefore changes no shell multipole. Choosing $B = |h_g| + w$ with any strictly positive integrable w makes B and $B + h_g$ two distinct positive spectra with the same complete shell covariance.

Proof. The spherical-Bessel addition theorem gives the displayed restriction identity. The map $\mu \mapsto R\sqrt{2-2\mu}$ covers exactly $[0, 2R]$. Radial Fourier inversion gives $\xi_{h_g} = g$ and is injective, so the kernel contains an infinite-dimensional copy of $C_c^\infty((2R,\infty))$. The positive pair differs by the same null element. □

Corollary (exact low-mode-removed kernel). For a continuous signed correlation perturbation ξ_h, invisibility to every shell multipole $\ell \geq 0$ is equivalent to $\xi_h(s) = 0$ on $0 \leq s \leq 2R$. Invisibility to every retained OPH multipole $\ell \geq 2$ is equivalent to

$$\xi_h(s) = a + b\left(1 - \frac{s^2}{2R^2}\right), \qquad 0 \leq s \leq 2R,$$

for constants a, b. Legendre completeness gives the first statement. Removing $\ell = 0, 1$ leaves the span of P_0 and P_1 invisible, which gives the second.

Theorem (one common bulk on unbounded round radii). Assume one homogeneous isotropic generalized field on a common Euclidean three-dimensional time slice, with well-defined spherical restrictions. Its covariance is represented off the diagonal by a locally integrable radial function $\xi(s)$, with no additional contact distributions. Suppose an unbounded sequence R_j of round radii has

$$C_\ell(R_j) = \frac{2\pi A_j}{\ell(\ell+1)}, \qquad \ell \geq 1, \quad A_j > 0,$$

in one common readout normalization. Then all $A_j = A$, and

$$\xi(s) = -A \log s + c \quad \text{for almost every } s > 0, \qquad P_3(k) = \frac{2\pi^2 A}{k^3} \quad (k > 0).$$

The ideal field is understood modulo constants, on smooth mean-zero test observables; it does not have finite point variance. The covariance identity is distributional, with the displayed logarithm as a chosen off-diagonal representative. Gaussianity is not needed for the covariance conclusion. If the hypotheses concern only $\ell \geq 2$, the residual freedom is $c + ds^2$, representing unobserved zero-wave-number or affine data.

Proof. The identity $\int_0^\infty j_\ell(x)^2\,dx/x = 1/[2\ell(\ell+1)]$, or Legendre expansion of the logarithm, identifies each mean-zero spherical kernel. Chord restriction and completeness give $\xi(s) = -A_j \log s + c_j$ almost everywhere on $0 < s < 2R_j$. Every two intervals overlap. Linear independence of $\log s$ and 1 forces their coefficients to agree. Unbounded radii cover all positive separations; radial Fourier uniqueness gives the stated spectrum away from zero. With the dipole removed, use independence of $\log s, 1, s^2$ instead. Contact terms are excluded by hypothesis, so off-diagonal equality does not hide an extra diagonal distribution. □

Separate screen marginals do not establish the common-bulk premise. The metrics, time slice, normalization and cross-screen records must be jointly compatible. For finitely many radii, unobserved separations allow the preceding null-space perturbations; a finite run cannot invoke unbounded coverage.

Example (positive alternatives to the scale-invariant baseline). For a unit-radius screen set

$$h(s) = (s-3)^2(4-s)^2\mathbf{1}_{[3,4]}(s), \qquad H(k) = 4\pi\int_3^4 s^2 h(s)\frac{\sin(ks)}{ks}\,ds.$$

Writing $g(s) = sh(s)$, both g and g' vanish at the support endpoints. Two integrations by parts and $\int_3^4 |g''(s)|\,ds \le 30$ imply $|k^3 H(k)| \le 120\pi$. Hence

$$P_\pm(k) = \frac{2\pi^2 A}{k^3} \pm \frac{\pi A}{120}H(k) \;\ge\; \frac{\pi^2 A}{k^3} > 0.$$

Both positive generalized Gaussian covariances restrict to exactly the same law as the baseline on the unit sphere, since $h = 0$ for all its chord separations. Their bulk covariances differ. This exhibits ambiguity around the scale-invariant baseline itself, without permitting a negative spectrum. The analytic bound proves positivity for all k; numerical quadrature is only a check. The source-pinned radial controls under `evidence/observer_dynamics_20260925/sim-analysis/codex/necessity/boundary_bulk/` check the forward integrals and this perturbation with analytic tail bounds, not formal interval arithmetic.

Producer theorem (source-refinement orbit to physical dilation). Let $\mathcal{H}_r^{\mathrm{src}}$ be the scale-labelled scalar mode space generated by the cofinal refinement orbit of the released screen, including its scale label. A single angular cut is insufficient. Let U_r map the orbit unitarily onto the physical rank-one adiabatic source subspace. If

$$D_{s,r}U_r = U_r R_{s,r}, \qquad C_{\zeta,r} = U_r C_{\mathrm{src},r} U_r^*,$$

then

$$D_{s,r}^{-1}C_{\zeta,r}D_{s,r} - u_r(s)C_{\zeta,r} = U_r\left(R_{s,r}^{-1}C_{\mathrm{src},r}R_{s,r} - u_r(s)C_{\mathrm{src},r}\right)U_r^*.$$

Thus the source and physical operator residuals have equal norm. For a bounded-operator limit on a common Hilbert space, strong covariance convergence with a uniform norm bound, strong convergence of the dilation maps and their inverses, $u_r(s) \to u(s)$, and vanishing norm residual imply $D_s^{-1}C_\zeta D_s = u(s)C_\zeta$ in the continuum. This sufficient criterion does not cover a nonzero tilted power law on all $0 < k < \infty$: unitary conjugation preserves the norm of a bounded operator, so $\|C_\zeta\| = |u(s)|\|C_\zeta\|$ would force $u(s) = 1$ for positive $u(s)$ and $C_\zeta \neq 0$.

For the global power-law branch, use instead a local limit receipt. After embedding on the common physical mode space, let $\mathcal{D}$ be the dense space of L^2 functions supported in compact $\ln k$ bands, and let D_s be the exact physical dilation, preserving $\mathcal{D}$. Require $C_{\zeta,r}f \to C_\zeta f$ for every $f \in \mathcal{D}$ and

$$(D_s^{-1}C_{\zeta,r}D_s - u_r(s)C_{\zeta,r})f \longrightarrow 0 \qquad (f \in \mathcal{D}).$$

If finite dilation maps are used to establish the latter condition, their replacement by D_s must also have vanishing error on this core. The resulting identity holds on $\mathcal{D}$, with a possibly unbounded covariance operator; it does not assert a global uniform covariance norm bound. The commuting square states that source refinement followed by the physical embedding agrees with physical embedding followed by wavelength dilation, and it transfers the source covariance error exactly to the physical covariance.

Proof. Conjugate the source covariance by U_r and use the commuting square; unitary invariance gives the finite residual identity. For the bounded limit, strong convergence and uniform boundedness allow passage through the products. For the local limit, both f and $D_s f$ belong to $\mathcal{D}$, so pass to the limit in each of the two terms using the fixed bounded unitary D_s^{-1}. □

Theorem (source-dilation uniqueness). Let Δ_ζ^2 be a nonnegative measurable radial function, locally essentially bounded in $\ln k$, and nonzero on a set of positive measure. Let $C_\zeta = M_{\Delta_\zeta^2}$ on the common physical $L^2(d\ln k\, d\Omega_k)$ mode basis, with its natural multiplication-operator domain. The operator may be unbounded; the dilation identity below is required on the common compact-band core $\mathcal{D}$. Define

$$(D_s f)(k,\hat{k}) = f(e^{-s}k,\hat{k}).$$

If the source refinement packet proves, for all real s,t and positive $u_q(s)$,

$$D_s^{-1} C_\zeta D_s = u_q(s) C_\zeta, \qquad u_q(s+t) = u_q(s)u_q(t), \qquad -\frac{d}{ds}\ln u_q(s)\Big|_{s=0} = \theta,$$

then $u_q(s) = e^{-\theta s}$ and there is $A > 0$ such that

$$\Delta_\zeta^2(k) = Ak^{-\theta} \quad \text{almost everywhere.}$$

For a continuous source representative and pivot $k_\star$,

$$\boxed{\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}}, \qquad A_\zeta = \Delta_\zeta^2(k_\star).$$

Proof. The stated logarithmic derivative at zero gives continuity there and solves the positive multiplicative Cauchy equation for u_q. Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s}\Delta_\zeta^2(k)$ almost everywhere for each s. In logarithmic coordinates, $e^{\theta t}\Delta_\zeta^2(e^t)$ is invariant under all rational translations on one common full-measure set; bounded-transform convolution makes it constant almost everywhere. Nonnegativity and the nonzero premise make that constant strictly positive. Continuity upgrades the identity to every k. The compact-band functions form a core of the stated locally bounded multiplication operator, so the identity also identifies its natural domains. □

Theorem (exact flat thin-shell lift). On the source-dilation branch, for $-2 < \theta < 4$ and $\ell \geq 2$,

$$\int_0^\infty \frac{dx}{x} x^{-\theta} j_\ell^2(x) = \frac{\sqrt{\pi}}{4}\frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)}\frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)},$$

so

$$C_{\ell,R_\star}^q = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)},$$

with the complete amplitude conversion

$$\boxed{A_\zeta = \frac{A_q}{\pi^{3/2} Z_q^2 (k_\star R_\star)^\theta}\frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.}$$

Proof. Insert the source power law into the exact forward map, set $x = kR_\star$, and apply the Weber–Schafheitlin square integral. The convergence strip follows from $j_\ell(x) \sim x^\ell/(2\ell+1)!!$ at zero and $j_\ell(x) = O(x^{-1})$ at infinity. □

At the source-native pivot $k_\star R_\star = 1$ with $Z_q = 1$ and the source-law tilt θ, the conversion factor evaluates to approximately 0.16080676. It is an arithmetic consequence of the symbolic relation and supplies no source amplitude by itself.

Corollary (source-family injectivity and multipole consistency). Fix $\theta, Z_q, k_\star$, and a source window for which the forward coefficient is positive and finite. The forward map is injective in A_ζ on the family $\Delta_\zeta^2 = A_\zeta (k/k_\star)^{-\theta}$. For each retained multipole,

$$A_\zeta^{(\ell,W)} = \frac{C_{\ell,W}^q}{4\pi Z_q^2 k_\star^\theta \int d\ln k\, k^{-\theta} |\Psi_{\ell,W}(k)|^2}.$$

Every $A_\zeta^{(\ell,W)}$ must agree. A disagreement falsifies the declared window/dilation packet and cannot be repaired with an ℓ-dependent amplitude. Once the source law fixes θ, the screen has one amplitude left to determine, and the unused multipoles become consistency checks.

Theorem (finite-window bound). Let $\ell \geq 1$ be an integer, $R_\star, k_\star > 0$, $A_\zeta > 0$, and let W be a nonnegative probability measure on positive radii with finite $\theta/2$ moment. For $0 < \theta < 2\ell$, define

$$I_\ell(\theta) = \int d\ln x\, x^{-\theta} j_\ell^2(x), \quad J_\ell(\theta) = I_\ell(\theta - 2) - \left[\ell(\ell+1) - \frac{\theta(\theta+1)}{2}\right] I_\ell(\theta),$$

or, equivalently by gamma recurrence,

$$J_\ell(\theta) = I_\ell(\theta) \left[\frac{\ell(\ell+1)}{\theta} + \frac{(\theta+1)(\theta-2)}{4}\right].$$

$$\eta_{\ell,W} = \frac{2\sqrt{J_\ell(\theta)}}{\theta} \int W(dr) |r^{\theta/2} - R_\star^{\theta/2}|.$$

Then

$$|C_{\ell,W}^q - C_{\ell,R_\star}^q| \leq 4\pi Z_q^2 A_\zeta k_\star^\theta \eta_{\ell,W} \left(2R_\star^{\theta/2} \sqrt{I_\ell(\theta)} + \eta_{\ell,W}\right).$$

Proof. In the Hilbert norm $\|f\|_\theta^2 = \int d\ln k\, k^{-\theta} |f(k)|^2$, $\|j_\ell(kr)\|_\theta = r^{\theta/2}\sqrt{I_\ell}$ and $\|\partial_r j_\ell(kr)\|_\theta = r^{\theta/2-1}\sqrt{J_\ell}$. The fundamental theorem of calculus, Minkowski, and $|\|f\|^2 - \|g\|^2| \leq \|f-g\|(\|f\|+\|g\|)$ give the bound. □

Normalization and positivity of W are premises: for example, $W = 2\delta_{R_\star}$ gives $\eta_{\ell,W} = 0$ while multiplying the shell spectrum by four. A signed window would require its total variation in the norm estimate.

The reference implementation evaluates this analytic bound in binary64; it does not provide an outward-rounded interval certificate. Near $\theta = 0$ it uses the recurrence form of J_ℓ and evaluates the radius difference as $R_\star^{\theta/2}\,\mathrm{expm1}[(\theta/2)\ln(r/R_\star)]$. Direct subtraction of rounded powers can incorrectly return a zero bound for a nontrivial window. Numerical promotion requires a separate rounding-error budget in addition to the exact inequality.

Theorem (finite singular-value and prior continuation). For a declared finite radial basis, write $C = Ap$. If $A \in \mathbb{R}^{N_\ell \times N_k}$ has rank r, then $\dim\ker A = N_k - r$, and every exact solution is $p = p_{\mathrm{part}} + V_0 a$ for a right-null basis V_0. Given a prior center p_0, $Q \succ 0$, and a feasible datum $C \in \mathrm{Ran}\, A$, the unique minimizer over real vectors of $\frac{1}{2}(p-p_0)^{\mathsf{T}} Q (p-p_0)$ subject to $Ap = C$ is

$$\boxed{p_* = p_0 + Q^{-1}A^{\mathsf{T}}(AQ^{-1}A^{\mathsf{T}})^{+}(C - Ap_0).}$$

If $C \notin \operatorname{Ran} A$, no constrained minimizer exists; the displayed pseudoinverse expression instead leaves a nonzero constraint residual. Even when feasible, it need not give a nonnegative spectrum. Positivity therefore requires a separate check or a declared constrained optimization, whose solution is not in general the displayed expression. The run publishes the singular values, rank threshold, right-null basis, resolution operator, unresolved projector, positivity active set, and prior sensitivity. The output type is ConditionalRadialContinuation.

Theorem (unrestricted tomographic uniqueness). For

$$C_\ell(r, r') = \frac{2}{\pi} \int_0^\infty k^2 P_\zeta(k) j_\ell(kr) j_\ell(kr')\, dk,$$

the covariance operator on $L^2(r^2 dr)$ obeys

$$Q_\ell = \mathcal{H}_\ell^{-1} M_{P_\zeta} \mathcal{H}_\ell.$$

Thus a complete radial cross-covariance operator determines P_ζ uniquely almost everywhere on its natural operator domain. Fixing one radius $r_0 > 0$, the family $C_\ell(r, r_0)$ known for all r suffices, provided that this section admits spherical-Hankel inversion in the declared function or distribution class: its spherical-Hankel transform is $\sqrt{2/\pi}\, P_\zeta(k) j_\ell(kr_0)$, and the Bessel zeros are discrete. A finite collection of auto-spectra does not supply this tomography theorem. The positive-radius condition is essential: $j_\ell(kr_0)$ is identically zero at $r_0 = 0$ for $\ell \geq 1$, so that section contains no spectral information.

Proof. Apply spherical-Hankel inversion in the first radial variable. Unitary conjugation exposes the multiplication operator; equal multiplication operators have equal multipliers almost everywhere. □

Theorem (branch-typed radial boundary). The ordinary spherical-Bessel and gamma-function formulas apply to FlatExact and FlatAssumed branches. Open and closed spatial branches use their declared hyperspherical eigenfunctions and spectral measures. An unresolved curvature branch blocks physical radial identification, and the flat thin-shell amplitude formula cannot certify a curved branch.

Physical source requirements. A source-derived primordial packet using either certified route above requires all source-stress, single-clock, freezeout, growing-mode, phase, physical-scale, source-amplitude, null-space, positivity, and non-fitting forward-residual receipts, together with its radial-dilation intertwiner or radial-tomography certificate. An alternative source construction must supply its own injectivity or orbit-separation theorem and the same physical and no-target-leak interfaces. A finite inverse selected before evaluation is a conditional radial continuation. It supplies no physical temperature or polarization spectrum.

Corollary (transfer firewall). Radial identification changes no temperature, polarization, lensing, recombination, foreground, nuisance, covariance, or likelihood conclusion. Those outputs require the independent transfer and likelihood system.

total stress closure receipt	$T_I^{ab}, Q_I^a, E_T, \nabla_a T^{ab}$
single clock normal form receipt	$\chi_r, \operatorname{rank} d\mathcal{F},$ connected fibers
entropy repair gap receipt	$L_\perp, H, \gamma,$ forcing/refinement bounds
curvature evolution receipt	$\zeta_a, \Gamma_a^{\mathrm{eff}},$ evolution residual, freeze bound
adiabatic mode receipt	$S_{IJ}, \Gamma_I, \Xi_I,$ growing-mode projection
isocurvature bound report	$\beta_{\mathrm{iso}}(k)$ and source covariance
primordial phase coherence receipt	$\lambda_2/\lambda_1,$ phase and decaying-mode defects
screen to radial lift receipt	$W(r), R_\star, \Psi_\ell(k),$ source hashes
radial null space report receipt	singular values, null basis, resolution kernels
forward projection residual receipt	$r_\ell,$ aggregate norms, quadrature and solver errors.

Source-provenance and pooled-reducer firewall. Let the source artifact DAG have parent-to-child edges and let M be the set of measurement, residual, likelihood, posterior, fitted-parameter, or measurement-calibrated nodes. Define

$$\tau(v) = \mathbf{1}_{v \in M} \vee \bigvee_{p \in \mathrm{Parents}(v)} \tau(p).$$

Then $\tau(v) = 1$ exactly when v has a forbidden measurement ancestor. The proof is induction over a topological ordering of the DAG. Every node claimed as source-derived for $\eta_R, \Gamma_{\mathrm{rec}}, A_\zeta, q_{\mathrm{IR}}, \ell_{\mathrm{IR}}, B_A(k,a), \rho_A(a)$, and N_{CRC} must have $\tau = 0$. Unknown source metadata fails closed.

For nonlinear estimates, shards may not average nonlinear quantities. Each row must emit additive sufficient statistics in a commutative monoid $(\mathcal{M}, \oplus)$, for example

$$\sum w_i, \qquad \sum w_i x_i, \qquad \sum w_i x_i x_i^{\mathsf{T}}$$

or, for weighted response regression,

$$X^{\mathsf{T}} W X, \qquad X^{\mathsf{T}} W y, \qquad y^{\mathsf{T}} W y.$$

Associativity, commutativity, complete coverage, and duplicate removal make the pooled statistic partition-invariant; the deterministic estimator is then evaluated only after pooling.

Three-dimensional homogeneous lift. A rotationally invariant screen covariance about one observer is not a statistically homogeneous three-dimensional primordial field by itself. A frozen lift

$$\mathcal{L}_r : V_r^{\mathrm{screen}} \to \mathcal{H}_r^{(3)}$$

must define

$$\Sigma_r^{(3)} = \mathcal{L}_r \Sigma_r^{\mathrm{screen}} \mathcal{L}_r^*$$

and certify positivity, refinement convergence, rotation covariance, translation covariance or an equivalent multicenter consistency theorem, and explicit null-space treatment. If the limiting covariance commutes with translations and rotations, Fourier diagonalization gives

$$\langle \zeta_a(\mathbf{k}) \zeta_b^*(\mathbf{k}') \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') P_{ab}(|\mathbf{k}|),$$

with $P_{ab}(k)$ positive semidefinite.

Transfer-ready initial data. The source-to-transfer map must emit a complete initial state

$$Y(k, \eta_i) = \mathcal{M}_{\mathrm{init}}(k; \mathfrak{B}_{\mathrm{src}}) \xi(k),$$

including metric, photon, polarization, baryon, neutrino, anomaly, recipient, and auxiliary response variables. Near a singular early-time point, regular Frobenius modes solve

$$Y' = \left(\frac{M_{-1}}{\eta} + M_0 + O(\eta)\right) Y, \qquad Y = \eta^s(v_0 + v_1\eta + \cdots), \quad M_{-1}v_0 = sv_0.$$

The branch must classify regular adiabatic growing modes, allowed isocurvature modes, decaying modes, gauge modes, and singular interaction modes. On the adiabatic branch,

$$S_{IJ} = 3(\zeta_I - \zeta_J) = 0$$

for all active sectors. The finite evidence receipt must bound the initial Hamiltonian and momentum constraints and the propagated residual

$$\epsilon_{\text{constraint}} = \sup_{k,\eta} \|\mathcal{C}(k,\eta)Y(k,\eta)\|.$$

Receipt contract. Every ensemble run must record its ensemble id, interpretation class, regulator, representative schema, gauge action, quotient canonicalizer, base measure definition, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler kernel, partition-invariant random-event schema, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are:

seed, proposal, or repair noise,
conventional reference ensemble,
OPH-native quotient ensemble,
OPH physical vacuum,
OPH primordial field,
observable cosmological prediction.

Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared. For bounded coarse observables O, if the implemented law satisfies $\|\widehat{\mu}_s - \mu_s\|_{\text{TV}} \le \epsilon_{\text{samp}}$ and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}),$$

which is the finite evidence accuracy statement for this theorem.

Theorem 2.3 (EC from regulated overlap gluing). For a collar B_δ around a cap boundary Σ, there is a canonical decomposition

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

and the sector-preserving collar algebra

$$A_{\text{EC}}(B_\delta) := \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha}))$$

with

$$Z(A_{\text{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha,$$

such that $\mathcal{A}(A_\delta B_\delta)$ acts only on $H_{b_L^\alpha}$ and $\mathcal{A}(B_\delta D_\delta)$ acts only on $H_{b_R^\alpha}$ within each block.

Proof. Split the collar into half-collars B_L and B_R meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_\alpha (V_\alpha \otimes H_{b_L^\alpha}), \qquad \tilde{\mathcal{H}}_{B_R} = \bigoplus_\beta (V_\beta^* \otimes H_{b_R^\beta}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha,\beta} (V_\alpha \otimes V_\beta^*) \otimes (H_{b_L^\alpha} \otimes H_{b_R^\beta}).$$

By Schur's lemma,

$$(V_\alpha \otimes V_\beta^*)^{G_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The induced sector-preserving collar algebra is

$$A_{\mathrm{EC}}(B_\delta) = \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on Σ. QED.

Remark. On the central-defect subbranch, replace G_Σ by its central extension. The sector label α then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

Exact Markov route. If $I(A_\delta : D_\delta \mid B_\delta) = 0$, the Hayden–Jozsa–Petz–Winter (HJPW) structure theorem yields a blockwise factorization of the state over *some* direct-sum/tensor decomposition of $\mathcal{H}_{B_\delta}$; that decomposition is state-dependent, and HJPW does not by itself identify it with the preselected EC factors b_L^α, b_R^α of Theorem 2.3. The exact identity used for literal Markov-modular equalities below is the **EC-aligned** normal form

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha (\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}), \qquad I_\omega(A_\delta : D_\delta \mid B_\delta) = 0,$$

and the identification of the HJPW factors with the EC factors is a genuine additional state condition, the **Markov-split alignment hypothesis (MSA)**. It is not implied by exact Markovity: with four qubits A, B_L, B_R, D and $\rho = \Phi_{AB_R} \otimes \Phi_{B_L D}$ (Bell pairs across the cut, one center block), one has $I(A : D \mid B) = 0$ but $I(A : B_R) = 2 \ln 2$, so the state is exactly Markov yet not of the displayed EC-aligned form, which would force $I(A : B_R) = 0$. All literal Markov-modular equalities below therefore assume exact Markovity *plus* MSA, or an explicitly stated idealized limit that supplies both; The spacetime and Einstein paper states MSA formally and records this counterexample. MSA is checkable state by state: The spacetime and Einstein paper proves it equivalent to a splitting of $\log \rho$ into commuting one-sided terms supported on $A b_L^\alpha$ and $b_R^\alpha D$, to the existence of a ρ-preserving conditional expectation onto either one-sided edge algebra (a Takesaki commuting square over the collar center [61]), and to blockwise vanishing one-sided mutual information

$I(Ab_L^\alpha : b_R^\alpha D) = 0$. On the local information-projection branch this reduces the derivation of MSA to a sharp condition on the effective Hamiltonian: every cross-cut coupling must act through the central sector label, as it does in lattice-gauge-type regulators where the interface energy is a function of the conserved flux. On that declared **central-interface branch** The spacetime and Einstein paper derives both MSA and exact collar Markovianity as theorems (descent of one-sided invariants and boundary charges through the edge-center quotient), and it adopts the central-interface normal form as an explicit named branch premise, so the literal Markov-modular identities hold on the declared branch without a further state hypothesis. This premise is independent of A1–A3: The spacetime and Einstein paper exhibits a finite regulator package that satisfies observer agreement, schedule-independent transactional repair, and an exactly closed refinement-channel family while its retained constraint family carries a boundary-invariant noncentral cross-cut coupling. Observer agreement and repair do not force the premise. Off the declared branch MSA is carried as an explicit hypothesis.

On the separately declared rate-controlled family obtained by pushing the optimizer forward, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized route. The following lemma and theorem supply the quantitative decay route when the approximation is kept explicit.

Lemma 2.6 (interior local information projection implies local Gibbs form). Under Axiom 3 on the subbranch with identity-proportional local references, state-determining aggregation equivalent to the declared entropy objective, homogeneous global-sum constraints $\langle \sum_x O_a(x) \rangle = C_a$, and the finite-dimensional regulator realization above, assume also that the constraint vector lies in the relative interior of the attainable moment set, so the selected state is faithful on the retained sector. Then the selected state has the finite-multiplier Gibbs form

$$\omega = \frac{e^{-H_{\mathrm{eff}}}}{\operatorname{Tr} e^{-H_{\mathrm{eff}}}}, \qquad H_{\mathrm{eff}} = \sum_x \sum_a \lambda_a O_a(x) + H_{\mathrm{sec}},$$

where the sum runs over UV cells x and density labels a, with one cell-independent multiplier λ_a per label. For the finite-range collar theorem, H_{sec} is either a sum of uniformly bounded finite-range charge densities or is central and fixed on the chosen superselection sector. A genuinely nonlocal noncentral term falls outside the theorem. Under this condition the effective Hamiltonian is finite-range in UV graph units, and the multiplier count matches the constraint count N_{con} plus the number of retained sector terms on every lattice.

Proof. On a finite-dimensional algebra, an interior entropy maximizer subject to linear constraints $\operatorname{Tr}(\rho O_i) = c_i$ is given by finite Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\operatorname{Tr} e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. Here the constrained operators are the N_{con} global sums $O_a := \sum_x O_a(x)$ plus the retained sector terms, so the exponent has one shared multiplier per density label. MaxEnt gives the Gibbs form on this faithful/interior branch; the displayed locality restriction on H_{sec} is separate and load-bearing. A boundary constraint can instead force a rank-deficient optimizer (for example $\operatorname{Tr}(\rho|1\rangle\langle 1|) = 0$ on $\mathbb{C}^2$); it has no finite full-algebra multiplier and must be

treated on its support or by infinite penalties. Such a boundary branch does not supply the stagewise faithfulness used below. Had the constraints instead been imposed per cell, $\langle O_a(x)\rangle = c_a(x)$, the multipliers would be cell-dependent fields $\lambda_a(x)$ and the family would grow with the lattice; that enlarged family belongs to the non-equilibrium approximation regime of the clarification above and is not the branch used in the refinement argument. QED.

Definition 2.6a (closure defect and induced refinement map). Fix successive regulator scales ℓ (finer) and L (coarser), let $\{\omega_L(\lambda')\}$ be the homogeneous global-sum exponential family at scale L, and let $\Phi_{\ell\to L}$ be an admissible coarse-graining channel. For fine-scale multipliers λ set $\sigma := \Phi_{\ell\to L}(\omega_\ell(\lambda))$ and define the **closure defect**

$$\varepsilon_{\ell\to L}(\lambda) := \inf_{\lambda'} D(\sigma \,\|\, \omega_L(\lambda')),$$

with D the quantum relative entropy. When the infimum is attained at a unique λ^*, set $R_{\ell\to L}(\lambda) := \lambda^*$.

Lemma 2.6b (I-projection residual bound). If the constrained operators at scale L together with the identity are linearly independent, then $\lambda' \mapsto D(\sigma\|\omega_L(\lambda'))$ is smooth and strictly convex; for faithful σ the minimizer $\lambda^* = R_{\ell\to L}(\lambda)$ exists, is unique, and is characterized by moment matching $\langle S_c\rangle_{\omega_L(\lambda^*)} = \langle S_c\rangle_\sigma$ for every constrained operator S_c; and

$$\|\sigma - \omega_L(R_{\ell\to L}(\lambda))\|_1 \le \sqrt{2\,\varepsilon_{\ell\to L}(\lambda)},$$

so the family reproduces every bounded expectation up to $\|B\|\sqrt{2\varepsilon_{\ell\to L}(\lambda)}$. The separately declared exact optimizer-pushforward interface is the statement $\varepsilon_{\ell\to L}(\lambda) = 0$ along the realized branch. In that case $\sigma = \omega_L(R_{\ell\to L}(\lambda))$, and the induced refinement map is this moment-matching I-projection.

Proof. Writing $\log\omega_L(\lambda') = -\sum_c \lambda'_c S_c - \log Z_L(\lambda')$, the objective is $-S(\sigma) + \sum_c \lambda'_c\langle S_c\rangle_\sigma + \log Z_L(\lambda')$, and the Hessian of $\log Z_L$ is the Duhamel (Kubo–Mori) covariance matrix of the constrained operators, positive definite under the stated independence; this gives strict convexity, uniqueness, and the moment-matching first-order condition, while attainment for faithful σ is standard finite-dimensional exponential-family duality [59]. The displayed residual bound is the quantum Pinsker inequality $D(\rho\|\tau) \ge \frac{1}{2}\|\rho-\tau\|_1^2$ [60] evaluated at λ^*, and $\varepsilon_{\ell\to L}(\lambda) = 0$ forces $\sigma = \omega_L(\lambda^*)$ by strict positivity of relative entropy off the diagonal. QED.

Acceptance check. A finite two-lattice test implementing this package (the constraint/multiplier count against the displayed N_{con}-dimensional family on successive lattices, the moment-matching projection, the residual bound, a generic decimation channel with strictly positive closure defect, and the exactly closed transverse-field subfamily where $R_{\ell\to L}$ acts as the identity) is included with the repository sources.

Numerical projection and closure are distinct. For faithful σ on $\mathbb{C}^d$, choose a Hilbert–Schmidt orthonormal basis of the observable space V modulo identity, and write its traceless Hamiltonian as $H(\theta) = \sum_j \theta_j Q_j$, with $\rho_\theta = e^{-H}/\operatorname{Tr} e^{-H}$. The invariant moment residual is $\eta = \|P_V(\sigma-\rho_\theta)\|_{\mathrm{HS}}$; a raw residual in caller units is not a convergence

criterion. Coercivity and convexity give

$$D(\rho_*\|\rho_\theta) \le B := \eta\left(\|\theta\|_2 + \frac{\log d}{\lambda_{\min}(\sigma)}\sqrt{\frac{d-1}{d}}\right), \qquad \|\rho_* - \rho_\theta\|_1 \le \min\{2, \sqrt{2B}\},$$

where ρ_* is the unique exact projection. This optimizer error is separate from the closure defect $D(\sigma\|\rho_*)$; the Pythagorean identity adds the two. The implementation evaluates these bounds in floating point, without outward-rounded enclosures or an exact-zero certificate. Invertible changes of observable coordinates and identity offsets preserve the state; scalar energy origins shift $\log Z$ and must be retained in partition and free-energy calculations. Compensated scalar accumulation and a stable relative-entropy remainder avoid cancellation of small finite effects; neither supplies a spectral support certificate or recovers input differences lost to rounding.

Strong conditional mixing premise. Let G_ℓ be the UV cell graph. Assume local dimension at most q, interaction degree at most Δ_0, interaction diameter at most r_0 graph layers, and

$$\sup_x \sum_{X \ni x} \|\beta\Phi_\ell(X)\| \le J_0$$

with constants independent of the cut and refinement stage. Let m_ℓ be the resolved collar width in graph layers and set $\delta = m_\ell \ell$. For the faithful Gibbs state, define

$$\mathbf{J}_\rho(A : D \mid B) := \log \rho_{ABD} + \log \rho_B - \log \rho_{AB} - \log \rho_{BD}.$$

The strong mixing premise is a boundary-anchor expansion

$$\mathbf{J}_\rho(A_\delta : D_\delta \mid B_\delta) = \sum_{z \in \partial_{r_0}^{\mathrm{UV}} C} E_{z,\ell,\delta}, \qquad \|E_{z,\ell,\delta}\|_\infty \le \kappa e^{-(m_\ell - r_0)/\zeta},$$

with κ, ζ uniform in the cut, boundary condition, and stage. This conditional or matrix mixing condition is stronger than ordinary two-point exponential clustering. Local Gibbs form, a Lieb–Robinson bound, or a Hamiltonian spectral gap does not imply it for a general noncommuting Gibbs state.

Theorem 2.5 (finite-range conditional Gibbs mixing gives collar recovery). Under Lemma 2.6 and the strong conditional mixing premise,

$$\boxed{I_\omega(A_\delta : D_\delta \mid B_\delta) \le \kappa |\partial C|_{\mathrm{UV}} e^{-(m_\ell - r_0)/\zeta} \le c |\partial C|_{\mathrm{UV}} e^{-\delta/\xi_\ell}}$$

with

$$\xi_\ell = \zeta \ell, \qquad c = \kappa e^{r_0/\zeta}.$$

Proof. Embedding the marginal logarithms in the tripartite algebra gives

$$I(A : D \mid B)_\rho = \mathrm{Tr}[\rho_{ABD} \mathbf{J}_\rho(A : D \mid B)]\,.$$

The boundary expansion and trace-norm duality bound this expectation by the sum of the boundary term norms. The second form follows from $\delta = m_\ell \ell$. QED.

For one uniform family, the sharp sufficient limit condition is

$$\frac{\delta}{\xi_\ell} - \log|\partial C|_{\mathrm{UV}} \longrightarrow +\infty.$$

If

$$|\partial C|_{\mathrm{UV}} \le a(\ell_0/\ell)^p, \qquad \delta(\ell) \ge \zeta(p+\eta)\ell\log(\ell_0/\ell), \quad p \ge 0, \quad \eta > 0,$$

then $\delta/\ell \to \infty$, and

$$I(A_\delta : D_\delta \mid B_\delta) \le ca(\ell/\ell_0)^\eta \longrightarrow 0.$$

This lower bound on δ does not imply $\delta \to 0$. Physical collar shrinking remains the separate double-scaling hypothesis stated above. It is realized, for example, by choosing $\delta(\ell) = \zeta(p+\eta)\ell\log(\ell_0/\ell)$, or by rounding its graph-layer width upward with an additional error at most ℓ. For a counterexample to inferring shrinking from the lower bound alone, take $\ell_n = 1/(4n)$, $\ell_0 = \delta_n = 1/4$, $p = 0$, and $a = \zeta = \eta = 1$. Then $\delta_n \ge \ell_n \log n$ and $\delta_n/\ell_n = n$, but $\delta_n = 1/4$ does not shrink. A zero-Hamiltonian product Gibbs chain has zero conditional matrix defect and CMI and satisfies all the local finite-range and mixing bounds in this example; it fails precisely the separately declared shrinking-collar hypothesis. The ratio $\delta/\ell \to \infty$ alone is insufficient. For $\ell_n = \ell_0 e^{-n^2}$, $\delta_n = \zeta n \ell_n$, and $|\partial C_n|_{\mathrm{UV}} = e^{n^2}$, that ratio diverges while the upper envelope grows as e^{n^2-n}.

A finite receipt records the interaction range and norm bound, treatment of global sector terms, boundary-cell count, graph separation, regional CMI in nats, matrix-defect norm, predeclared κ, ζ, log-envelope slack, Fawzi–Renner error, held-out cuts, and the rate margin $\delta/(\zeta\ell) - \log|\partial C|_{\mathrm{UV}}$. Fitted constants and local random-triplet CMI remain finite proxies. They do not certify a cofinal limit.

This bound is the quantitative hinge for constructive gluing.

2.0.2 Concrete UV realization: quantum link models

Section 2.3 internalizes the regulator package: the fixed-cutoff type-I algebra and boundary fixed-point structure are the realized form of the screen net plus overlap gluing. Quantum link models are included here only as an explicit microscopic example of that structure, not as a separate premise layer.

UV regulator. Triangulate S^2 at scale ℓ_{UV}, giving vertices v, oriented links ℓ, and plaquettes p. Refinement corresponds to $\ell_{\mathrm{UV}} \to 0$ with increasing lattice size.

Degrees of freedom. Attach to every oriented link ℓ a **finite-dimensional** Hilbert space $\mathcal{H}_\ell$. In ordinary Wilson lattice gauge theory, the continuum/refinement-limit edge description is modeled by $L^2(G)$ (infinite-dimensional for continuous G), but the microscopic OPH regulator is instead a **quantum link model** with finite-dimensional link Hilbert spaces that preserve gauge symmetry in operator form [79]. Optionally attach matter Hilbert spaces $\mathcal{H}_v$ at vertices. Then:

$$\tilde{\mathcal{H}}_{\mathrm{total}} = \bigotimes_\ell \mathcal{H}_\ell \otimes \bigotimes_v \mathcal{H}_v,$$

finite-dimensional on any finite lattice. This is a concrete realization of the extended type-I presentation used in Section 2.3.

Boundary gluing as Gauss-law invariants. Define a local gauge transformation group G_v at each vertex v acting on incident links (and matter at v). Physical states satisfy:

$$|\psi\rangle \in \mathcal{H}_{\mathrm{phys}} \quad \Longleftrightarrow \quad U(g_v)|\psi\rangle = |\psi\rangle \ \ \forall\, v,\ g_v \in G_v.$$

Equivalently: $\mathcal{H}_{\text{phys}} = \tilde{\mathcal{H}}_{\text{total}}^{\prod_v G_v}$.

Region algebras. For any region $R \subset S^2$, define an extended Hilbert space $\tilde{\mathcal{H}}_R$ from the links/vertices in R. The **boundary gauge group** $G_{\partial R}$ acts on the cut degrees of freedom (the "half-links" ending on ∂R). Define:

$$\mathcal{A}_{\text{inv}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the concrete quantum-link instance of the lifted fixed-point presentation used in Section 2.3. The same gauge constraints then induce the sector-preserving EC algebra on collars.

Why EC follows immediately. Take a cap C and a collar B_δ around ∂C. Because the *only* coupling between inside and outside is through the boundary gauge constraint, the collar Hilbert space decomposes into superselection blocks labeled by boundary irreps:

$$\mathcal{H}_{B_\delta} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with center generated by the projectors P_α. This is precisely the Schur-lemma mechanism of Theorem 2.3. The labels α are the familiar edge-mode / electric-flux labels appearing whenever one factorizes gauge theories across an entangling cut [80]. Exact Markovity depends on the state and is not forced by the decomposition alone; nor does exact Markovity align the HJPW factors with these edge-mode labels, which is the separate Markov-split alignment hypothesis of Section 2.3.

Dynamics and MaxEnt. The natural Hamiltonian is a 2+1D lattice gauge Hamiltonian on the screen worldvolume: plaquette ("magnetic") terms, electric terms on links, vertex Gauss terms as constraints, plus local matter couplings. In quantum link form this is finite-dimensional per link while behaving like gauge theory in the continuum limit. Then the MaxEnt assumption becomes concrete: the MaxEnt state is a Gibbs state $\rho \propto e^{-\sum_i \lambda_i O_i}$ with quasi-local O_i, precisely the local-Gibbs regime.

Geometry and G. This microphysics naturally supplies the emergent geometric objects:

- **Edge entropy / area operator:** $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$ becomes "log of boundary irrep dimension" in the gauge link model.
- **Newton constant G:** the conversion factor between edge entropy density per boundary UV cell and macroscopic geometric area.

Thus area is an operator living in the center of the boundary algebra, because in gauge systems the center is where the cut labels live.

Scope of this example. The quantum-link realization makes the fixed-cutoff matrix/fixed-point bookkeeping concrete. Modular flow on caps becomes geometric conformal dilation with the 2π KMS normalization through the support-visible BW scaling theorem on the geometric subnet. Viable architectures for this include holographic quantum error-correcting codes [81] and quantum double / string-net Hamiltonians [82], but any use of QECC distance, min-cut resilience, or Knill–Laflamme correction requires the corresponding code subspace, logical-operator, error-family, and recovery certificate.

2.0.3 Conformal-modular fixed-point microphysics

On the faithful/interior local finite-constraint branch of Axiom 3 with identity-proportional reference, the logarithm of the selected state is a quasi-local UV generator. A separately declared

optimizer-pushforward and refinement-tail interface places the branch inside one common finite-dimensional multiplier family. At each regulator stage the patch and cap algebras are type-I matrix algebras. The finite regulator class is not refinement-closed, so the scaling-limit observer algebra may leave that class. The fixed-cutoff analysis proves the local-Gibbs result and propagation. Endpoint-Lipschitz control of *reduced* modular Hamiltonians requires the additional exact-Markov/central-interface and endpoint-control branch stated below; it does not follow from finite-range locality alone. The cross-scale interface supplies refinement stability. The geometric cap pair with geometric modular flow is obtained in the support-visible sense: the target algebra is the extracted geometric subnet, the support-visible modular matrix elements converge under regularization, and weak-$*$/GNS extraction plus support-readable modular covariance and ordered cut-pair rigidity gives the geometric cap automorphism.

Local finite-constraint information-projection branch. The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{\mathrm{UV}})$, constrained through their homogeneous global sums $\langle \sum_x O_a(x) \rangle = C_a$. Axiom 3 selects the fixed-stage information projection. Retention of the same finite label set under refinement belongs to the separate optimizer-pushforward and refinement-tail interface.

Theorem 2.6 (Interior local constraints imply a local-Gibbs form). If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale ℓ_{UV}, and the prescribed moment vector lies in the relative interior of the attainable moment set so that the optimizer is faithful on the retained sector, then the entropy maximizer is

$$\omega \propto \exp\left(- \sum_a \lambda_a O_a \right),$$

so the MaxEnt generator $H_{\mathrm{MaxEnt}} = -\log \omega$ is a UV-range quasi-local sum. This is exactly the local-Gibbs form used later.

Proof. The finite-dimensional interior exponential-family theorem gives finite Lagrange multipliers λ_a. Boundary optima require restriction to their support or extended infinite multipliers and are not included in this faithful branch. QED.

Derived propagation control on the same branch. Because H_{MaxEnt} is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates [41] apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity v_{LR} and constants C, ξ such that for local observables A_X, B_Y,

$$\left\| [\tau_t(A_X), B_Y] \right\| \leq C \, \|A_X\| \, \|B_Y\| \, \min(|X|, |Y|) \, e^{-(d(X,Y) - v_{\mathrm{LR}}|t|)/\xi}.$$

Additional endpoint-control branch. Locality of the global Gibbs generator does not by itself make reduced interval modular Hamiltonians local; the four-qubit counterexample in Theorem 4.3d rules out that inference. On the exact-Markov/central-interface branch, assume in addition a specified endpoint-to-regulator-cell map and a uniform local norm/site-density bound such that, after subtracting the central endpoint block, the operator-norm sum of terms changed between allowed intervals I, I' is at most $L|I' \Delta I|$. On this explicitly named branch, the local split and that geometric norm bound give bounded-interval endpoint-Lipschitz matrix elements,

$$\left| \langle \psi, (K[I'] - K[I]) \phi \rangle \right| \leq C_{\psi, \phi, I_{\max}} \, |I' \Delta I|,$$

used later in the null-modular bridge. Thus quasi-local propagation follows on the faithful local-Gibbs branch, while reduced-modular endpoint control is a separate exact-Markov/central-interface receipt with the stated endpoint geometry and uniformity.

Refinement-stable multiplier branch. The separately declared optimizer-pushforward and refinement-tail interface preserves the same finite constraint family under coarse-graining and places the regulator states in one common finite-dimensional multiplier family. This is a substantive renormalization condition because coarse-graining a finite-range Gibbs family generically generates interactions outside a fixed density list. The "refinement-stable" language used later means persistence along the stable or fixed branch of this multiplier family. This state-side notion is enough to compare realized states across cutoffs. FixedCutoffBosonicSectorCategory constructs the fixed-stage tensor-generated category and its forgetful fiber. RefinementFunctorAndFiberDescent constructs the refinement pullbacks and compatible fibers only on a cofinal tail carrying the compact-gauge refinement receipt. The cofinal compact-gauge witness uses the same receipt. The separate finite Standard Model route uses complete reversible response and endogenous overlap transport to force the local Standard Model gauge Lie algebra. Incidence and target-blind readback derive the carrier module and $R = -J$. The finite conditional fixture supplies the charge-conjugate rank-15 projector pair, hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. Source derivation of that current, matter action, and global quotient is not supplied. Under the complete-band and operational-cost premises, the exact screen theorem selects the rank-three response band. Physical family attachment, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own premises. Axiom 3 supplies neither the finite current realization nor the sector-colimit data or laboratory attachment.

Scaling limit and algebraic type. The regulator presentation gives a family of finite type-I algebras

$$\mathcal{A}_\ell(C) \cong \mathcal{B}(\mathcal{H}_{C,\ell}).$$

A scaling limit of this family need not be type I. In the continuum-QFT case of interest one expects the local limit algebra $\mathcal{A}_\infty(C)$ to be non-type-I, typically type III. Accordingly the fundamental modular datum in the limit is the automorphism group of the pair $(\mathcal{A}_\infty(C), \omega_\infty^C)$, not a density matrix inside $\mathcal{A}_\infty(C)$.

Corollary 2.6 (Geometric modular action on paired-certified caps). For any support-visible extracted scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ emitted on a branch carrying both the finite cap-normal certificate and the independently complete same-branch multiresolution algebra-state comparison package, let $\alpha_{\lambda_{\widehat{C}}(s)}$ be the automorphism induced by the independently normalized BW-framed cap flow. Then

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit cap algebra happens to be type I, this may be written as

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}}.$$

In the generic continuum case, the automorphism identity is the complete statement.

Proof. This is The spacetime and Einstein paper's finite cap-net BW theorem applied to the extracted geometric cap pair. The finite cap-normal input supplies BW framing, support-order faithfulness, held-out oriented cross-ratio rigidity, the geometric support flow, independently normalized geometric 2π-KMS comparison, and wrong-normalization controls. The separate multiresolution algebra-state comparison input supplies the complete algebra-state reference tower, cap-family-uniform mixed-GNS convergence, comparison maps, and modular support covariance. QED.

Support envelope from net regularity (not an alternative derivation). The outer-regularity / minimal-support condition used later lets one read a support envelope for the modular

image from the scaling-limit net. By itself it does not establish modular covariance; that equality comes from the paired certificate in Corollary 2.6, or must be assumed as an additional two-sided net-automorphism property.

(NR) Outer regularity / minimal support. For any operator O, the intersection of all connected regions P with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\mathrm{supp}(O)$.

Proposition 2.7 (Modular support-envelope inclusion). Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \mathrm{supp}\Big(\sigma_t^{\omega,C}(O)\Big).$$

Then

$$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(f_t^C(R)).$$

Equality does not follow from (NR). It requires, for example, a two-sided order-isomorphism of the local net compatible with σ_t and σ_{-t}, together with the relevant additivity/local-generation property. Those are substantive covariance data, not consequences of knowing minimal operator supports.

Proof. For every $O \in \mathcal{A}(R)$, (NR) assigns $\sigma_t^{\omega,C}(O)$ a minimal support contained in $f_t^C(R)$; isotony therefore places that operator in $\mathcal{A}(f_t^C(R))$. Taking all such operators gives the stated inclusion. Surjectivity onto the entire envelope algebra is not supplied by the fact that σ_t is an automorphism of the global algebra. QED.

Null-surface modular structure. On the same support-visible scaling branch and extracted geometric-subnet branch, the null-surface modular machinery narrows as follows:

- **Fixed-cutoff null-strip bridge.** The null-strip package proves transferred cut-center data, the theorem-local inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, exact-or-controlled four-term strip additivity on one inherited strip model, and renormalized endpoint-Lipschitz control up to the weak tail generator. On the scaling branch the geometric action supplies the half-sided relation; the separately registered common-vector and relative-commutant standardness receipts make it a standard half-sided modular inclusion whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain with the affine half-line modular relation. The same half-line family fixes the generator's endpoint-derivative identity internally; its local null-stress reading consumes the separately supplied effective-locality and same-source bridge.

- **Derived half-sided modular relation.** Nested null half-line algebras satisfy the half-sided covariance and containment relation on the declared geometric scaling branch by Corollary 5.2e. To obtain a *standard* half-sided modular inclusion, the common reference vector must additionally be cyclic and separating for both half-line algebras and cyclic for their relative commutant. On that separately registered standardness receipt, Borchers–Wiesbrock identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation [45].

- **Weak continuity and finite variation.** The bounded-interval and half-line endpoint-Lipschitz control follows on the exact-Markov/central-interface branch only after the specified endpoint-to-cell map and uniform norm/site-density bound of the endpoint-control clause. It then defines the weak tail generator at fixed cutoff. The support-visible scaling theorem together with the extracted geometric cap pair supplies the continuum null-generator setting in which that weak-tail data can be matched to the geometric null modular action of the relativistic phase.

Constraint set specification. On the local finite-constraint branch, the "correct fixed-cap constraint set" becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.

2. **Gauge charges:** fix boundary flux or charge operators.

3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator B_C or its microscopic lattice approximation.

MaxEnt therefore selects the unique finite-stage invariant state compatible with those conserved charges. The support-visible BW scaling theorem is the continuum statement: if the limit algebra is non-type-I, the geometric modular action on that branch is outer rather than an inner density-matrix identity.

Focusing boundary. QNEC has rigorous QFT proofs in broad settings [55, 57], with hypotheses (null-surface locality and analyticity, modular-flow/causality/defect-OPE structure) that have not been verified on the reconstructed net, and QFC is strictly stronger than QNEC [56]. Quantum focusing is an optional gravity interface and is not consumed by the Einstein theorem:

- EC plus the declared state, edge-reference, and entropy-split premises identify $S_{\mathrm{gen}} = S_{\mathrm{bulk}} + \langle L_C \rangle$ on the finite branch (Section 5.4).

- The null-stress and Einstein theorem branch supplies the kinematic inputs required by an internal QNEC/QFC proof; that derivation is not supplied.

Summary. The CMFP package therefore separates the dependency structure into two layers:

- **Fixed-stage and cross-scale consequences:** Axiom 3 with the declared local constraint and reference package gives the local-Gibbs form and quasi-local propagation on the faithful/interior branch. Bounded-interval reduced-modular endpoint-Lipschitz control additionally consumes the exact-Markov/central-interface split and the specified endpoint-to-cell uniform bound. The separate optimizer-pushforward interface supplies the realized state-side refinement branch along which later transport questions are asked.

- **Support-visible scaling statement:** the scaling limit emits the support-visible geometric cap pair and ordered cut-pair rigidity holds on it. On that branch the limit algebra may be non-type-I, the modular action may be outer, and the 2π normalization follows. The later null half-sided-inclusion bridge additionally consumes its registered common-vector and relative-commutant standardness receipts.

The theorem route works on the support-visible quotient. The realized transported cap-local system packages the geometric cap-local test family, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The regularized transport theorem below replaces the unavailable full-algebra lower spectral floor. Local weak-$*$ extraction and GNS gluing then emit the support-visible scaling-limit cap pair, support-readable modular covariance reads the modular group as a cap-local support map, and ordered cut-pair rigidity collapses the residual cap-preserving conformal freedom to the unique hyperbolic subgroup.

Proposition 2.6a (Recoverability is not modular geometry: common-floor collapse countermodel). Exact or asymptotically exact Markov recovery at finite cutoff does not by itself

imply a full-algebra lower spectral floor along refinement. In a two-dimensional matrix algebra, let

$$\rho_n = \begin{pmatrix} e^{-n} & 0 \\ 0 & 1 - e^{-n} \end{pmatrix}.$$

Each ρ_n is faithful at finite n, and tensoring it with any fixed finite exact-Markov collar factor gives a full-rank exact-Markov collar family. Nevertheless $\lambda_{\min}(\rho_n) = e^{-n} \to 0$, so there is no positive lower bound along the refinement family. On the off-diagonal matrix unit E_{12}, the modular generator carries the logarithmic gap

$$[-\log \rho_n, E_{12}] = n\, E_{12} + O(1)E_{12},$$

and the unregularized modular transport has no finite common-floor limit on that direction. Thus collar Markovity and finite-stage faithfulness are not enough for the false stronger full-algebra BW lift. The observer-facing theorem uses the support-visible regularized replacement, with the exact-Markov comparison family checked only on fixed collars and made replacement-independent after weak-$*$/GNS extraction.

Theorem 2.6b (regularized support-visible modular transport). Fix a local collar model of finite dimension $d_{m,\delta}$. Let ρ_n be the transported physical collar marginal, $\widehat{\rho}_n$ the exact-Markov comparison marginal on the same collar model, and $\Delta_n = \|\rho_n - \widehat{\rho}_n\|_1$. For $a > 0$, set $K_a(\rho) = -\log(\rho + a\mathbf{1})$. For every bounded collar observable O in the support-visible algebra,

$$\left|\operatorname{Tr} \rho_n O\big(K_a(\rho_n) - K_a(\widehat{\rho}_n)\big)\right| \le \|O\| \left(\frac{4\Delta_n}{a} + d_{m,\delta} a + 4\Delta_n |\log a|\right).$$

Consequently, if $a_n \downarrow 0$ is chosen with

$$\Delta_n / a_n \to 0, \qquad d_{m,\delta} a_n \to 0, \qquad \Delta_n |\log a_n| \to 0,$$

then the regularized support-visible modular matrix elements converge on that fixed collar model.

Proof sketch. On the spectral interval $[a, \infty)$, the logarithm is operator-Lipschitz at the scale used above by its integral representation. Splitting the comparison into the support above a, the a-tail, and the trace-distance error gives the displayed bound. The three displayed conditions make the three error terms vanish. QED.

Definition 2.6c (explicit multiresolution reference branch). The finite-to-continuum bridge is evaluated on a declared reference branch, not on an unstructured sequence of finite plots. A regulator is

$$r = (m, L, b),$$

where m is the refinement depth, L is the finite support or volume scale, and b is a boundary/phase label. The screen branch uses nested icosahedral subdivisions of S^2; Euclidean branches use dyadic mesh $a_m = a_0 2^{-m}$, with an additional temporal spacing a_t when a transfer matrix is present. Each added cell, collar, edge-center, or support-visible channel j carries

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state τ_j. At regulator r,

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \qquad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit $\Gamma_r : \widetilde{M}_r \to M_r$ changes from multiresolution coordinates to the OPH patch, port, collar, and gauge coordinates. For $r \preceq s$, write

$$\widetilde{M}_s = \widetilde{M}_r \otimes D_{s\setminus r}, \qquad \tau_{s\setminus r} = \bigotimes_{j\in\mathcal{J}_s\setminus\mathcal{J}_r} \tau_j.$$

The canonical refinement and coarse-graining maps are

$$\iota_{rs}(A) = \Gamma_s\Big[\Gamma_r^{-1}(A)\otimes \mathbf{1}\Big],$$

and

$$Q_{sr}(X) = \Gamma_r\Big[(\mathrm{id}\otimes\tau_{s\setminus r})(\Gamma_s^{-1}(X))\Big].$$

The embedded conditional expectation is $E_{sr} = \iota_{rs}Q_{sr}$. The reference state is $\widehat{\omega}_r = \widetilde{\omega}_r \circ \Gamma_r^{-1}$.

Theorem 2.6d (exact modular-compatible reference tower). The data

$$(M_r, \widehat{\omega}_r, \iota_{rs}, E_{sr})_{r\preceq s}$$

form a directed tower. The maps ι_{rs} are unital injective $*$-homomorphisms with $\iota_{st}\iota_{rs} = \iota_{rt}$. The Q_{sr} are faithful state-preserving completely positive coarse-grainings with $Q_{tr} = Q_{sr}Q_{ts}$. The maps E_{sr} are faithful conditional expectations onto $\iota_{rs}(M_r)$, and

$$\widehat{\omega}_s \circ \iota_{rs} = \widehat{\omega}_r, \qquad \widehat{\omega}_r \circ Q_{sr} = \widehat{\omega}_s, \qquad \widehat{\omega}_s \circ E_{sr} = \widehat{\omega}_s.$$

The finite reference modular groups are also exactly compatible:

$$\sigma_t^{\widehat{\omega}_s}(\iota_{rs}A) = \iota_{rs}(\sigma_t^{\widehat{\omega}_r}(A)).$$

This theorem uses no full-algebra spectral floor uniform in r; in bare coordinates it is only the identity $A \mapsto A\otimes\mathbf{1}$, partial expectation against faithful detail states, and conjugation by Γ_r, Γ_s.

Theorem 2.6e (canonical renormalization and correlation Cauchy bound). Let $\mathfrak{A}$ be the C^*-inductive limit with embeddings $j_r : M_r \to \mathfrak{A}$. There is a unique state $\widehat{\omega}$ with $\widehat{\omega}\circ j_r = \widehat{\omega}_r$, and the maps Q_{sr} induce state-preserving conditional expectations

$$\mathbb{E}_r : \mathfrak{A} \to j_r(M_r)$$

such that $\|\mathbb{E}_r(A) - A\| \to 0$ for every $A \in \mathfrak{A}$. The finite-regulator realization of a continuum test observable is

$$\mathcal{R}_r(A) = j_r^{-1}(\mathbb{E}_r A).$$

For $\|A_k\| \le M$,

$$\left|\widehat{\omega}_r\left(\prod_{k=1}^n \mathcal{R}_r(A_k)\right) - \widehat{\omega}\left(\prod_{k=1}^n A_k\right)\right| \le M^{n-1}\sum_{k=1}^n \|A_k - \mathbb{E}_r A_k\|.$$

Thus the operator mixing matrix and additive subtraction at finite cutoff are outputs of $\mathcal{R}_r$, not free fits to the target continuum answer. Independent lattice-spacing, volume, and boundary sweeps must publish the corresponding martingale-tail or physical-error envelopes before finite regulator data are read as continuum evidence.

Theorem 2.6f (compact-time modular convergence on fixed local algebras). Let $\rho_{s\downarrow r}^{\mathrm{phys}}$ be the density matrix of the physical state at fine stage s, restricted to the embedded fixed local algebra M_r, and let

$$\epsilon_{s,r} := \|\rho_{s\downarrow r}^{\mathrm{phys}} - \widehat{\rho}_r\|_1.$$

For a regularizer $\eta_s > 0$, define

$$K_{s\downarrow r}^{(\eta_s)} = -\log(\rho_{s\downarrow r}^{\mathrm{phys}} + \eta_s \mathbf{1}).$$

If $\lambda_r = \lambda_{\min}(\widehat{\rho}_r) > 0$, then for every $A \in M_r$ and every $T < \infty$,

$$\sup_{|t|\le T} \left\| e^{-itK_{s\downarrow r}^{(\eta_s)}} A e^{itK_{s\downarrow r}^{(\eta_s)}} - \sigma_t^{\widehat{\omega}_r}(A) \right\| \le 2T\|A\| \left[\frac{\epsilon_{s,r}}{\eta_s} + \log\left(1 + \frac{\eta_s}{\lambda_r}\right) \right].$$

Hence $\eta_s \to 0$ and $\epsilon_{s,r}/\eta_s \to 0$ give uniform-on-compact-time convergence on each fixed support-visible local algebra. For a countable cofinal local test tower, pointwise $\epsilon_{s,r} \to 0$ admits a diagonal subsequence and one cutoff schedule. The phrase "finite-stage modular action" below means this transported, regularized local action; it does not mean the unregularized full finite-cap modular group.

Cap-family uniformity clause. For application to the finite cap-net BW theorem, the compact-time estimate is required uniformly over the declared finite nondegenerate cap family and the fixed separating support-visible test tower. The operator-norm estimate then implies the quadratic mixed-GNS Cauchy residual because $\|XD\|_{2,\omega} \le \|X\|\|D\|_{2,\omega}$. The reference tower supplies the analytic engine; the finite cap-normal certificate adds the support-order, BW-frame, oriented cross-ratio, geometric 2π-KMS, and wrong-scale controls.

Theorem 2.6g (finite positive transfer and Lorentzian handoff condition). On Euclidean repair branches, let $P_{C,r}$ be weighted resampling inside the fibers of the complete repaired visible datum for active collar C. Ref. [16] proves that this finite operator is the orthogonal conditional expectation and gives the exact support, equal-fiber-row, and weighted detailed-balance matrix receipt that recognizes it. A concrete active collar repair kernel is identified with $P_{C,r}$ only when its independently derived transition matrix passes that receipt; constructing the matrix from the target formula is not a verification. Choose $c_{C,r} \ge 0$, and set

$$L_r^{\mathrm{rep}} = \sum_C c_{C,r}(I - P_{C,r}), \qquad T_r(a_t) = e^{-a_t L_r^{\mathrm{rep}}}.$$

Then $L_r^{\mathrm{rep}} = L_r^{\mathrm{rep}\,*} \ge 0$, $0 < T_r(a_t) \le I$, $T_r(a_t)\mathbf{1} = \mathbf{1}$, and the stationary Euclidean path measure is reflection positive:

$$\mathbb{E}_r[(\Theta F)F] \ge 0$$

for positive-time cylinder functions F. A Lorentzian unitary statement requires this finite positivity plus either exact shell-additive direct-limit semigroup identities or a Mosco / strong-resolvent substitute, including vacuum-projection and intertwiner convergence. Weak-$*$ state convergence alone is not a transfer theorem.

BW-side closure boundary. Proposition 2.6a blocks the stronger unregularized full-algebra conclusion from the proved finite-cutoff package. Theorems 2.6c–2.6g supply the explicit reference-tower certificate needed for the finite-to-continuum reading: continuum correlations come from the conditional-expectation renormalization map, compact-time modular group convergence is only for transported regularized physical marginals on fixed support-visible local algebras, and Lorentzian transfer needs reflection positivity or an equivalent positive-transfer certificate. Theorem 4.2 then uses exactly the observer-facing support-visible content to close the BW/geometric cap-pair automorphism statement. Without the multiresolution regulator certificate, finite regulator data remain regulator evidence. A black-box AQFT/BW reconstruction route would require separately verifying conformal-net properties such as locality, additivity, duality, standardness, positive energy, and suitable modular inclusions.

2.1 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the finite-dimensional overlap algebras and their allowed chart-change groupoid are put in the block-complete spatial presentation specified below, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group G_Σ understood as shorthand for this spatialized compact boundary action, not as an automatic consequence of an arbitrary algebra representation. On the genuinely noncentral branch, the corresponding declared weak gluing data are encoded by a compact crossed-module change system. Block permutations and other groupoid data are retained rather than silently treated as inner automorphisms. A second structural point is UV underdetermination. If each patch Hilbert space is tensored with an inert finite ancillary factor and the observable patch algebras are embedded as $\mathcal{A}(P) \otimes \mathbf{1}$, then the physical observables, collar conditional mutual information, carried Markov errors, and quotient normal forms are unchanged. So the physical UV branch is fixed only modulo gauge or implementation hiding together with such ancillary stabilization, not a unique microscopic presentation; the unique theorem-grade object is the induced family of terminal expectation functionals on the declared physical observable algebras, not a preferred microscopic representative.

2.1.1 Constructive gluing on tree covers

Theorem 3.1 (tree gluing). Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization (A_k, B_k, C_k) at step k. If a target state ρ^* obeys $I(A_k : C_k \mid B_k) \leq \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho^*_{A_k B_k C_k} - (\mathrm{id}_{A_k} \otimes \mathcal{R}_k)(\rho^*_{A_k B_k})\|_1 \leq \delta_k,$$

with

$$\delta_k = 2\sqrt{1 - e^{-\varepsilon_k}} \leq 2\sqrt{\varepsilon_k},$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \leq \min\left(2, \sum_{k=2}^{n} \delta_k\right).$$

Proof. Induct on k. The recovery error contracts under CPTP maps, so the errors add. QED.

2.1.2 Gauge-as-gluing and loops

At finite regulator scale, the fixed-cutoff gauge-as-gluing package identifies the overlap-consistency redundancy of local chart presentations. Choose block-complete faithful finite-dimensional spatial presentations of the overlap algebras on a connected cut Σ, and require the allowed overlap-chart isomorphisms to be represented by their unitary normalizers. A finite-dimensional C^*-algebra is a direct sum of full matrix blocks: its star automorphisms permute isomorphic central blocks and act inner within each block. They are spatial in the canonical block-complete presentation, but need not be inner in the algebra itself (an abelian two-block swap is the elementary counterexample). Fixing a reference chart and retaining any allowed block permutations, the compact closure of the subgroup generated by the spatial recharting unitaries is the boundary gluing group K_Σ; before fixing the reference chart, the same data form a compact unitary groupoid.

Proposition 3.2a (Spatialized gauge-as-gluing at finite regulator). Let a finite regulator chart be chosen for the patches meeting along a connected interface Σ, together with the block-complete spatial presentations and unitary-normalizer realization of the allowed chart-change groupoid just stated. Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, after a reference chart is fixed, this transition system reduces to a compact boundary group K_Σ, and when the triple-overlap defect is central its projective composition law lifts to a genuine action of a compact central extension $\widehat{K}_\Sigma$. On this realized presentation, gauge names the overlap redundancy; the spatialization data are explicit hypotheses rather than consequences of finite dimensionality alone.

Proof. The structure theorem for finite-dimensional C^*-algebras decomposes each chart algebra into full matrix blocks. An allowed star isomorphism permutes isomorphic central blocks and is inner on each matched block; the chosen block-complete presentation implements the combined operation by a unitary normalizer. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

Lemma 3.2b (trees vs loops). If the patch adjacency graph is a tree, one can choose local charts h_i so that the overlap labels satisfy $g_{ij} = h_i^{-1} h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

transforms by conjugation at the chosen basepoint under local frame changes. Its conjugacy class, and therefore every class-function value (including every one-dimensional character value), is invariant. In particular, nontriviality is invariant and obstructs global trivialization. QED.

Proposition 3.2ba (Finite path ratio, character phase, and cyclic quantization). For reversal-compatible group-valued edge labels on a typed finite path system, the ordered transport ratio of two paths with common endpoints is exactly the holonomy of the loop obtained by following the first and reversing the second. Every one-dimensional unitary character sends this identity to an exact relative phase. A vertex rechart conjugates based holonomy, so holonomy itself is invariant for an abelian structure group and the character phase is invariant for an arbitrary group. There is an explicit four-vertex path/face control whose two declared triangular faces have trivial holonomy while a separate undeclared, unfilled loop and its two-arm relative character phase are nontrivial. Finally, if the loop holonomy factors through a supplied $\mathbb{Z}_n$ sector, every character phase is an nth root of unity.

Scope. The result is machine checked for supplied finite paths, edge labels, character, faces, and cyclic-sector factorization. The path/face control declares exactly two triangular faces; it neither claims flatness on undeclared triangles nor proves a puncture or noncontractibility statement. No theorem here constructs the connection from observer records or identifies the phase with a physical gauge flux, spacetime loop, charge, laboratory arm, detector, or interference fringe.

Corollary 3.2c (Collar consequence on the ordinary or central-defect branch). Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary Σ, and set $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A_{\mathrm{EC}}(B_\delta)) = \bigoplus_\alpha \mathbb{C}\, \mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ *together with* the Markov-split alignment hypothesis of Section 2.3, or the explicitly stated idealized recoverability limit supplying both; the block decomposition itself is kinematic and follows from the derived boundary action.

Proof sketch. Decompose the left boundary data into irreps ($V_\alpha \otimes H_{b_L^\alpha}$) of $\widehat{K_\Sigma}$ and the right boundary data into the dual modules ($V_\beta^* \otimes H_{b_R^\beta}$). Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

2.1.3 Loop obstruction class (central defect)

On the central-defect subbranch, identify the overlap centers with one fixed abelian unitary coefficient group Z_Σ and assume overlap transport acts trivially on it. Choose unitary overlap implementers normalized by $U_{ji} = U_{ij}^{-1}$ and define central defects z_{ijk} at the implementer level by

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

or equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. This implementer-level relation records the projective multiplier; the automorphism-level expression $\mathrm{Ad}(z_{ijk})$ alone would be the identity and would not determine z_{ijk}.

Then $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ, and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace K_Σ by its central extension $\widehat{K_\Sigma}$. The condition $[z] = 0$ is exactly the strictifiability of the projective triangle defect. It is not by itself global path independence: at least one resulting strict edge 1-cocycle must also have trivial represented holonomy around every closed overlap loop. With a nontrivial action on varying centers, the corresponding statement uses the twisted/local-system Čech differential instead. (A full proof for the fixed-coefficient subbranch appears in Section 6.4 below, in the algebra-net language.)

2.1.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \to G)$ with an action of G on H by conjugation. Here G is the reconstructed gauge group, and H is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \to G$.

A crossed module is a homomorphism $\partial : H \to G$ together with an action of G on H such that

$$\partial(g \triangleright h) = g\,\partial(h)\,g^{-1}, \qquad \partial(h) \triangleright h' = hh'h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \to G, \qquad h_{ijk} : P_{ijk} \to H,$$

obeying the 2-cocycle conditions

$$g_{ij}g_{jk} = \partial(h_{ijk})g_{ik},$$

and on quadruple overlaps,

$$h_{jkl}h_{ijl} = (g_{ij} \triangleright h_{ikl})h_{ijk}.$$

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology.

Write

$$q_\Sigma = [(g,h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$

for the full crossed-module gluing orbit. It classifies the equivalence orbit of the pair (g,h) but does not in general determine a unique ordinary G_Σ-valued 1-cocycle class after strictification, because the map below need not be injective. Let

$$\begin{aligned} \iota_\Sigma : \check{H}^1(N_\Sigma, G_\Sigma) &\longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma), \\ [g] &\longmapsto [(g,1)] \end{aligned}$$

be the natural strict-locus map. Define the separate $\{0,1\}$-valued strictification obstruction by

$$\begin{aligned} o_\Sigma^{(2)}(g,h) = 0 \quad &\Longleftrightarrow \quad q_\Sigma \in \mathrm{im}(\iota_\Sigma) \\ &\Longleftrightarrow \quad [(g,h)] \text{ has a representative } (g^{\mathrm{str}}, 1) \\ &\qquad\qquad \text{with } g^{\mathrm{str}}_{ij} g^{\mathrm{str}}_{jk} = g^{\mathrm{str}}_{ik}. \end{aligned}$$

and set $o_\Sigma^{(2)}(g,h) = 1$ otherwise.

Theorem 3.4 (non-central higher-defect strictification). The higher associator defect of (g_{ij}, h_{ijk}) is removable iff $o_\Sigma^{(2)}(g,h) = 0$. In that case a higher-gauge change gives $h_{ijk} = 1$ and a genuine G-valued edge 1-cocycle. Strict endpoint-only transport additionally requires that at least one allowed strict representative have trivial represented holonomy. The full orbit q_Σ need not determine a unique ordinary H^1 class.

Proof sketch. Removing the higher associator corresponds to $h_{ijk} = 1$ and $g_{ij}g_{jk} = g_{ik}$. Crossed-module coboundaries preserve the full orbit, so this strictification exists exactly when that orbit meets the image of the ordinary G-valued 1-cocycle locus, which is the definition of $o_\Sigma^{(2)} = 0$. The map ι_Σ need not be injective: H-valued edge changes can relate strict representatives and alter their ordinary holonomy through ∂. Thus the invariant residual test is whether at least one allowed strict representative has trivial represented holonomy, as in Theorem 3.4b. QED.

The central-defect case is the abelian truncation with H central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the point where the fixed-cutoff collar theorem upgrades from the ordinary-group package to the higher-gauge one below.

Corollary 3.4a (Fixed-cutoff higher-gauge EC and transportability). For a finite regulator chart on a connected cut Σ, the genuinely noncentral branch admits a compact crossed-module change system

$$\mathcal{T}_\Sigma = C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma).$$

The physical higher-gauge collar is

$$\mathcal{H}_B^{2g} = (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_\lambda (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda}),$$

with

$$Z(\mathcal{A}_{2g}(B)) = \bigoplus_\lambda \mathbb{C}\,\mathbf{1}_\lambda,$$

and the full orbit q_Σ is invariant under local rechartings and classifies the fixed-cutoff genuinely noncentral gluing datum. Its separate invariant property $o_\Sigma^{(2)} = 0$ holds iff the higher associator is removable. The resulting strict representatives can have different ordinary G_Σ-valued 1-holonomies, so transportability is the separate existential test of Theorem 3.4b rather than a uniquely defined holonomy attached to q_Σ.

Proof sketch. Pair-overlap rechartings are inner, while triple-overlap associators strictify to compact crossed-module data. Finite-dimensional unitary $\mathcal{T}_\Sigma$-modules decompose semisimply, and Schur matching leaves only diagonal left/right blocks. Crossed-module cohomology classifies the higher defect; Theorem 3.4b separately tests the ordinary loop holonomy that remains after strictification. QED.

Theorem 3.4b (TransportabilityFromOverlapGluing). Fix a connected cut Σ and a finite regulator charting nerve N_Σ. Transport of a collar charge is defined by the path composite in the overlap recharting groupoid: along $p = (i_0 \to \cdots \to i_m)$, the charge block is moved by $U_p = U_{i_{m-1}i_m} \cdots U_{i_0 i_1}$. This construction uses the overlap unitary transition system itself, not an added DHR transportability assumption.

After a central or crossed-module strictification, let U^{str} denote a resulting genuine edge 1-cocycle. Write $\mathcal{C}_\alpha$ for the full transported collar-charge block, including its boundary carrier and multiplicity/intertwiner data, and define

$$\mathrm{Hol}_{\alpha,U^{\mathrm{str}}} : \pi_1(|N_\Sigma|, i_*) \longrightarrow U(\mathcal{C}_\alpha), \qquad [\gamma] \longmapsto U^{\mathrm{str}}_\gamma\big|_{\mathcal{C}_\alpha}.$$

On the ordinary branch, transport is strictly path-independent iff every closed overlap loop has trivial holonomy on the collar-sector block. On the central branch, elementary triangle moves accumulate the central cocycle z_{ijk}. Strict path-independent transport exists iff $[z]_\Sigma = 0$ and there is a central 1-cochain strictification for which $\mathrm{Hol}_{\alpha,U^{\mathrm{str}}} = 1$. The first condition removes the projective triangle defect; the second removes the residual ordinary loop action. If $[z]_\Sigma = 0$ but the second condition fails, the lifts form a genuine 1-cocycle and transport is path dependent.

On the genuinely noncentral branch, path comparisons are H_Σ-valued 2-morphisms in the crossed module $H_\Sigma \to G_\Sigma$. Strict ordinary transport exists iff $o^{(2)}_\Sigma = 0$ and at least one crossed-module strictification gives a genuine G_Σ-valued 1-cocycle with trivial represented holonomy on $\mathcal{C}_\alpha$. If $o^{(2)}_\Sigma \neq 0$, the sector has higher transport rather than ordinary compact-group DR transport. If $o^{(2)}_\Sigma = 0$ but every allowed strict representative has nontrivial represented G_Σ-holonomy, the higher associator has strictified but ordinary path dependence remains. The full orbit $q_\Sigma = [(g, h)]$ need not determine a unique ordinary H^1 class and is not itself the strictification test. Thus "zero obstruction" below means the combined associator-strictification and trivial-holonomy criterion, not vanishing of triangle or higher-associator data alone. QED.

Triangle-free cycle check. Let $N_\Sigma = C_n$, $n \geq 4$, have no filled 2-simplices. On $\mathcal{C}_\alpha = \mathbb{C}^2$, put identity transport on every oriented cycle edge except one, where the sector action is the non-scalar unitary $V = \mathrm{diag}(-1, 1)$, and use inverse transports on reversed edges. All central 2-data and higher associator data are vacuously trivial, so $[z]_\Sigma = 0$ and $o^{(2)}_\Sigma = 0$. Its ordered cycle holonomy is $V \neq 1$, which no central rephasing removes. For a genuinely noncentral instance, take the crossed module

$$\mathrm{SU}(2) \hookrightarrow \mathrm{U}(2)$$

with conjugation action and the standard action on $\mathcal{C}_\alpha$. The band $\mathrm{U}(2)/\mathrm{SU}(2) \cong \mathrm{U}(1)$ is detected by the determinant, and the cycle has band holonomy $\det V = -1$. An SU(2)-valued edge change has determinant one and a vertex-frame change only conjugates the based loop, so no allowed strict representative has trivial holonomy. The two routes between the endpoints of the distinguished edge differ, and the combined criterion rejects both the central and noncentral assignments.

2.2 Modular Flow and Lorentz Kinematics

2.2.1 Modular additivity in the Markov collar limit

Consider a collar tripartition $A : B : D$ around a cap boundary, with the EC decomposition of Section 2.3 understood. Define, for a faithful reference state ω,

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Exact modular additivity is not a consequence of EC alone. It is available only on the EC-aligned exact Markov set, or along a controlled fixed-cutoff family that approaches that set on one fixed collar model. Four quantities must therefore be kept separate:

1. the raw collar conditional mutual information $I(A : D \mid B)$;

2. the constructive Fawzi–Renner comparison error

$$r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon};$$

3. the fixed-collar exact-Markov replacement modulus

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1 : I(A : D \mid B)_\rho \le \varepsilon \right\},$$

   where

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : I(A : D \mid B)_\sigma = 0\}.$$

4. the EC-aligned subset and a family-specific aligned replacement modulus,

$$\mathfrak{M}^{\mathrm{EC}}_{A:B:D} := \{\sigma \in \mathfrak{M}_{A:B:D} : \sigma \text{ satisfies MSA on the preselected EC split}\},$$

$$\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}} \|\rho - \sigma\|_1 : \rho \in \mathcal{F},\ I(A : D \mid B)_\rho \le \varepsilon \right\}$$

   for the declared controlled collar family $\mathcal{F}$. Convergence of the generic modulus in item 3 does not imply convergence of this aligned modulus; $\delta^{\mathrm{M,EC}}_{\mathcal{F}} \to 0$ is a separate family hypothesis.

Lemma 4.1a (EC-aligned exact Markov implies exact additivity up to a central term). If $I(A : D \mid B)_\omega = 0$ and ω satisfies the Markov-split alignment hypothesis of Section 2.3, then the state takes the EC-aligned HJPW block form

$$\omega_{ABD} = \bigoplus_\alpha p_\alpha\, \omega^{(\alpha)}_{Ab^\alpha_L} \otimes \omega^{(\alpha)}_{b^\alpha_R D},$$

and $\Delta K(\omega)$ is central. On the canonical HJPW block model one may take

$$\Delta K(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,ABD}(\omega) \in Z(\mathcal{A}(B))$ such that

$$K_{ABD}(\omega) = K_{AB}(\omega) + K_{BD}(\omega) - K_B(\omega) + K_{\partial,ABD}(\omega).$$

Proof. By alignment, the HJPW blocks are the EC blocks. On each block the modular Hamiltonians of ABD, AB, BD, and B are the logarithms of tensor-product states with the same

classical block weight p_α. The blockwise logarithms therefore cancel exactly in the combination $K_{ABD} - K_{AB} - K_{BD} + K_B$. If one keeps the blockwise endpoint-label bookkeeping explicit rather than absorbing it into the canonical block identification, the remainder is a direct sum of block constants, hence central in $Z(\mathcal{A}(B))$. Without alignment, exact Markovity cancels the four logarithms on the state's own HJPW blocks, but the resulting block constants need not lie in $Z(\mathcal{A}(B))$, so the central form used here is unavailable. QED.

Proposition 4.1b (Controlled exact-Markov replacement on a fixed collar). Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Thus small collar CMI converges to the exact HJPW normal form only in this controlled fixed-cutoff sense; the manuscript does not claim a dimension-free one-shot trace-norm bound from small $I(A : D \mid B)$ directly to an exact Markov state. The limit points are exact Markov states in the CMI-zero sense; identifying their factorization with the preselected EC factors is the Markov-split alignment hypothesis of Section 2.3, which is carried separately and is not produced by small CMI (the Bell-pair counterexample of Section 2.3 has zero CMI at fixed distance from every EC-aligned state).

Theorem 4.1b' (Finite-collar Markov replacement stability). On one fixed faithful finite-dimensional collar model, the qualitative modulus above can be made collar-local and quantitative. If all relevant marginals stay above a chosen floor $\lambda_* > 0$, then there are constants $C_{A:B:D,\lambda_*} > 0$ and $\theta_{A:B:D,\lambda_*} > 0$, depending only on that collar model and floor, such that

$$d_{\mathrm{M}}(\rho) \le C_{A:B:D,\lambda_*}\, I(A : D \mid B)_\rho^{\theta_{A:B:D,\lambda_*}}.$$

This is a fixed-collar Lojasiewicz-type rate for the analytic function $I(A : D \mid B)$ on the faithful state manifold. It is not a dimension-free stability theorem for arbitrary tripartite quantum systems.

Corollary 4.1c (Carried EC-aligned collar defect operator). Let $\omega_\varepsilon \in \mathcal{F}$ satisfy $I(A : D \mid B)_{\omega_\varepsilon} \le \varepsilon$, assume the declared family has an EC-aligned replacement modulus tending to zero, and choose $\sigma_\varepsilon \in \mathfrak{M}^{\mathrm{EC}}_{A:B:D}$ with

$$\|\omega_\varepsilon - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon) := \Delta K(\omega_\varepsilon) - \Delta K(\sigma_\varepsilon).$$

Then

$$K_{ABD}(\omega_\varepsilon) = K_{AB}(\omega_\varepsilon) + K_{BD}(\omega_\varepsilon) - K_B(\omega_\varepsilon) + K_{\partial,ABD}(\sigma_\varepsilon) + \mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon),$$

where $K_{\partial,ABD}(\sigma_\varepsilon) = \Delta K(\sigma_\varepsilon)$ is central. For every bounded observable X supported on $A \cup B$,

$$|\mathrm{Tr}[X(\omega_\varepsilon - \sigma_\varepsilon)]| \le \|X\|_\infty\, \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon).$$

If the relevant marginals of ω_ε and σ_ε are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon).$$

Independently, Fawzi–Renner recovery guarantees a channel acting on the collar and a comparison state $\omega^{\mathrm{rec}}_\varepsilon$ with squared Uhlmann fidelity at least $e^{-\varepsilon}$, for CMI ε in nats, and hence

$$\|\omega_\varepsilon - \omega^{\mathrm{rec}}_\varepsilon\|_1 \le r_{\mathrm{FR}}(\varepsilon), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

This $O(\varepsilon^{1/2})$ term bounds observable errors for that recovered state; it is not a guarantee for an arbitrary unrotated Petz map. The generic modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ justifies replacement only by some exact Markov state on its state-dependent HJPW split. The separately assumed $\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon)$ is the quantity that justifies the EC-aligned normal form used in later geometric modular arguments.

Accordingly, the manuscript's later exact collar formulas take one of two forms:

1. literal exactness when the reference state is exact Markov and EC-aligned on the relevant collar; or
2. a controlled collar family for which $\delta^{\mathrm{M,EC}}_{\mathcal{F};A:B:D}(\varepsilon_\delta) \to 0$, with the constructive Fawzi–Renner remainder and the carried operator defect $\mathfrak{D}_{A:B:D}$ kept explicit at finite stage.

Theorem 4.1d (Finite-stage modular-defect propagation and dimension quarantine). For any fixed downstream branch calculation that uses finitely many collar or strip modular-additivity identities, replacing each exact identity by its controlled finite-stage form changes every bounded downstream modular observable O by at most

$$\mathcal{P}_O\Big(\{r_{\mathrm{FR}}(\varepsilon_j)\}, \{\delta^{\mathrm{M,EC}}_{\mathcal{F}_j;j}(\varepsilon_j)\}, \{\eta^{\mathrm{reg}}_j\}\Big),$$

where $\mathcal{P}_O \to 0$ as all listed errors vanish. The polynomial-continuity modulus depends only on the declared fixed collar models, support-visible dimensions, bounded observable class, faithful floors or regularization schedules, and bounded modular-parameter intervals. No BW/Lorentz, null-modular, or Einstein scaling step uses a dimension-free trace-norm stability theorem from small CMI directly to exact Markov normal form.

2.2.2 Theorem: BW_{S^2} on the extracted geometric subnet

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a fixed-cutoff matrix-algebra identity and not the slogan "finite cells imply Lorentz invariance." Its target is the support-visible refinement-limit geometric cap pair $(\mathcal{A}^{\mathrm{geo,sv}}_\infty(C), \omega^{\mathrm{geo},C}_\infty)$. The branch theorem is the conditional implication

$$\mathsf{FiniteCapBWCertificate} \;+\; \mathsf{ModularAlgebraStateData} \Longrightarrow \sigma_t^{\omega^{\mathrm{geo},C}_\infty} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

The finite cap-normal certificate consists of cap-normal density and nondegeneracy, BW framing, support-order faithfulness, geometric support-flow group and continuity control, held-out oriented cross-ratio convergence, independently normalized geometric 2π-KMS comparison, and wrong-normalization separation. The independent multiresolution algebra-state comparison package supplies the algebra-state reference tower, common comparison maps, cap-family-uniform regularized modular transport, mixed-GNS convergence, and modular support covariance. The support-visible theorem below uses both inputs; it does not require a type-I continuum algebra or a full-algebra unregularized common spectral floor. Bare finite consensus implies neither input.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta^{\mathrm{M}}_\delta := 4\lambda_*^{-1}\, \delta^{\mathrm{M,EC}}_{\mathcal{F}_\delta;A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound for comparing modular Hamiltonians. All exact collar formulas in this subsection are therefore to be read either literally at EC-aligned exact Markovity or along a controlled collar family satisfying

$$\delta^{\mathrm{M,EC}}_{\mathcal{F}_\delta;A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η^{M}_δ carried explicitly.

For each cap C, let $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

Theorem 4.2 (Finite BW_{S^2} cap-net scaling). For a nondegenerate BW-framed cap $\widehat{C} = (C, n_C, p^-_C, p^+_C)$, consider an OPH realization on an observer-supporting refinement branch that satisfies Axioms 1–3 and carries the named fixed-cutoff regulator, collar, consensus, and optimizer-pushforward interfaces. Assume the finite cap-normal BW certificate above and, independently, the complete multiresolution reference tower and algebra-state comparison package of Theorems 2.6c–2.6f on the same branch. The support-visible geometric cap net then admits a weak-$*$/GNS scaling-limit cap pair $(\mathcal{A}^{\mathrm{geo,sv}}_\infty(C), \omega^{\mathrm{geo},C}_\infty)$, whose modular automorphism group is

$$\sigma_t^{\omega^{\mathrm{geo},C}_\infty} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

At finite regulator stage the nonadditive part of

$$K^{(\delta)}_C := -\log \rho^{(\delta)}_C$$

is confined to the shrinking collar up to the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$, the EC-aligned fixed-collar replacement bound η^{M}_δ, and the regularized support-visible modular transport bound. The support-visible modular limit is independent of the EC-aligned exact-Markov replacement sequence after projective compatibility and GNS quotienting, and support-readable modular covariance turns the limiting modular group into a cap-local support map before the certificate's framed-cap and oriented cross-ratio rigidity clauses are applied. No separate cap-isotropy/SO(2)-equivariance selector, finite-cell Lorentz-invariance premise, or full-algebra unregularized common floor is used. Record/pointer and interface-inert auxiliary registers are outside the extracted geometric subnet. If the scaling-limit cap algebra is type I, the same automorphism identity may be represented by

$$K_C = 2\pi B_C + Z_C, \qquad Z_C \in Z(\mathcal{A}(C))_{\mathrm{sa}};$$

otherwise the automorphism identity is the full statement and the action is outer. Consequently, under the paired same-branch finite cap-normal and complete multiresolution algebra-state comparison inputs, the transported regularized modular action on every fixed support-visible geometric local algebra converges uniformly on compact modular-parameter intervals to the geometric cap-dilation action. Without both inputs, this theorem makes no BW or finite-to-continuum modular claim. Even with them, it does not assert compact-time convergence of finite-regulator modular groups beyond the declared support-visible certificate.

Proof. Markov locality localizes the modular defect to the shrinking collar, with carried finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η^{M}_δ. Proposition 2.6a shows why a full-algebra unregularized floor is unavailable in general. Theorem 2.6b supplies the matrix-element replacement on fixed collar models, and Theorem 2.6f supplies the stronger compact-time group estimate only for transported regularized local actions with the stated reference-tower certificate. Projective replacement compatibility and the support-visible convergence theorem ensure that the emitted modular limit is independent

of the chosen EC-aligned exact-Markov comparison family. Support-visible cap-pair extraction on the local GNS support quotient uses weak-$*$ compactness of finite-stage state spaces and the consensus refinement mechanism to supply local limit states, then applies GNS construction to obtain the scaling-limit cap pair. Support-readable modular covariance reads the modular group as a cap-local support map. The finite cap-normal certificate supplies support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and compact-time equicontinuity, so the support flow is the framed cap-preserving conformal flow up to normalization:

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}$$

for some $\kappa_C > 0$, with the certificate-selected frame understood in λ_C. The certificate then passes the independently normalized geometric 2π-KMS condition to $s \mapsto \alpha_{\lambda_{\widehat{C}}(s)}$. KMS uniqueness, together with wrong-β separation and nontriviality, fixes $\kappa_C = 2\pi$. Thus

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit algebra is type I, this automorphism identity is represented by $K_C = 2\pi B_C + Z_C$ with Z_C central; otherwise the automorphism identity is the full statement and the action is outer. QED.

Definition 4.2a (BW-branch observer-relative modular ordering). On the branch satisfying the hypotheses of Theorem 4.2, the modular automorphism parameter t of the extracted cap pair $(\mathcal{A}_\infty^{\mathrm{geo,sv}}(C), \omega_\infty^{\mathrm{geo},C})$ supplies a dimensionless ordering for that observer's accessible algebra-state pair. Physical time requires an observer-readable transition, event correspondence, and calibrated clock instrument. No claim about arbitrary operational clocks, global time, or the full problem of time follows from the geometric modular-flow theorem alone.

Scope note. The theorem fixes the 2π normalization and the carried refinement limit on the support-visible extracted geometric cap pair under the paired finite cap-normal and independently complete same-branch multiresolution algebra-state comparison inputs. The UV-side scaffold is the realized transported geometric cap-local system, regularized support-visible modular transport on fixed local collars, support-visible cap-pair extraction on the local GNS support quotient, support-readable modular covariance, BW framing, oriented cross-ratio rigidity, and geometric KMS normalization. No separate cap-isotropy input is used, and the conformal support-map clause is the explicit scaling-limit branch condition rather than a finite-regulator Lorentz premise. Compact-time convergence of finite-regulator modular groups is available only through the transported-local certificate of Theorem 2.6f; a black-box conformal-net BW reconstruction would be an additional validation route, not a hidden premise.

2.2.3 Theorem: cap-normal H^3 reconstruction

Theorem 4.3 (Cap-normal H^3 reconstruction, imported from the spacetime and Einstein derivation). On the support-visible round-cap BW branch, write the celestial null section as

$$q(\Omega) = (1, \Omega), \qquad \Omega \in S^2.$$

For a nondegenerate oriented round cap

$$C(\mathbf{c}, \alpha) = \{\Omega : \mathbf{c} \cdot \Omega \geq \cos\alpha\}, \qquad 0 < \alpha < \pi,$$

define

$$n_C = (\cot\alpha, \csc\alpha\, \mathbf{c}).$$

Then, for the time-first Lorentz form $\eta = -dt^2 + d\mathbf{x}^2$,

$$\eta(n_C, n_C) = 1, \qquad \eta(n_C, q(\Omega)) = \frac{\mathbf{c} \cdot \Omega - \cos\alpha}{\sin\alpha}.$$

Thus $\eta(n_C, q(\Omega)) = 0$ exactly on ∂C, with positive sign in the cap interior and negative sign outside. For $g \in \mathrm{Conf}^+(S^2)$ with Lorentz representative Λ_g,

$$\Lambda_g q(\Omega) = \omega_g(\Omega) q(g\Omega), \qquad n_{gC} = \Lambda_g n_C.$$

Consequently

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1),$$

oriented round caps form the de Sitter normal space

$$\mathrm{Cap}^{\mathrm{or}}_{\mathrm{round}}(S^2) \cong dS_3^{\mathrm{cap}} \cong \mathrm{SO}^+(3,1)/\mathrm{SO}^+(2,1),$$

and future unit timelike observer frames form the distinct hyperbolic space

$$H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3), \qquad \dim H^3 = 3.$$

Each cap normal determines the geodesic half-space

$$\mathcal{H}_C^+ = \{u \in H^3 : \eta(n_C, u) \geq 0\},$$

not a preferred observer point.

Proof. This is the cap-normal reconstruction theorem in The spacetime and Einstein paper, which supplies the projective-null-cone model, the cap-normal bijection, Lorentz equivariance, the H^3 observer-frame hyperboloid, the ideal-boundary identification, and the cap/half-space duality. The BW theorem supplies the cap-preserving conformal subgroup and its 2π modular normalization; the cap-normal theorem supplies the explicit projective Lorentz representation and does not rederive BW from bare finite consensus. QED.

Corollary 4.3a (Frame-space boundary). The theorem proves the exact canonical Lorentz/H^3 observer-frame hyperboloid on the stated branch. Its points are future unit timelike frames, not event positions. It does not populate a physical spacetime, construct a chart-blind neutral bulk, derive a physical curvature radius R_H, produce stress-energy, or close the Einstein branch-entry umbrella. Record-conditioned cap responses can estimate frame data under the certificate imported next; event location, object families, neutral bulk, scale, stress, and Einstein branch entry are separate receipts. Theorems 4.3c–4.3h below import The spacetime and Einstein paper's geometry-producer, null-net-standardness, source-causal precursor and continuum, stress, entropy-bridge, and composed branch-entry packets. Their composition is conditional on one source-derived common-domain tower with certified cofinal tails, universal coupling, a vacuum reference, and independent physical scale readouts. No source-derived tower carrying that complete packet is constructed here.

Theorem 4.3b (Conditional record-conditioned H^3 frame estimate). Fix a quotient-visible record token i and clock slice t. On the frame space of Corollary 4.3, suppose its record-conditioned modular cap responses

$$R_i(C,t,O) = \omega_{i,O}\Big(\sigma_t^{C,O}(M_{C,0,O})\Big), \qquad \omega_{i,O}(A) = \frac{\omega_O(P_{i,O} A P_{i,O})}{\omega_O(P_{i,O})},$$

admit the calibrated frame-local factorization

$$y_{i,j} = F_j(X_i(t)) + e_{i,j}, \qquad F_j(X) = \psi_j\left(\frac{\eta(X, n_j)}{R_H}\right),$$

on a compact frame region $\Omega \subset H^3_{R_H}$. Assume the finite response map has quantitative observability

$$\alpha \bar{d}(X, Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X, Y), \qquad \alpha > 0,$$

and the total error obeys $|e_i|_W \le \sigma_i$. If $\mathcal{N}_\varepsilon$ is an ε-net of Ω and $\widehat{X}_i(t)$ is a residual minimizer up to tolerance τ_i, then the conditioned feasible frame support obeys the conservative enclosure

$$S_i(t) \subseteq B_H\left(\widehat{X}_i(t), R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i + \frac{1}{\alpha}\tau_i\right]\right).$$

The theorem does not assert that every point of this output ball fits the data. Exact continuous data have a unique minimizer. Finite noisy data produce the displayed frame ball; a unique finite frame value is licensed only when the certified residual gap $\Delta_{\mathrm{loc},i}(t) > 0$. If that gap fails, the fail-closed protocol returns AMBIGUOUS, meaning that no unique frame value is certified; it is not a proof that the mathematical data have multiple minimizers. This theorem does not locate an event, populate the event base, resolve unlabeled mixtures, identify particle species, produce stress-energy, or establish chart-blind neutral bulk.

Theorem 4.3c (Quotient-intrinsic geometry producer and dimension-selection boundary). On a refinement tower of repaired quotient normal forms, form at each level the support-visible incidence complex: vertices are the quotient-visible patch classes and simplices include every jointly supported patch set, including all higher simplices. A filled tetrahedron is not its spherical boundary. Each level is computable from its normal form and is invariant under repair schedule and gauge. Refinement generally changes the literal complex. Topology preservation requires a certified subdivision or homeomorphism: a complete midpoint subdivision has such a realization, whereas its endpoint-valued simplicial projection is only homotopic to it. Simplicial maps or CPTP transport alone do not establish topology preservation. If this compatible tower carries the computable receipts (i) spherical incidence (connected closed orientable combinatorial surface with Euler characteristic 2), (ii) disk/mesh nondegeneracy, (iii) modular cross-ratio Cauchy convergence, and (iv) independently normalized geometric 2π-KMS comparison with wrong-normalization separation, then the refinement limit produces the round S^2, its conformal structure, the oriented round-cap family with cap normals, and every clause of the finite cap-normal BW certificate, with quantitative error envelopes. Conversely, six finite repair systems with one rewrite signature (three patches in the seed record, one repair step) have incidence complexes realizing S^2, T^2, the real projective plane, the Klein bottle, the 2-skeleton of $\partial\Delta^4$, and a wedge of two spheres, and the scalar KMS receipt admits any modular temperature β: repair confluence alone does not select topology, orientability, dimension, or modular normalization. On a connected closed surface the orientability clause of (i) follows from Euler characteristic 2, so the two nonorientable systems also fail (i) at that clause. Orientation framing enters as a separately typed receipt: the two coherent orientations of the S^2 system share every confluence-level datum, and a mirror pair of embeddings carries complex-conjugate cross-ratios. Finite cap reconstruction also requires a resolved spacelike boundary-circle plane and an inside witness or equivalent oriented boundary data to distinguish the two sides, including major caps. Cross-ratios determine only conformal data up to Möbius transformations; a declared observer frame is needed for angular radii, and a physical scale remains separate. The MGNS state enters as a separately typed receipt, and no countermodel artifact is exhibited for it. The screen S^2 is therefore *produced on the*

receipt branch; the receipt set is one sufficient certificate, but no semantic-minimality theorem or full-tower countermodel matrix is supplied. Producers that reach an oriented S^2 support limit by another computable function of the normal forms, such as the source-derived incidence nerve of the carrier-support bridge, are untouched by these countermodels. (Spacetime and Einstein paper, geometry-producer packet; machine receipts accompany the public geometry code.)

Theorem 4.3d (Null-net standardness, derived translations, and the four-translation assembly). On the producer branch, the null half-line/interval blow-up net at a produced cap boundary point lives on one common GNS space, and: (i) the reference vector is cyclic and separating for every half-line used, given faithful finite-stage states on the interior MaxEnt branch, the mixed-GNS Cauchy clause, and a cofinal cyclicity receipt uniform over every half-line in the construction; boundary MaxEnt solutions require a support-restricted treatment and do not supply faithfulness by themselves; (ii) the produced geometric cap flow derives the half-sided covariance relation without assuming translation unitaries, but the Borchers–Wiesbrock step additionally consumes standardness for every nested pair—cyclicity and separation for both half-line algebras and cyclicity for their relative commutant; interval cyclicity is therefore a registered standard-relative-commutant receipt, not a consequence of one half-line receipt; (iii) weak additivity of the null net is a separate cofinal receipt used for the generated net, rather than a consequence of finite-stage faithfulness; (iv) on the exact-Markov/central-interface branch, with a specified endpoint-to-cell map and a uniform local norm/site-density bound, the reduced interval modular Hamiltonians split into local terms plus uniformly bounded interface blocks (endpoint-Lipschitz), while an explicit four-qubit nearest-neighbour Gibbs witness shows finite-range locality alone does *not* imply this (machine-verified); (v) on the standard half-sided-inclusion and modular-intersection receipt branches, the derived positive null translations and modular dilations generate Möbius covariance and the projective bounded-interval action; and (vi) the per-direction positive generators commute, transform with the null-vector weight under the produced Lorentz action, satisfy the dependent-family linearity relation, and assemble into one four-parameter translation representation with joint spectrum in the closed future cone: a positive-energy Poincaré representation. (Spacetime and Einstein paper, null-net standardness packet.)

Theorem 4.3e (Source-derived finite $1+3$ precursor and effective-continuum criterion). An authenticated semantic-event log supplies its finite event carrier and strict generated read-from order. Its canonical source height is computed from that parent relation: roots have height zero and every nonroot has one plus the maximum height of its direct parents. Lean proves strict increase on parent edges and generated ancestry, bounds every authenticated direct-parent chain ending at an event by that event's height, and constructs an attaining chain. Thus source height is exactly the longest authenticated-parent-chain length. Let V_{src} be the exact rank-three positive source Gram quotient. Then $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ has real dimension four, and $Q(t,x) = t^2 - g_{\mathrm{src}}(x,x)$ has inertia $(1,3)$. Every source-unit direction gives the future-null vector $(1,u)$, realizing the exact equivalences $\mathsf{SourceUnitDirection} \simeq S^2 \simeq \mathsf{FutureNullRay}$. For a supplied spatial readback $x(e)$ and positive height scale τ, the placement $F(e) = (\tau h_L(e), x(e))$ carries authenticated parenthood, and hence generated ancestry, into the future cone whenever every parent displacement obeys the source-metric speed bound. If every future-cone comparison between placed events is supported by reflexive generated ancestry, order and cone agree exactly. Lean derives this support from two concrete finite checks: spatial readback separates each equal-height layer, and every increasing-height pair unsupported by ancestry obeys the strict spacelike inequality. Antisymmetry then derives injectivity, every source interval agrees with the corresponding cone interval on the placed image, and composition with the source-carrier equivalence gives a separated one-chart finite causal-chart interface. This global finite chart is not an open manifold atlas. Thus the ambient target-carrier dimension and Lorentz signature are constructed without a freely

declared rank-four chart, fitted cone, or supplied placement rank; the intrinsic poset dimension, spatial placement, speed/converse conditions, physical interpretation, and clock remain separate data. The exact-embedding route to a smooth effective manifold uses one cofinal source-selected family carrying physical event/link identification, exact two-way cone/order agreement, refinement compatibility, dense and isotropic source-S^2 link directions, a calibrated $\#I/\rho \to \mathrm{Vol}(I)$ law, independent dimension/manifoldlikeness and height/profile tests, stable thickened-antichain topology or homology, and a specified distinguishing Lorentzian limit with uniqueness control. On that certificate, causal order fixes conformal geometry and the calibrated measure fixes scale. Smooth Einstein promotion additionally requires either same-family tensor-curvature reconstruction converging to the Einstein tensor or the independently premised continuum small-ball/null-balance identification, together with physical stress, Ward/Bianchi, coupling, and remainder data on the same family. Scalar-curvature convergence alone is diagnostic. No physical instance of this certificate is supplied. (Spacetime and Einstein paper; Lean theorems and independently verified simulation evidence.)

A source-record order and count limit. In the continuous rank-three carrier, conservative Fibonacci records give q^3 distinct spatial sites. Repeat this population at every layer and let each event read all preceding-layer sites inside a radius a_q, including itself. With spatial fill distance $h_q = o(a_q)$, $a_q \to 0$, and model duration a_q/c, actual read paths approach the speed-c causal cone. Equal-volume cell assignments with vanishing displacement error make normalized raw counts converge to $dt\, d^3x$; the ordering fraction in interior timelike diamonds tends to $1/10$. This is a flat effective spacetime theorem for a specified population, read law and model clock (Spacetime and Einstein paper). It supplies neither their native dynamical selection nor their identification with a matter action or laboratory clock.

The same histories admit a retrospective clock. For intervals approaching fixed interior timelike diamonds, the fourth root of their inclusive count ratio converges to their proper-time ratio. A reference interval fixes the unit, and complete ancestry access supplies the count. This readout needs no oscillator or numerical timestamps. Operational covariance gives a separate cone criterion: a nonzero closed convex pointed cone invariant under the source icosahedral rotations and all boosts along one axis must be one of the two Lorentz cones. A time orientation selects the future cone. The covariance must act on possible influences and readouts. Algebraic direction maps alone do not establish that operational premise[17].

For Einstein promotion there are two distinct sufficient routes at this step: an explicit same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or a separately premised continuum small-ball/null-balance identification. Convergence of a scalar-curvature estimator is an independent diagnostic and is insufficient by itself.

Every finite semantic log also admits an explicit auxiliary separated one-way placement in the source (1+3) carrier. Enumerating events along one source axis and taking height scale one larger than the event count makes the event map injective, sends generated precedence into the future cone, and makes distinct same-height events spacelike. The enumeration is arbitrary and generally introduces cone comparisons unsupported by ancestry; it is not an order-reflecting, source-selected, or physical placement.

The provenance grammar has the complementary exact expressivity theorem. Every finite decidable irreflexive transitive relation R compiles into an abstract authenticated semantic log whose direct parent relation and generated strict order are both exactly R; predecessor cardinality supplies the rank. The generic parent relation is all of R rather than its Hasse reduction alone. The relation is supplied to the compiler, and the per-event snapshots are not proved to arise from one threaded executor. This is a universal finite-poset representation result, not a dynamics, manifoldlikeness, or continuum theorem.

An exact four-event Boolean diamond inhabits the order-faithful finite interface: its parent

edges are null and its two independent branches are spacelike. This is a machine-checked non-chain causal-set-like witness, not a physical continuum or an empirical dimension measurement.

The finite provenance order is causal-set-like rather than a physical causal set. Lean packages reflexive authenticated ancestry as a partial order and proves every interval finite at each finite cutoff, exactly realizing the abstract order-theoretic causal-set axioms. Standard causal set theory starts from a locally finite order interpreted as proto-causality [1, 2]. The exact-embedding continuum route requires an order embedding whose image has the approximately Poisson count density needed for a causal-set faithful embedding, together with manifoldlikeness and dimension [3, 4, 5]. Under the relevant distinguishing hypotheses, continuum causal order fixes the conformal geometry [14, 15]; the volume datum fixes the conformal factor. The Lorentz-invariant causal-set approximation uses Poisson sprinkling [13]; the OPH commit log is not a Poisson sprinkling, and its event count is not a spacetime volume. The finite source-causal theorem supplies the (1+3) precursor, not a continuum geometry. It derives neither a physical order embedding nor the count-to-volume density law that together constitute a causal-set faithful embedding. It also establishes neither continuum existence nor the causal-set Hauptvermutung from the finite OPH order. The finite causal/Lorentz precursor instead constructs the ambient carrier from an independent real axis and the exact rank-three source carrier. A spatial event readback, canonical source height, and positive `timeScale` then give a concrete placement; a local speed bound proves one-way cone compatibility, while equal-height spatial separation plus strict spacelikeness of unsupported increasing-height pairs supply converse cone support and exact two-way agreement. Antisymmetry then derives injectivity, while the two-way equivalence preserves every source interval exactly on the placed image. The event-count measure is distinct from public capacity N. The finite join does not constitute spacetime. Physical event/link identification, refinement, manifoldlikeness, topology, independent dimension tests, count–volume calibration, and uniqueness are required before the continuum can be read as emergent rather than fundamental. The exact source-direction bridge gives $\mathsf{SourceUnitDirection} \simeq S^2 \simeq \mathsf{FutureNullRay}$ for the rank-three source Gram quotient and fixes the finite (1+3) carrier and cone after adjoining an independent real axis. Positive `timeScale` times canonical source height enters only through event placement; supplied event-local Lorentz frames preserve the rest metric. This is an algebraic direction and causal-carrier result, not a physical sky or signal theorem. The effective reading additionally requires the physically faithful images of authenticated local read-from links to cover this S^2 densely and isotropically; that coverage is not supplied. The measure calibration separately requires $\#I/\rho \to \mathrm{Vol}(I)$. Refinement must preserve order, directions, and count density while soldering rank-three neighborhoods with $|B_{\mathrm{car}}(r)| \asymp r^3$ over a scaling window. This is compatibility of the four inputs, not a fifth one. Neither the local rank-three carrier nor the seeded global finite wiring determines event-base dimension.

The theorem-status composition is exact at the finite interface and conditional thereafter. The source-native causal-chart interface uses semantic commits as the event carrier and the generated reflexive order definitionally; it uses no separate population map or duplicate precedence field. The finite compatibility theorem constructs the ambient (1+3) carrier, proves its rank and source-unit/null identities, and proves the one-way or two-way cone result under the explicit placement conditions. Supplied chart transports retain their bounded algebraic covariance role. No physical placement, count-density field, or refinement family is constructed. Physical faithfulness, link-direction coverage, manifoldlikeness, topology, volume, and refinement conditions remain independently testable.

The exact same-event-type source-order Einstein theorem, in a separately supplied $3+1$ tensor interface, replaces an all-null balance hypothesis by nine supplied balances on fixed algebraic source-direction representatives. Given supplied symmetric geometry/stress fields indexed by the finite event type, four discrete step maps, Ward/Bianchi identities, step connectedness, a base event,

and an algebraic coupling, exact nine-direction tomography derives all-null balance, the pointwise metric ambiguity, its constancy, and the Einstein-form relation. The minimal premise bundle contains no normalization, vacuum-calibration, Newton-constant, or universal-source field. The generated order and the independently derived $1 + 3$ carrier use the same event type, but no provenance link selects a tensor field, step, or balance. The theorem does not identify tensor-coordinate differences with those of the constructed carrier or identify the supplied fields with smooth curvature or physical stress-energy. That reading requires the source-causal continuum, count–volume, convergent curvature/stress, small-ball, coupling, and scale receipts on one family.

Proposition 4.3e0 (Bounded algebraic event-frame soldering). Let raw record representatives carry an actual coincidence setoid and a readback into the intrinsic Hermitian Lorentz module that is constant on coincidence classes. The readback descends uniquely. Given one time-oriented affine Lorentz overlap cocycle and, separately, a supplied chart-coordinate family satisfying its affine overlap law, coordinate displacements transform through the cocycle's linear part, their Lorentz intervals are chart independent, and the reverse transition and triple-overlap laws follow from the same cocycle. Future-null displacements descend to covariant celestial directions. One base-chart future-unit frame selection induces compatible frames in every chart, and Lorentz-orthogonal rest projections transport by exact rest-metric isometries. The rank-three source Gram quotient is linearly and isometrically equivalent to the standard internal rest fiber, as a candidate local readout. The standalone quotient-descent theorem does not construct this atlas coordinate family or identify the descended readback with it. A finite two-event/two-chart control has nonzero affine translation and a nonzero future-null displacement. A separate three-record control proves that a reflexive symmetric pairwise-overlap relation need not be transitive, so it cannot replace the coincidence setoid. This bounded algebraic adapter assumes the supplied chart-coordinate overlap law, coincidence setoid, separation, physical cone attachment, refinement naturality, and operational clock. It can be used after a source-causal continuum certificate has selected an effective atlas, but it does not construct the event carrier and does not supply physical causality, manifoldlikeness, topology, count–volume calibration, or smooth curvature.

Proposition 4.3ea (Finite signed-seam operator realization). A declared sign-twisted derivative on a finite seam complex with unit-counting measure gives a positive semidefinite kinetic operator. Its kernel consists of parallel sections; on a connected frustrated component that kernel is zero. The finite realization with 8,662 sites and 11,816 seams has 38 frustrated triangles. Exact boundary-rank elimination gives lift-ambiguity rank 3,117, and deterministic integer sections check adjoint, kinetic, covariance, gluing and boundary identities. The separate rank-three response and declared fifteen-state charge assumptions define a conditional 45-dimensional internal factor. The declared operator $D_\sigma \otimes I_{45}$ retains the gap of D_σ with multiplicity 45. These are properties of the specified finite operator and matter table. They select neither a spacetime chart nor a physical matter action, and do not identify the domain with the twelve-port Spin construction. The microphysics paper gives the finite algebra and independent replay [28].

Theorem 4.3f (Finite null tomography and conditional local-stress upgrade). A symmetric bilinear form on $(\mathbb{R}^4, \eta)$ is determined by its null-null values up to a multiple of η, and a directional scalar family arises from a tensor iff it satisfies the finite dependent-family linearity relations; nine null directions with resolved rank nine reconstruct the trace-free part. Nine readings always interpolate and therefore do not test tensoriality. Additional dependent directions test consistency within a declared total measurement and numerical error budget; they do not constrain unsampled rays without a family-level premise. For a full-rank design using rays normalized to $k^0 = 1$ and a Frobenius-orthonormal trace-free basis, the tensor Frobenius error from charge noise is bounded by $\delta/s_{\min}$, separately from geometry and solve errors; floating-point singular values are estimates, not interval certificates. The released code verifies this finite linear algebra on declared

and synthetic charge families. Values violating a dependent-family relation admit no exact rank-two reconstruction, and scalar collar CMI alone cannot define a rank-two source.

The physical upgrade is conditional. If one common source-derived endpoint-charge family on the event region satisfies the linearity relations, admits the declared same-source identification with local null-stress charges, carries one common normalization and Poincaré-covariant local transformation law, and satisfies the weak Ward identity, then tomography defines a symmetric semiclassical $\langle T_{ab}\rangle$, with null projections equal to those charges and ambiguity ϕg_{ab} plus the declared improvement terms. Locality, covariance, the charge/stress identification, and weak conservation are separate weak-Ward and null-stress matching supplies; they are not consequences of finite tomography or of the synthetic reconstruction receipt. Universal coupling to the geometric branch is a further named premise. (Spacetime and Einstein paper, stress packet; finite tomography receipts accompany the public geometry code.)

Theorem 4.3g (Repaired entropy bridge and absolute Einstein equation). On the central-interface branch, write the state as a direct sum of bulk states tensored with maximally mixed edge factors. Its entropy splits exactly into $S_{\mathrm{bulk}} = H(p) + \sum_\alpha p_\alpha S(\rho_\alpha^{\mathrm{bulk}})$ and $S_{\mathrm{edge}} = \sum_\alpha p_\alpha \log d_\alpha$. At a faithful reference σ, the tangent first law is $\delta S = 2\pi\delta\langle B_C\rangle + \delta\langle Z_C\rangle$; for a finite change at fixed $K_\sigma = 2\pi B_C + Z_C = -\log\sigma$, the exact identity is $S(\rho) - S(\sigma) = 2\pi\Delta\langle B_C\rangle + \Delta\langle Z_C\rangle - D(\rho\|\sigma)$, with $Z_C = \sum_\alpha (\log d_\alpha) P_\alpha$. Hence $\delta S_{\mathrm{bulk}} = 2\pi\delta\langle B_C\rangle$ and $\delta S_{\mathrm{edge}} = \delta\langle Z_C\rangle$ throughout the declared central-interface class, including variations of the sector weights. The supplied nonconstant Gibbs family has the analytic MaxEnt envelope identity $dS/dt = \lambda$; setting $\lambda = 2\pi$ and identifying t with physical energy require the declared branch normalization, giving the Clausius balance $\delta S_{\mathrm{edge}} = \lambda\delta t - 2\pi\delta\langle B_C\rangle$ on the coupled variation class. Reaching the geometric balance additionally requires the declared fixed-cap generalized-entropy stationarity interface, the conditional same-source local-stress and weak-Ward bridge, the continuum diamond kernel, and the fixed-volume small-ball area identity. One common scaling family must carry the conditional-mixing, boundary-count, fixed-collar, sharp-rate, and $o(\ell^4)$ remainder bounds. Coverage of all local timelike directions follows from population plus the produced Lorentz orbit. The vacuum-reference premise fixes the first-variation integration constant. Independent scale readouts with trivial positive-rescaling stabilizer identify the structural coupling. Under all of these premises, $G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$ per connected component. Without the vacuum reference, the result fixes the residue tensor only up to one covariantly constant metric term. (Spacetime and Einstein paper, entropy-bridge and small-diamond packets.)

Theorem 4.3h (Conditional Einstein branch composition). Bare finite consensus is not Einstein-complete: the same finite-consensus reduct admits extensions in which the Einstein equation holds and extensions in which it fails. The valid gravity result is conditional. Let one source-derived repaired quotient tower carry deterministic geometry, modular, causal-order, stress, entropy, and scale readouts on a common domain, with commuting refinement diagrams. Assume the finite source-causal precursor, geometry and null-net receipts; cap-interior modular data; one physical source-causal continuum certificate on a cofinal family with either same-family tensor-curvature reconstruction or the independently premised continuum small-ball/null-balance identification, for which every relevant remainder is $o(\ell^4)$; the universal-coupling and vacuum-reference receipts; and two independent scale readouts with trivial positive-rescaling stabilizer. The stress-tomography, entropy, curvature-identification, timelike-polarization, Ward, and Bianchi results then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$$

on each connected event component. At the curvature-identification step, the causal-set route requires a same-family tensor-curvature reconstruction converging to G_{ab}; the alternative continuum

route instead premises the small-ball/null-balance identification independently. Convergence of a scalar-curvature estimator alone establishes neither route. The exact finite layer includes the nine-direction tomography frame with determinant 8192/27 and the entropy and MaxEnt identities. Conditional on the continuum diamond-kernel formula and the smooth fixed-volume small-ball area identity, exact arithmetic gives the coefficients $8\pi^2\ell^4/15$ and $-4\pi\ell^4/15$; the continuum formulas themselves remain explicit analytic premises. The finite operator of Proposition 4.3ea does not supply the geometric modular, stress, entropy, scale, or asymptotic data of this theorem. No source tower satisfying the joint continuum antecedent is constructed here, and no semantic-minimality theorem for that antecedent is claimed. (Spacetime and Einstein paper, composed branch-entry theorem.)

The register surface of this composition is inhabited. For every declared repair law and every horizon record over it, an explicit value of the typed premise bundle is machine-checked: its modular flow supply is the finite thermodynamic first-law datum on the simplex tangent space at the declared reference, with the boost and central charges of the record, so the first-law clause of the composed conclusion at this value is the first-order entanglement first law of the horizon-thermality surface on the same variation space. The other supplies are synthetic witnesses at the demo tier: the two-patch consensus tower with six typed readouts extending the committed event and geometry fragment, a sixteen-point four-direction torus chart with a nonconstant divergence-free stress field and the geometry defined as $\kappa T_{ab} + \Lambda\eta_{ab}$, a strict audit manifest evaluated on that chart, a shrinking family with ℓ^5 error channels, and small-ball reals solved from the three small-ball identities. The Einstein clause, the small-ball arithmetic, the universal-source clause, and the vacuum reference hold at this value by definition of those witnesses and select nothing. A nonvacuity receipt exercises the clauses the data can fail on nonzero data, and a counterexample shows that the split of the horizon record is the load-bearing input: with a uniform reference and zero central charge in place of the split, the same boost charge fails the modular-generator clause on the same variation. The value certifies joint satisfiability of the premises and places the composed branch on the thermodynamic variation space; it attaches no physical metric, stress tensor, horizon, temperature, or Newton constant, and realized nonemptiness on a source-generated tower is not established.

2.3 Gravity from Fixed-Cap Generalized-Entropy Stationarity

2.3.1 Cap first law

Section 4.2 fixes the cap modular statement first at the automorphism level. On the extracted geometric subnet, the scaling-limit cap modular flow is geometric with the standard 2π normalization,

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the realized scaling-limit cap algebra happens to be type I, one may choose a density matrix $\rho_C^{\omega_\infty^{\mathrm{geo},C}}$ and modular Hamiltonian

$$K_C := -\log \rho_C^{\omega_\infty^{\mathrm{geo},C}},$$

and then for a perturbation $\rho(\varepsilon)$ with $\rho(0) = \omega_\infty^{\mathrm{geo},C}$ the first-law identity reads

$$\delta S_C = \delta\langle K_C\rangle = 2\pi\,\delta\langle B_C\rangle.$$

This displayed variation is the special type-I/fixed-central-sector form of $K_C = 2\pi B_C + Z_C$, where the central term has no first-order contribution on the chosen sector. In the generic continuum/non-type-I case, Section 4.2 does not supply an inner operator identity; the theorem there is the geometric modular automorphism statement, so the cap first-law formula is restricted to this special realization.

2.3.2 Null-surface modular bridge

This subsection extends the fixed-cutoff weak-tail-generator boundary to the derived positive null-translation stage, the exact endpoint-derivative identity for its half-line generator, and a conditional local-charge reading. At fixed cutoff one first transfers cut-center data to regulated null strips, then imposes the extra inherited split condition needed for the spatial-collar-type tensor decomposition, obtains exact or controlled strip additivity on one inherited strip model, and derives a weak tail generator for the renormalized half-line family. On the scaling-limit geometric-cap branch of Theorem 4.2, the null half-line blow-up net then inherits geometric dilation and therefore the half-sided covariance relation. Together with the separately supplied common-vector and relative-commutant standardness receipts, this is a standard half-sided modular inclusion, so Borchers–Wiesbrock supplies an explicit positive null-translation generator on its Stone domain. Bounded-interval formulas are discharged by the E0.5 affine/projective kernel below, and the tensor upgrade carries the null-invisible metric ambiguity. Accordingly the null modular bridge consists of:

- null-cut center transfer yielding the central cut-label decomposition, together with the extra inherited split condition needed for the spatial-collar-type tensor decomposition;
- exact or controlled four-term strip additivity on one fixed inherited strip model;
- endpoint-Lipschitz control for the renormalized half-line family and the resulting weak tail generator;
- the geometrically derived half-sided relation plus the registered standardness receipts on the null half-line blow-up net, and the resulting Borchers positive translation generator.

The later density-upgrade template is kept below as an explicit downstream template rather than as a fixed-cutoff theorem proved in this subsection. Lemma 5.2f makes the positive null-translation generator itself explicit, and Theorem 5.2g records the Borchers unitary group, positivity and self-adjointness on the Stone domain, the corrected affine commutator relation, and the exact half-line modular-Hamiltonian identity $K_a = K_0 - 2\pi a P_\Omega$. Identifying that endpoint derivative with a local null-stress charge additionally consumes an effective-locality and same-source matching premise. Bounded-interval formulas are downstream.

Proposition 5.2a (Null-cut center transfer and inherited split). Fix a regulated null tripartition

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts $\Gamma_- := \{v = v_2\}$ and $\Gamma_+ := \{v = v_3\}$. Under the fixed-cutoff realized presentation, assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data used in the spatial collar branch, so that in a compatible type-I regulator presentation one has

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W_{\alpha_-}^* \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W_{\alpha_+}^* \otimes \mathcal{H}_{i_+^{\alpha_+}},$$

where opposite sides of each cut carry inverse transport. If the strip algebra is the commutant of the transported cut actions, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+}.$$

If, in addition, each multiplicity space factors as

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}),$$

and

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Proof. Complete reducibility gives the displayed decompositions. Commuting with the two transported cut actions forces strip operators to preserve the sector pair (α_-, α_+) and act only on the multiplicity space; Schur's lemma kills the off-diagonal intertwiners and leaves the direct-sum algebra above. Taking invariants across both cuts in the glued tripartition again uses Schur's lemma and leaves exactly the matching sector pairs. The left/right split of the multiplicity spaces is additional structure; it is not forced by the center transfer alone. QED.

Corollary 5.2b (Exact or controlled four-term null modular relation on an inherited strip model). On one fixed finite-dimensional strip model satisfying Proposition 5.2a, define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

and

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \le \varepsilon \right\}.$$

Also define the EC-aligned subset $\mathfrak{M}^{\mathrm{EC}}_{I_-:J:I_+}$ by requiring the strip MSA on the inherited split. For a declared controlled strip family $\mathcal{F}_J$, let

$$\delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon) := \sup\left\{ \inf_{\sigma \in \mathfrak{M}^{\mathrm{EC}}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : \rho \in \mathcal{F}_J, \ I(I_- : I_+ \mid J)_\rho \le \varepsilon \right\}.$$

The generic exact-Markov modulus need not control this aligned one; convergence of the latter is a separate controlled-family hypothesis. For any strip state η, let

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the strip reference state ω is exact Markov and EC-aligned on the inherited split (the strip form of the Markov-split alignment hypothesis of Section 2.3), then $\Delta K_J(\omega)$ is central, and on the canonical inherited HJPW model one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$ such that

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $\omega \in \mathcal{F}_J$ and $I(I_- : I_+ \mid J)_\omega \le \varepsilon$, assume the aligned modulus tends to zero and choose $\widetilde{\omega}_J \in \mathfrak{M}^{\mathrm{EC}}_{I_-:J:I_+}$ with

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon).$$

Define

$$\mathfrak{D}_J(\omega,\widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega,\widetilde{\omega}_J),$$

with $K_{\partial,J}(\widetilde{\omega}_J) = \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J))$. Every bounded observable on $I_-\cup J$ or $J\cup I_+$ then differs from the exact-Markov reference by at most

$$\|O\|_\infty\, \delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon).$$

If the relevant strip marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega,\widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon).$$

Independently, Fawzi–Renner gives a recovered comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Thus the exact four-term strip relation is available only at EC-aligned exact Markovity or in a controlled strip family on one fixed inherited strip model with $\delta^{\mathrm{M,EC}}_{\mathcal{F}_J;I_-:J:I_+}(\varepsilon_J) \to 0$, while $\mathfrak{D}_J$ is carried explicitly at finite stage.

Proof. On the inherited split of Proposition 5.2a, the EC-aligned exact-Markov case is the same HJPW block calculation as in the spatial collar theorem: alignment identifies the HJPW blocks with the inherited blocks, the four logarithms cancel blockwise on the canonical model, and any residual bookkeeping term is central. Without alignment, HJPW factorizes the state only over its own state-dependent split, and the centrality of ΔK_J in $Z(\mathcal{A}(J))$ is not available. The controlled replacement is the same fixed-model compactness argument used for ordinary collars, after relabeling $A, B, D \mapsto I_-, J, I_+$; the operator-norm bound on $\mathfrak{D}_J$ is the corresponding relabeling of the modular-transport estimate. QED.

Definition (Renormalized null modular functional). For a null interval I on generator Ω, let $K_\partial(I,\Omega)$ denote the central endpoint-label term singled out by Corollary 5.2b on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define

$$\widetilde{K}[I,\Omega] := K[I,\Omega] - K_\partial(I,\Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a,\infty),\Omega].$$

Proposition 5.2c (Endpoint-Lipschitz null modular families and weak tail generator). Work on the exact-Markov/central-interface inherited-strip branch, and assume the endpoint-to-regulator-cell map and uniform local norm/site-density bound of the endpoint-control clause in Section 2.7, uniformly on the allowed compact endpoint window. For renormalized modular Hamiltonians on one null generator,

$$\widetilde{K}[I,\Omega] := K[I,\Omega] - K_\partial(I,\Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a,\infty),\Omega],$$

the additional branch-internal endpoint-control gives:

$$\big|\langle\psi, (\widetilde{K}[(a',b'),\Omega] - \widetilde{K}[(a,b),\Omega])\phi\rangle\big| \le C_{\psi,\phi,\Omega}\big(|a'-a| + |b'-b|\big)$$

for bounded intervals in a compact endpoint window, and

$$\big|\langle\psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle\big| \le C_{\psi,\phi,\Omega}|a'-a|$$

for half-lines. Hence

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

is locally Lipschitz and therefore absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle \psi, q(a,\Omega)\phi\rangle := -\frac{1}{2\pi}\,\partial_a f_{\psi,\phi}(a).$$

For $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi\rangle\, dv.$$

$q(a,\Omega)$ is a weak tail generator rather than a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after Corollary 5.2e and Lemma 5.2f.

Proof. The assumed endpoint-control estimate applies to the renormalized family after removal of the central endpoint term. For bounded intervals, the symmetric-difference length is bounded by $|a' - a| + |b' - b|$; for half-lines it is exactly $|a' - a|$. Therefore the matrix-element functions are Lipschitz in the endpoints. A Lipschitz function on a compact interval is absolutely continuous, so $f_{\psi,\phi}(a)$ has an L^∞_{loc} derivative and obeys the fundamental theorem of calculus. Defining q as the rescaled negative derivative gives the displayed integral relation. QED.

Downstream c.12–c.13 boundary. The fixed-cutoff bridge established above is the end of this subsection's proved content.

Lemma 5.2d (Downstream density-upgrade template). If, in addition, the weak tail generator $q(a,\Omega)$ is weakly differentiable in a and both $f_{\psi,\phi}(a)$ and $q_{\psi,\phi}(a)$ vanish at $+\infty$, then one may define a density

$$\langle \psi, p(a,\Omega)\phi\rangle := -\partial_a \langle \psi, q(a,\Omega)\phi\rangle = \frac{1}{2\pi}\,\partial_a^2 \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

and obtain the half-line formula

$$\langle \psi, \widetilde{K}_a(\Omega)\phi\rangle = 2\pi \int_a^\infty (v-a)\,\langle \psi, p(v,\Omega)\phi\rangle\, dv.$$

This is a downstream density-upgrade step and is not proved from the fixed-cutoff strip arguments above.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v = 0$. For $a \geq 0$, write

$$H_a := (a,\infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

By isotony of the null interval net,

$$a \leq b \quad\Longrightarrow\quad \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 5.2e (Derived half-sided modular relation on null half-lines). On the scaling-limit geometric-cap branch of Theorem 4.2, assume that the common reference vector is cyclic and separating for every half-line algebra used. The blow-up modular action near a smooth entangling cut acts on the null coordinate by

$$v \mapsto e^{-2\pi t} v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^{\omega}(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \qquad (t \le 0),$$

so the inclusion $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$ obeys the half-sided modular relation. After the harmless convention change $t \mapsto -t$, this has the standard positive-time orientation. The relation is derived from the null-interval structure, isotony, and the scaling-limit geometric action. It becomes a *standard* half-sided modular inclusion only when the reference vector is also cyclic for $\mathcal{M}_a(\Omega)' \cap \mathcal{M}_0(\Omega)$; that relative-commutant standardness is a separate receipt, not a consequence of the geometric action.

Proof. Blow up the cap modular flow of Theorem 4.2 near a smooth cut and restrict to the chosen null generator. In the tangent limit the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c,d) \subset H_a$, this sends $\mathcal{A}((c,d),\Omega)$ to $\mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d),\Omega)$, and taking the von Neumann closure of the interval net yields the displayed identity for $\mathcal{M}_a(\Omega)$. If $t \le 0$, then $e^{-2\pi t}a \ge a$, so $H_{e^{-2\pi t}a} \subseteq H_a$, and isotony gives $\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega)$. QED.

Lemma 5.2f (Positive null-translation generator). Fix $a > 0$ and assume that the common reference vector is cyclic and separating for both algebras below and cyclic for $\mathcal{M}_a(\Omega)' \cap \mathcal{M}_0(\Omega)$. For the resulting standard half-sided modular inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \qquad (a > 0),$$

let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define $K_0(\Omega) := -\log \Delta_0(\Omega)$. Borchers–Wiesbrock then yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\mathcal{M}_b(\Omega)U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \ge 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

In addition,

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \qquad \Delta_0(\Omega)^{it} P_\Omega \Delta_0(\Omega)^{-it} = e^{-2\pi t} P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω,

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

If $\Delta_a(\Omega)$ denotes the modular operator of $(\mathcal{M}_a(\Omega), \omega)$ and $K_a(\Omega) := -\log \Delta_a(\Omega)$, then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. Equivalently,

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi \rangle = -2\pi(b-a)\langle \psi, P_\Omega \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the canonical normalization of Corollary 5.2b, the weak endpoint derivative from Proposition 5.2c is therefore exactly the Borchers generator. Bounded-interval modular-Hamiltonian formulas are downstream and require the separate interval-preserving branch recorded in Theorem 5.2g.

Proof. Corollary 5.2e gives the half-sided modular relation, and the hypotheses of this lemma supply standardness of the inclusion. Borchers–Wiesbrock then yields the unique strongly continuous unitary group $U_\Omega(a)$, its positive self-adjoint generator P_Ω, and the affine covariance relation with the modular group; Stone's theorem gives the displayed domain formula. Differentiating

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a)$$

first at $a = 0$ and then at $t = 0$ on the common analytic core gives

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and $U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^* = \mathcal{M}_a(\Omega)$, uniqueness of modular data for the translated standard pair gives the displayed formulas for $\Delta_a(\Omega)$ and $K_a(\Omega)$. Differentiating $K_a(\Omega) = U_\Omega(a) K_0(\Omega) U_\Omega(a)^*$ in a yields $dK_a/da = -2\pi P_\Omega$ as a quadratic-form identity, and integrating from 0 to a gives the stated half-line modular-Hamiltonian relation. QED.

Theorem 5.2g (Half-line generator and conditional EFT null-stress reading). In the canonical normalization of the preceding half-line construction,

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega) \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. If that same renormalized half-line family admits an effective local-stress description and a same-source identification, its null-stress charge is represented by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega),$$

so

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on the common domain. The OPH half-line theorem fixes P_Ω; reading it as a local null-stress charge consumes the effective-locality and same-source identification premise rather than deriving that premise. Bounded-interval transport through the affine-covariant kernel $g_I(v)$ is discharged by E0.5 below, and reconstruction of a full tensor from the directional charges is subject to the null-invisible metric ambiguity.

Boundary of the null bridge. The null bridge is an explicit theorem branch rather than an automatic fixed-cutoff identity. Its required data are:

- transferred cut-center data alone give the central sector decomposition of the null strip rather than the left/right HJPW split;
- the extra decomposition-inheritance condition of Proposition 5.2a is exactly what upgrades those sectors to the same collar-type block structure used in the spatial branch;
- the fixed-cutoff null-strip bridge is exact for exact Markov strip states on that inherited decomposition, and otherwise uses a controlled limit with the carried defect operator $\mathfrak{D}_J$;
- the exact-Markov/central-interface branch, together with its specified endpoint-to-cell map and uniform norm/site-density bound, gives endpoint-Lipschitz control strong enough to define a weak tail generator for renormalized half-lines; the scaling-limit geometric-cap branch of Theorem 4.2 then derives the half-sided relation on the half-line blow-up net, while common-vector and relative-commutant standardness remain separate receipts;

- with those standardness receipts, the Borchers–Wiesbrock theorem supplies $P_\Omega \geq 0$, its Stone domain, and $K_a = K_0 - 2\pi a P_\Omega$;
- the half-line generator and its endpoint-derivative identity are fixed inside the bridge itself; the local null-stress reading requires the separately supplied effective-locality and same-source bridge, while bounded-interval transport and the later tensor upgrade are downstream.

This is the route from null-strip factorization to the Einstein branch, with the positive half-line generator proved inside the null modular bridge and its physical stress reading conditional on the named EFT bridge.

2.3.3 Conditional modular-energy stress-tensor reading on null half-lines

The effective-theory input here is the standard stress-tensor representation that names the operator fixed by the null bridge. In the canonical normalization of the preceding null-half-line construction, the positive null generator is the endpoint derivative of the renormalized half-line modular family:

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$.

If that same half-line family admits the supplied effective local-stress description and same-source identification, its local null-stress charge is represented by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega),$$

so on the common quadratic-form domain

$$P_\Omega = Q_{kk}(\Omega).$$

Thus the null generator has the displayed local null-stress reading conditional on an effective description that admits a local stress tensor and identifies the endpoint derivative with the same source family. The algebraic half-line construction alone does not supply that effective-locality or source-identification bridge. In a UV CFT regime, or another local Lorentzian regime with local modular Hamiltonians, the conditional reading is the standard null-stress charge associated with the chosen null generator.

For comparison, in a CFT vacuum on a ball one has the familiar local formula

$$H_\zeta = \int_\Sigma T_{ab} \zeta^b \, d\Sigma^a,$$

so, under that EFT premise, the conditional identification is the null-half-line version of the same modular-energy locality.

The downstream boundary is narrower. The E0.5 bounded-interval kernel transports this half-line statement to bounded null intervals with affine-covariant kernel $g_I(v)$, and the null-to-tensor upgrade determines T_{ab} only up to the familiar ambiguity

$$T_{ab} \mapsto T_{ab} + \phi g_{ab}.$$

The generator is fixed inside the null bridge; its physical local-stress interpretation remains conditional on the stated EFT bridge.

2.3.4 Localized generalized entropy from Markov + MaxEnt

Using the collar decomposition, exact Markovity, and MSA, the declared central-interface branch has the following edge-aligned Markov normal form. On the nonexact branch, Theorem 2.5 supplies a recovered comparison state and a vanishing CMI envelope only under strong conditional mixing; convergence to an exact Markov state is a fixed-collar statement, and alignment with the displayed edge factors remains a separate MSA condition. Under those conditions MaxEnt selection within each edge sector gives

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^\alpha_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^\alpha_{\mathrm{edge}}}{d_\alpha} \right).$$

The entropy splits as

$$S(\rho_C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\mathrm{bulk},C}) + \sum_\alpha p_\alpha \log d_\alpha.$$

Convention: Throughout this paper, "log" denotes the natural logarithm (ln), so entropies are measured in **nats** (1 nat $= 1/\ln 2 \approx 1.443$ bits). This is standard in thermodynamics and QFT; the Bekenstein-Hawking formula $\mathrm{S} = \mathrm{A}/4\mathrm{G}$ uses nats. When clarity requires it, we write $\log_2$ explicitly for bits.

Define

$$S_{\mathrm{bulk}}(C) := H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\mathrm{bulk},C}),$$

and the central area operator

$$L_C := \sum_\alpha (\log d_\alpha) P_\alpha.$$

Then

$$S_{\mathrm{gen}}(C) := \mathrm{Tr}(\rho L_C) + S_{\mathrm{bulk}}(C).$$

Newton coupling as an edge-entropy area-law readout.

In the collar double-scaling limit, the edge contribution becomes extensive along the entangling surface $\Sigma = \partial C$:

$$\mathrm{Tr}(\rho L_C) \approx N_\Sigma \cdot \bar{\ell}(t), \qquad \bar{\ell}(t) := \sum_\alpha p_\alpha \log d_\alpha,$$

where N_Σ is the number of UV cut elements covering Σ and $\bar{\ell}(t)$ is the **single-cell edge entropy** from the heat-kernel distribution (Theorem 6.20). Similarly, the geometric area is extensive:

$$A(C) \approx N_\Sigma \cdot a_{\mathrm{cell}},$$

where a_{cell} is the area per UV cut element in the emergent metric.

Matching these expressions gives the geometric area-law coupling:

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\,\bar{\ell}(t)}$$

where:

- a_{cell} is fixed operationally from the UV correlation/mixing length ξ via $a_{\mathrm{cell}} \sim \xi^2$ (from the exponential mixing hypothesis),
- $\bar{\ell}(t) = \sum_R p_R(t) \log d_R$ is computed from the heat-kernel edge distribution with $p_R \propto d_R e^{-t\lambda_R}$.

Explicitly:

$$\bar{\ell}(t) = \frac{\sum_R d_R e^{-t\lambda_R} \log d_R}{\sum_R d_R e^{-t\lambda_R}}.$$

On the OPH gravity branch this formula does not back-solve the scale. The selected no-G scale certificate supplies $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. The local cell readings are

$$a_{\mathrm{cell}} = P\ell_\star^2, \qquad \bar{\ell}_{\mathrm{shared}} = P/4,$$

so the Newton area-law readout gives

$$G_{\mathrm{geom}} = \frac{P\ell_\star^2}{4(P/4)} = \ell_\star^2.$$

The SI display is $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$.

2.3.5 Finite MaxEnt identities and the fixed-cap generalized-entropy stationarity interface

The proved finite MaxEnt envelope identities hold within the declared state family. The calculation does not, by itself, identify a variation of the state constraints with a geometric area variation or prove stationarity of their sum. The Einstein branch therefore assumes the following joint state–geometry interface: for its admissible variations preserving cap labels (fixed size and charges),

$$\delta S_{\mathrm{gen}}(C) = 0.$$

Once this balance is supplied, the split above and the tangent first law for the bulk term give

$$\delta S_{\mathrm{gen}}(C) = \delta\langle L_C \rangle + \delta\langle K_{\mathrm{bulk}} \rangle.$$

2.3.6 No-smuggling dependency discharge for the Einstein bridge

Theorem E0 (Einstein bridge dependency discharge). Write E for the Einstein-branch premise package: the premises A1–A3 together with the collar-recovery, edge-normalization, and generalized-entropy interfaces consumed by this branch, each an explicit hypothesis outside A1–A3. Let one source-derived cofinal OPH regulator tower carry the typed common-domain geometry, modular, event, stress, entropy, and scale readouts of Theorem 4.3h, together with its finite, asymptotic, and physical-identification premises. On that branch, E and the additional receipts supply the inputs used by the Einstein bridge:

- geometry readout through quotient normal forms via the screen fold $\chi_{S,r} \circ n_r \circ \pi_r$;
- Lorentz/H3 readout from round cap pairs, since $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$;
- 2π-normalized support-visible BW flow on the extracted geometric subnet;
- the positive null generator on the same branch and, under the separately supplied effective-locality and same-source bridge, its local null-stress charge reading;

- the bounded-interval kernel of Lemma E0.5;
- fixed-cap generalized-entropy stationarity from the separately declared stationarity interface;
- the standard fixed-volume small-ball area identity after geometry readout;
- $o(\ell^4)$ remainder control by Lemma E0.6; and
- all local timelike directions by the cap-pair coverage of Lemma E0.7.

Thus the finite consensus reduct supplies quotient normal forms. The typed branch receipts supply the geometric, modular, stress, entropy, coverage, tail, coupling, vacuum, and scale inputs needed by the Einstein bridge. No source-derived tower carrying all of those inputs is constructed here. Bare finite consensus does not imply the Einstein equation.

Lemma E0.5 (Conditional bounded-interval kernel). On the null blow-up of the support-visible geometric branch, assume the effective-locality and same-source bridge that identifies the proved half-line endpoint derivative with a local null-stress charge. Together with endpoint-Lipschitz renormalized interval control and affine/projective covariance, this gives

$$K^{\mathrm{null}}_{(a,b)} = 2\pi \int_a^b \frac{(v-a)(b-v)}{b-a}\, T_{kk}(v)\, dv + R_{(a,b)}.$$

The remainder $R_{(a,b)}$ is the carried endpoint/collar term of the same refinement family. In the small-ball limit this is the ball modular kernel used below.

Lemma E0.6 (Uniform remainder control with its transfer bounds). Let (r, ℓ_r, δ_r) be one declared cofinal family with $\ell_r \to 0$ and faithful Gibbs states ρ_r, on which the strong conditional matrix-mixing premise of Theorem 2.5 holds uniformly over the event region and cap family. Set

$$\varepsilon_r = c|\partial C_r|_{\mathrm{UV}} e^{-\delta_r/\xi_r}.$$

The boundary-anchor expansion directly controls the operator norm:

$$\|\Delta K_r(\rho_r)\|_\infty = \|\mathbf{J}_{\rho_r}(A_r : D_r \mid B_r)\|_\infty \le \varepsilon_r.$$

Here the four canonical marginal logarithms define ΔK, so $\Delta K = -\mathbf{J}$. Exact faithful Markov references have zero canonical four-log defect. This direct bound requires no uniform spectral floor; it does not identify an EC splitting or control each individual modular Hamiltonian separately.

Assume separately that the needed EC-aligned replacements obey one modulus $d_r^{\mathrm{M,EC}} \le C_{\mathrm{M}} \varepsilon_r^{\theta}$, with $C_{\mathrm{M}} < \infty$, $\theta > 0$, and that for some

$$s > \max\{8, 4/\theta\}, \qquad \omega_r \longrightarrow +\infty,$$

the same family satisfies

$$\frac{\delta_r}{\xi_r} - \log\bigl(c|\partial C_r|_{\mathrm{UV}}\bigr) \ge s \log(\ell_0/\ell_r) + \omega_r.$$

Then

$$2\sqrt{\varepsilon_r} = o(\ell_r^4), \qquad d_r^{\mathrm{M,EC}} = o(\ell_r^4), \qquad \|\Delta K_r(\rho_r)\|_\infty = o(\ell_r^4).$$

Indeed $\varepsilon_r \le (\ell_r/\ell_0)^s e^{-\omega_r}$, and the exponents are $s/2 > 4$, $s\theta > 4$, and $s > 4$, respectively. For a downstream remainder bounded by a stage-independent constant times these three quantities, plus a separately controlled $o(\ell_r^4)$ kernel residual, the remainder is $o(\ell_r^4)$. Any growing transfer constant or further remainder must instead be included in the same-family tail receipt; the rate calculation does not bound it by itself.

On a weaker route that has only a trace-norm replacement estimate, the matrix-mixing bound above is unavailable. The logarithm comparison instead requires the complete rate

$$4\lambda_r^{-1} d_r^{\mathrm{M,EC}} = o(\ell_r^4),$$

where $\lambda_r > 0$ bounds the relevant marginals of both states below. For example, $\lambda_r \geq \lambda_0(\ell_r/\ell_0)^p$, with $\lambda_0 > 0$ and $p \geq 0$, gives the sufficient strengthened margin $s > \max\{8, (4+p)/\theta\}$, under uniformly bounded downstream transfer constants. Stagewise faithfulness alone supplies no such margin. This weaker route is not a replacement for the stronger hypotheses used by Theorem 2.5. Mere pointwise collar convergence also supplies neither common-family rate, and no per-radius diagonal choice is used.

Lemma E0.7 (Cap-pair timelike coverage). On the cap-normal H^3 branch, every future timelike vector v has $r = \sqrt{-\eta(v,v)}$ and $u = v/r \in H^3$. For any unit $s \in u^\perp$, the vectors

$$q_\pm = u \pm s$$

are future null and

$$v = \frac{r}{2}(q_+ + q_-).$$

Thus normalized future null directions in observer skies cover every local timelike direction once the observer-frame chart is supplied. This does not say that two unnormalized projective rays alone select a unique observer frame; the observer normalization or equivalent causal-diamond scale data is also required.

Proposition E2 (Conditional recovered-core Ward closure). Suppose the finite tomography hypotheses reconstruct a tensor from one compatible family of null and timelike charges on the recovered core. Assume separately that its same-source local charge family obeys the weak Ward balance: signed fluxes on opposite faces of infinitesimal recovered-core diamonds cancel with a controlled remainder, and this balance converges in the recovered charts to the distributional divergence law. Then $\nabla^a \langle T_{ab} \rangle = 0$ weakly, and pointwise on a smooth branch. Charge compatibility and tensor tomography alone imply neither the face cancellation nor its continuum limit.

2.3.7 Einstein equation from fixed-cap generalized-entropy stationarity

This subsection is a branch theorem. The finite consensus reduct supplies quotient normal forms; Theorem E0 discharges the recovered-core premise package E needed by the Einstein bridge: geometry readout, support-visible BW modular covariance, the conditional local null-stress reading under effective-locality and same-source matching, bounded-interval transport, the declared fixed-cap generalized-entropy stationarity interface, the fixed-volume small-ball area variation, carried-remainder control, and the timelike scalar-to-tensor upgrade.

In the d=4 scaling regime, the small-ball bridge follows conditionally once the effective-locality, same-source, endpoint-control, and affine/projective hypotheses of E0.5 are supplied. The additional standard geometric input is the fixed-volume area-variation identity for a small geodesic ball. The kernel shape then follows on the extracted geometric subnet: a sufficiently small cap is assumed to admit the tangent causal-diamond description D_ℓ, whose preserving conformal Killing field is

$$\xi_{D_\ell} = \frac{1}{2\ell}\Big((\ell^2 - r^2 - t^2)\,\partial_t - 2t\,x^i \partial_i\Big).$$

On the t=0 slice its lapse is

$$\xi_{D_\ell} \cdot n = \frac{\ell^2 - r^2}{2\ell}.$$

The bounded-interval kernel combines the geometric cap generator with the affine/projective transport of the half-line stress bridge, so the cap-modular theorem and the stress bridge together yield

$$\delta S_{\mathrm{bulk}}(C) = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle \, d^3x + \delta\langle E^{(\eta)}_{C,\ell}\rangle.$$

If $\delta\langle T_{00}\rangle$ is approximately constant across the ball and the carried remainder is $o(\ell^4)$, then

$$\delta S_{\mathrm{bulk}}(C) = \frac{8\pi^2 \ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5 \partial T) + o(\ell^4).$$

Stationarity of generalized entropy therefore gives the observer-covariant scalar relation

$$\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] = 8\pi G\, u^a u^b\, \delta\langle T_{ab}\rangle$$

for the local diamond rest-frame four-velocity u^a. In the adapted rest frame $u^a = (1,0,0,0)$, this is

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

2.3.8 Overlaps supply all timelike directions

This scalar-to-tensor upgrade is internal once the Lorentz branch is in place. Overlapping observers through the same bulk point realize every local timelike four-velocity u. Define

$$Y_{ab} := G_{ab} + \Lambda g_{ab} - 8\pi G\, \langle T_{ab}\rangle.$$

The local-diamond rest-frame relation says

$$u^a u^b\, \delta Y_{ab} = 0$$

for every such u. In local inertial coordinates,

$$u^a = \gamma(1, v^i), \qquad |\vec{v}| < 1,$$

so

$$0 = u^a u^b\, \delta Y_{ab} = \gamma^2\Big(\delta Y_{00} + 2v^i \delta Y_{0i} + v^i v^j \delta Y_{ij}\Big)$$

for all $|\vec{v}| < 1$. The polynomial therefore vanishes coefficientwise, hence

$$\delta Y_{00} = 0, \qquad \delta Y_{0i} = 0, \qquad \delta Y_{ij} = 0.$$

So the full tensor variation vanishes:

$$\delta Y_{ab} = 0.$$

Along any connected scaling branch of reference states, Y_{ab} is therefore constant. The only purely local freedom left by the null reconstruction is the metric term, and fixing that term on one maximally symmetric reference state gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab}\rangle.$$

Thus the tensor upgrade does not require a separate EFT handoff once the geometric-cap theorem, the null bridge, and the overlap-complete timelike family are in place.

2.3.9 Non-tunable numerical constants

The gravity chain yields specific numerical constants as rigid outputs of its declared premises.

The 2π KMS normalization. From Theorem 4.2, cap modular flow on the support-visible scaling-limit geometric cap pair is fixed by the KMS condition to the standard 2π normalization. This is the same rigidity that fixes Unruh/Hawking temperature normalization.

The geometric coefficient $\Omega_{d-2}/(d^2-1)$. This coefficient appears in both (a) the CFT-ball modular Hamiltonian weight integral and (b) the geometric area-variation identity. It is an exact integral identity:

$$\int_{B_\ell^{d-1}} \frac{\ell^2 - r^2}{2\ell}\, d^{d-1}x = \frac{\Omega_{d-2}\,\ell^d}{d^2-1}.$$

In $d = 4$:

$$\frac{\Omega_2}{4^2-1} = \frac{4\pi}{15} \approx 0.8377580409572781.$$

This is the reason prefactors cancel cleanly when going from $\delta S_{\text{gen}} = 0$ to the Einstein equation (leaving $8\pi G$ with the 2π fixed by Theorem 4.2).

What is predicted. The framework cleanly separates:

- **Non-tunable constants**: 2π (KMS period), $\Omega_{d-2}/(d^2-1)$ (geometric coefficient), the existence of the Einstein form.
- **Global closure constants**: G is the conversion between edge entropy and geometric area on the emitted local gravity branch. Λ is not fixed by the local reference-state variation data alone; stable correctable-record closure plus identification of the record carrier with de Sitter horizon entropy fixes $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$, while the SI curvature scale also uses the selected OPH scale certificate.

2.3.10 Quantitative Markov error and controlled corrections

The role of this subsection is to keep three distinct quantities separate:

1. the raw collar conditional mutual information
$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega;$$
2. the constructive recovery error
$$r_{\text{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta};$$
3. the exact-Markov replacement error on one fixed faithful collar model,
$$\eta_\delta^{\text{M}} := 4\lambda_*^{-1}\, \delta^{\text{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$
where $\lambda_* > 0$ is the lower spectral bound needed to compare modular Hamiltonians.

The exact identity for the modular defect expectation is unchanged:

$$\langle \Delta K_\delta \rangle_\omega = -I(A_\delta : D_\delta \mid B_\delta)_\omega = -\varepsilon_\delta,$$

with

$$\Delta K_\delta := K_{A_\delta B_\delta D_\delta} - K_{A_\delta B_\delta} - K_{B_\delta D_\delta} + K_{B_\delta}.$$

What changes is the interpretation of later "exact" collar formulas. For each fixed finite-dimensional collar one may choose an exact Markov reference state σ_δ with

$$\|\omega_\delta - \sigma_\delta\|_1 \le \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

and a constructive recovered comparison state $\omega_\delta^{\mathrm{rec}}$ with

$$\|\omega_\delta - \omega_\delta^{\mathrm{rec}}\|_1 \le r_{\mathrm{FR}}(\varepsilon_\delta).$$

These two comparisons do different jobs:

Bounded-observable control. For every bounded observable X supported on $A_\delta \cup B_\delta$,

$$|\mathrm{Tr}[X(\omega_\delta - \omega_\delta^{\mathrm{rec}})]| \le \|X\|_\infty \, r_{\mathrm{FR}}(\varepsilon_\delta),$$

and

$$|\mathrm{Tr}[X(\omega_\delta - \sigma_\delta)]| \le \|X\|_\infty \, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta).$$

The first bound is constructive and dimension-free; the second is the fixed-collar route to the exact Markov set.

Modular-additivity control. If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\Delta K_\delta(\omega) - \Delta K_\delta(\sigma_\delta)\|_\infty \le \eta_\delta^{\mathrm{M}}.$$

Since $\Delta K_\delta(\sigma_\delta)$ is central by exact Markovity, the finite-stage modular defect is within η_δ^{M} of the exact splice or additivity value.

Therefore the manuscript's exact collar identities arise as limits with a carried remainder

$$\eta_\delta := r_{\mathrm{FR}}(\varepsilon_\delta) + \eta_\delta^{\mathrm{M}},$$

together with the separate long-wavelength derivative remainders present in the small-ball expansion. Exact identities at finite collar width require exact Markovity; otherwise one works with η_δ-controlled approximations and then lets $\eta_\delta \to 0$ in the controlled collar limit.

Finite-cutoff Einstein remainder bound. In the small-ball rest-frame calculation, write

$$\delta S_C = \frac{8\pi^2 \ell^4}{15}\, \delta\langle T_{00}\rangle + \delta\langle E_C^{(\delta)}\rangle + O(\ell^5 \partial T),$$

where $E_C^{(\delta)}$ packages the carried collar remainder. Then the declared fixed-cap generalized-entropy stationarity interface, together with the proved finite MaxEnt identities, gives

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle + \mathcal{E}_\delta,$$

with

$$\mathcal{E}_\delta := \frac{15G}{\pi \ell^4}\, \delta\langle E_C^{(\delta)}\rangle + O(\ell\, \partial T),$$

and, by the bounds above,

$$\left|\delta\langle E_C^{(\delta)}\rangle\right| \le C_C\, \eta_\delta$$

for some collar-dependent constant C_C on the fixed faithful model. Equivalently, before the controlled collar and small-ball limits, one may write

$$|\mathcal{E}_{\ell,\delta}| \le C_1 r_{\mathrm{FR}}(\varepsilon_\delta) + C_2 \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) + C_3 \eta_\delta^{\mathrm{reg}} + C_4 \ell \, \|\partial T\| + o_\delta(1) + o_\ell(1),$$

with constants depending only on the fixed collar model, the bounded observable class, and the small-ball chart. Thus the exact Einstein relation is the controlled limit, not a finite-cutoff identity inherited from approximate Markovity.

If one wishes to package this carried remainder as an effective anomalous energy density, the natural definition is

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle.$$

Then

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\,\delta(\langle T_{00}\rangle + \langle T_{00}^{\mathrm{anom}}\rangle) + O(\ell\,\partial T).$$

This is a bookkeeping definition of the finite-stage remainder, not an upgrade of the recovered-core theorem. Its significance is exactly that it is controlled:

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le \frac{15\,C_C}{8\pi^2\ell^4}\,\eta_\delta.$$

On a separate conditional dark/anomaly branch, this carried scalar remainder may be identified with a repair stress only after a quotient-invariant source-localization receipt identifies the same collar operator with the small-ball Einstein remainder, and after a positive causal packet lift supplies the full tensor. In SI units that continuation has rest-energy normalization

$$T_A^{ab} u_a u_b = \frac{15\hbar c}{8\pi^2\ell^4} R_{r,\ell} + O(\epsilon_{\mathrm{loc}}),$$

where $R_{r,\ell}$ is measured in nats and ℓ is the proper geodesic small-ball radius. The scalar quantity $I(A:D\mid B)$ alone is not a stress tensor: packet factorization, causal directions, an allocation rule, a rest frame, source-localization map, and transport law are additional continuation data.

The framework carries the Markov error through three distinct controls:

1. $r_{\mathrm{FR}}(\varepsilon_\delta)$ controls bounded observables at one stage;
2. $\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta)$ controls convergence to the exact Markov normal form on a fixed collar;
3. η_δ^{M} controls modular-Hamiltonian replacements once faithfulness is assumed;
4. the Einstein-branch remainder is a carried $O(\eta_\delta)$ term, not a silent exact identity at finite cutoff.

This is the concrete bridge from "premises about screens" to "precision GR predictions plus a disciplined carried remainder".

2.3.11 Focusing/QNEC boundary on the null-modular branch

The focusing content does not follow from Axioms 1–3. The two focusing statements on the null-modular branch require the additional hypotheses stated below. Quantum focusing is an optional gravity interface and the Einstein theorem does not consume it.

What the branch supplies. The fixed-cutoff null bridge of §5.2 supplies transferred cut-center data and exact-or-controlled strip additivity on the inherited strip model. The geometric scaling limit supplies the half-sided relation; on the separately registered common-vector and relative-commutant standardness receipts, Borchers–Wiesbrock supplies its translation generator and the

affine relation $[K, P] = -i2\pi P$. These are kinematic inputs a QNEC statement consumes; they are not a proof of it or an identification of (P) with a local stress charge.

Why relative-entropy monotonicity does not yield QNEC. Monotonicity under restriction orders the relative entropies $S(\rho_{R(\lambda)}\|\omega_{R(\lambda)})$ along nested null cuts; it fixes the sign of the first λ-derivative and says nothing about the second. A monotone function of the cut need not be convex, so the step from monotonicity to $d^2S_{\mathrm{rel}}/d\lambda^2 \geq 0$ is unavailable. The entanglement first law $\delta S = \delta\langle K\rangle$ is a first-order state variation at fixed algebra and reference state; the QNEC second null shape variation is a different object, and expanding the first law to second order in the cut does not produce it. The published QNEC proofs supply exactly the missing machinery: null-surface locality and analyticity hypotheses in the free and super-renormalizable settings [55], and modular-flow, causality, and defect-OPE inputs in the general QFT setting [57]. None of those hypotheses is verified on the reconstructed patch net.

Why QFC is not derived from QNEC. The Quantum Focusing Conjecture is strictly stronger than QNEC: the foundational QFC statement obtains QNEC as its locally parallel limit [56], and no implication runs the other way. QNEC together with the Einstein branch (Theorem 5.1) and the Raychaudhuri identity controls the diagonal, stationary-cut part of the generalized-expansion variation and leaves the off-diagonal, shear, counterterm, and general nonstationary-cut content of QFC untouched. The optional quantum-focusing interface therefore carries no independent internal support from this branch.

Claim boundary. The null-modular branch supplies the half-sided modular structure but not an internal QNEC proof. Such a proof additionally requires the QFT-side hypotheses named above on the reconstructed net and the off-diagonal QFC content. Any theorem that consumes focusing must name the optional quantum-focusing interface explicitly.

Theorem 5.1 (OPH Einstein-branch closure in the scaling regime). Under the premise package E on a recovered-core, support-visible scaling branch satisfying the E0 dependency discharge above, the fixed-cap generalized-entropy stationarity condition implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

If this relation holds for all local directions and reference states in the locally Lorentzian scaling regime, the timelike scalar-to-tensor theorem yields the semiclassical Einstein equation modulo the expected Λg_{ab} ambiguity. The capacity branch closes that metric term globally under stable whole-fiber correctable-record closure, the unique finite-size slack zero, and the horizon–record identification.

Proof sketch. The recovered-core branch supplies the exact hypothesis package required by the Jacobson-type bridge: quotient-safe geometry readout, 2π-normalized BW cap flow, the conditional same-source local null-stress charge, bounded-interval ball kernel with $o(\ell^4)$ remainder, the declared fixed-cap generalized-entropy stationarity interface, the fixed-volume small-ball area identity, and all timelike directions. The cap first law identifies δS_C with $\delta\langle K_C\rangle$, while the null bridge together with its effective-locality and same-source premise and the bounded-interval kernel relates the bulk modular variation to the required null stress-tensor charge on the diamond. Since the reference state is maximally symmetric, the fixed-volume small-ball area identity enters at first order in the perturbation, and the declared joint state–geometry balance for admissible fixed-cap variations, combined with the proved MaxEnt envelope identities on the cap-label-preserving state family, yields the displayed rest-frame relation. The timelike-coverage lemma supplies all local timelike directions, the scalar-to-tensor theorem upgrades the relation to the tensor equation modulo Λg_{ab}, and the capacity branch closes the remaining metric term under its separately typed stable-capacity

and horizon-record premises. QED.

2.3.12 Black-hole structural statements and continuation-level spectroscopy

This subsection separates two result classes. The structural black-hole package is a four-step chain: fixed-cutoff edge-center collar decomposition on the exact-Markov or idealized exact-recoverability horizon carrier, the small-CMI recoverability reading of interior encoding on that same carrier, Hawking/KMS normalization on the geometric modular branch, and the discrete area spectrum together with the leading semiclassical Schwarzschild first-law relation for an actual sector change. Finite same-boundary records and finite reconstruction thresholds do not by themselves supply physical radiation entropy, exterior time, evaporation flux, a geometric island, a radiative quotient, or an asymptotic waveform readout. Any clean comb structure, QNM selector, Page-type linewidth estimate, PBH burst template, Kerr/LIGO horizon spectroscopy template, or Page-curve/island closure is continuation-level and uses continuation inputs stated below.

Derived structural statements.

Area eigenvalues from edge sectors. The central area operator (Section 5.4) is

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha,$$

where $d_\alpha \in \mathbb{N}$ is the dimension of the edge Hilbert space in sector α. With the normalization $\mathrm{Tr}(\rho L_C) = \langle A \rangle / 4G_{\mathrm{geom}}$, the area eigenvalues are

$$A_\alpha = 4G_{\mathrm{geom}} \log d_\alpha = 4\ell_\star^2 \ln d_\alpha,$$

where $\ell_\star^2 = \hbar G_{\mathrm{SI}}/c^3$ is the scale-readout area displayed as the Planck area. Since d_α is a positive integer, **areas are discretely spaced** with logarithmic gaps.

Hawking emission energy quantization. For a Schwarzschild black hole with $A(M) = 16\pi G^2 M^2/c^4$, define the signed change of the hole for a transition from d_{before} to d_{after} by

$$\Delta_{\mathrm{BH}} A = 4\ell_\star^2 \ln(d_{\mathrm{after}}/d_{\mathrm{before}}).$$

The differential first law, evaluated on the initial background, gives the leading semiclassical relation

$$\Delta_{\mathrm{BH}}(Mc^2) = k_B T_H \ln(d_{\mathrm{after}}/d_{\mathrm{before}})$$

with $T_H = \hbar c^3/(8\pi G k_B M)$, whose 2π normalization is fixed by the BW_{S^2} tangent-limit normalization of Theorem 4.2. This is not an exact finite-difference identity: using the exact Schwarzschild mass–area relation, an emission with entropy loss $\ln k$ has $E_{\mathrm{emit}} = Mc^2[1 - \sqrt{1 - \ln k/S_{\mathrm{BH}}}]$, whose expansion is $k_B T_H \ln k[1 + O(\ln k/S_{\mathrm{BH}})]$.

Continuation-level spectroscopy templates. The remainder of this subsection is not part of the recovered core. It records what follows only if one adds extra discrete-horizon selection rules and, where stated, standard semiclassical inputs such as evaporation-power models, QNM/transition identifications, or greybody matching.

Integer transitions. If one imports the additional rule that dominant emission steps reduce the edge dimension by an integer factor k, then $d_{\mathrm{before}} = k d_{\mathrm{after}}$, so k must divide the initial dimension. The signed black-hole entropy change is $-\ln k$, while the positive entropy loss is $\ln k$. To leading semiclassical order the emitted spectrum becomes a discrete comb:

$$\Delta E_k = k_B T_H \ln k, \qquad \Delta f_k = \frac{c^3}{16\pi^2 GM} \ln k.$$

Structural condition: comb vs. generic discreteness. The log-integer *comb* structure requires the imported dynamical assumption that such integer-ratio emission steps dominate. A fixed $d_{\rm before}$ admits only its divisors; the displayed all-k ladder is a union of candidate continuations, not a source-produced transition set. If generic transitions between arbitrary integers dominate instead, the set of $|\ln(d_{\rm after}/d_{\rm before})|$ values becomes a dense log-rational set that may appear quasi-continuous after folding in linewidths and astrophysical effects. What the recovered finite edge-sector structure supplies, under the stated generalized-entropy and horizon-area dictionaries, is the *discrete area spectrum*; the clean comb pattern follows only with that selection rule.

Continuation-level template (Discrete Hawking spectrum). If the additional integer-transition selection rule is realized, the Hawking emission spectrum consists of discrete lines with spacing $\Delta E_k = k_B T_H \ln k$, where k is an integer characterizing the dominant sector transitions, instead of a continuous thermal profile.

Mass-independent linewidth-to-spacing ratio (continuation-level estimate). Using Page's semiclassical calculation for emission power $P(M) = p_0 \hbar c^6/(G^2M^2)$ with $p_0 \approx 2 \times 10^{-4}$, the emission rate is $\dot{N} \approx P/\langle E\rangle$ where $\langle E\rangle = a\, k_B T_H$ with $a \sim \mathcal{O}(1-10)$. The natural linewidth $\Gamma \sim \hbar \dot{N}$ divided by the level spacing gives:

$$\frac{\Gamma}{\Delta E_k} \approx \frac{64\pi^2 p_0}{a \ln k},$$

a function of (a, k). At $k = 2$ the declared range $a \in [1, 10]$ spans 1.8% to 18%; at $k = 3$, 1.1% to 11%. A few-percent value therefore requires the particle content and greybody model that pin a inside a narrower interval (for instance $a \approx$ 4–6 at $k = 2$); no such model is selected here. The fraction is mass-independent at every (a, k).

Connection to quasinormal modes (interpretive continuation). The highly-damped Schwarzschild quasinormal modes have asymptotic real part (Motl, 2002):

$$\mathrm{Re}\,\omega \to \frac{c^3}{8\pi GM} \ln 3.$$

This matches exactly the $k = 3$ transition frequency $\Delta E_3/\hbar$. If one adopts a Bohr-type identification between quantum transition frequencies and asymptotic QNM frequencies, this selects

$$\Delta A = 4\ell_\star^2 \ln 3 \approx 4.39\, \ell_\star^2$$

as the fundamental area quantum.

Scope statement. The area quantization follows from the edge-sector structure (derived). The $k = 3$ selection requires the additional interpretive identification with QNM frequencies (not derived from A1–A3). The linewidth prediction uses standard semiclassical inputs.

Numerical examples. For Δf_k = (c^3/16π^2GM) ln k:

- **M = 30 M$\odot$**: k=2 at 29.7 Hz, k=3 at 47.1 Hz
- **M = 1 M$\odot$**: k=2 at 891 Hz, k=3 at 1412 Hz
- **M = 10^{12} kg (primordial)**: k=2 at 7.3 MeV, k=3 at 11.6 MeV

Within the integer-transition continuation, these frequencies track $k_B T_H \ln k$ exactly and are in principle distinguishable from a continuous thermal spectrum.

Continuation-level PBH burst search template.

Within the discrete-horizon continuation, the Hawking comb would provide a distinctive gamma-ray template. A template-level discriminant is **log-integer energy ratios**: if two emission lines are observed at energies E_2 and E_3, their ratio must satisfy

$$\frac{E_3}{E_2} = \frac{\ln 3}{\ln 2} \approx 1.585$$

exactly, independent of black hole mass. Within that continuation template, the ratio is fixed once the log-integer rule is assumed.

Available instruments and energy coverage. The $k = 2$ line energy $E_2 = k_B T_H \ln 2$ determines which instruments can see a given BH mass:

Instrument	Energy band	BH mass range (k=2 in band)
Fermi GBM (BGO)	0.15–40 MeV	2×10^{11}–5×10^{13} kg
Fermi LAT	0.1–300 GeV	2×10^{7}–7×10^{10} kg
H.E.S.S.	0.1–100 TeV	7×10^{4}–7×10^{7} kg
LHAASO-WCDA	1–15 TeV	5×10^{5}–7×10^{6} kg

Detector resolution vs. intrinsic linewidth. The continuation-level Schwarzschild linewidth-to-spacing estimate is $\Gamma/\Delta E_k = 64\pi^2 p_0/(a \ln k)$, spanning 1.8%–18% at $k = 2$ over the declared $a \in [1, 10]$ (mass-independent at every (a, k)). Published detector energy resolutions:

- Fermi GBM: $< 10\%$ (0.1–1 MeV), $\sim 4\%$ at 10 MeV (BGO)
- Fermi LAT: $< 10\%$ (1–100 GeV)
- H.E.S.S.: $\sim 15\%$ (TeV)
- LHAASO-WCDA: $\sim 33\%$ (TeV)

Within this template, the comb could be resolvable with GBM/LAT; at TeV energies it would appear as moderately broad bumps rather than sharp lines.

Search protocol. A dedicated OPH-comb search would:

1. Select burst-like candidates (10–120 s time windows, matching existing PBH burst search protocols).
2. Fit each candidate with null model (smooth continuum) vs. OPH comb model (peaks at $E_k = E_0 \ln k$ convolved with detector response).
3. Scan over the single scale parameter $E_0 = k_B T_H$ (equivalently, BH mass).
4. Require at least two lines satisfying log-integer ratio to claim detection.
5. Correct significance for trials (time windows $\times$ sky positions $\times$ E_0 scan).

Observational context. Dedicated PBH burst searches (H.E.S.S., LHAASO) report **no significant bursts**. An OPH-specific comb-template analysis of archival data would:

- Set upper limits on OPH-comb PBH burst rates
- Demonstrate direct testability of the discrete spectrum prediction
- Provide constraints comparable to or stronger than generic PBH burst limits

Data availability. Fermi GBM provides public Time-Tagged Event (TTE) burst data; Fermi LAT provides public photon event lists with documented analysis workflows. H.E.S.S. has a small public test data release.

Continuation-level GW horizon spectroscopy template for Kerr remnants.

The same continuation-level discrete-horizon branch extends to gravitational wave observables. For Kerr black holes, the thermodynamic first law fixes the leading small-transition relation. For

emission the hole changes by $\delta E_{\rm BH} = -\hbar\omega$ and $\delta J_{\rm BH} = -\hbar m$, while the imported integer-division rule has $\delta S_{\rm BH} = -\ln k$. Hence

$$\hbar(\omega - m\Omega_H) = k_B T_H \ln k, \qquad k \in \{2, 3, 4, \ldots\}$$

up to finite-step corrections from the changing Kerr background. Under those additional discrete-horizon assumptions, this gives the **GW horizon spectroscopy comb**: a continuation-level set of candidate resonant frequencies. The integer-division selection and its all-k union are imported, not selected by the edge-sector spectrum alone. A physical QNM or ringdown claim needs a separate exterior-background bridge, a radiative quotient, a finite operator converging to the continuum perturbation generator, future-horizon and future-null-infinity boundary conditions, and an asymptotic readout map. A finite repair spectrum or fitted clock scale is not that bridge.

Kerr line frequencies. For a remnant with mass $M > 0$ and dimensionless subextremal spin $\chi = a_*/M$, $|\chi| < 1$, define the spin correction factor:

$$g(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}}, \qquad \Omega_H(M,\chi) = \frac{c^3}{2GM} \cdot \frac{\chi}{1+\sqrt{1-\chi^2}}.$$

The line frequencies are:

$$f_{k,m}(M,\chi) = \frac{m\,\Omega_H(M,\chi)}{2\pi} + \frac{c^3}{16\pi^2 GM}\, g(\chi)\, \ln k$$

The subextremal restriction makes the thermal scale positive. At $|\chi| = 1$, $g(\chi) = 0$, so all candidate teeth collapse to the rotation line; the rescaled coordinate and offset-subtracted ratios below are undefined.

Within this continuation template, once LIGO/Virgo infers (M, χ) for a remnant, the line pattern is fixed by the inferred remnant parameters and the assumed discrete-horizon rule. The line pattern belongs to a continuation template outside recovered core.

Within-line KMS response, not a cross-line prior. In this continuation template, the GR greybody absorption response may be combined with a declared bin width $\Delta\omega_k \approx \omega_T \ln(1 + 1/k)$, where $\omega_T = k_B T_H/\hbar$. KMS detailed balance then supplies the net absorption-minus-stimulated-emission factor at each candidate frequency:

$$W^{\rm net}_{k,\ell m} = \Gamma^{\rm GR}_{\ell m}(\omega_{k,m}) \cdot \Delta\omega_k \cdot \frac{k-1}{k}$$

where $\Gamma^{\rm GR}_{\ell m}$ is the standard GR greybody factor and the $(k-1)/k$ factor follows from KMS detailed balance with $e^{(\omega - m\Omega_H)/T_H} = k$. This factor does not determine transition rates across different k, and it cannot be normalized into a cross-k prior without an additional rate model.

Universal stacking coordinate. Define the dimensionless rescaled frequency:

$$x := \frac{GM}{c^3 g(\chi)}(\omega - m\Omega_H).$$

Then the predicted line locations collapse to universal constants:

$$x_k = \frac{\ln k}{8\pi} \qquad (k = 2, 3, 4, \ldots)$$

Numerically: $x_2 = 0.02758$, $x_3 = 0.04371$, $x_4 = 0.05516$, $x_5 = 0.06404$.

Stacking test. Multiple BBH events can be mapped to this universal x coordinate and stacked only after fixing a common frame. Observed hertz must use $M_{\rm det} = (1+z)M_{\rm source}$ in both Ω_H and

the thermal offset. Under a prospectively normalized comb submodel, its peaks then align across events with different (M, χ); detector noise need not stack coherently.

Comparison to existing work. Prior area-quantization searches [83] used parameterized models with one free spacing constant. The imported continuation considered here has multiple candidate lines with exact $\ln k$ ratios. The $(k-1)/k$ detailed-balance factor is a within-line net-response factor, not a derived hierarchy of transition probabilities.

Numerical example (GW170608). Remnant parameters: $M_f \approx 18.0 M_\odot$, $\chi_f \approx 0.69$. For $m = 2$, the horizon rotation frequency is $m\Omega_H/(2\pi) \approx 719$ Hz. The **thermal comb spacing** (the part that encodes the area quantization) is:

k	$\Delta f_k := \frac{c^3 g(\chi)}{16\pi^2 GM} \ln k$ (Hz)	KMS net-response factor $(k-1)/k$
2	41.6	0.500
3	65.9	0.667
4	83.2	0.750
5	96.5	0.800
6	107.5	0.833

The displayed values are source-frame frequencies. Detector-frame frequencies are lower by $1+z$, equivalently obtained by replacing M_f with $M_{\rm det} = (1+z)M_f$ throughout. A stacking analysis uses the rescaled coordinate $x = GM(\omega - m\Omega_H)/(c^3 g(\chi))$ with all quantities in one frame, which maps the thermal spacing to universal constants $x_k = \ln k/(8\pi)$ regardless of the rotation offset.

Template-matching criterion. After rescaling by (M, χ), spectral features in this continuation template must satisfy the offset-subtracted ratio

$$\frac{f_{k,m} - m\,\Omega_H/(2\pi)}{f_{2,m} - m\,\Omega_H/(2\pi)} = \frac{\ln k}{\ln 2}$$

exactly, equivalently $x_k/x_2 = \ln k/\ln 2$ in the universal coordinate defined above, independent of a common scale and offset. The full frequencies $f_{k,m}$ do not satisfy $f_k/f_2 = \ln k/\ln 2$ at nonzero Ω_H: the additive rotation offset breaks the ratio, and only the thermal comb pieces carry it. A null result can challenge only the finite support or normalized prior fixed before comparison. It cannot exclude the unrestricted integer family merely by ruling out $2 \le k \le 12$, and it does not by itself test the derived area-spectrum statement.

2.3.13 Classical mechanics on the conditional GR branch

If the physical source-causal continuum, curvature/stress, scale, and convergence premises promote the finite Einstein-form identity to the standard smooth Einstein equation, its ordinary GR consequences apply. This section records the textbook dust and weak-field reductions on that conditional branch; it is not an additional derivation from the informational order.

Stress-energy conservation is automatic. The contracted Bianchi identity is geometric:

$$\nabla^a G_{ab} = 0.$$

Combined with the Einstein equation, this implies:

$$\nabla^a \langle T_{ab} \rangle = 0.$$

Geodesic motion from dust limit. For pressureless classical matter ("dust"), $T^{ab} = \rho\, u^a u^b$. Conservation yields:

$$\nabla_a(\rho u^a u^b) = 0 \quad \Rightarrow \quad u^b \nabla_a(\rho u^a) + \rho\, u^a \nabla_a u^b = 0.$$

Projecting orthogonally to u^b using $h^b{}_c = \delta^b{}_c + u^b u_c$ kills the first term, giving:

$$\rho\, u^a \nabla_a u^b = 0 \quad \Rightarrow \quad u^a \nabla_a u^b = 0.$$

This is the geodesic equation: free classical bodies follow spacetime geodesics.

Newtonian limit from weak-field GR. Take the weak-field, slow-motion limit with metric:

$$g_{00} \approx -(1 + 2\Phi/c^2), \qquad g_{0i} \approx 0, \qquad g_{ij} \approx \delta_{ij}(1 - 2\Phi/c^2),$$

and velocities $|\mathbf{v}| \ll c$. Then $G_{00} \approx 2\nabla^2\Phi/c^2$ (leading order), and $T_{00} \approx \rho c^2$. The Einstein equation reduces to:

$$\nabla^2\Phi = 4\pi G\rho.$$

Geodesic motion reduces to:

$$\ddot{\mathbf{x}} = -\nabla\Phi.$$

These are Newton's gravitational law and Newton's second law. They are controlled limits of the supplied smooth GR branch, rather than consequences of the finite provenance poset alone.

One finite piece of this limit is machine-checked on the committed carrier. On the three-dimensional completion of the exact record frame, the ball volume scales as radius to the carrier dimension by the generic finite-dimensional volume theorem, the shell content at positive radius is $4\pi r^2$ with the exponent equal to the carrier dimension minus one, and any declared spherically symmetric strength profile whose product with the shell content is a constant charge equals $Q/(4\pi r^2)$ exactly, with $f(r_1)r_1^2 = f(r_2)r_2^2$ shell-independent (`Lean/Geometry/InverseSquareShellLaw.lean`). The inverse-square exponent is therefore derived from the committed carrier dimension rather than assumed, while the radial readout and the shell-flux normalization are declared premises on the audited register. No join of this shell law to the composed Einstein branch is supplied because the screen-carrier-to-physical-position attachment is absent. The weak-field reduction above otherwise remains a standard prose derivation without a finite receipt.

This shell law lands on the rest space of every declared observer frame. For each future unit-timelike frame of the declared Hermitian Lorentz module, an explicit boost-shear map is a machine-checked linear equivalence from the completion carrier onto the frame's Lorentz-orthogonal rest space, and the pullback of the positive-definite rest metric along it is exactly the Euclidean coordinate form; at the standard frame the map reduces to the candidate bridge that places the source Gram quotient in the standard rest fiber (`Lean/Geometry/RestFiberShellTransport.lean`). Under the same two declared premises the radial readout pulled back through the equivalence is the carrier norm, the strength at a nonzero rest vector is $Q/(4\pi r^2)$ with r the rest radius, the exponent reads as the rest-space dimension minus one with the dimension proved through the transport equivalence, and every oriented Lorentz chart transport carries the strength field exactly. One composed identity carries the screen chain into the fiber: the finite source Gram of two integer record controls equals the rest metric of their transported difference at every declared frame. The frame is a declared choice and no uniqueness of the identification is claimed. No attachment of the rest fiber to physical position or corresponding Einstein-branch join is supplied.

Precision classical predictions. On a branch that identifies the finite tensor fields with smooth physical curvature and stress and supplies the required source-causal continuum, scale, and curvature-convergence receipts, the standard GR precision toolbox applies (post-Newtonian expansion, lensing, time delay, etc.). These are inherited consequences of that physical identification, not independent predictions of the informational order.

Selected precision predictions (in the regime where the GR derivation applies):

Light bending by mass M: For impact parameter b,

$$\Delta\theta = \frac{4GM}{c^2 b}.$$

For the Sun with $b \approx R_\odot$: $\Delta\theta \approx 1.751$ arcsec.

Mercury perihelion advance: Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

Using Mercury's orbital parameters: $\Delta\varpi \approx 42.98$ arcsec/century.

Gravitational redshift: Between two radii in a static potential,

$$\frac{\Delta\nu}{\nu} \approx \frac{\Delta\Phi}{c^2}.$$

For the Sun (surface to infinity): $z \approx 2.12 \times 10^{-6}$.

These predictions are fixed functions of G and known source parameters, and are confirmed observationally to high precision. The framework contains them automatically on the derived Einstein branch.

2.3.14 Precision gravity predictions and experimental bounds

The pure Einstein linearization makes classical dispersion and polarization statements that can be confronted with the tightest available experimental bounds. This section translates those branch-scoped statements into the observables that experiments constrain; it does not infer a quantum graviton pole from the classical equation.

Speed of gravitational waves. On a suitable background, the two transverse-traceless classical modes of the pure Einstein linearization propagate on the same invariant null cone as the Maxwell modes:

$$\frac{c_{\mathrm{GW}} - c_\star}{c_\star} = 0 \text{ exactly on this action/background branch.}$$

Published bound (GW170817 + GRB 170817A multi-messenger):

$$-3 \times 10^{-15} < \frac{c_{\mathrm{GW}} - c}{c} < +7 \times 10^{-16} \quad (90\% \text{ credibility}).$$

For a source at ~ 40 Mpc, this fractional difference corresponds to only a few seconds of propagation-time mismatch across $\sim 10^8$ years of travel.

Einstein quadratic mass parameter. On the additional pure two-derivative Einstein–Hilbert action branch, the transverse-traceless quadratic operator has no Fierz–Pauli hard-mass parameter:

$$\mu^2_{\mathrm{FP,hard}} = 0.$$

This is an action-content statement, not a graviton-particle mass prediction from diffeomorphism redundancy alone. A quantum mass or pole claim requires Definition 6.18, while diffeomorphism- invariant Stueckelberg/bimetric completions and extra-field theories lie outside the pure-Einstein field-content branch.

Published bound (GW dispersion analysis, PDG 2025):

$$m_{\mathrm{grav,disp}} \leq 1.76 \times 10^{-23}\ \mathrm{eV}/c^2 \quad (90\%\ \text{credibility; model-dependent dispersion fit}).$$

This corresponds to a reduced Compton wavelength $\bar{\lambda}_C \gtrsim 1.6 \times 10^{16}$ m, i.e., order ~ 1.6 light-years.

No dipole radiation. Many modified gravity theories predict extra channels (scalar/vector) producing dipolar radiation at (-1)PN order. The derived GR limit predicts no such channel.

Published bound (GW170817 inspiral phasing, PDG 2025):

$$-4 \times 10^{-6} < \delta\hat{p}_{-2} < 2 \times 10^{-5} \quad (90\%\ \text{credibility}).$$

Tensor polarizations on the selected field-content branch. The pure Einstein linearization contains two tensor polarizations. It does not prove that every broader diffeomorphism-invariant EFT lacks additional scalar, vector, or massive modes. Pure non-tensor hypotheses are disfavored by observational constraints, and mixed tensor-scalar/vector models are tightly constrained.

Equivalence principle tests. Additional null checks from the derived GR structure:

- Universality of free fall (space tests): precision ~10^{-15}
- Nordtvedt parameter ($\eta = 4\beta$ - γ): $(0.47 \pm 0.55) \times 10^{-4}$
- Binary pulsar radiative damping (PSR J0737-3039): 0.999963 ± 0.000063

2.3.15 State–observable error propagation and the dynamical bridge

On the additional pure-Einstein action/background branch, the framework states the classical equalities recorded in Section 2.3.14. Finite Markov/recovery control supplies a separate and more limited quantitative statement. Two error classes must remain separate. The *state–observable error class* concerns expectation values of specified bounded local observables in a recovered state; the Markov/recovery machinery controls this class. The *dynamical-parameter error class* concerns spectral and dispersion properties of an evolution generator: propagation speeds, particle-pole masses, and correlator poles. The trace-distance chain alone does *not* control those quantities.

The key quantitative hook. From Theorem 3.1, if the target state satisfies

$$I(A_k : C_k \mid B_k) \leq \varepsilon_k,$$

then, writing ρ for the target state and σ for its recovered comparison state, the recovery map gives the trace-norm bound

$$\|\rho - \sigma\|_1 \leq \delta_k := 2\sqrt{1 - e^{-\varepsilon_k}} \leq 2\sqrt{\varepsilon_k}.$$

State–observable bound. Trace distance gives immediate bounds on expectation-value errors for the declared operator class: bounded operators supported in the recovered region, compared at the time of recovery. Using the standard dual norm inequality:

$$|\langle O\rangle_\rho - \langle O\rangle_\sigma| \leq \|O\|_\infty \|\rho - \sigma\|_1 = 2\|O\|_\infty D(\rho, \sigma),$$

where $D(\rho,\sigma)=\frac{1}{2}\|\rho-\sigma\|_1$ is the trace distance. This inequality is exactly as strong as it looks and no stronger: the norm $\|O\|_\infty$ is the Lipschitz constant only for bounded O, and the comparison is an equal-time statement about states.

Exponential decay from the mixing hypothesis. Writing w for collar width, so it is not confused with the recovery error δ_k, the mixing assumption (Section 2.3) provides

$$I_\omega(A_w : D_w \mid B_w) \le C_{\mathrm{mix}}|\partial C|_{\mathrm{UV}} e^{-w/\xi}.$$

Combining these gives an explicit precision dial for bounded local observables:

$$\delta_{\mathrm{step}} \lesssim 2\sqrt{C_{\mathrm{mix}}|\partial C|_{\mathrm{UV}}}\cdot e^{-w/(2\xi)}.$$

For an operator normalized by $\|O\|_\infty \le 1$, an independently chosen absolute expectation-error target τ_O is guaranteed at one step if $\delta_{\mathrm{step}} \le \tau_O$; the theorem therefore gives the sufficient CMI condition $\varepsilon \le (\tau_O/2)^2$. The target τ_O applies only to the state–observable bound and cannot be set by importing a gravitational-wave speed tolerance.

What the dial does not certify: dynamical parameters. The gravitational-wave propagation speed and any graviton pole or dispersion-mass parameter are not bounded local observables. They are spectral/dispersion properties of the dynamical generator (poles of retarded correlators, or long-time wave-packet phase and group velocities), so the trace-distance inequality supplies neither their norm nor their Lipschitz constant. Two states can be close in trace distance at one time while being vacua of generators with different dispersion relations, and a small equal-time expectation error does not control phase accumulation over the $\sim 10^8$-year propagation baseline of GW170817. Consequently, $\delta \lesssim 10^{-15}$ is *not* a theory-side certificate for the multimessenger bound $|c_{\mathrm{GW}}/c-1| \lesssim 10^{-15}$ of Section 2.3.14, and the recovery error must not be presented as the precision dial for that published bound. The ideal classical relation $c_{\mathrm{GW}} = c_\star$ and the vanishing pure-Einstein hard parameter $\mu^2_{\mathrm{FP,hard}} = 0$ belong to the additional action-level branch recorded there; neither stability away from that branch nor a quantum graviton pole follows from the trace-distance chain.

Required dynamical-parameter bridge. A sufficient conditional theorem can be stated precisely, but its hypotheses have not been derived from the Markov remainder in this paper. Let $\mathcal{B}=[k_{\min},k_{\max}]$, with $0<k_{\min}<k_{\max}$, be the declared experimental wave-number band. For each tensor helicity, assume a C^1 self-adjoint finite-regulator generator $H_\nu^{\mathrm{TT}}(k)$ and the GR reference generator $H_0^{\mathrm{TT}}(k)$, expressed in angular-frequency units ($\hbar=1$) on one specified common finite-dimensional helicity fiber or after a declared C^1 unitary identification of the two fibers. Assume that the reference TT tensor-mode eigenvalue $\omega_0(k)=c_\star k$ is simple in the helicity block and is separated from the rest of the spectrum by a uniform gap $g_{\mathcal{B}}>0$. Define

$$M_{\mathcal{B}} := \frac{1}{c_\star}\sup_{k\in\mathcal{B}}\left\|\partial_k H_0^{\mathrm{TT}}(k)\right\|_\infty$$

and the dimensionless dynamical error

$$\eta_{\mathrm{dyn},\nu} := \max\left\{\frac{1}{g_{\mathcal{B}}}\sup_{k\in\mathcal{B}}\left\|H_\nu^{\mathrm{TT}}(k)-H_0^{\mathrm{TT}}(k)\right\|_\infty, \frac{1}{c_\star}\sup_{k\in\mathcal{B}}\left\|\partial_k H_\nu^{\mathrm{TT}}(k)-\partial_k H_0^{\mathrm{TT}}(k)\right\|_\infty\right\}.$$

For $\eta_{\mathrm{dyn},\nu}<1/4$, let $\omega_\nu(k)$ be the continued isolated eigenvalue and define its group velocity by $c_{\mathrm{GW},\nu}(k) := \partial_k\omega_\nu(k)$. The standard isolated-eigenvalue projector estimate and the Hellmann–Feynman formula then give, uniformly over $\mathcal{B}$,

$$\sup_{k\in\mathcal{B}}\left|\frac{c_{\mathrm{GW},\nu}(k)}{c_\star}-1\right|\le L_{\mathcal{B}}\,\eta_{\mathrm{dyn},\nu},\qquad L_{\mathcal{B}}:=1+4M_{\mathcal{B}}.$$

Indeed, if $P_\nu(k)$ and $P_0(k)$ are the rank-one spectral projectors, then $\|P_\nu-P_0\|_\infty\le 2\eta_{\mathrm{dyn},\nu}$, hence $\|P_\nu-P_0\|_1\le 4\eta_{\mathrm{dyn},\nu}$. Differentiating the isolated eigenvalue therefore bounds the group-velocity change by the derivative perturbation plus $4M_{\mathcal{B}}c\,\eta_{\mathrm{dyn},\nu}$. This displays a dimensionless constant derived from a specified generator and uniform over the declared band. If $q:=L_{\mathcal{B}}\eta_{\mathrm{dyn},\nu}<1$, then stationary propagation at fixed group velocity over path length D obeys the group-delay bound

$$|\Delta t|\le\frac{D}{c_\star}\frac{q}{1-q}.$$

The same bound applies to varying propagation only if it holds pointwise along the entire path; on a cosmological path, the redshifted wave number must remain inside $\mathcal{B}$. A wave-packet phase bound likewise requires uniform control of the inverse dispersion $k_\nu(\omega)$ on the corresponding frequency band. A model-dependent particle dispersion-mass bound is a separate spectral estimate at $k=0$; it is not a corollary of the band-limited speed estimate.

The OPH-specific bridge requires three additional conditions:

1. H_ν^{TT}, or an equivalent retarded two-point kernel, is derived from the finite and scaling OPH dynamics;
2. a norm-resolvent, quadratic-form, or correlator estimate has the form $\eta_{\mathrm{dyn},\nu}\le C_{\mathcal{B}}r_N$, where $r_N:=\min(2,\sum_{k=2}^N\delta_k)$ is the accumulated trace-norm recovery bound of Theorem 3.1 for the associated N-step construction and $C_{\mathcal{B}}$ is derived rather than assumed; and
3. the band-uniform dispersion estimate applies to the detector band and propagation baseline, including phase accumulation where used by the likelihood.

Only after the second item is proved may one combine the estimates to obtain

$$\sup_{k\in\mathcal{B}}\left|\frac{c_{\mathrm{GW},\nu}(k)}{c}-1\right|\le L_{\mathcal{B}}C_{\mathcal{B}}r_N.$$

No estimate $\eta_{\mathrm{dyn},\nu}\le C_{\mathcal{B}}r_N$ is proved here. Substituting δ_k, δ_{step}, or any equal-time trace-distance error for $\eta_{\mathrm{dyn},\nu}$ without that bridge is invalid.

Precision summary. The framework provides:

1. The ideal pure-Einstein linearization statements $\mu^2_{\mathrm{FP,hard}}=0$ and $c_{\mathrm{GW}}=c_\star$, without a quantum graviton-pole conclusion.
2. Translation of those ideal equalities into the specific observables experiments constrain (Section 2.3.14).
3. Explicit bounds on how far expectation values of bounded local observables in the recovered state can drift, using the conditional mutual information → trace distance → observable error chain.

It therefore provides quantitative error control on recovered states and bounded observables. No theory-side certificate is supplied for the published GW-speed or model-dependent particle dispersion-mass bounds, and no derived dynamical-parameter certificate is supplied.

2.3.16 Dark-sector response from the modular anomaly

The conditional dark-sector continuation maps scalar repair occupation to a canonical pair (n, θ). Its lattice action is a proposed dynamical completion rather than a recovered-core theorem.

The candidate anomaly-tensor slot T_{ab}^{anom} is available only after the finite covariant source packet of Section 5.9 supplies source localization, CMI-to-modular-source matching when that route is used, rank-two reconstruction, conservation, normalization, and universal coupling. On that packet it supplies one structural ingredient for a possible dark-sector continuation without introducing additional particle species. Raw scalar collar CMI supplies only a recovery diagnostic.

Conditional source identification. Once the source packet has identified an anomalous local modular energy, its rest-frame normalization is

$$\langle T_{00}^{\mathrm{anom}} \rangle = \frac{15}{8\pi^2} \cdot \frac{\delta \langle K_C^{(\mathrm{anom})} \rangle}{\ell^4}$$

The quantity can be called dark only on a branch whose packet also establishes gravitational coupling and absence of an electromagnetic coupling. These properties motivate the dark-sector interpretation; neither they nor the observational identification follow from the collar-CMI theorem.

Conditional repair-charge action. The phase equation supplies a conserved repair current with explicit boundary flux. Its dilute homogeneous phase has $\rho_R \propto a^{-3}$, while the cubic condensed link energy gives $a_R = \sqrt{a_b a_0}$ and $v^4 = GM_b a_0$ on the spherical deep branch. The same action couples coherent matter through $q_\star \chi_\nu^{\mathrm{can}} S_{\mathrm{coh}}^{\mathrm{can}}$ and carries the field stress required for momentum closure.

Scope. The canonical pair, compact-phase completion, baryonic source map, dimensional couplings, full constitutive law, relativistic refinement limit, abundance, and physical likelihoods are not supplied. A compact neutral coherent source has no monopole and therefore cannot close the declared single-monopole source ansatz. Bound neutral composites, dipole or higher-multipole sources, and topological or solitonic architectures are not excluded.

Cosmology/Boltzmann contract. The conditional background scaling above is not an FLRW perturbation kernel. A cosmological dark/anomaly claim must instead expose the variables needed by an Einstein–Boltzmann implementation:

$$\bar{\rho}_A(a), \quad \bar{\rho}_{A,\mathrm{eq}}(a), \quad w_A(a), \quad c_{s,A}^2(k,a), \quad \sigma_A(k,a), \quad Q_A^\mu, \quad B_A(k,a), \quad \Gamma_{\mathrm{rec}}(k,a).$$

The first, source-side no-go is that $\bar{\rho}_A(a)$, $\bar{\rho}_{A,\mathrm{eq}}(a)$, and $B_A(k,a)$ alone do not define a physical Boltzmann source. A physical source must be emitted by a finite covariant collar-packet parent

$$\mathcal{P}_r[X, g] = (C_r, Z_r, A_r, R_r, G_r, \pi_r[X], L_r[X], Q_r, D_r),$$

whose finite packet states split the anomaly and any repair recipient, whose repair generator and reaction channels close the total stress tensor, and whose causal response data set the pressure, sound speed, anisotropic stress, exchange current, and repair rate. The current registered Boltzmann contract accepts either of two non-conflated local source routes: fixed-reference anomalous modular energy, or nonlinear CMI stress with a CMI-to-modular-source matching theorem or hypothesis and explicit residual bounds. A hybrid or different mechanism lies outside those two routes and must provide its own typed source contract and closure receipts. Raw collar CMI is otherwise a recoverability diagnostic, not the $15/(8\pi^2)$ BW source. B_A is gauge-invariant only when built from the anomaly-frame baryon density $n_b^{(A)} = -u_{A\mu} J_b^\mu$. If the exchange branch is nonzero, explicit recipient stress and an equal-and-opposite exchange current are mandatory; otherwise the repair term is not a closed stress-energy source. A transition spectral number is only $\gamma_{\text{repair step}}$ without

active-fiber, physical-clock, and common-parent response-pole receipts for a physical Γ_{rec}. A physical likelihood comparison also requires declared source, solver, dataset, covariance, and nuisance provenance.

Canonical finite covariant collar-packet parent. For a regulator r and background (X, g), the source object is

$$\mathcal{P}_r[X, g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \mathsf{Read}_r).$$

Here $\mathcal{C}_r$ is a declared physical-causal regulator complex, not the source-derived informational read-from poset. It carries oriented faces, cell four-volumes, causal adjacency, and finite parallel transports $U_{F\to c}$; e_r is the tetrad/local-frame data; $Z_r = \bigsqcup_s Z_{s,r}$ is the packet state space for anomaly, recipient, radiation, standard matter, and any interaction sector; ω_r, p_r, f_r are invariant weights, local momenta, and occupations; Φ_r is the face flux; $\mathcal{R}_r$ is the reaction channel list; $\mathcal{G}_r$ is the finite gauge/quotient data; π_r is the restriction/refinement map; and Read_r is the local observable readout. With signature $(-+++)$, every packet used in a kinetic stress branch must satisfy $g_{ab}p_z^a p_z^b = -m_z^2$, $p_z^0 > 0$, and $\omega_z f_{c,z} \geq 0$, with the invariant measure convention declared explicitly.

Scalar-to-stress nonuniqueness. A scalar collar row, even a finite row for $\rho_A(a)$, $\rho_{A,\mathrm{eq}}(a)$, or $B_A(k, a)$, does not determine a stress tensor. Many inequivalent packet parents can have the same scalar density while differing in pressure, sound speed, anisotropic stress, exchange current, gauge-invariant perturbation variables, and causal response. Scalar rows are therefore insufficient as physical CMB, lensing, growth, or S_8 inputs unless the parent supplies the missing stress and response variables.

Finite MaxEnt lift. When only finitely many one-cell source readings $e_x(u_i) = T^A_{ab}(x)u_i^a u_i^b$ are available, a stress lift is not unique until the branch supplies a finite selection rule. The admissible MaxEnt lift minimizes no hidden observational residual: it maximizes the declared packet entropy subject to the source readings, mass shells, conserved charges, local frame covariance, and refinement restrictions. The lift is physical only when held-out timelike probes pass a quadraticity/tomography test and the variational stress agrees with the packet-moment stress in the predeclared norm.

Packet stress and exchange closure. For a kinetic parent,

$$T^{ab}_{I,r}(c) = \frac{1}{V_c}\sum_{z\in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a p_z^b, \qquad J^a_{I,r}(c) = \frac{1}{V_c}\sum_{z\in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a.$$

More general packet parents may add an internal stress Σ_z^{ab}, but then Σ_z^{ab} is part of the primitive parent data and must obey the same local-frame and refinement checks. The finite transport/reaction equation must imply

$$\mathrm{Div}_r\, T^{ab}_{I,r} = Q^b_{I,r}, \qquad \sum_I Q^b_{I,r} = 0.$$

If the repair exchange is nonzero, a row labelled "recipient" is not enough: the recipient stress T^{ab}_R, exchange current Q^b_R, and closure residual $Q^b_A + Q^b_R$ must be emitted and certified. If exchange is off, the parent instead emits a repair-exchange-off receipt.

Response, rate, and gauge receipts. The finite causal response must provide a finite domain of dependence, subluminal characteristic bound, retarded support residual, and response stability residual. The transition-matrix diagnostic may be called $\gamma_{\text{repair step}}$ until the active fiber, conserved-sector decomposition, physical clock $\Delta\tau_{\text{physical}}$, and common parent response pole are certified. Only then may one write

$$\Gamma_{\text{rec}} = -\operatorname{Re}\lambda_{\text{active}}/\Delta\tau_{\text{physical}}.$$

Gauge checks must be stated in gauge-invariant or rest-frame variables; raw Newtonian/synchronous agreement is a diagnostic, not a gauge-independence receipt.

Certificate hierarchy. The finite parent certificate is the conjunction of primitive evidence, not a producer-declared Boolean. The source side must certify the route and entropy unit, modular nonadditivity, localization and stress tomography, finite-packet kinematics and mass shell, covariant transport and four-momentum, stress readout and variational agreement, local-frame and quotient invariance, total stress and exchange-current closure, cosmological gauge invariance, finite domain of dependence, subluminal and retarded response, stability under refinement, and the cold-dark-matter limit. Frozen source, solver, likelihood, and data hashes belong to the separate comparison contract. They do not belong to the parent theorem and cannot rescue a failed parent.

The same source-provenance requirement applies to the CMB-source inputs

$$\eta_R, \quad \gamma_{\text{repair step}} \text{ or certified } \Gamma_{\text{rec}}, \quad A_\zeta, \quad q_{\text{IR}}, \quad \ell_{\text{IR}}, \quad B_A(k,a), \quad \rho_A(a), \quad N_{\text{CRC}}.$$

Each one must have a source-provenance dependency node recording the source report, parents, no-CMB-data flag, and measurement-use flags. Contradictory provenance, such as declaring no CMB data use while also fitting to Planck, fails closed. Reducers must pool additive sufficient statistics globally before any nonlinear estimate of tilt, amplitude, inverse precision, rank, condition number, isocurvature leakage, B_A, ρ_A, or Γ_{rec}; raw transition diagnostics are $\gamma_{\text{repair step}}$. Shard-local nonlinear averages and shard-local `any()` likelihood rollups are diagnostic only. N_{CRC} is the stable capacity-closure invariant unless a separate additive capacity schema proves disjoint coverage. The cold transported limit is the check case: when exchange, pressure, sound-speed, and anisotropic stress corrections are turned off, the anomaly slot must reduce to a CDM-like component before recombination. Any nonzero-field response, late-time growth suppression, or S_8-relief branch has to be emitted, under this registered contract, by the finite-collar parent evaluator through $B_A(k,a)$ and $\Gamma_{\text{rec}}(k,a)$ after the physical-clock and response receipts pass, not fitted as a free environmental kernel to CMB, weak-lensing, SPARC, or cluster data. The compressed H_0, Ω_m, σ_8, and S_8 rows are plumbing diagnostics for a low-H_0, Planck-like branch, not theorem-grade cosmology.

Finite cosmological source and prediction eligibility. The physical cosmology condition is the conjunction

$$\text{CMB ready} = \mathsf{SOURCE} \wedge \mathsf{SCALE} \wedge \mathsf{PARENT} \wedge \mathsf{KERNEL} \wedge \mathsf{INIT} \wedge \mathsf{TRANSFER} \wedge \mathsf{FREEZE} \wedge \mathsf{LIKE}.$$

SOURCE is the source-only DAG with transitive measurement-taint rejection and globally pooled sufficient statistics; SCALE is the physical mode/clock/lift bridge; PARENT is the finite covariant dark/anomaly parent with packet kinematics, mass shell, packet stress readout, reaction-channel four-momentum, recipient stress and exchange-current closure for nonzero exchange, total stress closure, local-frame/quotient invariance, cosmological gauge invariance, finite domain of dependence, subluminal retarded response, response stability, refinement, and CDM-limit recovery receipts; KERNEL emits physical ρ_A, $\rho_{A,\text{eq}}$, B_A, and certified Γ_{rec}; the source-only ρ_A row additionally requires the anomaly abundance selector

$$\mathcal{E}_{A,r} = \mathbb{E}_{\mu^{\text{rel}}_{A,r}}[\mathsf{L}_{A,r}]$$

and its no-data-use and refinement receipts; INIT supplies regular initial modes and Einstein constraints; TRANSFER supplies Boltzmann well-posedness and numerical convergence; FREEZE fixes the model generation; and LIKE is the official likelihood execution. While this conjunction is false, physical CMB, BAO, lensing, growth, RSD, and S_8 outputs are diagnostic or conditional, not recovered-core predictions.

This branch is a conditional completion contract, not a physical dark-matter theory.

Falsifiability. The repair-charge condensate requires a source-derived action, relativistic limit, and joint galaxy, Solar-System, lensing, cluster, and cosmological likelihood. These constructions are not supplied, so the falsification program contains no cosmological target or verdict.

2.3.17 De Sitter holography: static patch vs boundary-at-infinity

A natural question arises: how does this framework relate to the "unsolved problem" of de Sitter holography?

What the usual dS holography problem is. When people say "dS holography is unsolved," they typically mean that we do not have anything as sharp as AdS/CFT, where the bulk has a timelike asymptotic boundary supporting a well-defined dual CFT with a precise dictionary. For de Sitter, there is no asymptotic timelike boundary in the static patch where one can simply place the dual theory. The classic dS/CFT proposal at future infinity has familiar difficulties, including non-unitarity worries and complex conformal weights.

Static-patch/horizon-screen setup. The framework begins with an observer's static patch and its horizon screen S^2, building a net of subregion algebras on that screen. At finite cutoff those algebras are type-I regulators. The Lorentz statement, when invoked, is a support-visible scaling-limit statement about the refinement-limit observer net, and that limit may leave the regulator class. By Theorem 2.2.2, the realized scaling-limit cap modular action on the extracted geometric cap pair is geometric and, in the non-type-I case of interest, generally outer.

This is therefore a fundamental fork away from AdS/CFT-style holography:

AdS/CFT	This framework
Codimension-1 boundary at infinity	Codimension-2 horizon screen (S^2)
Specified boundary field theory	Observer patches with typed overlaps
Conformal dual in specified constructions	Patch algebras plus geometric and physical premises
Negative Λ in AdS	Positive Λ on the declared de Sitter branch

This aligns with the static-patch/complementarity intuition in the dS literature, where the fundamental description is patch-based and different static patches are related by consistency rules, not by a single global boundary theory.

The mechanism: Λ as global capacity, not local physics. Once the separately supplied effective-locality and same-source bridge identifies the directional null charges with one symmetric local stress tensor, finite null tomography determines that tensor only up to an additive metric term. Vacuum-energy or cosmological-constant shifts are therefore invisible to this conditional local null-data route. The conditional Einstein equation obtained under the declared fixed-cap generalized-entropy stationarity interface is correspondingly fixed only up to Λg_{ab}. Accordingly, this null-data route cannot fix Λ without independent trace-sensitive or global input. A source-selected vacuum potential or non-null local equations could supply such input.

Conditional theorem boundary: correctable public-record closure. Freeze a capacity carrier $\mathcal{H}_{\mathrm{cap},r,D}$ with $D = \dim \mathcal{H}_{\mathrm{cap},r,D}$ and $N = \log D$. For every terminal world q, compatible record atoms, endogenous reachability, a frozen publicness policy, and the complete source-supplied global checkpoint family define a compound confusability graph G_q. The exact public capacity and whole-fiber readback are

$$M_0(q) = \alpha(G_q), \qquad \mathfrak{F}_{r,0}(D) = \{M_0(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

When the entire nonempty terminal fiber has one common readback, define $M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N)$. The universe-level closure is then

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

The notation is typed: M_0 is always a multiplicative record count, while N is its logarithm. Stable direct closure and its exact finite-size selector are

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad s(D) := \log D - \log M_0(D), \qquad s(D_\star) = 0 < s(D) \quad (D \neq D_\star).$$

On the exact reversible branch every checkpoint generator is injective, so

$$M_0(q) = |X_{\mathrm{reach}}(q)|,$$

reducing the finite computation to exact CSP or model counting. This supplies an exact implementation route for the source-only finite public checkpoint packet $\mathcal{C}^{\mathrm{pub}}_{r,D}$.

The fixed-$D = 24$ source packet realizes the complete declared terminal fiber inside its declared finite source category. A separate all-rung counterfamily shares base agreement, positivity, the carrier bound, and finite incidence, reversible-action, semantic-projection, composition, extension, and oriented-record fiber-product controls. Reversible, copy-collapse, two-class, and hidden-spectator completions give different exact fixed sets. Lean proves the all-rung arithmetic consequence of the base, positivity, and carrier-bound structure. The executable certificate checks the remaining finite controls, and an independent simulator reconstructs held-out finite rungs. This does not lift the complete A1–A3 terminal fibers, atom maps, joint kernels, meaning maps, feasible sets, and regulator controls across capacity, so it is not a verdict on the complete source class.

The local null-data route is blind to metric-term shifts, so the Einstein branch is fixed locally only modulo Λg_{ab}. Conditional on stable direct closure and the independent horizon–record identification,

$$N_\star = \log D_\star = \frac{A_{\mathrm{dS}}}{4\ell_\star^2}, \qquad \Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}.$$

With the selected scale certificate the same branch has the display

$$G_{ab} + \frac{3\pi}{GN_\star}\, g_{ab} = 8\pi G\, \langle T_{ab} \rangle.$$

The independent common screen/electroweak load-carrier identification tests the same closed N against the weak/Higgs load through

$$R_{\mathrm{EW}}(P, N) = \alpha_U(P) \log\!\left(\frac{N}{\pi}\right) - \frac{6\pi}{P} = 0, \qquad N_{\mathrm{bridge}} = \pi \exp\!\left[\frac{6\pi}{P\alpha_U(P)}\right].$$

Neither physical bridge constructs the direct capacity map.

Capacity readout. The branch uses N_{scr} as the entropy capacity. The bare radius-squared ratio is $N_{\mathrm{patch}} = (r_{\mathrm{dS}}/\ell_P)^2$, and

$$N_{\mathrm{scr}} = \pi N_{\mathrm{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

The observed late-time scale gives $N_{\mathrm{patch}} \simeq 1.05 \times 10^{122}$, with a Planck-$\Lambda$ central entropy capacity $N_\Lambda = 3.3126\ldots \times 10^{122}$. The conditional source-forward electroweak bridge value $N_0 = 3.5321315434\ldots \times 10^{122}$ is about 6.6 percent higher. The separately premised finite-presence and Poisson candidates are $3.2920978773\ldots \times 10^{122}$ and $3.3000722254\ldots \times 10^{122}$. Exact positive composition countermodels select neither action from the finite source. Their comparisons are retrospective.

Fixed-capacity shock sign. The pure de Sitter shock parameter obeys

$$\mu^2 = (d-2)\kappa r_c = d - 2 = \lambda_{\ell=1}(S^{d-2}),$$

so the cosmological constant cancels and the smooth isometry modes lie at zero shock eigenvalue. For finite sector dimensions, maximizing

$$S_{\mathrm{gen}}(p, d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i$$

over p gives $p_i = d_i/M$ and $S_{\mathrm{gen}}^{\max} = \log M$. The associated logarithmic area observable $\mathcal{A} = \sum_i (d_i/M) \log d_i$ has symmetric-point Hessian

$$\operatorname{Hess} \mathcal{A}|_{d_i=d} = \frac{1}{nd^2}\left(I - \frac{2}{n}J\right)$$

on the positive-real relaxation of the integer dimensions. Its curvature is positive on fixed-M tangent directions and negative in the homogeneous direction; the gradient at the symmetric point is nonzero. The physical sign comes from a one-sided transfer. Assigning a fraction f of a fixed total horizon–observer capacity to the observer uniformly depletes the horizon and gives

$$\Delta S_{\mathrm{gen}}^{\max} = \Delta \mathcal{A} = \log(1-f) < 0.$$

The finite identity applies to admissible integer depletions, and its positive-real interpolation is strictly decreasing. Under the independent horizon-area and fixed-budget dictionaries, this is the de Sitter time-advance sign studied in Ref. [53]. It is a boundary maximum along the transfer direction, rather than an interior maximum at fixed horizon capacity. Identifying the finite capacity transfer with observer mass is an additional physical dictionary.

On the icosahedral carrier, the spectrum

$$\{-2,\ 0,\ 1 + 3/\sqrt{5},\ 1 + \sqrt{5}\}$$

is conditional on two further statements: the A_5 rotation triplet is an exact gauge sector, and the shock kinetic term is the scaled nearest-neighbour port or edge-sector Laplacian. The regular line-graph identity makes the port and edge-sector low spectra equal. It adds the eigenvalue 10 with multiplicity 18 on the edge carrier. No shock coefficient, physical attachment, or generator for sector-dimension shocks is supplied. The finite heat-bath repair generator acts at fixed sector dimensions and therefore supplies none of these objects. The sign mechanism does not rescue the static-patch trace conjecture [27]. That obstruction tests the cited positive cyclic trace interpretation. Nontrace horizon-screen descriptions, separately constructed dimension-changing generators, and source-derived kinetic operators beyond the scaled nearest-neighbour graph remain outside these classes. None of these boundaries disproves OPH; the pure-de-Sitter normalization, finite entropy and transfer identities, and regular line-graph theorem survive.

What this solves vs. what it assumes. The model does **not** solve the classic "give me a unitary CFT at future infinity" problem. It does not aim there. It also does **not** prove that every refinement-stable MaxEnt branch lands in the static-patch geometric modular branch with emitted cap pair and standard modular action. What it does provide is a coherent route to **patch holography** in which de Sitter static patches are natural:

1. the fundamental object is a horizon screen in a static-patch description;
2. Λ is a capacity parameter tied to finite Hilbert-space dimension, not a locally reconstructible vacuum-energy term;
3. Einstein-like dynamics emerge up to Λg_{ab};
4. on the support-visible BW scaling branch, the realized scaling-limit cap modular action is geometric and may be outer on a non-type-I observer algebra.

BW-side boundary. Theorem 4.2 proves the support-visible Lorentz scaling statement. It avoids the false stronger route through a full-algebra unregularized common floor; the automorphism statement on the observer-facing geometric cap pair is the required static-patch content.

Many observers, one Λ. In this framework, each timelike observer is associated with a horizon patch rather than a single global description. On the conditional capacity branch, the dimensionless de Sitter capacity relation is the shared global constraint across overlap-consistent descriptions; its SI Λ display uses the selected scale certificate.

Summary. The model moves the holographic screen from "infinity" to an observer's horizon. Direct N closure is a correctable public-record conditional construction, not a checkpoint-fixed or marginal-information count. The fixed-$D = 24$ packet is exact inside its declared finite source category. The bounded completion class has incompatible fixed sets. No universal all-rung membership in the complete A1–A3 capacity-source contract or executable-to-Lean bridge is supplied, so direct N is not evaluable on the incomplete source antecedent. Horizon saturation and the weak/Higgs common-load equation are independent downstream commuting squares. The weak/Higgs bridge and the late-time de Sitter coordinate differ by about 6.6 percent. A finite presence correction and a separate Poisson correction give the retrospective comparison coordinates $3.2920978773\ldots\times10^{122}$ and $3.3000722254\ldots\times10^{122}$. Exact compositional countermodels select neither correction from the finite source, so the named-law branch is not evaluable there. Neither is a physical capacity until a stronger source law and both bridges share a certified carrier.

2.4 Einstein-chain theorem and falsification boundary

The finite support and algebraic packets establish repair-invariant incidence, the spherical branch receipts, cap-normal identities, the Lorentz action on the celestial screen, exact finite entropy splitting on the central-interface branch, and the finite tomography identities used to reconstruct a trace-free symmetric tensor from null projections.

Authenticated semantic read-from provenance supplies the finite event carrier and its generated precedence order. Its canonical source height is the attained longest authenticated-parent-chain length and strictly increases on generated ancestry. Separately, every finite decidable strict partial order compiles into an abstract authenticated log whose direct parent relation and generated order are exactly the supplied relation. Thus the generic parent relation contains every supplied predecessor rather than Hasse covers alone; this proves grammar expressivity, not dynamical or physical selection. Independently, the exact rank-three positive source Gram quotient and a real axis define the four-dimensional ambient target carrier

$$\mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t,x) = t^2 - g_{\mathrm{src}}(x,x),$$

with Lorentz inertia $(1,3)$; source-unit directions are exactly its future-null rays. Canonical source height enters only as the temporal coordinate of a positively scaled event placement. Every finite

log has an explicit auxiliary injective forward-causal placement obtained from a finite enumeration along one source axis and a scale larger than its diameter. It is not order-reflecting or source-selected. A supplied physical-candidate spatial readback and edge speed bound give one-way cone compatibility. Equal-height spatial separation and strict spacelikeness of unsupported increasing-height pairs imply converse support and exact two-way order–cone equivalence. Antisymmetry derives injectivity; the two-way equivalence gives exact interval preservation. Physical event identity, the spatial readback, the order-faithful-placement conditions, and operational clock calibration remain separately testable. A controlled effective route uses vanishing causal discrepancy and compatible count-volume convergence. Conservative Fibonacci-record populations with a specified complete-neighbour read law have such a flat $1+3$ limit when their fill error is small relative to the shrinking read radius. Equal-mass assignments with vanishing displacement error give normalized event counts converging to volume. On fixed interior timelike diamonds, fourth roots of interval-count ratios recover proper-time ratios, with a reference interval fixing the unit and complete ancestry access supplying the readout. Within the class of nonzero closed convex pointed displacement cones, operational invariance under source rotations and every boost along one axis forces the Lorentz cone up to time orientation. The covariance is an assumption about actual influences and readouts. Physical selection of the population, read law and common matter realization is separate from these conditional results.

The source-direction Einstein theorem works in a separately supplied $3+1$ tensor interface indexed by the same finite event type and takes balance on nine fixed algebraic inverse images of the coordinate null-tomography directions, symmetric geometry/stress fields in that tensor interface, discrete Ward and Bianchi identities, an algebraic coupling, four step maps, a base event, and connectedness. It derives all-null balance, the pointwise and constant metric ambiguities, and the Einstein-form tensor identity in that supplied tensor interface. The nine directions are not observed provenance links, physical rays, or sky samples, and the supplied fields are not thereby identified with curvature or physical stress-energy. The informational order shares the event type but is not used in the tensor calculation: no provenance link selects a direction, field, step, or balance. The distinguished nine-set is not selected by the source dynamics and is not invariant as a set under Lorentz or SO(3) transformations. The matrix fields and discrete step maps remain supplied tensor data, and their coordinate differences are not identified with those of the constructed carrier. Ambient target-carrier dimension and signature are constructed independently by the source-causal carrier theorem; intrinsic poset dimension remains a continuum diagnostic. Neither those fields nor the physical premise bundle is constructed.

On the exact-embedding route, an effective Lorentzian manifold requires the source-causal continuum certificate of Theorem 4.3e: one physical refinement family with exact two-way order/cone agreement, calibrated count–volume, independent dimension and manifoldlikeness tests, stable topology, and a unique distinguishing Lorentzian limit. The smooth Einstein reading additionally requires a same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or the separately stated continuum small-ball/null-balance identification, plus modular and scaling control, physical stress, Ward/Bianchi, universal coupling, a vacuum reference, and physical $8\pi G$ scale identification on that same family. A convergent causal-set scalar-curvature estimator is an independent diagnostic and is insufficient by itself. Finite certificate programs verify their stated algebra and controls; no source-derived family carrying the full continuum antecedent is constructed here.

The implication requires its receipts on one cofinal branch, a rank-three source Gram quotient, tensor-linearity of the directional charges, the stated curvature or small-diamond remainder rate, and scale readouts with trivial common rescaling. The chosen causal route must meet its exact order–cone criterion or its controlled causal and volume error bounds.

A Technical Details for the Specialist Derivations

This appendix carries the gravity details that support the spacetime and Einstein derivation. It sharpens the Lorentz, Einstein, and cosmological-constant surfaces without widening the recovered core beyond its stated branch conditions.

A.1 Port-record metric completion

Theorem A.1 (Exact repair-response three-dimensional carrier)**.** *Let A be the adjacency matrix of the declared twelve-port incidence, $L = 5I - A$, $T = I - L/60$, and let Q remove the constant port mode. Write P_{slow}, P_5, and P_{fast} for the exact nonconstant spectral projectors of ranks $(3, 5, 3)$. For the complete equally weighted port-probe census,*

$$C_n = QT^{2n}Q = r_{\mathrm{slow}}^{2n}P_{\mathrm{slow}} + (9/10)^{2n}P_5 + r_{\mathrm{fast}}^{2n}P_{\mathrm{fast}},$$

where

$$r_{\mathrm{slow}} = \frac{55+\sqrt{5}}{60}, \qquad r_{\mathrm{fast}} = \frac{55-\sqrt{5}}{60}, \qquad 0 < r_{\mathrm{fast}} < \frac{9}{10} < r_{\mathrm{slow}} < 1.$$

Trace-twelve normalization therefore gives

$$\frac{12C_n}{\operatorname{tr} C_n} \longrightarrow 4P_{\mathrm{slow}} =: G.$$

The intrinsic space $H = \operatorname{ran} P_{\mathrm{slow}}$ has dimension three, and the labeled vectors $v_p = 2P_{\mathrm{slow}}e_p$ satisfy $\langle v_p, v_q\rangle = G_{pq}$.

Let

$$M_{\mathbb{Z}} = \mathbb{Z}[P]/\langle e_{-p} + e_p\rangle \cong \mathbb{Z}^6$$

be the cumulative signed port-record module. The limiting form on its real six-control extension has a three-dimensional radical. Its real quotient is isometric to H. The integer module has no nonzero metric kernel, its image is dense in H, and its metric completion is an abstract continuous three-dimensional Euclidean vector group. The construction is unique up to port-label-preserving isometry and selects no Cartesian axes.

Proof. The spectral polynomial of A gives the four exact projector ranks $(1, 3, 5, 3)$ and the displayed decomposition of C_n. The strict response order makes the two faster ratios vanish after normalization by the slow response. Vertex transitivity fixes the unit diagonal and hence $G = 4P_{\mathrm{slow}}$. Projector symmetry and idempotence give the intrinsic Gram identity. The source Gram calculation identifies its real radical with the coordinate kernel. Irrationality of the golden ratio makes the integer map injective, and the raw coordinate image has finite index eight in $\mathbb{Z}[\varphi]^3$. Its density in $\mathbb{R}^3$ gives the completion. These finite identities, the limit, the radical calculation, injection, and density are machine checked. □

Corollary A.2 (Conditional local position carrier)**.** *Suppose cumulative signed port load is the carrier-position readout, the complete probe census and T belong to that same source record, and the normalized limiting Gram is operational distinguishability. Then the finite record architecture reconstructs one abstract continuous local three-dimensional Euclidean carrier instead of consuming a Cartesian three-space.*

Boundary. For finite n, C_n has rank eleven on the centered port space and rank six on the signed antipodal sector. The normalized limit must precede the radical quotient and metric completion. The theorem does not select the repair law, the cumulative record semantics, or the Gram topology from the premises A1–A3. It supplies no faithful isometric completion action, overlap/refinement gluing, global or physical spacetime, preferred frame, physical scale, clock, field sector, or observable. Ordered record history and repair cost remain a separate fiber.

A.2 Lorentz and Einstein Bridge

Theorem A.3 (Conformal group isomorphism)**.** *The orientation-preserving conformal group of S^2 is isomorphic to the connected Lorentz group:*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

Theorem A.4 (Support-visible geometric modular flow on caps on the extracted geometric subnet)**.** *Assume the premises A1–A3, the derived fixed-cutoff collar package, and the support-visible BW scaling theorem on the extracted geometric subnet. That theorem requires the finite cap-normal support/flow certificate and an independently complete multiresolution algebra-state comparison package on the same tower. Then the scaling-limit modular automorphism group of the cap pair is geometric conformal dilation on that subnet, with the standard 2π normalization supplied by the finite cap-normal certificate. If the emitted scaling-limit cap algebra is type I, this may be written as $K_C = 2\pi B_C + Z_C$ with Z_C central; in the generic continuum case the theorem is the automorphism statement on a non-type-I cap algebra with outer geometric modular action. The collar replacements used in this branch are exact only at EC-aligned exact Markovity or in controlled fixed-collar families with $\delta_{\mathcal{F}}^{\mathrm{M,EC}} \to 0$; small CMI supplies a Fawzi–Renner recovered comparison state, not a dimension-free one-shot exact Markov normal form.*

Definition A.5 (BW-branch observer-relative modular ordering)**.** *On the branch satisfying the hypotheses of the preceding geometric modular-flow theorem, the modular automorphism parameter t of the extracted cap pair supplies a dimensionless ordering for that observer's accessible algebra-state pair. Physical time requires an observer-readable transition, event correspondence, and calibrated clock instrument. No claim about arbitrary operational clocks, global time, or the full problem of time follows from the geometric modular-flow theorem alone.*

BW-side UV scaffold. The UV data on this branch are not a separate cap-isotropy or Euclidean-regularity selector. The fixed-cutoff cap algebras are type-I regulators, but the scaling-limit observer algebra may leave that class, and MaxEnt alone does not select the BW / canonical cap phase. The UV package is the realized transported geometric cap-local system together with the carried-collar schedule derived from the transported fixed-local-collar Markov/faithfulness datum on each fixed local collar model, followed by support-readable modular covariance and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair.

Null-strip completion boundary. The fixed-cutoff strip package is more specific than the generic inherited-strip summary. The null cuts first transfer the same cut-center data as the spatial collar branch, which fixes the central sector-pair decomposition of the strip algebra. The stronger left/right tensor decomposition used by the null modular bridge then requires the extra inherited strip-split condition on the multiplicity spaces, together with the exact-or-controlled Markov hypotheses on one fixed inherited strip model. On that same exact-Markov/central-interface model, a specified endpoint-to-cell map and uniform local norm/site-density bound make

the renormalized half-line family endpoint-Lipschitz; this is additional endpoint-control data, not a consequence of finite-range locality alone. It hence defines the weak tail generator, and on the scaling-limit geometric-cap branch the half-line blow-up net carries the derived half-sided modular pair. Borchers–Wiesbrock then supplies the positive null-translation generator on its Stone domain together with the affine half-line modular relation $K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$, and the same half-line family fixes the generator and its endpoint-derivative identity internally. Its reading as a local null-stress charge additionally requires the effective-locality and same-source identification premise. Bounded-interval formulas are downstream of the E0.5 affine/projective kernel in the Einstein bridge.

Lemma A.6 (Null data ambiguity)**.** *If a symmetric tensor X_{ab} satisfies*

$$X_{ab}k^a k^b = 0$$

for every null vector k, then $X_{ab} = \phi\, g_{ab}$ for some scalar ϕ.

Corollary A.7 (Conditional null-stress data leave the metric term invisible)**.** *If the supplied effective-locality and same-source bridge identifies the directional generators with one symmetric local stress tensor, their null contractions determine that tensor only up to ϕg_{ab}. Under the remaining Einstein-branch premises, the local equation consequently retains the corresponding Λg_{ab} ambiguity.*

Theorem A.8 (Jacobson-type rest-frame relation)**.** *In the local Lorentzian scaling regime, assume the declared fixed-cap generalized-entropy stationarity interface for admissible fixed-cap variations, together with the proved finite MaxEnt identities on the realized cap-label-preserving family. If the effective-locality and same-source bridge gives the half-line generator its local null-stress reading, the conditional bounded-interval transport and small-ball premises yield the rest-frame first-variation relation that drives the spacetime and Einstein paper's local Einstein branch; on the same scaling branch, if that rest-frame relation holds for all local observer four-velocities and all reference states, the spacetime and Einstein paper upgrades it to the full tensor equation by the explicit local quadratic-polarization argument in the main derivation.*

Theorem A.9 (Newton coupling from the scale certificate and edge entropy density)**.** *The geometric coupling read by the Newton area law is*

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

On the OPH gravity row the scale is supplied first by $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, as an independent scale certificate rather than a consequence of $P_\star$ and $N_\star$ alone. With $a_{\mathrm{cell}} = P\ell_\star^2$ and $\bar{\ell}_{\mathrm{shared}} = P/4$, this gives

$$G_{\mathrm{geom}} = \ell_\star^2, \qquad G_{\mathrm{SI}} = \frac{c^3\ell_\star^2}{\hbar}.$$

Here $\bar{\ell}_{\mathrm{shared}}$ is a shared-cut density. It is not the logarithm of an independent cell Hilbert-space dimension and does not select a fixed primitive observer capacity.

A.3 Cosmological-Constant / Screen-Capacity Closure

Lemma A.10 (Vacuum energy blindness)**.** *For any null vector k,*

$$T^{\mathrm{vac}}_{kk} = T^{\mathrm{vac}}_{ab}k^a k^b = -\rho_{\mathrm{vac}}\, g_{ab}k^a k^b = 0.$$

Vacuum-energy contributions therefore lie in the kernel of the null map $T_{ab} \mapsto T_{kk}$.

Proposition A.11 (Structural separation)**.** *Within the structural split used here:*

1. *local modular/null data fix T_{ab} only up to ϕg_{ab}; and*
2. *the dimensionless Λ-capacity relation is fixed by global screen capacity, not by local null data.*

So the large vacuum-energy bookkeeping of EFT is not itself the local quantity determining curvature on this branch.

Theorem A.12 (No Λ prediction from the declared local null data)**.** *Within the null-modular reconstruction used here:*

1. *all T_{kk} data are unchanged under $T_{ab} \to T_{ab} + \phi g_{ab}$;*
2. *therefore Λ cannot be fixed by these local null-modular data alone; and*
3. *the dimensionless Λ-capacity relation requires the global screen-capacity closure beyond the local null data. The stable closure target is $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_{\mathrm{CRC}} = \log D_\star$. The correctable public-record definition and conditional operational closure theorem close its operational public-section, correctable-code, capacity-bound, approximate-stability, and greatest-fixed-point implications under named premises. The full specification and receipt schema are carried by the flagship synthesis. The finite record restrictions, reachability, publicness, global coupling, carrier, scalarization, extension, refinement, and sewing receipts are exact at $D = 24$ inside their declared finite source category. A bounded all-rung counterfamily has different exact fixed sets under shared base, positivity, carrier, and executable finite controls. This proves nonidentifiability for the bounded completion class. Universal all-rung membership in the complete A1–A3 capacity-source contract and an executable-to-Lean bridge are absent. Direct N is not evaluable on the incomplete source antecedent, and the bounded result does not determine the complete source class. A positive cosmic value requires a complete source antecedent, one physical zero, a universe-level carrier attachment, and the horizon–record identification. The electroweak relation is an independent comparison after F and does not construct it. SI curvature values additionally use the selected scale certificate.*

A different typed local route carrying trace-sensitive or non-null overlap observables is not excluded and would require its own source and physical receipt.

Corollary A.13 (Benchmark cosmological readout on the capacity branch)**.** *On the conditional capacity branch defined above, inserting the observed value $\Lambda \approx 1.09 \times 10^{-52}\,\mathrm{m}^{-2}$ gives*

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}} \approx 1.66 \times 10^{26}\ \mathrm{m}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c} \approx 17.5\ \mathrm{Gyr},$$

$$N_{\mathrm{patch}} = \left(\frac{r_{\mathrm{dS}}}{\ell_P}\right)^2 \approx 1.05 \times 10^{122}, \qquad N_{\mathrm{scr}} = S_{\mathrm{dS}} = \pi N_{\mathrm{patch}} \approx 3.31 \times 10^{122},$$

$$\Lambda \ell_P^2 \approx 2.85 \times 10^{-122}.$$

Proof. Immediate from the displayed static-patch and capacity relations. □

Observed-age benchmark boundary. The branch quantity t_Λ is the de Sitter static-patch timescale. The usual cosmic age t_0 is a compare-only FLRW benchmark, with no additional capacity-theorem output. On the flat ΛCDM benchmark

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \sinh^{-1}\left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\right)$$

one gets the standard comparison value $t_0 \approx 13.8\,\mathrm{Gyr}$ at $H_0 \approx 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_\Lambda \approx 0.685$.

Lemma A.14 (FLRW curvature as visible scalar holonomy)**.** *On a homogeneous-isotropic spatial slice with constant sectional curvature K, small spatial loop holonomy obeys*

$$\mathrm{Hol}_{\square_{uv}} = \exp\Big(K A_\square J_{uv} + O(A_\square^{3/2})\Big)\,.$$

For oriented nondegenerate loops small enough that the logarithm branch is fixed, define the normalized scalar readout

$$\kappa_\square := \frac{\langle J_{uv}, \log \mathrm{Hol}_{\square_{uv}}\rangle}{A_\square \langle J_{uv}, J_{uv}\rangle}.$$

Then $\kappa_\square = K + O(A_\square^{1/2})$. On a visibly separated OPH refinement system, the limit of this normalized readout vanishes if and only if $K = 0$. The unnormalized holonomy itself tends to the identity for every finite K, so it is not a flatness test. Thus a flat FLRW branch is the zero-normalized-spatial-curvature-holonomy branch.

Flatness boundary. This holonomy statement names a conditional cosmology bridge only. It does not add a capacity-theorem output and does not solve the inflationary flatness or horizon problems. Selecting $K = 0$ requires an additional theorem or premise: a direct flatness theorem, a conditional cosmological-minimal-holonomy selector on a clocked FLRW boundary, or an explicit flat-branch assumption. No gauge- or matter-sector result supplied in this packet selects $K = 0$. Any gauge-, matter-, or dynamics-based flatness route needs its own theorem and physical receipt.

Theorem A.15 (Local/global cosmological-capacity closure)**.** *Assume the spacetime and Einstein paper's local Einstein branch, stable direct public-record closure and its capacity coordinate*

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star,$$

and the independent horizon–record identification readout

$$N_{\mathrm{CRC}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard static-patch formulas

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Then the cosmological-constant package is one local/global theorem chain: local null data fix the Einstein branch only modulo Λg_{ab}, and with the selected scale certificate the same branch has the global display

$$G_{ab} + \frac{3\pi}{G N_{\mathrm{CRC}}}\, g_{ab} = 8\pi G\, \langle T_{ab}\rangle,$$

while the capacity closure itself fixes the entropy and dimensionless capacity relations

$$S_{\rm dS} = N_{\rm CRC}, \qquad A_{\rm dS} = 4GN_{\rm CRC}, \qquad r_{\rm dS} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\rm dS}}{c},$$

and the observed cosmic age is a downstream FLRW benchmark, with no additional theorem output.

Scope boundary. The hypotheses are exactly the local Einstein branch, stable whole-fiber public-record closure $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, horizon–record identification $N_{\rm CRC} = S_{\rm dS}$, the standard de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed only on a discharged stable-public-record branch with horizon–record identification. Conditional on that branch, the capacity closure fixes $\Lambda_{\rm CRC}\ell_\star^2 = 3\pi/N_{\rm CRC}$; the SI static-patch scale additionally requires the selected scale certificate. CMB kernels, inflation-replacement claims, H_0/S_8 branches and dark/anomaly growth kernels require separate theorems or likelihood contracts. The baryogenesis anomaly/current theorem and gauge/deck no-go are separate from the global-capacity branch; the anomalous record attachment and physical CP-odd source generator are not supplied.

Theorem A.16 (Conditional OPH screen-spectrum theorem)**.** *Let $(\mathcal{S}_r)$ be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar q_r, removal of the background and dipole sector, and a target-free positive quadratic repair operator K_r. If $K_r \to K$, the source-selected finite scalar release energy satisfies $2E^{\rm src}_{q,r}/d_r \to A_q$, and local MaxEnt is imposed at fixed expected quadratic release energy, then q_r converges in finite harmonic distributions to the centered Gaussian screen field with covariance A_qK^{-1}. Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is $t^{\theta/2}\,dt/\Gamma(1+\theta/2)$, then*

$$K_{\rm asy} = (-\Delta_{S^2})^{1+\theta/2}, \qquad C^q_{\ell,{\rm asy}} = A_q[\ell(\ell+1)]^{-1-\theta/2},$$

as the large-ℓ asymptotic model. The theorem-grade finite-ℓ conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell+2+\theta/2)}{\Gamma(\ell-\theta/2)},$$

with

$$C^q_\ell = A_q\frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}.$$

If q is additionally certified as the thin-shell pullback of a homogeneous curvature field with $\Delta^2_\zeta(k) = A_\zeta(k/k_\star)^{-\theta}$, the corresponding inverse amplitude relation is

$$A_\zeta = \frac{A^{\rm shell}_q\Gamma(3/2+\theta/2)}{\pi^{3/2}Z_\star^2(k_\star D_\star)^\theta\Gamma(1+\theta/2)}.$$

Suppose the same source construction emits a strongly continuous full-collar survival cocycle with generator density $P_\star/24$, together with the orientation-reversal half-collar identity. Then

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa^{\rm edge}_{\rm rep} = \frac{P_\star}{48(P_\star-\varphi)}.$$

A finite one-step survival value determines θ through $-\log u_q(\log b)/\log b$; it does not supply the infinitesimal generator receipt. If a scale-natural source embedding transports this cocycle to the physical covariance through $D_s^{-1}C_\zeta D_s = e^{-\theta s}C_\zeta$, the source family is $\Delta^2_\zeta(k) = A_\zeta(k/k_\star)^{-\theta}$. A

single shell has an infinite-dimensional radial kernel, including positive ambiguities. Complete radial cross-covariances provide the independent tomography route. A Fawzi–Renner/Markov-collar observable estimate gives an upper bound on release observables, not an amplitude equality. No physical TT/TE/EE statement follows without independent Boltzmann transfer and likelihood closure.

Screen-spectrum boundary. The conditional theorem fixes the exact angular family. Its radial theorem gives physical source-dilation and cross-covariance tomography as separate uniqueness routes. A finite OPH source construction that emits the primitive collar ensemble, full-collar generator density, half-collar identity, conformal precision, and physical dilation-intertwiner or tomography receipt is not supplied. The thin-shell lift uses the gamma-ratio operator above; finite-width windows require a Bessel-kernel error certificate. Physical temperature and polarization transfer and a frozen likelihood contract are not supplied.

Separate pixel-side benchmarks. The companion pixel numbers

$$a_{\mathrm{cell}} \approx 1.63\,\ell_\star^2, \qquad \ell_{\mathrm{UV}} = \sqrt{a_{\mathrm{cell}}} \approx 1.28\,\ell_\star, \qquad \bar{\ell} \approx 0.408$$

belong to the separate pixel-closure/local-readout package, outside the cosmological-parameter corollary itself. After the selected scale certificate is supplied, $\ell_\star$ is displayed as the usual Planck length.

Local ports and the spherical support. A degree-one federation bridge and each carrier's local twelve-port boundary are separate data. In the declared source construction, the local oriented icosahedral boundary K and the level-zero support S_0 have the same ordered face table. Let $s_0 : S_0 \to N_0$ be the designated section of the nerve projection $b_0 : N_0 \to S_0$, and set $z_0 = (s_0)_*[S_0]$.

Proposition A.17 (Oriented seed attachments)**.** *The maps $h_0 = s_0 \circ p$, where $p : K \to S_0$ ranges over orientation-preserving simplicial isomorphisms, form a set of sixty elements with a free transitive A_5 action. Every such map satisfies*

$$(h_0)_*[K] = z_0$$

as an integer chain. For the fixed source row with fibre sizes $(2, 1, \ldots, 1)$, the designated section and the process link within the double fibre, the proper support relabelings preserving this decoration form C_5. Their action on the sixty maps has twelve orbits.

Proof. The shared oriented face table supplies an isomorphism, so every other one is its composite with a proper automorphism of the icosahedron. These automorphisms form A_5, of order sixty. Composition acts freely and transitively. Each automorphism permutes the twenty oriented triangles, giving $p_*[K] = [S_0]$ exactly; composing with the section proves the chain identity. The double fibre singles out one support vertex. Its stabilizer consists of the five rotations around that vertex and preserves the section and process link. This subgroup acts freely on the seed maps, so every orbit has five elements and there are $60/5 = 12$ orbits. □

The twelve orbits refer to this fixed marked presentation. They do not classify physical alternatives or equivalence after jointly transporting the source decoration and local port charts. Exact enumeration and a separate reconstruction check the sixty maps and their chain images in `code/a5_closure/port_gram_support_attachment_certificate.py` and `code/a5_closur`

`e/verify_port_gram_support_attachment_independent.py`. The support checks also verify $(b_r)_* z_r = [S_r]$ and the section/coarsening squares at four stages with $12, 42, 162, 642$ support vertices.

This seed classification does not supply a source-selected attachment. The analyzed source definitions prescribe each local boundary independently of the federation nerve. The local carrier construction scales barycentric coordinates within a single face, whereas the support tower uses midpoint subdivision and fibre coarsening. Those data do not specify a common local incidence tower and a refinement-commuting family h_r.

The two local Galois radial maps have degrees one and seven. A comparison with the degree-one support bridge requires a source-bound attachment and continuous realizations as maps between the same oriented S^2, with the same orientation convention. Under these hypotheses degree classifies their homotopy classes: the degree-one radial map is homotopic to the support composite, and the degree-seven map cannot. The equality of seed tables alone supplies neither those hypotheses nor a physical direction, metric or scale.

A.4 Causal order on a source-metric population

The continuous rank-three repair carrier admits finite populations chosen by its own metric. A covering property of such a population also supplies a constructive causal approximation, with an explicit local read law and model clock. Let Ω be a convex subset of a real normed space, and let $S \subset \Omega$ satisfy

$$\forall x \in \Omega \quad \exists s \in S : \quad \|s - x\| \le h, \qquad h \ge 0.$$

For an edge radius $a > 2h$, put an event at every (j, s), $j \in \{0, \dots, K\}$, $s \in S$. An event at $(j+1, t)$ reads every (j, s) with $\|t - s\| \le a$. This includes the same-site read, so waiting is permitted. The event order is the reflexive transitive closure of these reads. Reading every permitted neighbour, keeping the full population at each layer, and assigning a duration $\Delta = a/c$ with $c > 0$ are hypotheses of this construction. They are substantive choices of dynamics and clock.

Proposition A.18 (Covering constructs causal paths)**.** *For $s, t \in S$ and $k \ge 1$, with both layers present,*

$$\|t - s\| \le k(a - 2h) \quad \Longrightarrow \quad (j, s) \preceq (j + k, t) \quad \Longrightarrow \quad \|t - s\| \le ka. \tag{1}$$

If S is finite and nonempty, this layered poset has width exactly $|S|$. Every event at layer j has height j, counting edges from layer zero.

Proof. Divide the segment from s to t into k equal pieces, with division points $x_i = s + i(t - s)/k$. Convexity gives $x_i \in \Omega$. Keep the endpoints $p_0 = s$, $p_k = t$ and choose $p_i \in S$ with $\|p_i - x_i\| \le h$ at every interior division point. Then

$$\|p_{i+1} - p_i\| \le \|t - s\|/k + 2h \le a.$$

These are actual permitted reads. For any permitted path the triangle inequality gives the opposite bound in (1). A layer is an antichain, and the same-site waiting chains partition all events into $|S|$ chains. This proves the width statement. Every read increases the layer by one, and a waiting chain reaches every layer, proving the height statement. □

The path construction and outer displacement bound hold in arbitrary real normed spaces; `Lean/Geometry/SourceNetCausalCone.lean` formalizes them. The partial order, its equality with the reflexive transitive closure of the reads, the graph-distance characterization, and the width and height statements are formalized in `Lean/Geometry/SourceNetLayeredOrder.lean`. The volume

estimates below concern the three-dimensional Euclidean repair carrier. The vertical weighted bound and its family limit are formalized below; the separate general moving-tip finite estimate is analytic.

For model time $\tau = k\Delta$, the two radii in (1) are $c(1 - 2h/a)\tau$ and $c\tau$. Consider a family with

$$a_n \longrightarrow 0, \qquad h_n/a_n \longrightarrow 0. \tag{2}$$

The read law agrees with the speed-c continuum causal order away from its null boundary. More explicitly, take two continuum events with positive time separation whose positions lie in Ω. Sample positions within h_n and times on the layer grid within Δ_n. A strict timelike margin eventually exceeds the endpoint errors, the time rounding errors, and $2ch_n/a_n$ times the sampled duration; the inner implication then applies. A strict spacelike margin eventually exceeds the endpoint and time errors, and the outer implication forbids a path. This statement requires the sampled layers to exist in the chosen time window. It makes no assertion of exact order reflection for null-separated events at a finite resolution.

Volume from a controlled population measure. Suppose $\Omega \subset \mathbb{R}^3$ is bounded and convex, and S is finite. In addition to the h-covering population S, supply a measurable partition $(C_s)_{s\in S}$ of Ω, up to null sets, with positive masses $v_s = \mathrm{vol}(C_s)$ and assignment error

$$y \in C_s \quad \Longrightarrow \quad \|y - s\| \le H \quad \text{for almost every } y. \tag{3}$$

Clipped Voronoi cells allow $H = h$. Equal-volume cells with a different assignment bound are also admitted; no Voronoi hypothesis is required. For $x \in S$ and $T = K\Delta > 0$, let I be the inclusive discrete interval between $(0, x)$ and (K, x), and assign it spacetime mass

$$V_{S,\Delta}(I) = \Delta \sum_{(j,s)\in I} v_s.$$

The spatial assignment rule and the temporal factor are part of the measure. The limiting volume convention is $dt\, d^3x$. For the metric $c^2dt^2 - |dx|^2$, its metric volume is c times this coordinate volume.

Proposition A.19 (Weighted Alexandrov-volume limit)**.** *Assume the preceding hypotheses and $B(x, cT/2 + H) \subset \Omega$. Then*

$$\left| V_{S,\Delta}(I) - \frac{\pi c^3 T^4}{24} \right| \le 4\pi(T + \Delta)(cT/2 + H)^2 \left(H + \frac{cTh}{a} \right) + \frac{\pi c^3 T^3 \Delta}{2}. \tag{4}$$

Thus $H_n \to 0$ and (2) imply the continuum Alexandrov-volume limit on aligned, bounded interior time windows. If all cells have a common mass v, this is the raw event-count statement $v\Delta\,|I| \to \pi c^3 T^4/24$.

Proof. Write $u_j = \min(j\Delta, T - j\Delta)$ and $r_j = cu_j$. The causal sandwich puts every population point within $c(1 - 2h/a)u_j$ of x in the interval, and puts every interval point within r_j. At the endpoint layers the interval contains precisely the single endpoint. The partition assignment therefore gives, at every layer,

$$\frac{4\pi}{3}\big(c(1 - 2h/a)u_j - H\big)_+^3 \le \sum_{(j,s)\in I} v_s \le \frac{4\pi}{3}(r_j + H)^3.$$

For the lower estimate, a point in the smaller ball is assigned to a population point inside the inner causal radius. For the upper estimate, the union of cells assigned to interval points lies in the larger

ball. The stated interior buffer keeps these balls in the spatial window. The deviation of either radius from r_j is at most $H + cTh/a$. Since the derivative of $(4\pi/3)r^3$ is $4\pi r^2$, the error per layer is bounded by $4\pi(cT/2 + H)^2(H + cTh/a)$.

Finally, $F(t) = (4\pi/3)[c\min(t, T-t)]^3$ has Lipschitz constant $\pi c^3 T^2$, vanishes at both endpoints, and integrates to $\pi c^3 T^4/24$. Its left Riemann sum differs from its integral by at most $\pi c^3 T^3 \Delta/2$. The inclusive right endpoint contributes zero to this reference sum. Summing the spatial errors over $K + 1$ layers proves (4). □

The complete finite estimate, with almost-everywhere cell coverage, disjointness and assignment, is formalized by `SourceNetVolumeError.weighted_alexandrov_error` in `Lean/Geometry/SourceNetVolumeError.lean`. The zero-hop endpoint cases and all $K + 1$ layers are included. The general limit for varying, possibly unequal cell masses is proved by `SourceNetVolumeError.weighted_alexandrov_tendsto`: it assumes $a_n, H_n, h_n/a_n \to 0$, $K_n(a_n/c) \to T$ and the eventual interior buffer, and derives convergence of the actual weighted generated intervals. `Lean/Geometry/GoldenSourceVolumeLimit.lean` instantiates the bound with the actual golden sites and equal cell masses, proves its exact raw count interpretation, and derives the aligned weighted-volume limit as $K_n\Delta_n \to T$, under the stated interior buffer.

Proposition A.20 (Interior diamonds with moving tips)**.** *Let $x, y \in S$, $\|y - x\| < cT$, and $T = K\Delta > 0$. Suppose the continuum diamond between $(0, x)$ and (T, y) has a positive spatial buffer inside Ω. Write*

$$D(t) = B(x, ct) \cap B(y, c(T - t)), \qquad V(D) = \int_0^T \mathrm{vol}(D(t))\, dt.$$

For the inclusive discrete interval I between these tips,

$$|V_{S,\Delta}(I) - V(D)| \le 8\pi(T + \Delta)(cT + H)^2 \left(H + \frac{2cTh}{a}\right) + 4\pi c^3 T^3 \Delta. \tag{5}$$

In particular, the same refinement conditions give weighted interval-volume convergence for arbitrary fixed interior timelike diamonds.

Proof. At layer j, the union of assigned cells contains the intersection of the two balls with radii $(c(1 - 2h/a)j\Delta - H)_+$ and $(c(1 - 2h/a)(T - j\Delta) - H)_+$, up to null sets. It is contained in the intersection with radii $cj\Delta + H$ and $c(T - j\Delta) + H$. Zero-radius balls contribute zero volume. Changing either reference radius changes the intersection only within that ball's spherical shell. Each radial error is at most $H + 2cTh/a$, and each outer radius is at most $cT + H$. The sum of the two shell bounds gives the spatial term in (5). The reference section volume has Lipschitz constant at most $8\pi c^3 T^2$: over a time increment, each of the two radii changes at speed c, and each ball has surface area at most $4\pi c^2 T^2$. Both endpoint sections have zero volume. The same Riemann-sum argument gives the temporal term. □

A conservative-record family with a raw count law. The population and equal-mass hypotheses have an explicit realization in conservative source records. In the exact source-axis presentation, write $\varphi = (1 + \sqrt{5})/2$ and $L = 2/\sqrt{\varphi + 2}$. For consecutive Fibonacci numbers $q = F_n$, $p = F_{n+1}$, with $n \ge 4$, define

$$\xi_b = b\varphi - \lfloor b\varphi \rfloor, \qquad 0 \le b < q.$$

For a triple $b = (b_1, b_2, b_3)$, put $m_i = -\lfloor b_i\varphi \rfloor$ and take the integer record

$$z(b) = (b_2 - m_1, b_2 + m_1, b_3 - m_2, b_3 + m_2, b_1 - m_3, b_1 + m_3). \tag{6}$$

Its coordinate sum is $2(b_1+b_2+b_3)$. It therefore lies in the exact conservative seam-current image, whose integer section supplies a finite source word. Direct contraction with the source axes gives the readback

$$s(b) = L(\xi_{b_1}, \xi_{b_2}, \xi_{b_3}) \in \Omega = [0, L]^3.$$

These are actual source-record images. A regular mesh is used below only to prove a volume estimate; it does not replace the event positions.

Proposition A.21 (Source-record quadrature and four-dimensional counting)**.** *The population $S_q = \{s(b) : 0 \le b_i < q\}$ has exactly q^3 distinct sites. It admits a partition of Ω into cells of common volume $v_q = L^3/q^3$, assigned to the sites with error*

$$H_q = \frac{2\sqrt{3}L}{q}. \tag{7}$$

Consequently, for every Lipschitz function f on Ω,

$$\left| v_q \sum_{s \in S_q} f(s) - \int_\Omega f(x)\, d^3x \right| \le L^3 H_q \operatorname{Lip}(f).$$

Apply the preceding complete-neighbour read law on S_q, with $a_q = L/\sqrt{q}$ and $\Delta_q = a_q/c$. Its density in the declared $dt\, d^3x$ convention is

$$\rho_q = \frac{q^3}{L^3 \Delta_q}. \tag{8}$$

Normalized raw event counts converge to volume on fixed bounded regions with volume-null boundary. On intervals whose endpoints converge to a strictly timelike pair and whose limiting diamond has an interior buffer in the spatial and temporal windows, the generated order intervals converge to that same diamond in count measure. In particular, for vertical tips separated by model time T,

$$\frac{|I_q|}{\rho_q} \longrightarrow \frac{\pi c^3 T^4}{24}.$$

The local limiting order and measure are those of flat $1+3$ spacetime. This conclusion concerns this declared source-record family and update law.

Proof. The Fibonacci recurrence gives $\gcd(p, q) = 1$ and $p - q\varphi = (-1)^n \varphi^{-n}$, hence $|\varphi - p/q| < q^{-2}$. Let k_b be the residue of bp modulo q. These residues permute $0, \dots, q-1$. For $b = 0$ the two floors below are zero. For $0 < b < q$, bp/q is at least $1/q$ from either adjacent integer, whereas $|b(\varphi - p/q)| < 1/q$. Therefore

$$\lfloor b\varphi \rfloor = \lfloor bp/q \rfloor, \qquad \xi_b = \frac{k_b}{q} + b(\varphi - p/q), \qquad \left| \xi_b - \frac{k_b}{q} \right| < \frac{1}{q}.$$

Irrationality of φ makes the orbit points distinct; the source readback then distinguishes all q^3 records.

Assign to $s(b)$ the rectangular cell $\prod_i [Lk_{b_i}/q, L(k_{b_i}+1)/q]$, resolving cell-boundary ties arbitrarily. This is a partition up to null sets. A site need not belong to its assigned cell, but every coordinate displacement from a point of that cell is at most $2L/q$. This proves (7), the covering property and the Lipschitz estimate by integrating the displacement bound over each cell. Taking

products with time cells $[j\Delta_q, (j+1)\Delta_q]$ proves convergence of the normalized event measure: for a Lipschitz test function on an aligned time window of length T, the error is at most

$$TL^3(H_q \operatorname{Lip}_x(f) + \Delta_q \operatorname{Lip}_t(f)).$$

An extra endpoint layer contributes at most $L^3\Delta_q\|f\|_\infty$. Continuous approximation then gives the count statement for regions with volume-null boundary.

Use $h_q \le H_q$ in the causal sandwich. Here $a_q \to 0$ and $H_q/a_q = 2\sqrt{3}/\sqrt{q} \to 0$; the sufficient positive-inner-cone condition from this bound holds when $q > 48$. For converging timelike tips, the order and sampling discrepancies are confined to shrinking neighbourhoods of the two limiting null boundaries. Their volume is zero. The uniform cell displacement and temporal error also tend to zero, so the normalized number of events in these shrinking neighbourhoods tends to zero. This proves interval-count convergence; for vertical tips it also follows directly from (4). The formula for the flat diamond is its elementary ball-section integral. □

The spatial arithmetic, partition and quadrature statements are formalized in `Lean/Geometry/GoldenSourceAssignment.lean`; normalized event-count convergence on fixed measurable continuity sets for the declared time grid, with final-layer discrepancy at most $L^3\Delta_q$, in `Lean/Geometry/GoldenSourceCountLimit.lean`. Causal-frontier and pairwise null-separation nullity are proved by Fubini in `Lean/Geometry/SourceCausalBoundary.lean`, and eventual order agreement is derived from actual covering paths in `Lean/Geometry/SourceNetOrderLimit.lean`. Count convergence for moving generated intervals is proved in `Lean/Geometry/GoldenSourceCausalLimit.lean`; its reference volume is restricted to the supplied time and spatial window, and an uncut diamond requires the containment hypothesis. Strict generated-pair convergence to the geometric product measure with the $N_q(N_q-1)$ normalization is proved in `Lean/Geometry/GoldenSourcePairLimit.lean`, the explicit vertical volume error and the aligned weighted count-volume limit in `Lean/Geometry/GoldenSourceVolumeLimit.lean`, and the uncut interval volume together with the ordering-fraction limit of Proposition A.22, with the window containment hypothesis explicit, in `Lean/Geometry/FlatDiamondNormalization.lean`. The moving-tip finite error estimate (5) is an analytic estimate.

Proposition A.22 (The four-dimensional ordering fraction)**.** *For the contained timelike intervals in Proposition A.21, let $N_q = |I_q|$ and let C_q count their strict causally ordered pairs. Then*

$$\frac{2C_q}{N_q(N_q-1)} \longrightarrow \frac{1}{10},$$

the four-dimensional Myrheim–Meyer fraction [8, 9, 10].

Proof. The cell assignment gives convergence also for the product event measure. Interval-membership boundaries, the pairwise null-relation boundary and the coincidence diagonal have zero limiting measure. The causal sandwich confines disagreement to shrinking neighbourhoods of these boundaries. Hence the normalized strict-pair count tends to its continuum double integral; $N_q \to \infty$ removes the distinction between N_q^2 and $N_q(N_q-1)$.

In units $c = 1$, place the limiting tips at $(0,0)$ and $(T,0)$ and write $V = \pi T^4/24$. The volume to the future of (t,x) within the interval is $\pi[(T-t)^2 - |x|^2]^2/24$. Thus the strict oriented-pair integral is

$$\frac{\pi^2}{6}\int_0^T \int_0^{\min(t,T-t)} r^2\big((T-t)^2 - r^2\big)^2\, dr\, dt = \frac{\pi^2 T^8}{11520} = \frac{V^2}{20}.$$

The ordering fraction counts both orientations of distinct comparable pairs, giving 1/10. Constant changes of time and volume units cancel in the ratio. This is an interval statistic; the same value is not asserted for a whole spatial box times a time window. □

The strict-pair count converges to the geometric double-indicator measure, with the $N_q(N_q-1)$ normalization retained. The product endpoint clipping is bounded by $L^6\Delta_q(2T+\Delta_q)$ in `Lean/Geometry/SourceCountTransport.lean`, and the pair limit is proved in `Lean/Geometry/GoldenSourcePairLimit.lean`.

The spherical-section, Lorentz-volume and pair-integral modules derive the geometric measure by Fubini and radial integration. Their explicit Lorentz boost has determinant one and preserves the causal relation. The normalization theorem in `Lean/Geometry/FlatDiamondNormalization.lean` gives the $1/10$ limit for supplied $c>0$ and timelike limiting tips whose diamond lies in $[0,T]\times[0,L)^3$. Measure equality retains the null final-time boundary. These theorems select no population, read law, physical clock or physical measure, and they do not count additional instrument or routing operations.

A sharper finite covering constant is available without changing this construction. Sort the one-dimensional orbit as $0=\eta_0<\cdots<\eta_{q-1}<1$ and put

$$r_q=\max\left\{\frac{1}{2}\max_{i<q-1}(\eta_{i+1}-\eta_i),1-\eta_{q-1}\right\},\qquad h_q=\sqrt{3}Lr_q.$$

The product population has exactly this covering radius in the closed cube. The path condition $2h_q<a_q$ is the exact quadratic-field inequality $12qr_q^2<1$. This sharper h_q concerns paths; the assignment estimate H_q in the count proof is separate.

The source-word section has coefficients growing at most linearly in q; the finite implementation records their actual primitive lengths. Counting one distinguished event per site and layer is part of the stated law. Intermediate seam-word operations, read instructions and bookkeeping records are not silently included in the same density. The spatial populations are nested along increasing Fibonacci cutoffs. Their changing layer duration and edge radius do not make the spacetime histories induced prefixes; regulator convergence uses the common metric carrier and controlled cell assignments.

Source populations and physical interpretation. The finite-word source construction supplies arbitrarily fine metric nets in bounded convex windows with nonempty interior in the rank-three completion. A maximal separated subset retains a controlled fill distance and three-dimensional packing bounds. These statements supply the spatial hypotheses of Proposition A.18 under a specified resolution and population rule. Three-dimensional packing bounds alone do not supply equal-volume sampling. The count conclusion requires a partition satisfying (3) and, for unweighted counts, equal cell masses or a separate counting approximation.

The construction gives a causal approximation on source-metric points without inserting a Cartesian spatial table. The complete neighbour read law, repeated population, cell measure, and model duration are supplied. An implementation by accepted observer repairs must realize these inputs with its actual records and operation costs. A scalar action on the same population can have a different finite propagation relation; the causal cone proved here is not an identification of that action's stencil cone or its continuum response speed. Agreement with a field action and a physical clock is an additional common-model condition.

A.5 Protected spatial records and mutable repair states

A population attached to retained records and a population read from live repair amplitudes obey different evolution laws. The distinction can be characterized exactly for scalar seam averaging. On a finite port set I, let T_{uv} replace the two endpoint loads by their arithmetic mean and leave the other loads unchanged. For coefficients $w_i\in\mathbb{R}$, define $R_w(N)=\sum_i w_iN_i$.

Proposition A.23 (Linear records preserved by repair)**.** *For distinct endpoints u, v,*

$$R_w(T_{uv}N) - R_w(N) = \frac{(w_u - w_v)(N_v - N_u)}{2}.$$

Thus R_w is invariant under T_{uv} for every input precisely when $w_u = w_v$. For a specified seam graph, invariance under every allowed repair is equivalent to constancy of w on each connected component. On a connected graph every such record has the form $R_w(N) = w_* \sum_i N_i$.

Proof. Only the endpoint summands change. Substitution gives the displayed defect. Equality of their coefficients suffices for invariance. Conversely, use the input with value one at v and zero elsewhere. Along a graph path the endpoint equalities propagate, proving constancy on components. Grouping the resulting sum by components gives the claimed factorization. □

The defect, both directions of the invariance criterion, propagation along arbitrary finite paths, and connected-component factorization are proved in `Lean/Geometry/SourceRecordProtection.lean`. Apply the scalar theorem coordinatewise to a vector readout. An active seam changes that readout whenever its two coefficient vectors differ. This describes the specified linear readout and autonomous pair-mean law. It excludes neither nonlinear records nor preserved memory, driven evolution, or a different repair law.

For the twelve-port carrier, the live rank-three readback has distinct coefficient vectors at distinct ports. It therefore changes at every active seam. This does not change the rank or metric of the response space: those belong to the repair operator. It changes the point represented by the live load inside that space. If internal repairs converge to a constant load, the centered rank-three readback tends to zero.

An exact finite execution distinguishes the two record choices on the golden population with $q = 13$. For each of its 2197 sites, the initialized loads use the signed antipodal split of the conservative record. The first lexicographically ordered unequal intra-carrier seam is averaged once; the zero record requires no update. All 2196 active updates preserve total load and strictly decrease its quadratic potential. Every active carrier's live spatial address changes, while the 2197 resulting live addresses are distinct. For the squared dimensionless read radius $1/13$, the number of directed metric-neighbour pairs including self-pairs changes from 294191 to 32249; 135123 unordered adjacency decisions change. The exact rational event history and independent quadratic-field distance census are provided in `code/source_population/`. This execution uses intra-carrier repairs only; it is not an implementation of routed inter-carrier reads.

There is also a simple invariant population extension. Give each carrier an immutable initialized address register and a mutable port-load register; let each repair write only the latter. The address population and its metric-neighbour graph are then unchanged under every repair word, even though the live amplitudes relax. The same execution retains all 2197 prepared addresses exactly. The golden population, its initialization and the protected write footprint are supplied inputs. The causal/count limit applies to these retained addresses when its read law and measure hypotheses hold. Availability of a conservative source word alone does not establish the preparation or preservation of its live readback. A maintained source population requires a stated protected-record or driven-population law.

A.6 Source-population quadrature and detector errors

Cell masses connect a finite source population to an integral on its metric carrier. A covering radius controls nearest-point geometry; an assignment radius controls quadrature for a specified

measure. These radii can differ. For example, a golden-orbit point need not lie in its equal-volume permutation cell. The estimates below use the actual cell assignment and its measure.

Let Ω be a metric window with a finite positive Borel measure μ. Choose a finite measurable partition C_i, representatives $s_i \in \Omega$, and masses $v_i = \mu(C_i)$. A measurable convention assigns shared boundaries; null modifications do not affect the statements. Representatives need not belong to their cells. Let f take values in a real Banach space and be integrable on every cell. Set

$$Qf = \sum_i v_i f(s_i), \qquad D_i = \int_{C_i} d(s_i, x)\, d\mu(x).$$

Proposition A.24 (Integrated cell-transport estimate)**.** *Suppose $\|f(s_i) - f(x)\| \le K_i d(s_i, x)$ for almost every $x \in C_i$, where $K_i \ge 0$ and the displacement is integrable. Then*

$$\left\| Qf - \int_\Omega f\, d\mu \right\| \le \sum_i K_i D_i. \tag{9}$$

If $d(s_i, x) \le H_i$ almost everywhere on C_i, the right-hand side is at most $\sum_i v_i K_i H_i$. In particular, for a globally K-Lipschitz function and a uniform assignment bound H,

$$\left\| Qf - \int_\Omega f\, d\mu \right\| \le \mu(\Omega) K H. \tag{10}$$

For a sequence of such partitions of the same measured window, $H_n \to 0$ therefore gives $Q_n f \to \int_\Omega f\, d\mu$ for each integrable Lipschitz test function. Neither nesting nor a fixed number of representatives is required.

Proof. The difference equals $\sum_i \int_{C_i} [f(s_i) - f(x)]\, d\mu(x)$. The norm of a Bochner integral is bounded by the integral of the norm. Apply the local Lipschitz bound and sum the resulting inequalities. The uniform estimate uses $D_i \le v_i H_i$ and $\sum_i v_i = \mu(\Omega)$. Its right-hand side tends to zero under the stated refinement. The first-moment coefficient is sharp: on $[0, H]$, with Lebesgue measure, one representative at zero and $f(x) = x$, the error and displacement integral both equal $H^2/2$. □

The Lean module `SourcePopulationQuadrature` proves the actual Bochner-integral estimates for finite cell measures, their specialization to measurable disjoint partitions, and convergence under a vanishing assignment bound. It also checks that a constant detector exposes any error in total assigned mass. The source coordinates, partition geometry and their displacement bounds must be established for each population. These hypotheses are not replaced by an assumed quadrature limit.

For the golden permutation cells of side L/q, the established assignment bound $H_q = 2\sqrt{3}L/q$ and equal masses L^3/q^3 instantiate (10). Product time cells of length Δ_q contribute the independent temporal error $TL^3\Delta_q \operatorname{Lip}_t(f)$ on an aligned window of duration T. The spatial term is $TL^3 H_q \operatorname{Lip}_x(f)$. An unmatched endpoint layer contributes at most $L^3\Delta_q \|f\|_\infty$. This application retains the declared event population and time-cell convention; extra routing or bookkeeping events cannot be inserted into its density without another count estimate. The module `GoldenSourceAssignment` proves the Fibonacci approximation and floor agreement, constructs the residue-permuted rectangular partition, and derives its equal Lebesgue masses and Euclidean assignment bound. For each fixed globally Lipschitz Banach-valued detector, it proves the actual golden-population quadrature estimate and its refinement limit; the closed and half-open cube integrals agree almost everywhere.

Proposition A.25 (Actual event counts on continuity sets)**.** *Fix $T \geq 0$, $L, c > 0$, $q_n = F_{n+2}$ and $\Delta_n = L/(c\sqrt{q_n})$. Let $C_n(B)$ count the distinct points $(j\Delta_n, s_n(b)) \in B$ with $0 \leq j \leq \lfloor T/\Delta_n \rfloor$ and $0 \leq b_i < q_n$. If B is measurable and its frontier is null for coordinate volume restricted to $\Omega = [0,T) \times [0,L)^3$, then*

$$\Delta_n \frac{L^3}{q_n^3} C_n(B) \longrightarrow \mathrm{vol}(B \cap \Omega).$$

The discrepancy between equal event weights and the corresponding clipped time-cell weights is at most $L^3\Delta_n$, including an aligned final endpoint layer whose clipped cell is empty.

Proof. On each cell, replace the indicator of B by its value at the assigned event. Away from the null frontier these indicators converge pointwise, because the assignment distances vanish. They are bounded by one, so dominated convergence gives the clipped-cell count limit. The sum of the nonnegative equal-weight excesses is at most $L^3\Delta_n$; this also bounds the excess over any selected cells. The golden and time-coordinate maps are injective, so the label sums are point counts. □

These steps, the actual Δ_n specialization and the endpoint control are kernel-checked in `Lean/Geometry/GoldenSourceCountLimit.lean`. Causal frontiers have zero volume by the Fubini argument in `Lean/Geometry/SourceCausalBoundary.lean`. The actual count measure converges on moving generated-order intervals; the finite window is part of the reference volume. Strict-pair counts converge to the geometric product measure with the pair endpoint error retained. The corresponding proofs are in `Lean/Geometry/GoldenSourceCausalLimit.lean` and `Lean/Geometry/GoldenSourcePairLimit.lean`.

The geometric pair measure is evaluated by Fubini and radial integration in `Lean/Geometry/FlatDiamondPairIntegral.lean`. For a timelike diamond contained in the supplied window, the ordering fraction tends to 1/10; this normalization is proved in `Lean/Geometry/FlatDiamondNormalization.lean`. The explicit vertical Alexandrov error and the general weighted-family limit are proved in `Lean/Geometry/SourceNetVolumeError.lean`, and the golden specialization in `Lean/Geometry/GoldenSourceVolumeLimit.lean`. Extra routing events require another count estimate. These theorems select no physical measure or clock.

Proposition A.26 (Localized radial-kernel quadrature)**.** *For $\epsilon > 0$, write*

$$w_x(y) = (1 - d(x,y)/\epsilon)_+, \qquad g_x(y) = w_x(y)[f(x) - f(y)].$$

If f is a real K-Lipschitz function, then g_x is K-Lipschitz and vanishes outside the closed radius-ϵ ball about x. With an assignment map $P(y) = s_i$ on C_i satisfying $d(P(y), y) \leq H$, where $H \geq 0$, its quadrature error obeys

$$\left| \sum_i v_i g_x(s_i) - \int_\Omega g_x(y)\, d\mu(y) \right| \leq K \int_{\Omega \cap B(x,\epsilon+H)} d(P(y), y)\, d\mu(y) \leq KH\, \mu(\Omega \cap B(x, \epsilon + H)). \quad (11)$$

The same statement holds with closed balls; their boundaries are irrelevant for Euclidean volume.

Proof. The tent weight lies in $[0,1]$ and satisfies $|w_x(y) - w_x(z)| \leq d(y,z)/\epsilon$. Order the two weights so that $w_x(y) \leq w_x(z)$. If both vanish the claim is immediate; otherwise $d(x,z) < \epsilon$. The identity

$$g_x(y) - g_x(z) = w_x(y)[f(z) - f(y)] + [w_x(y) - w_x(z)][f(x) - f(z)]$$

is bounded by $Kd(y,z)[w_x(y) + d(x,z)/\epsilon]$. The bracket is at most one because $w_x(y) \leq w_x(z) = 1 - d(x,z)/\epsilon$. The same proof with y, z exchanged covers the other order. If $d(x,y) \geq \epsilon + H$, both kernel terms $g_x(y)$ and $g_x(P(y))$ vanish. Integrate the pointwise bound over the remaining region. The tent Lipschitz estimates have Lean proofs; the displayed support-restricted integral applies the cell-transport argument. □

For three-dimensional volume, multiplying (11) by $45/(\pi\epsilon^5)$ gives the quadrature contribution $60KH(\epsilon+H)^3/\epsilon^5$ in the scalar operator estimate, and hence $480KH/\epsilon^2$ when $H \le \epsilon$. Actual local cell masses or integrated displacements can give smaller bounds than the enclosing-ball volume. They must be computed or bounded for the same assignment. A small fill radius does not justify replacing the equal-volume assignment radius. This estimate alone yields no vanishing operator error when $H = O(q^{-1})$ and $\epsilon = O(q^{-1/2})$.

Dual cells for a local tensor action. A different supplied action can use sorted source coordinates $0 = x_0 < \cdots < x_N = L$, with interval lengths $\ell_j = x_{j+1} - x_j$. Half-interval endpoint weights and interior weights $(\ell_{j-1} + \ell_j)/2$ are the masses of the one-dimensional dual cells. Their tensor products satisfy the preceding transport theorem. For $f \in C^2([0, L])$, linear interpolation on each interval also gives the sharper analytic trapezoidal bound

$$\left| Qf - \int_0^L f(x)\,dx \right| \le \frac{\|f''\|_\infty}{12} \sum_j \ell_j^3 \le \frac{L\ell_{\max}^2}{12} \|f''\|_\infty.$$

Indeed the pointwise interpolation error is at most $\|f''\|_\infty (x - x_j)(x_{j+1} - x)/2$; integrate it on each interval. Successive application in three coordinates gives the tensor bound $L^3 \ell_{\max}^2 \sum_{a=1}^3 \|\partial_a^2 f\|_\infty / 12$. This applies to detector products and sampled mode Gram entries with their stated derivative bounds. Dual-cell weights and equal-volume source-count weights are different finite measures, each with its own comparison to volume. The trapezoidal argument does not improve the radial action by changing its weights implicitly, and it supplies no source-selection law, primitive timing rule or physical calibration.

A.7 Readback through destructive seam averaging

A seam mean destroys an endpoint difference in its mutable state, but this alone does not obstruct communication when an observer retains local readback records. Consider a simple path of d seams with arbitrary real initial port values $(x_0, \ldots, x_d)$. No other operation touches the path during the following declared schedule. For $k = d-1, d-2, \ldots, 0$, apply canonical pair means successively on seams $(k, k+1), (k+1, k+2), \ldots, (d-1, d)$. The destination records its own initial value and its own value after each sweep, giving $d+1$ local samples. Every intermediate repair and readback is retained; the record decoder changes no port value.

Theorem A.27 (Destination-local path reconstruction)**.** *The destination samples determine all $d+1$ initial path values uniquely. They also determine the changed port state after every completed suffix calibration. The schedule uses exactly $d(d+1)/2$ pair-mean evaluations, of which $d(d-1)/2$ calibrate the relays and d perform the final source sweep.*

Proof. For a relay state $B = (b_1, \ldots, b_m)$, a forward sweep has endpoint map

$$E_B(x) = \Phi_{b_m} \circ \cdots \circ \Phi_{b_1}(x), \qquad \Phi_b(x) = \frac{x + b}{2}.$$

Its inverse applies $y \mapsto 2y - b_m$, then $y \mapsto 2y - b_{m-1}$, and so on. The destination initially knows its singleton suffix. Suppose it has reconstructed the initial and current states of the calibrated suffix. The next endpoint sample and that *current* suffix state therefore recover the next source value by E_B^{-1}. Replaying the same means gives the new current suffix state. Induction reconstructs the entire initial vector using only destination samples. Summing the sweep lengths $1 + \cdots + d$ gives the operation count. □

This construction also exposes its precision cost. With the calibrated relay state fixed, changing x_0 by δ changes the final destination sample by exactly $2^{-d}\delta$.

Theorem A.28 (Exact conditioning of the local readback map)**.** *For exact seam dynamics and decoding, let O_d map initial port values, ordered by distance from the destination, to chronological samples $y_0, \dots, y_d$. Then*

$$\kappa_\infty(O_d) = \|O_d^{-1}\|_\infty = g_d, \qquad g_d = \frac{(1+\sqrt{2})^{d+1} + (1-\sqrt{2})^{d+1}}{2}.$$

Independently bounded additive sample errors of size $\epsilon \geq 0$ give sharp worst-case reconstruction error $g_d\epsilon$ in the maximum norm.

Proof. Let u_r^k be the value at distance r after the length-k sweep, and write $Sy_k = y_{k-1}$. For $k \geq 1$, the actual means give

$$u_0^k = u_1^k = y_k, \qquad u_r^k = 2u_{r-1}^k - u_{r-2}^{k-1} \quad (2 \leq r \leq k).$$

For $k \geq 1$, the initial value at depth k is $2u_k^k - u_{k-1}^{k-1}$; at depth zero it is y_0. Define $H_0 = H_1 = 1$ and $H_r(z) = 2H_{r-1}(z) - zH_{r-2}(z)$. Induction gives $u_r^k = H_r(S)y_k$ and the recovered initial value $H_{k+1}(S)y_k$. Thus inverse-row delay polynomials $F_k = H_{k+1}$ obey $F_0 = 1$, $F_1 = 2 - z$ and $F_k = 2F_{k-1} - zF_{k-2}$. Their coefficients alternate in sign; their magnitudes combine by addition. Consequently their row norms satisfy $g_0 = 1$, $g_1 = 3$, $g_k = 2g_{k-1} + g_{k-2}$, giving the displayed formula. Every observation is a convex combination of initial values, so $\|O_d\|_\infty = 1$; the largest inverse-row norm is g_d. Sign-matched sample errors attain the bound. □

At fixed range and reconstruction tolerance, the additional sample precision scales as $\log_2 g_d = 1.27155\ldots\, d + O(1)$ bits, linearly in path length. This supplies no physical precision or clock guarantee.

Corollary A.29 (Bounded gate and retained-sample errors)**.** *Use the same fixed schedule. Let each complete pair-mean map have additive maximum-norm error at most η, let each retained sample (including memory drift) have error at most σ, and let decoding add error at most δ. Starting from identical initial port values, the reconstructed vector obeys*

$$\|\widehat{x} - x\|_\infty \leq g_d\left(\frac{d(d+1)}{2}\eta + \sigma\right) + \delta.$$

Proof. Every exact mean is nonexpansive in the maximum norm, so state error accumulates by at most η per mean. Apply the inverse norm to the retained samples. □

This sufficient bound assumes the stated noise limits; it is not a sharp gate-noise bound or a measurement of physical hardware errors.

Independent matrix inversion on embedded W12 paths of lengths $1, 4, 12, 24$ gives respectively $g_d = 3, 41, 47321, 1855077841$. The length-24 baseline uses 300 pair means, 25 protected samples and 351 total initialization, repair, local-readback and decoding events; every retained mean is active. Equal-endpoint evaluations in the general theorem are identity updates and assert no strict progress. Its largest written scalar register payload uses 111 sign, numerator and denominator bits under the declared convention; event metadata and temporary arithmetic storage are excluded. Source and nonsource interventions are independently replayed, together with wrong-law, missing-calibration and remote-read controls.

The module `SourceSeamPathTomography` formalizes the exact recursive inverse, injectivity, seam count and attenuation. Independent verification of the finite observation matrices and complete versioned histories is supplied in `code/source_routing/`. Initial values, wiring, schedule and protected readback memory are supplied. This is destructive finite readout transport; it neither selects the metric-neighbour read law nor supplies repeated immutable-version reads, an intervention-preserving refinement of the full layered order, bounded physical precision or a physical clock.

A.8 Reusable classical readout with local feedback

Destructive path reconstruction can be replaced by reusable transport when local feedback is admitted explicitly. Each participating patch has a mutable scratch port, a protected zero record and immutable scalar version records. In addition to the canonical pair mean, allow a patch to export one of its own records into its own scratch port, reset that port from its own zero record, and archive twice its own port readout. These local copy/reset and decoding operations are supplied classical software semantics; the pair-mean law alone does not select them.

For a support seam $u \sim v$ and a version value x held at u, perform the following six events: export x into u, reset v to zero, average the pair, archive twice the readout at v, and reset u and v from their respective local zeros. The final archive at v equals x, both scratch ports return to zero, and no protected version is changed. Every local operation reads and writes only its own patch; the mean is the sole operation spanning the seam. Counts include identity mean evaluations for a zero payload; they assert no strict repair progress. All means in the retained episodes are active.

Theorem A.30 (Reusable version transport and semantic order)**.** *On a supplied connected support, the stated local operations transport an immutable scalar version along any finite path of d seams using d means, d exports, d captures and $3d$ resets. The transport uses exactly $6d$ events, $7d$ scalar reads and $7d$ scalar writes, with non-interleaved six-event hops. It remains valid under repeated reads, reuse of scratch ports and arbitrary changes of logical values at their publication, with fixed paths, schedule and read interfaces.*

For a finite acyclic scalar read program with total local commits, compile each requested read to such a path and let every logical commit consume only its delivered records and its stated local inputs. The induced semantic read-from order on logical commits is exactly the program's read-from order. All intermediate events remain in the refinement.

Proof. The received value is $2(x+0)/2 = x$, independently of the previous scratch values. Induction along the path transports the same immutable version. Count the six events and their explicit interfaces: the mean has two reads and two writes; each of the other five events has one of each.

Label each payload event by its originating logical writer and label local initialization and constant resets by a bottom symbol. A semantic edge out of a constant has no logical ancestor; other transport edges retain their source label until a logical commit consumes the payload. Every such edge therefore maps to equality or a logical read path. Conversely, every requested read has its retained export–mean–capture path from its writer to its consumer. Path composition gives both inclusions. Reset writers depend only on protected zeros, so a prior scratch payload creates no additional semantic ancestor. The argument holds for every choice of scalar values and hence commutes with the stated interventions. □

This is a projection of semantic data dependencies. It does not identify the projected order with physical causal order: shared-resource execution, scheduler/control order, write-after-write constraints and custody-chain sequencing remain separate. Version identity is bound by the fixed compiled schedule and authenticated writer chain; a physical address/header channel is not supplied.

Feedback can change the live loads used by a carrier's instantaneous readback; maintained addresses are separate protected records here.

Theorem A.31 (Additive error under restored transport)**.** *For each hop, bound the export, receiver reset, receiver mean coordinate, readback and decoder errors by $E, Z, M, R, D \geq 0$, respectively. Include archive retention/read/write error in E. If the initial archive error is at most E_0, the final depth-d payload error satisfies*

$$|\widehat{x} - x| \leq E_0 + d(E + Z + 2M + 2R + D).$$

The bound is sharp when the stated additive errors may be chosen separately.

Proof. Writing one hop's errors as e, z, m, r, a, the receiver obtains

$$2\left(\frac{\widehat{x} + e + z}{2} + m + r\right) + a = \widehat{x} + e + z + 2m + 2r + a.$$

The incoming error has coefficient one. Sum over the hops and apply the triangle inequality. Equal-sign extremal errors attain the bound. Fresh export and receiver reset precede every mean, so cleanup errors from an earlier hop are overwritten before they can enter a later payload. □

The noise limits are hypotheses, not measurements of hardware. The exact event receipt introduces no noisy dynamics or physical precision guarantee.

The independently replayed W12 construction in `code/source_feedback_transport/` retains 16 complete episodes, 740 seam hops and 4,928 events at path depths $1, 4, 12, 24$. Its routes are reused and intersect; an intervened branch leaves subsequent reads of an older version unchanged. At depth 24, five requests use 108 hops and 706 total events, including initialization and logical commits, with 141 protected scalar registers and 25 scratch registers. Version identifiers, event metadata and arithmetic workspace are additional storage costs. The Lean module `SourceFeedbackTransport` proves exact hop and route identities, error bounds and the finite semantic-projection criterion; the verifier instantiates that criterion from every consumed writer.

This supplies a classical implementation of declared remote scalar reads. The complete scalar field execution is not compiled into these episodes. The intended field update, source selection of feedback and read rules, quantum operations, outcome maps and physical clock remain additional inputs. The unicast census accompanying these episodes applies their cost to the declared-family read counts at $q = 13, 21$; it is not a full execution at those sizes or a lower bound for other compiler designs. Section A.9 gives separate complete multicast executions of the layered record recurrence on the wired support, with all intermediate operations retained. That recurrence is distinct from the scalar field Hamiltonian and its quantum instrument. Its local feedback, metric read menu, placement and controller are supplied.

A.9 Conditional full-family routing on the wired support

Fix the prepared golden record family at $q = 13$ or $q = 21$. Supply protected local addresses, payload versions and zeros; local record export to an owned port, zero reset, factor-two capture, integer accumulation and immutable commits; an injective host placement, the complete preceding-layer metric read menu, and a fixed serial schedule. These are supplied classical operations and control data. The construction does not derive their selection from A1–A3.

The public simulator's geometric assignment applied to all shared-vertex cell pairs reproduces the captured W12 level-three fixture exactly. Its level-four and level-five extensions have respectively

5120 and 20480 carriers. The source sites are assigned injectively by exact golden-coordinate Morton order to the first q^3 hosts. This is a declared placement, not an identification of the spherical host coordinates with the source metric.

For each immutable source version, form an ascending-neighbour breadth-first tree and retain the union of its paths to all required receivers. Traverse the retained tree in breadth-first order. Each seam exports its source archive, resets its receiver from a protected local zero, performs one pair mean, archives twice the receiver readout, and resets both endpoints. Every recipient then adds its delivered value to its local accumulator. New versions are committed after the complete layer. Paths and interfaces are independent of payload values.

Finite compilation theorem. Under these local feedback and scheduling hypotheses, every completed value equals the value of the declared metric-read recurrence. This equality holds for arbitrary initial integer values and integer local-commit interventions in exact arithmetic. The induced semantic read-from order on the distinguished publication/commit events is exactly the layered order. All intermediate operations are retained.

Proof. The source archive remains protected. Export and receiver reset prepare $(x, 0)$, the mean gives $(x/2, x/2)$, and local factor-two capture stores x. Cleanup restores the used ports. Induction along the tree transports the same immutable version to every recipient, including branching relays. The recipient set is the complete metric menu, with self-read represented by the source's own archive. Induction over layers proves the value claim; the local biases and initial values may be changed throughout this induction.

Label each transported payload by its actual originating logical version; label accumulator operations by their future commit, and constant preparations/resets by no logical payload. Every consumed-writer edge respects the logical order. Conversely, every required read has a retained writer path through export, mean and capture (or the direct self-read) into the receiver's accumulator and commit. Path induction gives both inclusions for the full induced order. The independent verifier checks these edge and path hypotheses from the actual register writers at every primitive. The finite integer six-operation word, its cost, the instantiated history induction and the abstract order arguments are kernel checked in `Geometry/SourceReadRouting.lean`. Native execution is separately checked code, not a formally verified interpreter. The exact real hop and sharp additive error law are in `Geometry/SourceFeedbackTransport.lean`. The universal intervention statement uses unbounded integers; only the retained bounded executions are claimed to fit the native 64-bit format.

Executed resource census. For C hosts, n source sites, K rounds, R logical reads and H retained multicast hops, the event count is

$$13C + 8n + 6H + R + 2Kn.$$

Scalar reads number $7H + 2R + 2Kn$ and writes number $13C + 8n + 7H + R + 2Kn$. Protected storage is $C + 6n + (K + 1)n + H$ scalars and mutable storage is $12C + n$. A depth-d read traverses $6d$ transport events plus a local accumulation; shared prefixes are counted once in H.

At $q = 13$, each full run has $H = 2972512$ and 19113548 events; at $q = 21$, $H = 46288315$ and 292722053 events. Maximum requested shortest-path depths are 44 and 91. Baseline and centre-source intervention runs retain 1176764 and 14559225 logical reads respectively. These are executed multicast costs, distinct from one-hop unicast estimates. The four full histories contain 623,671,202 primitive events in total. The reproduction procedure in `code/source_read_routing/`

regenerates each complete event stream from pinned inputs and source. It checks every decoded phase and whole-stream hash, independently reconstructs the metric menu, and replays every operation, consumed writer and completed logical value. This authenticates the full stream without retaining a large production tape in the repository.

Each explicit event occupies 64 bytes, including its owner, register and consumed-writer metadata. The archival producer and verifier retain history-sized register arrays with respective cell sizes 24 and 32 bytes; routing buffers, arithmetic workspace and runtime overhead are additional. The run receipts record native allocation requests and a producer peak resident-memory observation from a separate timed execution. Event replay does not remeasure that historical host-memory observation or bound the memory of the combined regeneration pipeline. These software resources do not identify physical patch capacity.

The $q = 13$ baseline comparison also checks the complete metric-read menu and all 10,985 logical preparation and commit values against the separately executed record-metric diagnostic. Equality concerns this logical projection. It does not identify the full primitive order with the diagnostic's order or with its canonical-only pair-mean trajectory. The diagnostic's lag-three interval is interior, whereas its lag-four interval is clipped; compiling its reads preserves this distinction.

Support-distance bound. At triangular subdivision level ℓ, join two distinct carrier faces when they share a vertex, and let D_ℓ denote the graph diameter. There are $C_\ell = 20\,4^\ell$ faces. The four children of one face intersect pairwise in a midpoint, and every old vertex belongs to a corner child. A shared vertex of two coarse faces therefore supplies an edge between suitable children. Lift a coarse path of d edges by moving within its initial parent to a crossing child, crossing to the next parent, and repeating, then moving to the requested final child. This uses at most $2d + 1$ edges. The twenty-face base graph has diameter three, so induction gives

$$D_0 = 3, \qquad D_{\ell+1} \le 2D_\ell + 1, \qquad D_\ell \le 4\,2^\ell - 1.$$

The base graph, the four-child refinement condition, the abstract diameter induction and the closed recurrence are separately kernel checked in `Lean/Geometry/SourceRoutingHierarchy.lean`. Their composition with the concrete triangle graphs is the analytic argument above; no Lean theorem instantiates those graph identifications at every level. Exact integer reconstruction checks the faces and undirected carrier seams against all three captured levels, three through five. Physical support selection and an all-level geometric assignment to twelve distinct ports are separate from this combinatorial bound. For $n > 20$, choosing the smallest level with $n \le C_\ell$ gives $n \le C_\ell < 4n$, since the preceding level contains $C_\ell/4 < n$ faces. The route bounds at $q = 13, 21$ are respectively 63 and 127 hops.

Proposition A.32 (Sufficient local working storage)**.** *Suppose each layer reads only the preceding payload layer, and each source's pruned multicast tree and recipient accumulations finish before the next source starts. A store with 23 scalar slots per carrier suffices for these admitted reads. Slots may be reused after the final admitted read of their logical version; every primitive event and its consumed writer remain in the separate history. The bound counts scalar values; precision, metadata and controller storage are additional resources.*

Proof. Give each carrier twelve mutable transport slots, one protected zero, six address slots, one accumulator, two payload slots indexed by layer parity, and one relay slot. The live payload layers $k, k+1$ have distinct parities. Only the current source's relay can be live at each carrier: its pruned tree visits that carrier at most once, and its complete episode precedes the next source's episode. All other slot kinds are disjoint. The slot map is therefore injective on this live set.

Maintain the relation that the physical cell at a live register's slot equals its logical value, including its writer identifier when required. A write preserves this relation because no other live register uses that slot. Retirement restricts the relation to a smaller live set and changes no admitted read. Induction over writes and retirements proves the claim. The live-set injectivity and both preservation statements are kernel checked in `Lean/Geometry/SourceRoutingStorage.lean` for an arbitrary abstract value type. The program's lifetime discipline is a hypothesis; its attachment to native execution is checked by code and finite replay. □

With n occupied hosts among C carriers, addresses, accumulators and payload banks need only $9n$ slots, while ports, zeros and relay slots use at most $14C$. This gives the sufficient occupied-host bound $14C + 9n \le 23C$. The Lean live set permits all slot kinds at every carrier, so its layout bound is $23C$; its sharper displayed inequality is an arithmetic corollary, not a formal occupancy theorem. The allocator in `code/source_routing_refinement/` enforces the occupied-host census and checks each reuse against a separate last-use pass. Both bounds exclude scalar precision, writer metadata, controller state and retained audit history. Two payload banks are sufficient; necessity and optimality are not asserted. Requests for a retired version lie outside this lifetime interface, even though its historical event remains recorded.

The four complete $q = 3$ mapped controls retain 91,272 events. Each uses 16,919 physical scalar slots, below the bound 18,163, and reuses 863 of 17,782 logical allocations. Verification decodes the committed target tapes, compares every source and target opcode, value, owner and consumed writer, preserves each transport port's identity, and independently replays the physical store. The source oracle checks the preparation, complete read menu, interventions and all 6,561 logical pairs per control. Applying the bound to the supplied $q = 13, 21$ programs gives respectively 91,453 and 370,069 working scalar slots. Those two numbers are bound applications, not executed mapped production histories or measured process-memory peaks. The archival full-family executions and their history-sized allocations are distinct from these small mapped controls.

Proposition A.33 (Raw-operation count and serial duration)**.** *Let $N = (K + 1)n$ count the distinguished logical publications and commits, and let $E = 13C + 8n + 2Kn + R + 6H$ count all primitive events of the stated compiler. Then*

$$Kn \le R \le Kn^2, \qquad H \le Kn(C - 1), \qquad E \ge 3N + 13C + 5n.$$

If $w \ge 0$ and $|wN - V| \le \varepsilon$, assigning the same weight to every primitive event gives

$$|wE - V| \ge 2V - 3\varepsilon.$$

For a serial layer of $m > 1$ primitive events, a positive tick τ and duration Δ satisfying $m\tau \le \Delta$ obey $\tau < \Delta$.

Proof. Each site's read menu includes itself and has at most n members. Each source's pruned tree visits at most $C - 1$ nonroot carriers, with shared prefixes counted once. Summing over sites and rounds proves the read and hop bounds. Substituting $R \ge Kn$ into E proves the event lower bound. Consequently

$$wE - V \ge 3wN - V \ge 2V - 3\varepsilon,$$

which implies the absolute-error inequality. Positivity gives $\tau < m\tau \le \Delta$. These arithmetic statements are kernel checked in `Lean/Geometry/SourceRoutingBudget.lean`; identifying their counts with the program uses the explicit operation census. □

The raw-mass inequality concerns the whole execution window. It supplies no count for each interval of the primitive order and excludes no other event weighting. In particular, the logical count-volume limit cannot be assigned unchanged to every transport event. Parallel duration requires a separate scheduling analysis, and neither duration convention identifies a physical clock.

Scope of the additional law. Every closed finite word of pair means preserves total live load. A reset that changes that total therefore cannot be such a closed word; the finite invariant obstruction is also kernel checked. This excludes this particular reset realization on that closed register set. It does not exclude tomography, changed ancillary state, other feedback, other decoders, or all admissible realizations of the metric order.

Serial execution, write hazards and custody remain present even when a constant reset has no prior scratch payload as a semantic parent. The serial unit-event duration counts every primitive; no physical time scale is inferred. Physical header/address transport is not implemented by the scalar channel. Routing events are not the distinguished count-volume events of the continuity-set theorem. Selection of the supplied record, feedback and routing law from the source architecture is not derived. These conditional finite receipts implement neither a quantum instrument nor a physical continuum.

B Recovery from finite federation logs

The local repair carrier and the complete glued federation have different identification problems. Fix the supplied twelve-port register interface, protected records, exact half-unit arithmetic and operation grammar of the routed read program. A blind decoder receives only the eight integer fields of each authenticated event. It checks the actual consumed writers and operation values, reconstructs the observed port uses, initial address records, logical reads and publication ancestry. The decoder cannot read the generator, its input file, or the complete support graph. Archive decoding and authentication take place outside this interface.

The retained level-three control contains 1,280 prepared carriers, 27 populated hosts, 22,818 events and 81 logical publication or commit records. Its 622 logical reads induce 1,331 strict ancestry pairs among those records. Its 872 mean events use only 86 distinct seams. These are finite observed quantities. The six recorded address coefficients, opcode meaning, arithmetic convention and authenticated completeness of the supplied log are premises; neither a general population law nor a physical clock is inferred from them.

Indistinguishable complete gluings. Write a seam as (a, p, b, q), connecting port p of carrier a to port q of carrier b. In the supplied level-three support replace

$$(120, 5, 123, 4), \quad (132, 9, 135, 11) \quad \text{by} \quad (120, 5, 135, 11), \quad (123, 4, 132, 9).$$

Both graphs are connected, have 7,650 seams and preserve every carrier's degree and occupied ports. All 27 complete ascending-neighbour breadth-first parent maps and queue orders agree. None of the changed seams occurs in the retained executions. Running the same native producer with the two input graphs gives byte-identical event histories for the baseline and the source, branch and scratch interventions. The carrier graphs contain 14,000 and 13,990 triangles respectively, so no graph isomorphism identifies them. Consequently, no function of these histories alone can recover the complete graph's triangle count on this candidate class. Full gluing is a failed identification invariant even when carrier count, seam count, degrees, occupied ports and the complete source traversals are given.

The candidate class here is the declared connected finite twelve-port gluing interface. The second graph is not asserted to be a canonical icospherical refinement. Restricting to that geometric family adds information outside the log. The counterexample neither excludes recovery under an explicit such restriction nor excludes a richer probing protocol. The supplied feedback word includes export, reset, capture and protected publication; its execution is not a derivation of that word from pair means. Re-execution of the observed finite program cannot establish recovery of unvisited topology or a universe-level fixed point. The executable evidence, independent verifier and wrong-law and wrong-wiring controls are in `evidence/federation_recovery/`.

B.1 Native scalar records and the selection boundary

Consider bounded self-reading patches whose scalar ports undergo pair averaging, with local readback, retained records and a declared controller. The following constructions distinguish operations available in that grammar from selection of the controller and its observer interface.

Passive records and reset. On a finite connected support, every state-only function invariant under each allowed pair mean on all nonnegative inputs factors through total load on those inputs. No continuity or linearity assumption is needed. Indeed, moving a nonnegative addend between adjacent coordinates gives two states with the same mean image. Invariance identifies their readouts. Moving each addend along a path to a fixed root proves the factorization. This equal-image argument does not execute inverse repairs. Restricted encodings and recorded histories are outside its quantifiers.

A positive raw port initially equal to a retains at least $a/2^m$ after m incident means with nonnegative ancillary loads. Fresh zero neighbours attain this bound and retain the removed load. Thus finite raw-zero reset is impossible in this class. The exact identity

$$\sum_i x_i^2 - \sum_i (P_{uv}x)_i^2 = (x_u - x_v)^2/2$$

also excludes an exact closed cycle of the whole register state that contains a nontrivial mean, and a full-amplitude copy of a nonzero two-rail balanced record, with the original intact, while restoring every ancillary coordinate. This quadratic functional has no physical-energy identification. The proofs are in `Lean/Geometry/SourceNonlinearRecord.lean`, `Lean/Geometry/SourcePassiveReset.lean` and `Lean/Geometry/SourcePassiveMemoryBudget.lean`.

Encoded copying and reusable reads. Encode a logical sign by $(b + a, b - a)$, with $b \geq |a|$, and prepare a blank pair (b, b). Averaging corresponding rails produces two pairs with amplitude $a/2$; averaging either pair clears its encoded amplitude. Repeated copy-and-clear transport costs $3n$ means and attenuates the signal to $a/2^n$. For positive amplitude a, if initial sup-norm error is at most E_0, every rail is disturbed by at most δ after each mean, and readout error is at most ρ, the sufficient sign condition is

$$E_0 + 3n\delta + \rho < a/2^n;$$

the opposite sign follows by exchanging rails. Nonexpansiveness of pair averaging proves the accumulated-error bound. The construction supplies neither amplitude restoration nor an exact unit-gain analog archive. Its proofs and independent execution replay are in `Lean/Geometry/SourceEncodedMemory.lean` and `code/source_encoded_memory/`.

On twelve ports of a captured 15,360-port support, two prepared records can be read in the order A, B, A through a four-cell paired-rail bus. All 16 declared input pairs retain their complete

histories, totaling 27,264 means. Raw residual influence survives cleanup; the finite codebook and positive error margins under supplied preparation, rounding, disturbance and readout allowances certify the decoded records. A seven-mean extension writes an encoded sum $C = A + B$ at one quarter of the input amplitude while halving the old records (`Lean/Geometry/SourceNativeRecords.lean`), and reads C, A, C, B with the record scales tracked explicitly. Independent replay covers all 36,464 means over its 16 input pairs. The smallest certified read margin exceeds 0.0032996 under the stated preparation, rounding, disturbance and readout budgets. The complete controllers, supported seams, consumed writers and precision controls are specified in `code/source_reusable_bus/` and `code/source_native_updates/`.

For a paired-rail chain of $n \geq 1$ edges with amplitudes initially bounded by $A \geq 0$, clear its first pair and sweep the adjacent rail copies forward. After $(n+1)^2 k$ sweeps the maximum residual is at most

$$A \frac{(n+1)^2}{2n+1} \, 2^{-k},$$

at a cost of $(1+2n)(n+1)^2 k$ scalar means; archives outside the chain are untouched. A weighted sup-norm estimate with weights $(n+1)^2 - (n-i)^2$ gives one-sweep contraction at most $1 - (n+1)^{-2}$, and blocks of $(n+1)^2$ sweeps give the displayed bound. The zero-edge case is exact clearing. Forward read attenuation is a separate cost. `Lean/Geometry/SourceBusScaling.lean` proves these statements for an abstract paired-rail chain; the rail embedding into a captured support, the preparation and the schedule are supplied.

Reusable accumulation from a retained unit. Represent an integer input v_i by the amplitude $a_i = g v_i / 2^{e_i}$ on rails $(b + a_i, b - a_i)$, with $g > 0$ and nonnegative integer scales e_i. The baseline can be chosen so that the prepared rails are nonnegative. Reserve record 0 as a mutable accumulator, record 1 as workspace and record 2 as a unit input $v_2 = 1$; reserve additional blank bus records for the final read. The other input values are supplied once. All subsequent data writes in the following construction are scalar pair means; the requests and scale metadata are supplied controller data.

Four means across the accumulator–workspace rectangle clear both encoded amplitudes: first average corresponding rails, then average the two opposite rail pairs. This identity holds for arbitrary initial amplitudes with a common baseline. Two more means copy the unit into the cleared accumulator. The unit and accumulator amplitudes both halve, and their updated scales represent the same logical value 1. The previous accumulator value is retired; it is not archived by this reset.

For an input $s \geq 2$, set $t = \max(e_0, e_s + 1)$. Halve the accumulator $t - e_0$ times and input s another $t - e_s - 1$ times, using the three-mean copy-and-clear construction. Then copy s into the blank workspace, copy the accumulator into that workspace and clear the workspace. These last five scalar means leave

$$a_0' = \frac{g(v_0 + v_s)}{2^{t+1}}, \qquad a_s' = \frac{g v_s}{2^t}, \qquad a_1' = 0.$$

Every other input amplitude is unchanged. Thus finite induction over the actual concatenated words preserves the original logical input values, including under repeated requests, and gives the final episode value $1 + \sum_{s \in \mathcal{R}} v_s$, with multiplicity in the supplied request list $\mathcal{R}$. For $S \geq 1$ episodes containing N additions and initial scales at most M, the proved bounds are

$$e_i \leq M + S + 2N, \qquad W \leq 6S + N\big[8 + 3(M + S + 2N)\big].$$

Here W counts all scalar means before the final read. A prepared four-edge paired bus adds eight means and four to the receiver's decoding scale. It attenuates the final accumulator amplitude by $1/16$.

Let p, m be the final local rail samples. Preparation error E_0, per-mean sup-norm disturbance δ and sample error ρ give

$$\left|\frac{p-m}{2} - \frac{gv}{2^e}\right| \le E_0 + W\delta + \rho.$$

In this inequality W includes the final transport. Clearing a record does not restart the error count. Normalize the observation by $2^e/g$. For a public magnitude bound B obtained from the input caps, intersect its closed error interval with $[-B, B] \cap \mathbb{Z}$. The clipped ceiling and floor endpoints agree exactly when this set contains one integer; the decoder then publishes that integer and otherwise abstains. The actual sum is a feasible candidate by the transport bound and the triangle inequality for its input caps. A normalized radius below $1/2$ therefore proves publication of the sum, rather than assuming the desired answer. The decoder uses only the two samples, the scale, the cumulative mean count and the public cap.

The captured realization prepares fourteen scalar ports with $b = 2$, $Q = 2^{256}$ and $g = (Q-3)/Q$. Under the declared allowances $E_0 = \rho = 1/(4Q)$ and $\delta = 5/(8Q)$, the normalized radius is

$$R = \frac{(4 + 5W)2^e}{8(Q-3)}.$$

Thus $(4 + 5W)2^e < 4(Q - 3)$ suffices. Exact replay covers 1,280 histories and 78,255 means, including five histories of fifteen episodes and 108 additions each. It checks supported seams, consumed values and current writers, 12,182 odd-sum rounding cases and 67,720 scalar samples; the largest decoding scale is 104. The full tapes are regenerated from the compact evidence in `code/source_native_accumulator/`. The arbitrary-finite-word, error, cost and exact decoder theorems are in `Lean/Geometry/SourceNativeAccumulator.lean`, `Lean/Geometry/SourceAccumulatorProgram.lean` and `Lean/Geometry/SourceAccumulatorDecoder.lean`. Logical input retention in this accumulator does not give immutable amplitudes or committed layer archives. Conditional stored computation with two protected banks is established in Section B.2. The bound charges native mean work, with controller computation, metadata, physical isolation and a physical publication instrument supplied separately.

Conditional publication and stopped histories. For any declared observation model, associate an observation with its set of compatible payloads. On feasible observations, every sound deterministic publication must identify a singleton set. Publishing that unique value is maximal among sound policies on feasible observations. The implemented policy abstains on ambiguous sets and on empty sets, where soundness alone would be vacuous. This derives a policy relative to an observation model, not a physical instrument or a requirement that an observer execute it.

For a positive source amplitude A, blank other records and t paired copies, let $E_t = E_0 + 2t\delta + \rho$. The ideal unsigned receiver amplitude lies in $[0, A]$; when nonzero it is at least $A/2^t$. The two signed payloads therefore admit the respective observation intervals

$$I_+ = [-E_t, A + E_t], \qquad I_- = [-A - E_t, E_t].$$

Their singleton policy publishes the positive sign only for $E_t < z \le A + E_t$, the negative sign only for $-A - E_t \le z < -E_t$, and otherwise abstains. This policy is maximal for these declared gain-and-error intervals, which need not equal the discrete set of observations of a particular word family. It is sound for every native word under the error assumptions. Once influence reaches the receiver, the strict margin $2E_t < A/2^t$ proves availability as well. Words that avoid the source cut can leave the receiver uninformative while incurring their full update cost.

On a path of $d \geq 1$ paired edges, nonzero influence occupies a prefix; exactly one of the d edge letters advances an unfinished prefix. Under the supplied uniform law on all d^H complete words of horizon H, let T be the first checkpoint with influence at the receiver. Then

$$\Pr(T > H) = d^{-H} \sum_{j=0}^{\min(d-1,H)} \binom{H}{j} (d-1)^{H-j}, \qquad \mathbb{E}[\text{scalar means}] = 2 \sum_{t=0}^{H-1} \Pr(T > t).$$

The convention $0^0 = 1$ includes $d = 1$. With a sufficient margin at the declared horizon, the implemented stopping time is $\min(T, H)$. A first arrival at t retains the probability mass of its d^{H-t} complete-word suffixes, although those suffixes are not executed. Failed attempts and deadline aborts are included; the expectation is not conditioned on publication. Uniform one-time marginals alone do not imply this complete-word law.

For four edges, $Q = 2^{14}$ and source amplitude $(Q-3)/Q$, exhaustive replay through horizon eight checks 174,762 word–payload cases and 2,642,144 means. At that horizon publication has probability 7459/65536, abort has probability 58077/65536, and the unconditional expected mean count is 32245/2048. Preparation, update and sample allowances are the same grid-scaled bounds used above. Finite precision does not establish indefinitely repeated availability. The general singleton, native error, frontier and complete-history counting recurrences are proved in `Lean/Geometry/SourceReadAcceptance.lean` and `Lean/Geometry/SourceReadAcceptanceSchedule.lean`. The binomial closed form is an analytic consequence of those recurrences, independently checked for the finite cases in `code/source_read_acceptance/`.

Selection of successful histories. Let K be a convex family of nonnegative finite vectors, with strictly positive reference coordinates and entropy weights. If the weighted relative entropy attains its minimum at p_*, then, as for classical information projections, every coordinate positive in any feasible q is positive in p_*. Mixing a purported zero coordinate with q produces a negative $t \log t$ term that dominates the change of first order in t. Consequently a nonnegative failure observable of the scored atoms vanishes at the minimum only if it vanishes throughout K. Applied with unit weights to complete repair histories, a guaranteed remote read requires every feasible schedule to exclude words that avoid the source cut: such words leave the exterior transcript identical under different source payloads.

This criterion concerns the atoms actually scored. An injective observer cover is insufficient to transfer it to every history event. For $p = (t, 1/2, 1/2 - t)$, $0 \leq t \leq 1/2$, and reference $r = (1/4, 1/4, 1/2)$, full-history relative entropy has unique minimizer $(1/6, 1/2, 1/3)$. Scoring the cover $(p_0 + p_1, p_2)$ against its pushed-forward reference instead gives $(0, 1/2, 1/2)$. Both covers determine the state on this family, but reconstruction of p_0 from the coarse cover subtracts $1/2$ and is not a nonnegative readout. The support theorem, both minima and the native cut argument are proved in `Lean/Geometry/SourceConstrainedSelection.lean`, `Lean/Geometry/SourceSelectionControls.lean` and `Lean/Geometry/SourceConstrainedRead.lean`; exact controls are in `code/source_read_selection/`.

These results neither select the feasible history grammar, reference, cover and stopping policy nor derive the complete immutable scalar-read service. The finite classical controllers supply no quantum instrument, physical precision, clock calibration or count-volume interpretation of their additional operations.

B.2 Conditional stored computation by supported pair means

Pair means can store an intermediate answer, use it in subsequent layers and reuse its record after the last admitted read. The construction assumes a prepared unit and integer payloads, a

supplied request list for each output, two protected record banks and verified paired routes on a finite twelve-port support. A supplied controller executes the compiled word and carries its public scale metadata. The reference loads encoding an integer v are $(b + gv/2^e, b - gv/2^e)$, with public exponent e and gain $g > 0$. Attenuation changes e while preserving the decoded integer. The logical recurrence is

$$v_{t+1,i} = 1 + \sum_{j \in \mathcal{M}_{t,i}} v_{t,j},$$

where the finite list $\mathcal{M}_{t,i}$ permits repetitions and empty sums. A public cap follows the same recurrence with absolute-value bounds; it uses no realized payload.

Transfers, retirement and composition. Each record occupies two ports on its bank carrier. A separate core contains a unit, an accumulator, workspace and an operand. The placement requires $2N + 1 \le C$ for N sites and C carriers. A paired route has disjoint scalar paths and avoids other live records; either admitted rail polarity is checked against the captured seams. In the reference evolution, copying a source into a blank export pair halves its amplitude and increments its scale. A route with $n + 2$ paired cells, $n \ge 1$, delivers the pre-transfer source amplitude divided by 2^{n+2}. Actual deviations from the blank reference loads enter the propagated error budget. Repeated root clearing and forward sweeps clean the bus without touching the destination. After $(n+1)^2 k$ sweeps, its maximum residual is bounded by

$$A \frac{(n+1)^2}{2n+1} \, 2^{-k},$$

where A bounds the reference amplitudes. Projection of this residual to zero is used only for error comparison; no physical reset to a prescribed value occurs. The read and store words contain, respectively,

$$4 + 2n + (1 + 2n)(n + 1)^2 k, \qquad 5 + 2n + (1 + 2n)(n + 1)^2 k$$

scalar means.

Restart, aligned addition, operand retirement, read and store compose into a finite word of supported scalar means. All records in the input bank remain readable while the other bank receives the next layer. Banks switch roles only after that layer finishes. Induction over the outputs and layers gives the stated recurrence, including repeated requests and reuse of retired records. A kernel-checked abstract bank compiler derives the instruction stream, recurrence, public caps, clean workspace and protected lifetimes. Its lowering theorem constructs the native paired-cell word and cumulative error ledger from route annotations, without assuming a completed correctness certificate. The scalar work is identified with the computable word length, including alignment and every cleanup repetition.

An explicit depth-three star bus supplies every required route by finite permutations. For every finite program with admissible public amplitude bound, the resulting existence theorem selects finite cleanup and grid precision independently of the payload. Every execution within the declared preparation, operation and sample allowances then publishes the recurrence; neither the answer nor a final strict-margin hypothesis is supplied. The declared star bus is not an embedding into the captured twelve-port support. Identification of the captured scalar-port embeddings and general Python compilation with the abstract construction uses analytic arguments and independent finite certificates.

A separate finite Lean certificate checks the signed two-site control: all 42 logical instructions, observed scales, stage lengths, their total of 1,798,154 means and all nine final output pairs. It

does not encode the captured scalar edge lists or every intermediate stored output. Native verification authenticates the full packet before checking freshness of this projection. This finite control certificate proves no corresponding kernel replay of the 27-, 2,197- or 9,261-site histories.

A single error budget. For initial integer bound P, put $A = \max(1, P)$ and choose normalized baseline $b = A + 1$. Normalized loads are represented in grid units by multiplication by Q; in particular p, m below are grid-unit samples. For the captured words, let J count restart and transfer stages, let D bound their bus size, and let E bound every observed exponent. For the complete scalar word length W, sufficient choices are

$$\begin{aligned} k &= E + \lceil \log_2(64AJD^2) \rceil, \\ R &= AJD^2\, 2^{-k}, \\ B &= E + \lceil \log_2(64(W+1)) \rceil, \\ Q &= 2^B, \qquad g = (Q-3)/Q. \end{aligned}$$

Here R bounds all reference cleanup discrepancies in normalized-load units. The disturbance allowances below use those same units. With preparation and sample allowances $1/(4Q)$ each and maximum-norm error at most $5/(8Q)$ per scalar mean, including disturbances on idle ports, the local samples p, m in grid units give the normalized observation and radius

$$z = \frac{(p-m)2^e}{2(Q-3)}, \qquad r = \frac{((4+5W_{\text{prefix}})/8 + QR)2^e}{Q-3}.$$

The error clock includes all preceding means and never restarts at a record boundary. The true integer lies in the intersection of $[z-r, z+r]$ with the public integer cap. The sufficient budget proves $r < 1/2$, so that intersection contains exactly one integer. The decoder receives the samples, scale, cap and error allowances, rather than the expected answer.

Finite evidence and its scope. Nine signed three-layer histories exercise cancellation, repeated and empty requests, stored intermediate values and bank reuse. Two complete 27-site, two-layer histories use the supplied golden metric menu, with a baseline and a centre-site intervention. Together they charge 1,409,653,226 means: 292,557,295 are explicitly evaluated and 1,117,095,931 have their exact rounded effect certified by deterministic stationary-block checks. Acceleration requires an executed whole block to leave its complete touched state unchanged; every omitted repetition remains charged in the word length and error clock. This is not a retained event-by-event expansion of those repetitions.

Independent checks validate all 45,940 two-bank routes at populations 27, 2,197 and 9,261, together with every metric-menu decision. For the two larger populations, sufficient bounds are 27,466,293,578,568,192 and 11,563,547,398,990,878,570 means, with 11,309 and 52,720 grid bits. Their native words are not replayed. These bounds establish finite compilability on checked routes under the supplied inputs; they make no physical-efficiency claim. Route search has no all-level completeness theorem.

The bank placement is separate from the geometric address layout. Runtime payload computation does not prepare, preserve or read those geometric addresses to select its menu. Agreement of the decoded recurrence does not identify the raw write history with the separate protected-address routing history. The controller, physical record protection, observer cover, isolation, attainable precision and physical publication instrument remain supplied. Scalar operation counts define neither a spacetime volume nor a physical clock. Host control, metadata storage and observation costs are

additional. Reproduction is specified in `code/source_native_programs/`; the composed proof and its standard-axiom audit are in `Lean/Geometry/SourceBankCompiler.lean`, `Lean/Geometry/SourceBankExecution.lean` and `Lean/Geometry/SourceNativeProgramsAxiomAudit.lean`.

B.3 Temporal acceptance, native continuation and preservation guards

The scalar seam interface permits a complete information-recovery theorem without a supplied successful route. Its hypotheses are explicit: a finite connected graph of scalar ports, the pair mean on each supported edge, exact local samples at one receiver, retained classical observations and readout arithmetic. This is an exact mathematical interface. It does not instantiate the full A1 response and endogenous-transport architecture or its physical instruments. The result complements the protected-bank compiler: tomography changes the physical state and does not itself implement immutable archives.

Acceptance from native observations. Let V be a real vector space of source preparations, P a linear map to the port scalars, A_e the supported pair-mean map, and r_i a local coordinate readout. A sample after k events has response

$$L_{k,i} = r_i A_{e_k} \cdots A_{e_1} P.$$

These forms are generated independently of the requested linear meaning $f : V \to \mathbb{R}$. Write L for all retained sample rows. Equal observations imply equal meanings exactly when

$$\ker L \subseteq \ker f \quad \Longleftrightarrow \quad f \in \operatorname{rowspan}(L) \quad \Longleftrightarrow \quad f = cL \text{ for some } c.$$

Indeed, agreement on observation fibers gives the kernel containment by comparing z to zero. Finite linear duality gives the row-span factorization; the converse follows by evaluating cL. If factorization fails, a vector z has $Lz = 0$ and $f(z) \neq 0$. The source possibilities x and $x + z$ are indistinguishable, so every sound partial publication rule abstains on their observation. This proves completeness for all linear meanings on this domain, rather than for an enumerated output menu. Retaining samples makes observability monotone and defines the earliest sound prefix. For sample errors $|\eta_i| \leq \epsilon_i$, a decoder certificate gives $|c\eta| \leq \sum_i |c_i|\epsilon_i$; attainable precision is a separate input.

Connected support constructs complete native recovery. Calibrate the receiver's own scalar first. Suppose a connected set K has its evolving values determined by the retained receiver samples. A proper K has an edge to its complement, since otherwise every graph walk from the receiver remains in K, contradicting connectivity. Choose a fresh vertex u and a simple path through K to the receiver. If the relay pre-values are $b_1, \ldots, b_d$, a forward sweep has the nested endpoint

$$y_0 = x_u, \qquad y_j = (y_{j-1} + b_j)/2.$$

The receiver observes y_d and reverses $y_{j-1} = 2y_j - b_j$ to recover x_u. It can then compute every changed relay value. Thus K grows by one vertex; no unintroduced vertex was touched earlier. Induction constructs a complete supported plan, recovers every initial scalar and identifies the evolved state on the calibrated region. No desired payload enters the path choice. For N ports there are exactly $N - 1$ phases and N receiver samples. A simple phase has at most $|K|$ means, hence

$$B \leq \sum_{j=1}^{N-1} j = \frac{N(N-1)}{2}.$$

The native endpoint identity, induction, lowering to pair means, sample count and work bound are kernel checked in `SourceTemporalTomography` and `SourceTemporalConnected`.

The exact continuation constraint. At a checkpoint let Lx be the retained observations and Mx the remaining complete scalar state. A linear meaning can be recovered by a native continuation exactly when

$$\ker \begin{bmatrix} L \\ M \end{bmatrix} \subseteq \ker f.$$

Necessity follows because equal records and equal states remain indistinguishable under every common continuation. For sufficiency, the connected protocol recovers Mx from receiver-local samples, after which the factorization evaluates f from (Lx, Mx). The corresponding fiber-factorization statement holds for arbitrary meanings and preparation domains, with equality of meanings in place of linear kernel notation. In particular, an unobserved record may remain recoverable; an erased record cannot be restored by waiting, even for an infinite common word.

A maximal guard and an unconditional proposal bound. For a specified source-record domain X, a checkpoint consists of functions $s : X \to \mathbb{R}^{\text{ports}}$ and $o : X \to \text{List}(\mathbb{R})$. Call it protected when (o, s) is injective. Admit a proposed mean exactly when the proposed checkpoint remains protected, retaining the resulting receiver sample. A rejected proposal leaves the scalars unchanged. On a linear domain this is exactly a joint-kernel test on source response matrices, independent of the realized payload. The additional receiver sample is determined by the proposed scalar state.

The topology-derived completing word works from every protected checkpoint whose record determines the receiver's current value. The initial receiver sample establishes this hypothesis; retained samples after admitted means and unchanged data after rejections maintain it. Every prefix of that word is admitted: an erasing prefix would give two equal records and states which its common remaining suffix cannot separate, contradicting complete recovery. Consequently an occurrence of the completing word among arbitrary other proposals guarantees full receiver information. The kernel proof identifies the actual guarded and unguarded native runs, rather than assuming that a selected route succeeds.

For a separately specified uniform full proposal alphabet of m event identities, encode the completing word as a block of length B. At most $(m^B - 1)^k$ of the m^{Bk} proposal words avoid that block at all k disjoint positions. Thus the incomplete fraction obeys

$$\Pr(T > kB) \le (1 - m^{-B})^k.$$

The finite count and its composition with guarded native execution are kernel checked in `SourceTemporalAttempts`. Under the consistent product proposal law, the geometric-tail argument additionally gives the analytic bound $\mathbb{E}T \le Bm^B$ for $B > 0$. The empty word represents immediate completion. Rejected proposals remain in the denominator and in proposal work. This is not uniform sampling of successful or admitted words. Computing response coefficients, the guard, proposal metadata and decoder arithmetic are explicit interface resources outside the scalar-mean count. Canonical A3 alone does not imply this uniform/IID proposal law.

Meaning requirements and A3. For an arbitrary function of finitely many independently variable records, call coordinate j essential when changing it alone can change the public meaning. A direct-read set S determines the meaning exactly when it contains every essential coordinate. Necessity is the isolated-coordinate intervention. For sufficiency, replace unobserved coordinates one at a time; each replacement is inessential and preserves the meaning. Thus this set is the unique minimal direct-read requirement, including nonlinear meanings. For $b + \sum_j a_j x_j$ it is precisely the support of the coefficients. Correlated source domains require a different fiber analysis.

In an attained convex finite full-history KL problem with positive weights and a faithful reference, the support theorem for the minimizer retains every history admitted by any feasible law. Therefore a linear meaning is certainly completable under the selected law exactly when every such history satisfies the joint-kernel condition. This is the composed theorem in `SourceTemporalSelection`. A state-determining coarse cover alone does not imply full-history support. The actual A1 grammar, A2-visible constraints, A3 observer cover, reference and weights remain inputs to this specialization; the result does not compare unrelated policy spaces.

Finite evidence and the M1 boundary. The captured 15,360-port, 46,050-edge support has a completely checked topology plan with 15,359 phases, 425,979 means, 851,958 primitive reads and as many primitive writes, plus 15,360 preparation writes and receiver samples. The largest phase has 51 edges. This is a topology certificate, not a large-plan payload replay or an attainable-precision claim. Eight five-port histories, including all five basis states, independently replay native execution and reconstruct original and evolved values from receiver samples alone. Separate captured controls distinguish aggregate recovery, individual recovery and wrong temporal order. Merging the two unprotected source ports first erases their difference on the entire support, so no common future repairs it. Under the stated uniform first-seam law this cylinder has probability 1/46050.

The guarded three-edge chain exhausts all 9,841 proposal words through horizon eight. At that horizon 857 of 6,561 attempts complete both source records; none loses their distinction irreversibly. The census retains 51,750 stopped proposals, 45,445 executed means, 6,305 rejected proposals and 52,006 samples. Dual witnesses, decoder coefficients, supported paths, complete coverage, costs and scope are independently checked. The transitive trust gate uses only the standard Lean axioms.

Finite recovery and temporal feasibility under these hypotheses do not select M1's metric menu or its particular meaning. Native recoverability extends to the whole connected component. The causal-limit condition $h_q/a_q \to 0$, with h_q proportional to L/q, also holds for every $a_q = Lq^{-\alpha}$ with $0 < \alpha < 1$ when the declared layer clock is a_q/c. That conclusion alone cannot select the square-root exponent. Choosing an objective that balances these errors would be an additional selection rule, not the canonical A3 optimization. The interpretation map, record domain, physical guard, protection, precision and clock require their source derivations. Full-state tomography also does not identify its raw write history with M1's semantic read-from order or replace the separate version-preserving bank construction. Physical spacetime identification is a separate premise. These results do not constitute a countermodel to the full premise set A1–A3.

B.4 Finite checkpoint selection and the shortest native two-record order

The preceding continuation theorem can be strengthened from an occurrence bound under supplied IID proposals to an explicit finite information projection and its causal implementation. The interface remains declared: the public record map, preparation domain, allowed native moves, retained samples, finite deadline, full-history entropy cover and reference are inputs. A1's availability of a checkpoint does not imply mandatory publication of this record by this deadline. This section changes the feasible family by imposing that requirement; it does not call the result the original unconstrained proposal law or a full A1–A3 derivation of M1.

Complete record constraints and finite selection. For a public map $r : X \to R$ and observation $o : X \to O$, agreement of every test $g(r)$ is equivalent to $o(x) = o(y) \Rightarrow r(x) = r(y)$. Equality predicates prove the reverse implication. For a tuple of records, testing all primitive coordinates is therefore complete for its generated test algebra, including nonlinear tests. This is not a completeness assertion about the entire A1 operational algebra. Preservation by the joint

old-record/current-state map is necessary and sufficient for bounded native continuation on connected support, provided the current receiver scalar is determined by the retained record. The topology-derived completion supplies sufficiency even after irrelevant preparation distinctions have been lost.

At horizon T, let $\mathcal{W}_T$ contain every word in the declared finite move alphabet and let G_T be the words whose native observations determine r. On the full simplex supported on G_T, a faithful inherited reference R_T has the unique information projection

$$Q_T(w) = \frac{R_T(w)\mathbf{1}_{G_T}(w)}{Z_T}, \qquad Z_T = \sum_{w \in G_T} R_T(w) > 0.$$

For every feasible p, its entropy gap is

$$D(p\|R_T) = -\log Z_T + \sum_{w \in G_T} Q_T(w) \left[\frac{p(w)}{Q_T(w)} \log \frac{p(w)}{Q_T(w)} + 1 - \frac{p(w)}{Q_T(w)} \right].$$

Nonnegativity and the strict equality case prove uniqueness, including candidate laws with zeros. The information cost is $-\log Z_T$.

For positive reference move weights $a(c, e)$, backward continuation masses and the resulting online transition law are

$$H_0(c) = \mathbf{1}_{\{c \text{ publishes } r\}}, \qquad H_{t+1}(c) = \sum_e a(c, e) H_t(A_e c), \qquad K_t(c, e) = \frac{a(c, e) H_{t-1}(A_e c)}{H_t(c)}.$$

Here c is a response checkpoint over all admissible preparations, so neither an unknown payload nor a future sample is read by the controller. The recursion sums every native continuation exactly once. Its positive mass is equivalent to existence of a publishing continuation; connected topology supplies a finite feasible deadline. Products of the normalized transition probabilities telescope to Q_T. Every positive selected history publishes, and every good history retains positive probability. When the alphabet covers the graph edges and every alphabet move is permitted on that graph, the composed theorem also ensures that every selected history stays on the declared support. For arbitrary record maps the publication predicate is classical; no general decision algorithm is claimed. In the rational linear controls, exact elimination decides it. There is no retrospective rejection of failed physical runs. Exact planning, retained records, random choice and decoding remain host interfaces whose physical implementation cost is unsupplied.

A sharp path theorem. For the isolated path $0, 1, \ldots, d$, $d \geq 1$, put independent unknown deviations x, y at ports $0, 1$, prepare all other deviations at zero, and sample receiver d initially and after every mean. Requiring both records forces at least $2d - 1$ means. Edge 0 must transmit x. Each other cut separates both unknowns from the receiver and needs at least two crossings: one crossing supplies at most one independent scalar to the initially calibrated downstream region. A successful word at equality therefore uses edge 0 once and each other edge twice.

Label the first crossings $F_1, \ldots, F_{d-1}$ and the second wave $S_0, \ldots, S_{d-1}$. The successful minimal words are exactly the linear extensions of

$$F_1 < \cdots < F_{d-1}, \qquad S_0 < \cdots < S_{d-1}, \qquad F_{j+1} < S_j \quad (0 \leq j \leq d - 2).$$

A first crossing out of order averages zero deviations and wastes one of the two required transmissions. If S_j precedes F_{j+1}, the cut behind it is exhausted while the downstream region holds only one source scalar; for $j = 0$, this is immediate erasure of the independent record difference. The

second crossings must occur in outward order to convey x before each cut is exhausted. These arguments give necessity. For sufficiency, incomparable events have disjoint edges and their means commute, so every linear extension has the final state of the serial two-wave word. Its two informative receiver samples are

$$u = \frac{y}{2^{d-1}}, \qquad v = \frac{x}{2^d} + \frac{(d+1)y}{2^{d+1}}.$$

They identify both records. At $d = 1$, u is the initial receiver sample. Deleting the final second-wave event gives Dyck words with $d-1$ events in each wave, hence exactly $C_{d-1} = \binom{2d-2}{d-1}/d$ histories.

Every positive time-homogeneous edge-product reference assigns each of these shortest words the same numerator $a_0 \prod_{i=1}^{d-1} a_i^2$. Thus its selected law is uniform over the Catalan histories, independently of those edge weights. This does not extend automatically to longer deadlines or arbitrary A3 reference rules. The decoder is

$$y = 2^{d-1}u, \qquad x = 2^d v - (d+1)2^{d-2}u.$$

Its sharp independent sample-error gains are 2^{d-1} and $(d+5)2^{d-2}$. The linear operation count therefore coexists with exponential precision amplification; no physical precision claim follows.

Evidence and scope. The finite grammar, continuation, entropy and online-policy theorems have kernel-checked proofs, as do native commutation and the decoder and error identities. The audit checks their transitive axioms. The full path-language classification and Catalan necessity proof above are analytic, not kernel-formalized. The independent executable packet checks every shortest-horizon word through $d = 5$ (1,969,761 words), and all 6,918 positive histories through $d = 10$, using native scalar basis replay. Producer Dyck enumeration and verifier prerequisite-poset enumeration are independent.

On the captured seams $0, 1, 14, 23, 45$, both records require seven means and exactly five shortest histories exist. All selected histories execute seven means and retain eight samples. The sum alone takes four means and does not determine either record. At eight means, 104 of 65,536 reference words succeed and the first-move probabilities are $(0, 94, 5, 5)/104$; the seven-mean law starts on edge 1 with certainty. Independently selected deadline optimizers therefore need not form a compatible temporal family. The control packet also retains tilted references, every signed intervention and consumed writer in thirty online trajectories, complete reference denominators and all selected native work. Every captured preparation charges 15,360 writes. Expanded enumeration tapes are recomputed from compact commitments.

Selecting the public record interface and metric menu, compatible refinement and geometric identification, physical storage/control/precision, and the immutable-version semantic event-order bridge remain separate source obligations. This theorem does not select $L/\sqrt{q}$ or provide a physical spacetime identification. Its finite temporal selection replaces a supplied successful schedule inside the stated interface.

B.5 Eventual publication and the source-selection problem

A finite deadline can be eliminated from the declared native record model. This gives compatible temporal laws and, on paths with separated source coordinates, an unconditional rank invariant for every finite native word. The result does not identify the public meaning, population or radius with those selected by the complete A1–A3 source architecture.

Constructed continuation potential. Let $T_a s$ advance a response checkpoint by a permitted native move and let $p(s,a) > 0$, $\sum_a p(s,a) = 1$, be its declared finite-alphabet reference. Write $f(s)$ for the indicator that retained observations determine the required public meaning. Retention makes publication persistent. The exhaustive finite continuation masses satisfy

$$H_0 = f, \qquad H_{n+1}(s) = \sum_a p(s,a) H_n(T_a s), \qquad h(s) = \sup_n H_n(s).$$

They increase from f to a limit in $[0,1]$. Finite summation commutes with the limit, so $h(s) = \sum_a p(s,a) h(T_a s)$. Induction proves that h is the least harmonic majorant of f. Its positivity is equivalent to a finite publishing continuation; the connected native completion theorem identifies this with preservation of the required meaning by the joint retained-record/current-state map. The initially sampled receiver hypothesis is essential.

For $h(s) > 0$, the cylinders and transition probabilities

$$Q_s[w] = \frac{R_s[w] h(T_w s)}{h(s)}, \qquad K(s,a) = \frac{p(s,a) h(T_a s)}{h(s)}$$

are normalized and obey $\sum_a Q_s[wa] = Q_s[w]$. This is the standard Doob transform, with its potential constructed from native publication.[2] The actual length-n prefix marginal of the finite deadline-$n+k$ information projection is

$$\frac{R_s[w] H_k(T_w s)}{H_{n+k}(s)} \longrightarrow Q_s[w].$$

An unpublished but recoverable prefix has zero mass at its own deadline when that deadline is feasible, and positive limiting mass. An infeasible deadline has no conditioned probability law. This limiting construction resolves the finite-deadline incompatibility without identifying time restriction with regulator refinement.

Analytically, the finite-alphabet reference defines a probability on infinite words. Publication at some finite time is a countable union of cylinder events, of probability $h(s)$. Its conditional law has exactly the displayed cylinders and publishes almost surely. For any competing path law V supported on this event and absolutely continuous with respect to the reference,

$$D(V\|R) = D(V\|Q) - \log h(s).$$

Singular laws have infinite cost. Hence Q is the unique information projection on this declared infinite history algebra. This is an analytic extension of the finite schedule problem, not a replacement of canonical A3's required source grammar by a chosen history algebra.

Quantitative completion. Suppose there are $N \geq 2$ ports and all reference move probabilities are at least $\eta > 0$ at every reachable checkpoint. Connected completion supplies a word of at most $B = N(N-1)/2$ means from every viable checkpoint. Padding preserves publication, so its reference probability is at least $\delta = \eta^B$. Published, dead and viable unpublished states give

$$h - H_B \leq (1-\delta)(h-f), \qquad 0 \leq h - H_{kB} \leq (1-\delta)^k (h-f).$$

The second inequality follows by applying the positive transition operator and iterating. Consequently the selected first-publication time satisfies $Q(T > kB) \leq (1-\delta)^k$ and $\mathbb{E}_Q T \leq B/\delta$. For

[2] For the standard transform see D. A. Levin and Y. Peres, with E. L. Wilmer, *Markov Chains and Mixing Times*, second edition, section 17.6, https://pages.uoregon.edu/dlevin/MARKOV/markovmixing.pdf.

the effective result, assume rational preparation and meaning matrices on a finite-dimensional real parameter space, an exact rational evaluator for the reference probabilities and a supplied positive rational uniform lower bound η. Finite enumeration and kernel elimination then compute H_n exactly, and

$$H_{kB} \le h \le \min\{1, H_{kB}/[1-(1-\delta)^k]\}, \qquad k \ge 1,$$

give convergent rational enclosures. Refining these enclosures and random bits implements exact transition sampling almost surely. No efficient planning bound, fixed physical precision or physical random source follows. Positivity alone does not supply effectiveness: the four-port reference $((1-r)/2, r/2, 1/2)$, for noncomputable $1/3 < r < 2/3$, has lower bound $1/6$ but publication potential $h = r$. The real-valued existence and tail results do not imply its computability. For $N = 1$, viability and the initial receiver sample imply publication at time zero.

Permanent native rank on a path. On the path $0, \dots, d$, $d \ge 2$, prepare independent unknown deviations x, y at $0, 1$ and calibrated zeros elsewhere; retain samples at receiver d. For the two response columns define the ordered minors $\Delta_{ij} = x_i y_j - x_j y_i$, $i < j$. All are initially nonnegative. An adjacent pair mean sends a minor containing both averaged rows to zero, leaves one containing neither unchanged, and otherwise averages two ordered minors. Nonnegativity is preserved. A fixed nonadjacent minor is therefore at least half its previous value after every move.

If edge 0 acts before edge 1, it erases the independent difference before any receiver sample. If edge 1 acts first, then $\Delta_{02} = 1/2$ and after every t subsequent native means $\Delta_{02} \ge 2^{-(t+1)} > 0$. Both coordinates remain encoded in the evolving scalar state forever. Connected completion and a uniform positive reference bound give eventual publication with probability one on this safe class. For constant probabilities $p_i > 0$, the initial potential and selected law are therefore

$$h = \frac{p_1}{p_0 + p_1}, \qquad K_0 = 0, \quad K_1 = p_0 + p_1, \quad K_i = p_i \ (i \ge 2).$$

After entry into the safe class the selected law is the reference. For $d = 1$ the receiver initially records y and $h = 1$ instead. Local deletion and renormalization of the destructive move gives a different law: on the uniform four-port path it is $(0, 1/2, 1/2)$, whereas the derived law is $(0, 2/3, 1/3)$.

There is an analytic extension to k independent preparation coordinates at nonadjacent path vertices I. The initial N-by-k response has nonnegative maximal ordered minors and determinant one on I. Adjacent pair-mean matrices are totally nonnegative. Cauchy–Binet preserves nonnegativity; in the update of the I minor, its own coefficient is one or one half, since no adjacent pair lies in I. All other contributions are nonnegative. Thus this minor is at least 2^{-t} after every finite word of length t. The state retains full preparation rank, and every prefix admits connected completion. Under uniformly positive references, eventual publication of the full preparation has probability one: $h = 1$ and the publication constraint changes no path probabilities. This removes the preservation guard and successful-word restriction in this exact prepared path model. Extra non-path means and fixed physical noise tolerance are outside the theorem; the determinant bound allows exponential attenuation.

Selecting the demand on one common algebra. Requiring only $x + y$ on the same path gives $h = 1$: native means conserve that total and it remains viable from every prefix. Include both this aggregate demand and the two-record demand as labels m in a single declared joint history algebra, with faithful prior π_m. Constrain publication of the labelled meaning. The unique joint information projection has marginal

$$Q(m) = \frac{\pi_m h_m}{\sum_j \pi_j h_j}.$$

Every recoverable demand retains positive mass. For equal priors in the uniform four-port control the two-record and aggregate probabilities are $1/3$ and $2/3$. Strict convexity on this common algebra therefore does not force the stronger demand. The complete A1-generated grammar must identify the required meaning or exclude the competitors; the declared label algebra is not asserted to be that grammar. For separated source coordinates the conclusion is stronger: every demand that is a function of the full preparation has $h_m = 1$, hence $Q(m) = \pi_m$. In particular, competing metric-neighbourhood read subsets keep exactly their prior weights at finite population. Publication supplies no preference among them in this native model.

Population and radius do not follow from the limit tests. For the golden sites, every fixed $0 < \alpha < 1$ gives radii $a_q = Lq^{-\alpha}$ and durations $\Delta_q = a_q/c$ with

$$a_q \longrightarrow 0, \qquad \frac{2\sqrt{3}L/q}{a_q} = 2\sqrt{3}q^{\alpha-1} \longrightarrow 0.$$

The existing covering-cone theorem and continuity-set counting theorem therefore apply throughout this family, under their stated boundary hypotheses. The grid Lb/q, $0 \le b_i < q$, has equal-volume cells and assignment error at most $\sqrt{3}L/q$, giving the same limiting tests. The finite read menus differ. At $q = 21$, for exponents $1/4, 1/2, 3/4$, the golden ordered read counts including self are respectively 20376393, 2911845, 327151; the grid counts are 20141913, 3118389, 277255.

The square root as an error balance. The proved generated-volume estimate gives a concrete reason to use a square-root scale. In units with $c = 1$, on a window $0 \le T \le 1$, let both covering and cell-assignment error be bounded by $0 < h \le 1$ and let $2h < a \le 1$. Under the existing partition and buffered-domain hypotheses, the actual interval-volume error obeys

$$\begin{aligned} E &\le 4\pi(T + a)(T/2 + h)^2(h + Th/a) + \pi T^3 a/2 \\ &\le 18\pi h + \pi(a/2 + 18h/a). \end{aligned}$$

The time quadrature contributes the a term, while accumulated spatial error contributes h/a. The radius-dependent bound has its unique minimum at $a = 6\sqrt{h}$, because

$$a/2 + 18h/a - 6\sqrt{h} = (a - 6\sqrt{h})^2/(2a) \ge 0.$$

For $h < 1/36$ the minimizer is admissible. With $h = C/q$ its exponent is one half, but its coefficient is $6\sqrt{C}$, not the prescribed unit coefficient. This optimizes a certified upper bound, not necessarily the actual error or canonical A3's entropy. Restoring a reference length gives the geometric-mean form $\sqrt{Lh}$. The exact radius is therefore dispensable for the stated continuum target and has a defensible role as a regulator; it has not been derived as a physical source length.

Removing locality fails; a fixed stencil also need not suffice. If every event reads every site in the preceding layer, the existing generated read relation satisfies

$$e \preceq f \quad \Longleftrightarrow \quad \text{layer}(e) < \text{layer}(f) \text{ or } e = f.$$

A fixed positive spatial separation is crossed in one vanishing layer duration, violating every fixed finite speed. An interval with two singleton tips and K full interior layers of N sites has ordering fraction

$$F = 1 - \frac{KN(N-1)}{(KN+2)(KN+1)}, \qquad 1 - 1/K \le F \le 1.$$

It tends to one uniformly in $N \geq 1$, not the four-dimensional Minkowski interval value $1/10$. The count includes both tips and compares actual intervals, not rectangular windows.

Nor does an arbitrary microscopic local stencil force a round cone. On the unit cube grid with spacing $1/q$, radius $a = 1/q$ and duration a, only axial neighbours and self are read. Reaching $(1/2, 1/2, 0)$ from zero takes time one, although it is strictly Euclidean-timelike at time $3/4$. This persists for every q divisible by four: the cone uses the ℓ^1 norm. At radius $1/q^2$ only self reads remain. These controls explain the separated scales of the successful family; they do not rule out every radius-free architecture. A source-derived read relation could replace the ball prescription if its actual propagation and counting limits were proved. The exact finite M1 menu and that continuum target are distinct obligations. None of these geometric controls is a full A1–A3 countermodel.

The full finite A3 state problem. Let $\mathcal{K}$ be A3's nonempty convex family of compatible local states, with finite observer cover $\mathcal{G}$, weights $w_P > 0$, faithful local references τ_P, and attained objective

$$F(\rho) = \sum_{P \in \mathcal{G}} w_P D(\rho_P \| \tau_P).$$

The algebras may be noncommutative, and the local family need not extend to a global state. Every minimizer ρ_* satisfies

$$\operatorname{supp} \sigma_P \subseteq \operatorname{supp}(\rho_*)_P \quad (\sigma \in \mathcal{K},\ P \in \mathcal{G}). \tag{12}$$

Thus for every positive effect E in a scored algebra,

$$(\rho_*)_P(E) = 0 \quad \Longleftrightarrow \quad \sigma_P(E) = 0 \text{ for every } \sigma \in \mathcal{K}. \tag{13}$$

An actual positive restriction pullback transfers this result to a public record; injectivity of a cover alone does not provide that pullback.

For $\rho_t = (1-t)\rho + t\sigma$, $0 < t < 1$, trace expansion gives

$$\begin{aligned}(1-t)D(\rho_P\|\tau_P) + tD(\sigma_P\|\tau_P) - D((\rho_t)_P\|\tau_P)\\ = (1-t)D(\rho_P\|(\rho_t)_P) + tD(\sigma_P\|(\rho_t)_P).\end{aligned}$$

The logarithms on the right are restricted to the mixture support, which contains both endpoint supports. If Q projects onto the kernel of ρ_P and $b = \operatorname{Tr}(Q\sigma_P) > 0$, binary measurement data processing bounds this gap below by $-tb\log t - tH(b)$, with $H(b) = -b\log b - (1-b)\log(1-b)$. All other local gaps are nonnegative. Consequently

$$F(\rho_t) - F(\rho) \leq t\big[F(\sigma) - F(\rho) + w_P H(b)\big] + tw_P b \log t.$$

At a minimum the left side is nonnegative, whereas the right side is negative for small enough t. This proves (12) without simultaneous diagonalization. Zero expectation of a positive effect means it annihilates the state support, proving (13). The imported data-processing theorem is Theorem 5.35 of Watrous, *The Theory of Quantum Information*, https://cs.uwaterloo.ca/~watrous/TQI/TQI.pdf.

On the established common supports, the derivative at $t = 0$ gives

$$\sum_P w_P D(\sigma_P\|(\rho_*)_P) \leq F(\sigma) - F(\rho_*).$$

For $b = \sigma_P(E) > 0$, binary data processing and $d(b\|p) \geq -H(b) - b\log p$ therefore imply

$$(\rho_*)_P(E) \geq \exp\left[-\frac{F(\sigma)/w_P + H(b)}{b}\right] > 0. \tag{14}$$

This uses a feasible witness, without solving the optimizer. It is not uniform across refinements without uniform witness-cost and weight bounds, so it does not exclude an asymptotically vanishing error. For a candidate with zero effect probability, set $a = w_P b$, $C = F(\sigma) + w_P H(b)$. The mixture $t = \exp(-1 - C/a)$ improves its objective by at least $a \exp(-1 - C/a)$; an approximation tolerance below this value cannot justify the candidate. For singular references, compress to their supports and apply the result to the nonempty finite-objective face of $\mathcal{K}$. A zero in a chosen reference can exclude an otherwise feasible failure, but its source derivation then requires proof.

A selected state does not determine its record process. Exact noncommuting controls use $R = |0\rangle\langle 0|$, $S = |v\rangle\langle v|$, rational unit vectors $v = (3/5, 4/5), (5/13, 12/13), (8/17, 15/17)$, and reference $I/2$. The unitary $\left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$ exchanges the endpoints, so strict convexity gives the unique midpoint optimizer, with determinant $b^2/4 > 0$ and last-coordinate probability $b^2/2$, although that probability is zero at R.

Identity and complete depolarization both preserve $I/2$ and are covariant under every unitary, but the mismatch probabilities of their uniform classical two-time records are zero and $1/2$. On the declared common set of normalized channel Choi states with input marginal $I/2$, reference $I/4$ selects depolarization. The bit-flip channel is a deterministic-failure witness of cost $\log 4$, giving the lower bound $1/4$ in (14). Adding the zero-mismatch face instead selects $\mathrm{diag}(1/2, 0, 0, 1/2)$, the dephasing channel. This preserves classical records while discarding phase; it is not the identity quantum channel. This process state space and constraint are inputs, not a reconstruction of A1's instruments or a full A1–A3 countermodel. The complete proofs are in `code/source_publication_selection/STATE_SELECTION.md`.

For an identified positive failure effect, (13) settles the finite A3 selection step: it enforces zero failure precisely when the source feasible family excludes failure. Establishing that family and the meaning of its record effects remains necessary. Conditioning on desired success cannot derive those inputs.

Proof and source boundaries. The modules under `Lean/Geometry/SourcePublicationAxiomAudit.lean` check the continuation potential, compatible cylinders, finite-marginal limit, native viability, block contraction, all-word two-column invariant, joint-demand marginal, radius/count formulas, complete-read order, interval-fraction bounds and the volume-error balance. Infinite-path measure and entropy arguments, the uniform native block instantiation and general k-column Cauchy–Binet argument are analytic. The independent controls in `code/source_publication_selection/` replay complete finite word trees and all maximal minors, and certify golden/grid metric counts by different exact arithmetic methods. Further controls independently reconstruct complete-read interval closure and exact grid-stencil paths. The noncommutative support, derivative and quantitative entropy arguments are analytic; `SourceStateSelection.lean` kernel-checks their real inequality reductions, with the matrix gap estimate as an explicit hypothesis. It does not formalize Umegaki data processing. Exact matrix and channel replay supplies independent regression controls. A full A1–A3 extraction of the golden population, required immutable-version meaning and particular radius, compatible with actual source response, transport and regulator refinement, remains necessary for literal finite M1. The continuum causal target can dispense with that exact radius, but requires a source-derived local propagation law. The results here do not discharge that obligation.

B.6 From registered source operations to numerical reads

A finite source can be tested before assigning it a metric read menu. We use the simulator's registered all-port capture, whose operations differ from both the shared-vertex W12 routing interface and the group-valued repair engine. Its local $\mathbb{C}^{12}$ unitary propagation precedes a separate scalar intensity-ledger repair loop. Each of the twelve port labels has a perfect carrier matching. On N even carriers, the resulting port-slot graph consists of $6N$ disjoint pairs, although the carrier graph is connected. Snapshot records expose twelve numerical ledger entries; the subsequent observer feedback chooses a displayed port without changing the earlier source evolution.

Let μ pair the $12N$ slots and let A be a union of pairs. Define $(P_A x)_i = (x_i + x_{\mu(i)})/2$ for $i \in A$ and x_i otherwise. Then

$$P_A P_B = P_{A \cup B}, \qquad P_A^2 = P_A, \qquad P_A P_B = P_B P_A.$$

These identities classify every finite ideal repair word, with exactly 2^{6N} distinct maps. Full coverage gives an ambient projection of rank $6N$. The native thresholded binary64 maps also commute and become idempotent after their first visit, but are not linear. Each pair commits at most once, reading version zero and writing version one. At the configured repair fraction $1/16$, a cyclic scheduler with $k = \lceil 6N/16 \rceil$ attempts per cycle covers all pairs within $\lceil 6N/k \rceil$ cycles; later attempts are no-ops. There are no committed repair-to-repair dependencies at any duration.

Identifiable records and an explicit failure. Give one carrier all its initial coordinates and retained local samples, an information upper bound for its actual snapshot records. A visited pair reveals its initial partner through $x_b = 2x_{\text{after}} - x_a$ in exact arithmetic. No further slot can influence this channel under any repair word. On the product of normalized preparation simplexes, eleven known coordinates determine the twelfth; two unobserved coordinates admit a balanced perturbation. The registered alternating matchings ensure that the single-carrier observation rank after full coverage is 23 for every even $N \geq 4$, leaving $11N - 23$ hidden dimensions. In the retained capture, observer zero has two support carriers and six snapshots; their normalized operator rows, numeric payloads, feedback ports and committed writer edges are separately reconstructed.

Choose two unobserved coordinates p, q on one remote carrier. Starting from $1/12$ everywhere, exchanging the values $5/48$ and $1/16$ at these coordinates gives two strictly positive normalized preparations with the same receiver payload under every common word. Every binary decoder has error $1/2$ under the balanced prior. The corresponding diagonal failure effect is positive. This is a witness in the native operations' open preparation interface, not a claim that a fixed integer seed realizes it. Global seeds and custody hashes can distinguish preparations and are not part of this local numeric channel. Applying finite A3 support preservation additionally requires membership in the canonical feasible family and a positive pullback to a scored algebra; the capture supplies neither.

Propagation and selector controls. The intensity ledger is insufficient for recurrent local propagation. For adjacent ports a, b, the states $(|a\rangle \pm i|b\rangle)/\sqrt{2}$ have equal intensities but different intensities after the actual $U(t) = \exp(-itL)$. Since $\|L\| \leq 10$, the difference at port a has magnitude at least $2t - 400t^2$. At $t = 2^{-17}$ this lower bound is $16359/1073741824 > 0$. No transition kernel depending only on current intensities reproduces both outputs. Phases or a specified instrument are required; this does not obstruct a full quantum instrument. An independent rational Taylor enclosure also proves all 144 entries of $U(2^{-17})$ nonzero. Generator adjacency is therefore distinct from exact finite-step dependence.

The native feedback provides a separate covariance control. At a uniform snapshot and current port zero it returns port two. An order-five icosahedral rotation fixes the input port and uniform state but moves the output port. Thus this feedback function is not an equivariant protected port selector. Covariance of the local Laplacian alone does not establish covariance of the record process.

Evidence and scope. Seven complete captures vary $N = 4, 8, 16$, duration and support level. An independent standard-library checker replays every attempt and native log commitment, reconstructs actual numeric observer records, and checks the exact rank and phase controls. Live verification binds these results to a pinned simulator commit and source bytes. Lean kernel-checks matching composition, all-word dependency, recovery and decoder obstruction, and the scalar phase-bound arithmetic. The normalized rank, matrix Taylor argument and executable identification are analytic with independent finite checks. The source population remains an input; changing its support-level field leaves this operation family unchanged. Longer runs past coverage cannot produce additional transport. No spatial coarsening, canonical A1–A3 family or M1 radius follows from this driver. The result identifies the exact read relation it does implement, without inserting the target geometry or filtering histories for successful reads.

B.7 Repair instruments, recurrent influence, and erased records

The current repair rule has explicit channel obstructions. A quantum channel must preserve convex mixtures. For the exact binary64 threshold $\epsilon = \texttt{1e-15}$ and $\delta = 2^{-51}$, consider the positive diagonal states

$$p_j = (1/2 + j\delta, 1/2 - j\delta), \qquad j = 0, 1, 2.$$

The native rule leaves p_0, p_1 unchanged and averages p_2 to p_0, since $2\delta \le \epsilon < 4\delta$. Its first output coordinate violates affinity at $p_1 = (p_0 + p_2)/2$ by exactly δ. Thus this thresholded rule is not the exact diagonal restriction of a quantum channel on these inputs. A numerical approximation to an ideal channel is a different claim.

The terminal lift also fails ensemble equivalence. Use the explicit density encoding $\rho(\psi) = N^{-1} \bigoplus_c |\psi_c\rangle\langle\psi_c|$. Keep other carriers in the uniform amplitude state and compare at carrier zero the equally weighted ensembles $\{|0\rangle, |1\rangle\}$ and $\{(|0\rangle + |1\rangle)/\sqrt{2}, (|0\rangle - |1\rangle)/\sqrt{2}\}$. Their average input densities agree. After full repair, the first ensemble has local intensities $(13/24, 1/24)$ or $(1/24, 13/24)$ at these ports. The native phase-preserving lift, including its zero-phase convention at zero amplitude, gives coherence $\sqrt{13}/(24N)$ for both. The second ensemble has intensities $(7/24, 7/24)$ and opposite coherences $\pm 7/(24N)$, averaging to zero. The output ensemble gap exceeds $1/(8N)$. The lift therefore has no affine extension on this density encoding. Retaining an additional classical preparation label would change the input state; it would not establish a channel on the stated densities. Individual carrier normalization is also not preserved by a generic seam: its sum becomes $1 + (x_b - x_a)/2$, while the global sum is conserved.

Two valid extensions with different later reads. On the declared global mode space $\mathbb{C}^{12N}$, let S_e exchange the endpoints of seam e. The map

$$\Phi_e(\rho) = \tfrac{1}{2}(\rho + S_e \rho S_e^*)$$

is CPTP, with Kraus operators $I/\sqrt{2}, S_e/\sqrt{2}$, and its diagonal is the ideal unthresholded pair mean. Disjoint seam twirls commute and are idempotent. Let E be their product and let $D(\rho) = \sum_i P_i \rho P_i$ be coordinate dephasing. Both E and DE are CPTP and have identical immediate diagonal reads

for every input. Every convex mixture of them has the same property. These are covariant under simultaneous relabeling of modes and seams; none is source-selected.

For a phase state $(|0\rangle + i|b\rangle)/\sqrt{2}$ at one carrier, with b adjacent to zero, and maximally mixed other blocks of weight $1/N$, E retains one quarter of the original local coherence while DE erases it. The next native unitary read distinguishes them by at least $(2t - 400t^2)/(8N)$. At $t = 2^{-17}, N = 4$ this is $16359/34359738368 > 0$. Native execution and an independent rational reconstruction of the full sparse matrices check the witness. Scalar repair data therefore do not determine the recurrent instrument.

The coordinate probes also cannot simply be renamed A1's central records. An accessible algebra in $\mathbb{C}^{12}$ containing all P_i and the native time translate $Q = U^* P_0 U$ contains

$$P_i Q P_j = \overline{U_{0i}} U_{0j} |i\rangle\langle j|.$$

Every coefficient is nonzero by the preceding propagator bound, so the algebra is $M_{12}(\mathbb{C})$, whose center is scalar. A separate classical record register can have central outcome labels, but those stored labels are distinct from these coherent instantaneous probes.

A recurrence cone and its exact information loss. The explicit measured cycle $DE\,\mathrm{Ad}_U$, restricted to diagonal inputs, has population matrix $T = RB_{\mathrm{local}}$, where R averages matching partners and $B_{pq} = |U_{pq}|^2 > 0$. If G is the actual carrier graph and $m(c,p)$ the partner carrier at slot (c,p), then for $k \geq 1$

$$(T^k)_{(c,p),(d,q)} > 0 \quad \Longleftrightarrow \quad \min\{d_G(c,d), d_G(m(c,p),d)\} \leq k-1.$$

The first step reads the two permitted input carriers at every port; induction takes unions over neighboring carriers, with no cancellation between nonnegative terms. Thus a carrier's outputs have exactly a radius-k graph *influence* ball. Independently checked matrix powers give carrier diameters $1, 2, 2$ at $N = 4, 8, 16$. This is neither a Euclidean radius nor a proof of exact record decoding.

Indeed, on the normalized preparation tangent V of dimension $11N$, the kernel of R consists of opposite endpoint values whose oriented seam weights have zero carrier divergence. The connected incidence matrix has rank $N - 1$, giving

$$\dim(\ker R \cap V) = 5N + 1, \qquad \mathrm{rank}(R|_V) = 6N - 1.$$

At the stated step, put $\eta = 20t$. The bounds $\|U - I\| \leq \eta$ and $\|B - I\|_\infty \leq 4\eta = 5/8192 < 1/2$ make B_{local} invertible on V, with inverse norm below two. Consequently the same dimensions hold for T. A circulation z around the source's alternating port-zero/port-one carrier cycle satisfies $Rz = 0$ and has zero sum on each carrier. The distinct inputs $x_\pm = 1/12 \pm B_{\mathrm{local}}^{-1} z/48$ are normalized, componentwise at least $1/24$, and obey $Tx_+ = Tx_-$. No later global population trajectory can recover that difference. Exact erasure is analytic; the executed inverse and complete output vectors are checked to residual 2^{-40}. An archive of initial records or suitable retained instrument outcomes is additional information and is not excluded by this obstruction.

Carrier totals do not close the measured dynamics either. A Markov map on those totals exists exactly when total transition into every output carrier is independent of the input port within each input carrier. Choose $d = m(0,0)$ and a port q with $m(0,q) \neq d$. Their probabilities of reaching d differ by at least $(1 - 2\eta - 10\eta^2)/2 = 536706947/1073741824 > 0$. This rejects that specific coarsening, not every possible refinement map.

Constructive recovery and its record cost. For deterministic ideal means $y = Rx$, orient each seam (a, b) and retain $d_e = (x_a - x_b)/2$. Normalization gives $\partial d = \mathbf{1} - Cy$, where C sums a carrier's slots. It suffices to retain differences on the $5N + 1$ non-tree seams of any spanning tree: subtract their divergence, then recover tree differences by deleting leaves. This gives $x_a = y_a + d_e$, $x_b = y_b - d_e$ for every normalized input. Any linear archive allowing recovery must be injective on $\ker R \cap V$, so $5N + 1$ real coordinates are also necessary. This is a minimal linear archive, not a finite-bit capacity bound. Exact rational execution and independent Gaussian elimination check it at all three populations. The same archive after B_{local}, followed by its inverse, also reverses the measured cycle on its normalized inputs. It requires global access to the means and extra records, and supplies no local decoder or canonical spanning tree.

There is a different quantum completion. For all binary seam words s, let S_s be the indicated product of swaps and $p_s = 2^{-6N}$. The flagged channel

$$J(\rho) = \bigoplus_s p_s S_s \rho S_s^*$$

has Kraus maps $\sqrt{p_s}|s\rangle \otimes S_s$ and is CPTP. Forgetting the flag gives E; applying its conditional inverse before forgetting gives $\sum_s S_s^*[p_s S_s \rho S_s^*]S_s = \rho$ for every input. Known interleaved unitary steps can likewise be undone in reverse order. This instrument retains $6N$ bits per sweep and its quantum state. Its individual branch populations are swapped or unchanged: only their expectation equals the deterministic mean. The flag marginal is uniform and independent of the preparation, so the flags alone are not its public read. These two constructions establish that extra records can prevent erasure, while distinguishing their costs and operational meanings.

Lean kernel-checks permutation conjugation, twirling and dephasing as channels, the diagonal and matrix-unit identities, labelled-branch recovery, affinity obstruction and scalar gap reductions. The ensemble, generated-algebra, all-time cone, incidence-rank, minimal-archive, flagged-channel and inverse arguments are analytic, with independently replayed finite controls. These results identify concrete changes or additional state needed by a recurrent source. They do not promote a constructed extension to canonical A1–A3 membership, selected M1 records, population or radius.

B.8 Source selection and a sufficient record-transport law

The source-selection question has two different meanings. A unique finite golden population and radius are stronger requirements than a common operational causal/count limit. Here we prove a sufficient source law for the latter, and compare M1 as one regulator of it. The proposed law is *additional* to A1–A3. It is neither adopted as a fourth core axiom nor asserted to hold in the registered simulator. Its several substantive requirements are stated together; no claim of minimality or unique necessity is made.

Why the existing data-access clauses do not supply a channel. There is a scalar-seam construction with a complete port response, rather than a response-free scalar toy. At each vertex of the normalized midpoint sphere S_n put

$$\mathcal{A}_i = \bigoplus_{p=1}^{12} M_6(\mathbb{C}), \qquad \mathcal{R}_i = Z(\mathcal{A}_i) = \mathbb{C}^{12}.$$

Its twelve block units are primitive central ports. Distinct-carrier seams are $\mathbb{C}$, with scalar inclusion and the unital normalized trace $E_i(a) = \sum_p \mathrm{Tr}(a_p)/72$ as visible data restriction. Triple restrictions are identities. Local chart overlaps are distinguished from these inter-carrier seams. Their proper

action groupoid implements all sixty carrier rechartings; it does not transmit a private record to another carrier.

An exact response on the same accessible internal $\mathbb{C}^6$ is

$$D_p = \operatorname{diag}\Big(\widehat{v_p/2} + iQ_p, \widehat{\sigma(v_p)/2}\Big), \qquad Q_p = \frac{v_p v_p^{\mathsf{T}}}{2} - \frac{r^2}{6} I + \frac{1}{12} I,$$

where v_p is an icosahedral calculation frame, $r^2 = \|v_p\|^2$, σ is its Galois conjugation and $\widehat{v}w = v \times w$. Even and odd antipodal fields span respectively the six imaginary symmetric directions and the two three-dimensional real skew spaces. Thus the response is injective, commutator closed and has image $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Full M_6 effects observe its relative block phase; restricting to block-diagonal effects loses one observable response direction. For each proper action a, $U_a = \operatorname{diag}(R_a, \sigma(R_a))$ satisfies $U_a D_p U_a^* = D_{ap}$. Executed Cayley factors in the original response span realize these matrices, their covariance and their compositions. The central permutation is the chart map; an inner automorphism of a carrier algebra is not claimed to permute its center.

The admitted instruments are local: central readout, destructive internal effect readout into two central ports, requested-block trace reset, prediction from a supplied preparation and local conditioning, and in-place checkpoint continuation. Only the declared response span generates infinitesimal reversible port operations. Irreversible read/reset channels have kernels and add no reversible direction. Independent ticks carry no record-dependent controller information between carriers.

The open-star nerve is S_n, the bridge is the identity and its fundamental cycle has degree one. Midpoints coarsen to declared endpoints, transported under relabeling; fundamental chains push forward exactly and spherical mesh contracts. Coarse local states average their descendant fibers. The instrument lift runs independently on every descendant, with each branch's own outcomes, then uses one fixed local tag through continuation. Selective conditioning updates tag weights; longer refinements use the composed weights. The tagged direct-sum experiment is not misidentified with the descendants' joint tensor algebra or with a communication channel.

The feasible local-state family is unconstrained except for normalization and chart covariance. Its complete grammar contains all 431 independent Hermitian state coordinates per carrier. The declared reference is $I_{72}/72$, the cover contains every carrier, and the weight is $1/N_n$. The weighted Umegaki objective has the unique tracial minimizer by nonnegativity and its equality condition. Every omitted coordinate admits two positive trace-one perturbations invisible to the remaining coordinates. This is an explicit A3 problem, not invariance used to choose a reference among all possible source ontologies.

Induction on every admitted local operation shows that changing an independent private source port leaves another carrier's entire local transcript unchanged. The induction permits arbitrary receiver-local memory and stopping rules, so no decoder of that transcript reads both interventions correctly. For a uniform unknown port its success probability is at most 1/12. Hence the stated typed data-access clauses, with local chart holonomy distinct from inter-carrier record transport, do not entail M1's remote read service. The detailed full-clause interpretation and all-level refinement proof are in `code/source_selection_model/DERIVATION.md`; finite receipts do not replace that analytic model-membership argument. If reversible overlap holonomy is strengthened to require inter-carrier record transport, this particular countermodel is excluded by that additional requirement.

Proposed law RG: response-natural classical record transport. The complete-response theorem for A1–A2 gives $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ at a complete reference carrier. Its unique three-dimensional simple ideal E, with the response trace form, has connected adjoint action $SO(E)$.

This is an internal Euclidean response space. RG says that the central-record source admits a complete, intervention-faithful, refinement-compatible realization of the following generated process theory, graded by a source-observable duration:

1. Objects are arbitrary finite independent classical registers at addresses in the affine space modeled on E. Finite tensor products, local preparation, finite deterministic classical maps, nondestructive read/copy, discard and retained versions are realizable at finite cutoff. Their ideal displacement and local duration are zero. Their actual implementations and storage costs are retained.

2. One retained-record flight has displacement e, $\|e\| = 1$, and duration $\tau > 0$. Translation, connected response rotation and positive scaling generate actual flights $(\lambda Re, \lambda\tau)$ for $R \in SO(E)$ and $\lambda > 0$. Waiting preserves the record. These processes transmit retained records between carriers.

3. Sequential and finite parallel composition are realized. Time adds on dependency paths; independent branches share elapsed time. Every native dependency, including controller, timing and metadata channels, factors through the graded local/flight/wait grammar. Conversely every finite generated process has an admitted native realization preserving all independent record interventions and immutable versions, with every intermediate event and its resource bill retained. Realizations commute with the declared source refinements.

Intervention faithfulness includes local interventions at every designated intermediate record output before publication, with all other registers and cached copies held fixed. Lowering must commute with insertion of these operations and subsequent reads. The version is committed once after its intervention. Merely agreeing on final outputs for all initial preparations is weaker: two versions can both contain $x + 1$, making a substituted read indistinguishable until one version alone is perturbed.

The exact version realizes these ideal grades. The vanishing-overhead version requires, for every complete finite experiment and every $\epsilon > 0$, a finite realization with exact record values and endpoint addresses, no faster native dependence, and each logical event time in $[t, t + \epsilon]$ relative to its ideal schedule. This uniform whole-program requirement is part of RG. Nonnegative primitive overheads summing to at most ϵ on every complete path suffice, by induction on the earliest-time recurrence using addition and maximum. A fixed error at each layer is not a substitute.

RG supplies flat homogeneous operational addresses, record protection, cofinal workspace, parallel composability and a physical connection from the internal response to record motion. None follows just from internal rotation covariance. Translations and scales act on a family of finite experiments, possibly changing cutoff; they are not extra infinitesimal port generators on one carrier. A proposed implementation must obey A1's full response constraint. A3 can select states inside such a source but is not used to derive RG or compel a particular observer program.

Proposition B.1 (RG suffices for the operational M1 limit)**.** *Under the complete-response conclusion of A1–A2 and RG, there exist cofinal finite independent central-record experiments whose native ancestry on their distinguished records has the round* $3 + 1$*-dimensional causal limit, continuity-set volume counts, Alexandrov interval counts and four-dimensional ordering fraction. M1's golden population with* $q = F_n$ *and observation duration* $L/(c\sqrt{q})$ *is one regulator with these limits, where* $c = 1/\tau$*. Its finite radius is not a uniquely selected microscopic length.*

Proof. Each primitive flight has norm c times its duration. The triangle inequality bounds every complete native dependency by $\|y - x\| \leq c(s - t)$, including paths through auxiliary events.

Conversely, for $\|v\| \le cT$, response transitivity rotates e to $v/\|v\|$. Scaling by $\|v\|$ and waiting for $T - \tau\|v\|$ realizes the read. The zero displacement uses waiting. The exact process has precisely the closed round cone; vanishing overhead attains every strict timelike read and excludes every spacelike read. Finite tensor composition realizes any finite list simultaneously, with protected source versions.

In a calculation frame prepare records at $x_b = Lb/q$, $b \in \{0, \dots, q-1\}^3$. RG supplies their joint finite preparation, rather than a presumed native carrier census. Half-open cells have equal volume $(L/q)^3$ and assignment/covering error $h_q \le \sqrt{3}L/q$. Choose $\delta_q \to 0$ with $q\delta_q \to \infty$. Exhaustively observe the preceding population within duration δ_q, then commit the consumed source versions. The primitive latency calculation derives the complete read menu $\|x_b - x_a\| \le c\delta_q$; this menu is not input to the source law. The covering-path theorem constructs a k-layer read path whenever

$$\|y - x\| \le k(c\delta_q - 2h_q), \qquad \text{and every generated path obeys} \quad \|y - x\| \le kc\delta_q.$$

Independently variable records and retained versions make these actual causal dependencies. Compiling a primitive correctly on every input state extends to every finite word by induction; concatenating the native words retains the sum of their event bills.

For each bounded horizon, lower the entire finite experiment with uniform time error $\epsilon_q \le \delta_q/q$. Commits need not be perfectly simultaneous. Let P_q be the consumed-writer order and N_q full native ancestry restricted to those commits. For each consumed-writer edge $u \to v$, vary the record at u's output boundary while holding other records fixed. The corresponding tuple coordinate read at v changes. If no native path connected them, the ancestors of v would form a dependency-closed region unchanged by that local intervention. Induction on the native updates would leave its read unchanged, a contradiction. This proves $P_q \subseteq N_q$ using intermediate intervention faithfulness; the value-only compilation induction does not prove it. Complete native grading then gives

$$P_q \;\subseteq\; N_q \;\subseteq\; C_c \quad \text{at the actual native event assignments.}$$

Every fixed strict timelike comparison eventually belongs to P_q, while every fixed spacelike or backwards comparison is eventually excluded from N_q. Moving two positions by at most H and times by at most ϵ changes the causal margin by at most $2H + 2c\epsilon$; this tends to zero.

Space and time cells give distinguished-event weight $w_q = (L/q)^3\delta_q$. For any bounded continuity set A in the window, its pulled-back indicator converges almost everywhere and is bounded by one, hence $w_q \# A_q \to \int_A dt\, d^3x$. A clipped terminal layer has mass error at most $L^3\delta_q$. For intervals, use the order sandwich and the null measure of the endpoint cone boundaries. For strict pairs, apply the same argument on the product window to both interval indicators and the precedence indicator; the time diagonal is also null. Dominated convergence gives the limits for both P_q and N_q, without excluding additional native edges by assumption. An interior rest diamond of duration T has $V = \pi c^3 T^4/24$ and oriented-pair measure $V^2/20$. Consequently $2\,\#\text{strictPairs}/[N(N-1)] \to 1/10$.

For the golden regulator use the proved equal-volume assignment with $h_q \le 2\sqrt{3}L/q$ and $\delta_q = L/(c\sqrt{q})$. RG supplies the finite preparations and transmissions, and all previous bounds apply. The radius $L/\sqrt{q} = c\delta_q$ is the reach of the chosen observation clock. Both regulators thus converge to the same causal/count observables. For a K-Lipschitz observable, assignment errors (H, ϵ) change its weighted integral by at most $K(H + \epsilon)$ times the window volume, plus the clipped endpoint term. Comparing regulators adds their bounds; indicator comparisons use null boundary tubes instead. □

This conclusion concerns a realizable observer probe and its operational limit. It neither selects spontaneous execution of that probe nor equates its distinguished-event volume with a count of all

microscopic repair events. It proves no Lorentz covariance of matter, curved geometry or laboratory clock calibration. Finite graphs need not be isomorphic, and unrelated field dynamics do not inherit equivalence from count convergence.

Finite clearance, verification and deletion controls. The golden squared displacements in units of L^2 lie in $\mathbb{Z}[\phi]$. For integer $q > 1$ none equals $1/q$: irrationality would first force the coefficient of ϕ to vanish, leaving an integer equal to $1/q$. Thus every admitted nonself golden flight has positive finite slack, although controller/commit costs remain charged. The rational-grid reference uses $L = c = 1$ and $\delta_q^2 = 2/(2q+1)$; boundary equality would require $(2q+1)m = 2q^2$, impossible since $\gcd(2q+1, q) = 1$.

The reference prepares, forks, flies, waits, adds locally and commits two layers for $q = 2, 3, 5$. Baselines, every single-site initial intervention and every first-layer commit intervention retain 323 executions, totaling 5,344,106 primitive events. Compact commitments are checked against live streams and independent semantic replay, including all consumed writers and intervention path multiplicities. These are executions of RG's reference grammar, not of an asserted native simulator implementation. Source pins and hostile-input tests reject forged timing, values, missing events and filtered intervention denominators. The additive reference diagnostic is distinct from the proof's lossless tuple payload, constructed analytically from RG's finite classical maps. Its intervention tests certify that each consumed value affects the diagnostic; they do not certify arbitrary payload maps or native programs.

The Lean proofs cover the local information bound, the primitive cone and its covering paths, the lowering induction, the localized-intervention ancestry reduction, event bills, population size and margin stability. The full model interpretation and the join from RG to native ancestry and counts are analytic. Detailed proofs appear in `code/source_selection_model/RECORD_GLUING.md`.

Deleting transport gives the scalar-seam control; keeping only axis flights gives an ℓ^1 cone; an ungraded metadata channel defeats the upper cone bound; destructive transfer or absent parallel workspace defeats the finite experiment. These controls explain the content of the proposed law instead of concealing it in a successful M1 schedule.

B.9 Operational selection of the causal cone

The response-selected source Gram space E is three-dimensional. Its unit directions form a sphere, and adjoining a scalar line gives the quadratic form $Q(t, x) = t^2 - \|x\|^2$ on $\mathbb{R} \oplus E$. The finite proper carrier rotations act isometrically on E. Under natural feasibility and objective data with a unique normalized minimizer, the sixty directed seams have equal weight and select the corresponding translation average on the conservative record carrier. These are spatial and algebraic results. The following analytic theorem states an additional condition that selects an operational causal cone on that same carrier.

A displacement cone C contains zero and is closed under addition and nonnegative scaling. Suppose it represents a homogeneous operational order by $p \preceq q$ if and only if $q - p \in C$. Pointedness, $C \cap (-C) = \{0\}$, expresses antisymmetry. Operational covariance means that transformations of admissible histories, interventions and readouts preserve possible influence, with the corresponding linear action on event displacements. Merely changing a frame or relabeling celestial directions does not establish this condition. If C is instead the closed cone generated by finite operations, the conclusion concerns that closure; it does not assert that every displacement is a finite direct read.

For $u \in E$, define

$$X_u(t, x) = (\langle u, x \rangle, tu).$$

For a unit vector n, $B_n(s) = \exp(sX_n)$ is the boost of rapidity s: writing $x = x_\parallel n + x_\perp$,

$$t' = \cosh(s)t + \sinh(s)x_\parallel, \qquad x'_\parallel = \sinh(s)t + \cosh(s)x_\parallel, \qquad x'_\perp = x_\perp.$$

Theorem B.2 (Finite rotations and one boost direction)**.** *Let $G \subset \mathrm{SO}(E)$ be finite and irreducible, as for the proper icosahedral representation. Let $C \neq \{0\}$ be a closed convex pointed displacement cone invariant under* $\mathrm{diag}(1, R)$ *for $R \in G$. Suppose, for one unit n, that $B_n(s)C = C$ for every real s. Then*

$$C = \{(t, x) : t \geq \|x\|\} \quad \text{or} \quad C = \{(t, x) : t \leq -\|x\|\}.$$

A positive time-axis vector in C selects the first alternative. It is enough to assume boost invariance for a sequence of nonzero rapidities tending to zero.

Proof. Irreducibility gives $\mathrm{span}\{Rn : R \in G\} = E$. Conjugation by $\mathrm{diag}(1, R)$ gives invariance under every $\exp(sX_{Rn})$. Write $u = \sum_j a_j R_j n$. Since $u \mapsto X_u$ is linear,

$$\exp(sX_u) = \lim_{m\to\infty} \left(\prod_j \exp\left(\frac{sa_j}{m} X_{R_j n} \right) \right)^m . \tag{15}$$

Indeed, the one-step product and $\exp(sX_u/m)$ have equal constant and linear Taylor terms, so their difference in matrix norm is $O(m^{-2})$. A telescoping sum over m factors bounds the total difference by $O(m^{-1})$; all intermediate products have a uniform exponential norm bound. Every finite product preserves C, and closedness passes this inclusion to the limit. Repeating with $-s$ gives equality. Thus every boost preserves C.

The finite average $|G|^{-1} \sum_{R\in G} R$ projects onto the fixed vectors of G, which are zero by irreducibility. Convexity therefore gives $(t, 0) \in C$ whenever $(t, x) \in C$. If $\|x\| > |t|$, boosts parallel to x can make the time coordinate positive or negative. Averaging these two images puts both signs of the time axis in C, contradicting pointedness. Hence every nonzero vector in C is causal.

A nonzero future causal vector has positive time coordinate. Its finite rotation average supplies a positive time-axis vector. Boosts and positive scaling generate every future timelike vector; closedness adds the null boundary. A past vector would similarly supply the opposite time axis, contradicting pointedness. This gives the future cone. Starting with a past vector gives the past cone. Nontriviality ensures one case occurs.

Finally, if $s_k \neq 0$ tends to zero, for every $s \in \mathbb{R}$ choose integers m_k with $m_k s_k \to s$. Integer powers and inverses of $B_n(s_k)$ preserve C. Continuity and closedness give invariance under $B_n(s)$, reducing the last assertion to the preceding proof. □

The theorem uses only two finite representation facts: the boost-axis orbit spans E, and the group average on E vanishes. It does not infer continuous spatial covariance from the finite carrier action. Its conclusion is a restriction: rotation covariance alone permits every cone $\{(t, x) : t \geq 0,\ \|x\| \leq vt\}$ with $v > 0$, and the stationary time-axis cone. Operational boost covariance excludes those alternatives except the unit-speed cone in the specified scalar coordinate.

The scalar coordinate requires an additive clock readout on actual histories. Once that attachment and the normalized boost action are fixed, the cone slope is fixed relative to them. Constructing the clock attachment is a separate dynamical requirement. The theorem does not select a layer tick, read weights, signal amplitudes or a field equation. The count construction in Section B.10 supplies a separate geometric duration readout on the declared order/count family.

At finite resolution, a bounded event set cannot carry a nontrivial continuous boost action by permutations. A physical use of the theorem therefore needs an operational continuum description

or controlled comparisons across a varying family. Finite energy cutoffs and bounded windows must be transformed or restricted consistently. The existing algebraic Lorentz and celestial maps alone supply none of these intervention or clock attachments. The cone classification and product limit above are analytic proofs, rather than Lean formalizations of operational covariance.

B.10 A clock read from source interval counts

The equal-mass source populations of Proposition A.21 admit a clock defined by their records. Suppose an observer can read the complete ancestry of an upper event and identify a lower anchor. Its inclusive interval is $I(p,q) = \{e : p \preceq e \preceq q\}$. Choose a nondegenerate reference interval J in the same history and define

$$\mathcal{T}(I;J) = \left(\frac{|I|}{|J|}\right)^{1/4}. \tag{16}$$

This readout uses order and cardinality. It needs no numerical event labels, spatial coordinates, timestamps, absolute density or oscillator frequency. Complete ancestry access is a substantive readout hypothesis: a hash of an unavailable record does not provide its contents. The cost of collecting and counting the records is separate from the duration reconstructed by the readout.

Theorem B.3 (Count clocks and proper-time ratios)**.** *Consider the declared source histories with their controlled order and equal-mass count limit. Let I_n, J_n approximate fixed interior timelike diamonds, with one common event weight $w_n > 0$, and let their limiting proper-time separations be $\tau_I, \tau_J > 0$. Then*

$$\mathcal{T}(I_n; J_n) \longrightarrow \frac{\tau_I}{\tau_J}.$$

The reference interval fixes the time unit; the clock origin is free.

Proof. For coordinate measure $dt\,d^3x$ and metric $c^2dt^2 - |dx|^2$, a flat diamond of proper-time separation τ has volume $K\tau^4$, where $K = \pi c^3/24$. The vertical formula follows by integrating its ball sections. A Lorentz transformation has determinant one and sends a timelike tip displacement to its rest form, so the same formula holds for moving tips. The count-volume theorem gives $w_n|I_n| \to K\tau_I^4$ and $w_n|J_n| \to K\tau_J^4$. Taking their ratio and its positive fourth root proves the claim. Both the event weight and the volume constant cancel. □

The supplied relation $\Delta = a/c$ enters the proof of the order and volume limit. It is absent from the clock's definition. This distinction allows a retrospective clock to be read from the same history without assuming an oscillator Hamiltonian. It does not select the population, read law or laboratory unit from accepted microscopic repairs.

Proposition B.4 (Finite clock enclosure)**.** *Suppose $A = wN$, $B = wM$ and independently justified volume bounds obey $|A - V_I| \le E_I$, $|B - V_J| \le E_J$, with $B > E_J \ge 0$, $E_I \ge 0$ and $V_J > 0$. Then*

$$\left(\frac{(A - E_I)_+}{B + E_J}\right)^{1/4} \le \frac{\tau_I}{\tau_J} \le \left(\frac{A + E_I}{B - E_J}\right)^{1/4}. \tag{17}$$

The same formula uses counts N, M and errors $E_I/w, E_J/w$.

Proof. The true volumes belong to their stated intervals. Division by a positive denominator and the increasing fourth root preserve the two inequalities. The error bound in (5) supplies such volume errors when its geometric hypotheses hold. Counts alone do not supply those errors. □

For relative volume errors $A = V_I(1 + d_I)$ and $B = V_J(1 + d_J)$, with $|d_I| \le \eta_I < 1$ and $|d_J| \le \eta_J < 1$, the measured clock divided by the exact proper-time ratio lies in

$$\left[\left(\frac{1-\eta_I}{1+\eta_J}\right)^{1/4}, \left(\frac{1+\eta_I}{1-\eta_J}\right)^{1/4}\right]. \tag{18}$$

Proposition B.5 (Unit freedom within the volume clock class)**.** *A positive elapsed-time function $F(V)$ depending only on diamond volume and additive for successive collinear inertial intervals is $F(V) = \kappa V^{1/4}$ for some $\kappa > 0$.*

Proof. Set $g(t) = F(Kt^4)$ for $t > 0$. Inertial additivity gives $g(t + s) = g(t) + g(s)$. Positivity implies strict monotonicity. Additivity first fixes g on positive rationals; monotonicity and rational approximation give $g(t) = g(1)t$ for all positive real t. This is uniqueness within the stated class, not a premise about every physical pointer. □

Accumulating duration along a curve. Let $\gamma(t) = (t, x(t))$ be C^2 on $[0, T]$, with $|x'(t)| \le v < c$, $|x''(t)| \le A_0$, and a common spatial buffer for its small diamonds. Segment durations are comparable to $\delta_n \to 0$. Suppose

$$\frac{\Delta_n}{\delta_n} \to 0, \qquad \frac{H_n}{\delta_n} \to 0, \qquad \frac{h_n}{a_n} \to 0,$$

and sampled tips have spatial and temporal errors $O(H_n + \Delta_n)$. Summing (16) over successive segment intervals, using the same fixed reference interval, converges to

$$\frac{1}{\tau_J}\int_0^T \sqrt{1 - |x'(t)|^2/c^2}\, dt. \tag{19}$$

Indeed, the two-shell estimate is uniformly $O(\delta_n^4 h_n/a_n + \delta_n^3(H_n + \Delta_n))$ per segment. The strict speed margin bounds each exact diamond volume below by a positive constant times δ_n^4. Tip perturbations have the same relative error order. Equation (18) therefore controls all segment clocks by one vanishing relative error.

The continuum polygon error has an explicit bound. For $f(u) = \sqrt{1 - |u|^2/c^2}$ on $|u| \le v$, its Hessian norm is at most $M_0 = [c^2(1 - v^2/c^2)^{3/2}]^{-1}$. Taylor expansion about each segment's average velocity has zero integrated linear term. Since the velocity differs from that average by at most $A_0\delta_i$, the chord and curve proper times differ by at most $M_0 A_0^2 \delta_i^3/2$. The total is at most $M_0 A_0^2 T(\max_i \delta_i)^2/2$. This proves (19). Fixed one-layer intervals do not satisfy these resolution hypotheses.

Finite records and calibration scope. The finite executable control parses writer identities, immutable register versions and values in the $q = 5$ history. Its 500 events give the three centered inclusive counts $2, 41, 80$. The first two continuum diamonds lie inside the source window; the third does not. Exact dyadic fourth-root bounds, schedule permutation and opaque event relabeling check the readout. These coarse intervals provide no finite proper-time accuracy certificate. The source-pinned expected hash authenticates the replayed content relative to that commitment; it is not an external signature or a proof of physical provenance. Unrelated serial audit links are excluded from the semantic read order.

Lean formalizes the finite interval and root inequalities. The count-volume and curve limits above are analytic proofs. In this layered law every causal pair k layers apart has a path of exactly k reads, and every strict chain has at most k edges. Longest-chain height consequently reads

coordinate duration independently of displacement. It is not a proper-time clock for moving tips in this family. The count construction concerns geometric duration reconstructed from accessible records; it neither sets the time cost of primitive seam words nor identifies physical clock readings without a common physical interpretation of the event law.

B.11 Observations of the causal manifold

An observer inside the source-record family of Section A.4 has access to the event order and to event cardinalities. This section records what such an observer reads from those two inputs at finite regulator, what the carriers produce of the family itself, and which of the finite statements are theorems.

The finite poset. For a Fibonacci level q the events are the pairs (j, s) of a round $j \le K_q = \lceil \sqrt{q}\, \rceil$ and a site $s \in S_q$, and $(j, s) \preceq (j', t)$ exactly when $j \le j'$ and the graph distance on the neighbour graph satisfies $d(s, t) \le j' - j$. This relation is a partial order, it equals the reflexive transitive closure of the one-round reads, every round is an antichain, the width equals $|S_q| = q^3$, and every event at round j has height j (`Lean/Geometry/SourceNetLayeredOrder.lean`, theorems `layerPrec_isPartialOrder`, `layerPrec_iff_reflTransGen`, `walk_iff_walkDist_le`, `width_eq_card_sites`, `height_eq_layer`; the concrete read law of the family is the instance `precedes_iff_layerPrec`). The flat reference constants of the ordering fraction are exact: the radial-time double integral of Proposition A.22 equals $T^8/1920$, so the fraction equals $1/10$, and the Myrheim–Meyer values in dimensions 2, 3 and 4 are $1/2$, $8/35$ and $1/10$ [8, 9] (`Lean/Geometry/OrderingFractionFourDimensional.lean`). The finite clock enclosure of Proposition B.4 is formalized in `Lean/Time/SourceCountClockEnclosure.lean`. The family limit, beyond these finite constants, is proved in `Lean/Geometry/FlatDiamondNormalization.lean`: actual generated strict pair counts have ordering fraction tending to $1/10$ for an interior timelike limiting diamond, with the supplied clock, population and read law kept explicit. The weighted count-volume limit and its finite error are proved in `Lean/Geometry/GoldenSourceVolumeLimit.lean`.

What the observer reads. On the central vertical diamond of K_q rounds the inverted Myrheim–Meyer dimension reads 2.63, 2.93, 4.15, 3.61, 4.10, 4.09 and 4.07 at $q = 5, 8, 13, 21, 34, 55, 89$, with ordering fractions 0.306, 0.241, 0.088, 0.138, 0.092, 0.093 and 0.094 against the flat value $1/10$; levels with even K_q approach the value from below, 0.088, 0.092, 0.093, 0.094 at $K_q = 4, 6, 8, 10$, and levels with odd K_q from above. The outermost diamond of K_q rounds extends past the sampling window at every level, by 9.2%, 0.4%, 0.9%, 0.4%, 0.02%, 0.2% and 0.1% of its continuum volume, so these values read clipped diamonds; the largest interior diamonds, of $K_q - 1$ rounds, read 4.04, 5.02, 2.83, 4.38, 3.55, 3.84 and 3.92. The moving-tip diamonds of Proposition A.20 read 0.250, 0.135, 0.121, 0.115, 0.108 and 0.102 at $q = 8, 13, 21, 34, 55, 89$, dimension 2.89 to 3.97. The same read law on the two- and one-dimensional golden populations reads dimensions 3.02 and 2.01 at $q = 89$ against the constants $8/35$ and $1/2$, so the readout responds to the dimension of the population. The normalized count of the vertical diamond against $\pi c^3 T^4/24$ reads 0.193 against 0.198 at $q = 13$, 0.141 against 0.147 at $q = 34$, 0.172 against 0.177 at $q = 55$ and 0.163 against 0.165 at $q = 89$. The count clock $(|I|/|J|)^{1/4}$ of two nested vertical diamonds reads 1.705, 2.088, 1.907 and 2.051 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Every pair count through $q = 89$ is an exact all-pairs count; the structural fields at $q = 5, 8, 13$ agree byte for byte with the replay of Section A.4. A sampled level at $q = 144$, with 2,985,984 sites and twelve rounds, reads 0.0955 ± 0.0002 on the outermost diamond, so the even-round ladder continues 0.0600, 0.0803, 0.0887, 0.0932, 0.0955 at $K = 4, 6, 8, 10, 12$ with halving increments; its dimension reads 4.05, the largest interior diamond

3.95, the moving tips 0.102, the count clock 1.959 against 2, and the two- and one-dimensional controls 3.02 and 2.00. These are stratified estimates with standard errors, not exact counts. The bounds of Proposition A.19 exceed the diamond volume at these levels, and the finite values above are readings of the supplied law, not a demonstration of its limit.

What the carriers produce. The family is produced by the carriers themselves. One exact twelve-port carrier per site holds the record $z(b)$ as its port loads on the six antipodal axes and reads its position through the rank-three response $x = 2P_{\mathrm{slow}}N$ of the rank-three source metric. The readback metric equals the source-axis contraction $s(b)$ with scale exactly one, an identity in $\mathbb{Q}(\sqrt{5})$ checked on every pair through $q = 21$ and on 10^6 seeded pairs at $q = 34, 55, 89$. Each carrier decides its neighbours from its own readback; the resulting relation equals the family's neighbour graph edge for edge at $q = 5, 8, 13, 21, 34, 55, 89$. The carriers then read one another for K_q rounds, every read an authenticated read-after-write event, and the precedence generated from those events with no declared parent lists equals the layered order on the centre diamond at every level, with strict pair counts 968, 4,244, 102,990, 2,904,889, 47,881,819, 2,083,062,313 and 55,108,871,450. A unit intervention at the centre in round zero changes exactly the future cone in every round. The operation costs are recorded: reads per round $2E + n$ for E edges and n carriers, at $q = 34$ a total of 157,922,880 reads and 275,128 writes, at $q = 89$ a total of 21,845,550,850 reads and 7,754,659 writes by 704,969 carriers over ten rounds.

Inference and boundary. Order and count determine conformal geometry and scale under the continuum causality hypotheses [14, 15]. The finite readings above compare a supplied family with flat reference values; they do not establish those reconstruction hypotheses or a statistical acceptance of flat $1 + 3$ spacetime. The population, the read law, the round duration and one counted event per carrier and round are supplied. Their selection by accepted native repairs, a physical clock, and the continuum limit are separate conditions rather than results of this section. Two further receipts read the order with more than one number. At $q = 8$ to 34 the Myrheim–Meyer dimension is the same at the centre and at every admissible off-centre tip of the same size to within 0.04; the odd multipoles of the link directions vanish by the two-way reads and the $l = 4$ power falls from 0.036 to 0.0016 over the four levels; the abundance of intervals of 64 to 1024 events matches a flat sprinkling within 14 percent while links and small intervals do not, as a layered order requires. Sampled chain counts and interval spectra are retained at $q = 55, 89, 144$. For the central diamond, the three inverted dimensions, the maximum CDF difference D_{25} on the declared 25-point grid, and the interval-mean deviation from the exact flat value 1/105 are:

q	$d(C_2)$	$d(C_3)$	$d(C_4)$	D_{25}	Mean deviation
55	4.09590	4.03210	4.00605	0.075810	+5.0739%
89	4.09437	4.03413	4.01664	0.024725	+8.5836%
144	4.07556	4.01605	3.97098	0.032342	−20.9353%

Here D_{25} compares the weighted source CDF with the retained seeded Monte Carlo flat reference, which has its own sampling uncertainty. It is not an exact continuum CDF distance or a continuous Kolmogorov–Smirnov statistic. The absolute interval-mean discrepancy grows at both refinements, while D_{25} first falls and then rises. Uniform monotone improvement of the retained point estimates therefore fails. This descriptive result neither establishes nor statistically rejects asymptotic convergence.

The accompanying comparison retains all 96 regions across the three levels and the three spatial dimensions, including every moving and off-centre region, both lower-dimensional controls, and

comparisons with the five sampled flat references in spacetime dimensions two through six. At $q = 89$ the moving-interval mean is 23.13% above 1/105; at $q = 144$ it is 15.60% below it. The $q = 144$ moving-diamond dimensions are 3.99045, 3.99998 and 4.03391. Over the fourteen off-centre diamonds and their central reference, the $q = 89$ three-chain dimension has mean 4.078 and spatial standard deviation 0.016; these are regional dispersion statistics, not an error on a common dimension.

Marginal chain-count standard errors are retained historical estimates; their cross-covariances and the uncertainty of the self-normalized interval mean and CDF are unavailable. They cannot support independent joint significances, an uncertainty-qualified refinement test or a numerical interval-mean error bound. The central four-chain relative standard error is about 7.49% at $q = 89$ and 10.56% at $q = 144$, whose source sampling uses half as many draws. The original large-run draws and the chunk-parallel $q = 144$ driver are not retained. Its neighbour digest agrees with the archived family receipt, with recorded sequential versus chunked identities at $q = 21, 34$; these ties do not reproduce the missing computation. The independently checked arithmetic comparison in `evidence/source_net_causal_poset/manifold_refinement/` exposes signed deviations, available marginal errors and missing uncertainty. A statistical decision requires a declared bounded control, retained joint sufficient statistics, reference uncertainty and complete driver custody. Curvature estimators are not evaluated. The supplied population, read law and clock, and the separate small clock controls below, retain their stated boundaries.

Exact temporal quadrature control. For continuum spatial balls sampled at $K + 1$ equally spaced layers, $K > 0$, the layer-summed vertical diamond volume divided by its continuum volume is

$$B_K = \frac{32}{K^4} \sum_{j=0}^{K} \min(j, K-j)^3 = \begin{cases} 1 + 4/K^2, & K \text{ even}, \\ (1 - 1/K^2)^2, & K \text{ odd}. \end{cases}$$

Indeed, for $K = 2m$ the sum is $m^3 + 2\sum_{j=1}^{m-1} j^3 = m^2(m^2+1)/2$; for $K = 2m+1$ it is $2\sum_{j=1}^{m} j^3 = m^2(m+1)^2/2$. The sum-of-cubes identity gives the displayed formulas. For $J \geq 2$, dividing the raw nested count-clock ratio by $(B_K/B_J)^{1/4}$ isolates this temporal discretization effect. The exact finite sums and both parity factors are proved in `Lean/Geometry/FiniteLayerQuadrature.lean`.

q	Raw relative error	Layer-corrected relative error
55	−4.6395%	−0.6853%
89	+2.5509%	−0.5014%
144	−2.0289%	−0.1007%

The numerical control uses the same count receipts, with no fitted parameter. It is exact for continuum balls at discrete layers, not for finite graph diamonds. Spatial error and clipping are separate: all these outermost diamonds fail the stated finite interior-enclosure hypotheses, including $q = 144$, whose clipped fraction is 8.9748×10^{-8}. At $J = 1$ the continuum layer volume vanishes and the correction is undefined. The residuals therefore establish no physical calibration or uniform finite-graph convergence rate.

Uniform history length and interval clocks. For any related pair K layers apart in a read graph with self-waits, a strict chain has at most K edges. A shortest spatial path of length $d \leq K$, padded by $K - d$ waits, attains K. Hence a uniform positive duration per history edge gives moving/rest clock ratio one at fixed K, whereas the supplied Minkowski ratio is $\sqrt{1-\beta^2} < 1$ for $0 < |\beta| < 1$. Refinement cannot remove this discrepancy at fixed velocity. This excludes

that clock assignment for this layered family; it says nothing about state-dependent durations or independently selected records.

A finite comparison retains all 120 scheduled cases: four golden populations $q = 8, 13, 21, 34$, five boost targets, three directions, and two read laws. There are 89 related pairs and 31 timelike but graph-unrelated pairs. At $q = 34$ the nine primary moderate-boost Euclidean cases have median absolute relative errors 16.290% for uniform history length, 2.914% for the raw volume clock and 0.813% after temporal quadrature correction. The same corrected instrument on axis-only reads gives 16.149% across six related primary pairs; three further failures are retained. Actual velocities use the rounded endpoints. At target $\beta = 0.9$ the Euclidean axis pair is unrelated, and the other two directions have corrected errors -17.19% and -9.70%. The comparison tests read laws under supplied geometry. Its moving-diamond correction uses continuum intersections of spatial balls, so it is a calibration control rather than an intrinsic order-only clock. These deterministic errors are not confidence intervals. The specifications, witness histories, receipts and independent verifier are retained under `evidence/observer_dynamics_20260925/sim-analysis/codex/causal/`; the count and quadrature controls are reproduced by `code/observer_dynamics/verify.py`.

B.12 Spatial populations and a common scalar continuum action

The completed conservative source-record carrier is a Euclidean three-space under the stated response and metric hypotheses. Its whole dense record group is not a locally finite population. A finite-resolution selection and a transport law can be stated on that carrier without inserting a Cartesian lattice. This construction keeps both choices explicit.

Fix $r, \delta, \eta > 0$, a response-metric ball $\Omega = \overline{B}(0, r)$, and a finite directed seam-word endpoint menu $M \subset \Omega$ that is an η-net of Ω. Such menus exist: density approximates a slightly inward-shifted point by a source endpoint, compactness extracts finitely many approximants, and a finite maximum word budget contains all of them. This is algebraic endpoint availability, not a proof that a repair law accepts those words as moves. Choose a maximal δ-separated subset $S \subset M$, and set $h = \delta + \eta$. The resolution, window and tie rule are supplied; maximal means inclusion-maximal, not maximum cardinality or a selected energy minimum.

Proposition B.6 (Finite spatial population and metric transport)**.** *The selected points form an h-net, and*

$$(r/h)^3 \le |S| \le (1 + 2r/\delta)^3.$$

For $R \ge h$ and $B(x, R) \subset \Omega$,

$$((R - h)/h)^3 \le |S \cap B(x, R)| \le (1 + 2R/\delta)^3.$$

Connect distinct points whose source distance is at most $a > 2h$, weighting edges by that distance. The graph is connected, degrees are bounded by $(1+2a/\delta)^3-1$, and its shortest-path metric satisfies

$$|s - t| \le d_G(s, t) \le \left(1 + \frac{2h}{a - 2h}\right) |s - t| + 2h.$$

Consequently $a \to 0$ and $h/a \to 0$ give uniform metric convergence on the fixed window. With $\eta \le \delta$ and $a = 8\delta$, degrees instead stay at most 4912 and $d_G \le 2|s-t|+4\delta$; this latter estimate alone does not imply a Euclidean metric limit.

Proof. Maximality puts each menu point within δ of S, proving coverage. Disjoint radius-$\delta/2$ balls give the upper counts; covering the window, or $B(x, R-h)$, by radius-h balls gives the lower counts. For the graph, divide the segment from s to t into $k = \lceil |s - t|/(a - 2h) \rceil$ pieces. Convexity retains

the segment in Ω; approximate its interior division points within h, keeping its endpoints exact. Consecutive approximants are at most a apart and the path length is at most $|s-t|+2hk$. The triangle inequality gives the lower bound. These estimates are independent of the choice among maximal sets. □

The metric fixes relative three-volume. Let C_i be the Voronoi cell of $x_i \in S$, clipped to Ω, and $v_i = \mathrm{vol}(C_i) > 0$. Thus $\sum_i v_i = V = \mathrm{vol}(\Omega)$, and almost every point of C_i is at distance at most h from x_i. These are metric weights, not a physical event-count density. For $0 < h \le \epsilon$ supply

$$a_{ij} = \frac{45 v_i v_j}{\pi \epsilon^5}(1 - |x_i - x_j|/\epsilon)_+, \qquad (Lq)_i = v_i^{-1} \sum_j a_{ij}(q_i - q_j).$$

The weighted inner product is $\langle p, q\rangle_v = \sum_i v_i p_i q_i$. Symmetry gives $\langle q, Lq\rangle_v = \frac{1}{2}\sum_{ij} a_{ij}(q_i - q_j)^2 \ge 0$. The common finite action is

$$S_h[q] = \frac{1}{2}\int \left[\|\dot q\|_v^2 - c^2\langle q, Lq\rangle_v - m^2\|q\|_v^2\right] dt, \qquad c > 0, \quad m \ge 0. \tag{20}$$

Its equation is $\ddot q + c^2 Lq + m^2 q = 0$. The coefficient m has inverse-time units in this convention; neither physical mass nor time units are inferred.

Proposition B.7 (Uniform scalar and detector limit)**.** *Let $u_{tt} - c^2\Delta u + m^2 u = 0$ be a smooth reference on the full carrier. On $[0,T]$, its spatial support has distance greater than ϵ from the complement of Ω. Write $M_1 = \sup|\nabla u|$ and $M_4 = \sup\|D_x^4 u\|_{\mathrm{op}}$, with suprema over the full carrier and time window. Then*

$$\left|(Lu|_S)_i + \Delta u(x_i)\right| \le \frac{15}{112}\epsilon^2 M_4 + \frac{120 h(\epsilon + h)^3}{\epsilon^5} M_1 \le \frac{15}{112}\epsilon^2 M_4 + \frac{960 h}{\epsilon^2} M_1.$$

Put $B = c^2\sqrt{V}[(15/112)\epsilon^2 M_4 + 960 h\epsilon^{-2} M_1]$. Initialize the finite action with the sampled reference position and velocity. Then

$$\|\dot q - u_t|_S\|_v \le tB, \qquad \|q - u|_S\|_v \le \tfrac{1}{2}t^2 B.$$

For a fixed C^1 detector b, the field readout $D_h = \sum_i v_i b(x_i) q_i$ and its continuum counterpart $D = \int_\Omega bu$ obey

$$|D_h(t) - D(t)| \le \tfrac{1}{2}\sqrt{V}\|b\|_\infty t^2 B + Vh\|\nabla(bu(t))\|_\infty.$$

Thus $\epsilon \to 0$ and $h/\epsilon^2 \to 0$ yield a uniform finite-time observable limit. In normalized carrier units $\epsilon = h^{1/4}$ gives an $O(\sqrt{h})$ bound. Nested populations are unnecessary.

Proof. Compare L with the integral operator having kernel $45(1-|z|/\epsilon)_+/(\pi\epsilon^5)$. Odd Taylor terms cancel and

$$\int (1 - |z|/\epsilon)_+ z_1^2\, dz = \frac{2\pi}{45}\epsilon^5, \qquad \int (1 - |z|/\epsilon)_+ |z|^4\, dz = \frac{\pi}{14}\epsilon^7.$$

The second moment gives $-\Delta$; the fourth-order Taylor remainder, bounded by $M_4|z|^4/24$, gives $15\epsilon^2 M_4/112$. For nearest-cell projection P, compare $g_x(y) = (1 - |x-y|/\epsilon)_+(u(x) - u(y))$ with $g_x(P(y))$. Splitting the difference at the larger kernel value uses a point within ϵ of x; kernel and field Lipschitz bounds give $|g_x(P(y)) - g_x(y)| \le 2hM_1$. This difference vanishes outside $B(x, \epsilon + h)$, giving the stated quadrature term. If $u(x) \ne 0$, the support buffer retains the whole kernel ball in Ω; otherwise the integrand outside Ω is zero. No boundary term is omitted.

The sampled residual in the weighted norm is at most B. The positive operator $A = c^2 L + m^2 I$ has $\|\cos(t\sqrt{A})\| \le 1$ and $\|\sin(t\sqrt{A})/\sqrt{A}\| \le t$, with the zero eigenspace interpreted by continuity. Duhamel integration gives the two trajectory bounds. Cauchy–Schwarz and nearest-cell quadrature for bu give the detector bound. □

The same cell argument gives $\|L\|_v \le 960/\epsilon^2$, using twice the largest weighted degree. `MetricKernelEnergy` formalizes the finite positive-energy and mass-weighted degree inequalities; the geometric, integral and continuum arguments above are analytic proofs. A local discrete-action time integrator with a strict Courant bound may consume this operator and add its own time and initialization errors.

Smooth compact initial data admit the reference until their speed-c domain of influence reaches the window boundary. Local energy flux for the Klein–Gordon equation gives that speed; the nonnegative mass term changes energy density but not the characteristic cone. Applied to a smooth local preparation difference, the detector estimate bounds the finite response outside that cone. A nonzero continuum signal is resolvable when its lower bound exceeds the approximation and detector errors. Instantaneous small tails of the finite graph are not identified with detectable superluminal signals. This is an interior effective limit, not a theorem about arbitrary boundary conditions or exact microscopic order reflection.

The canonical variables $Q_i = \sqrt{v_i} q_i$ and $P_i = p_i/\sqrt{v_i}$, $p_i = v_i \dot{q}_i$, give the oscillator Hamiltonian $\frac{1}{2}(|P|^2 + Q^T \widetilde{A} Q)$, where $\widetilde{A} = M^{1/2}(c^2 L + m^2 I) M^{-1/2} \ge 0$ and $M = \operatorname{diag}(v_i)$. Supplied canonical quantization on $L^2(\mathbb{R}^{|S|}, dQ)$ gives the same field readouts as multiplication operators. For $m > 0$ every mode has a Gaussian vacuum; at $m = 0$ zero modes remain free particles unless a separate restriction is declared. This finite Hilbert construction is not an interacting quantum continuum or a derivation of the Standard Model.

Coarsening, the volume prescription, radial force law, inertia, action coefficients and accessible observer variables remain added assumptions. The kernel graph may have growing degree. Algebraic source words alone do not implement its couplings as accepted primitive repairs or calibrate their duration. The construction supplies one conditional scalar effective world on the response carrier; its selection and physical interpretation remain separate obligations.

Proposition B.8 (Controlled boost comparison for scalar detectors)**.** *Let Λ be a proper orthochronous Lorentz transformation of $c^2 dt^2 - |dx|^2$, and let u be a smooth Klein–Gordon solution on the full spacetime. Put $u^\Lambda = u \circ \Lambda^{-1}$. Suppose both references satisfy the support and derivative hypotheses of Proposition B.7 on the same $[0,T]$ and Ω, with a common positive interior buffer. Choose a real $f \in C_c^1(\mathbb{R}^{1+3})$ such that both f and $f^\Lambda = f \circ \Lambda^{-1}$ are supported in $(0,T) \times \Omega^\circ$. For $w = u, u^\Lambda$, let q_h^w solve the same action (20), with the same L, v_i, c, m, initialized by $w(0, x_i)$ and $\partial_t w(0, x_i)$. Define the spacetime-smeared readouts*

$$\mathcal{D}_h(f; q) = \int_0^T \sum_i v_i f(t, x_i) q_i(t)\, dt, \qquad \mathcal{D}(f; w) = \int_{\mathbb{R}^{1+3}} f w \, dt \, d^3x.$$

Using the preceding spatial derivative bounds $M_1(w), M_4(w)$, put

$$B_w = c^2 \sqrt{V} \left[\frac{15}{112} \epsilon^2 M_4(w) + 960 h \epsilon^{-2} M_1(w) \right],$$

and

$$\mathcal{E}_h(f, w) = \frac{\sqrt{V} B_w}{2} \int_0^T t^2 \|f(t, \cdot)\|_\infty \, dt + V h \int_0^T \|\nabla_x (f(t, \cdot) w(t, \cdot))\|_\infty \, dt.$$

Then

$$\left| \mathcal{D}_h(f^\Lambda; q_h^{u^\Lambda}) - \mathcal{D}_h(f; q_h^u) \right| \le \mathcal{E}_h(f^\Lambda, u^\Lambda) + \mathcal{E}_h(f, u). \tag{21}$$

The right-hand side tends to zero for $\epsilon \to 0$ and $h/\epsilon^2 \to 0$. The conclusion is uniform over a bounded boost class when the common window, buffer and displayed derivative and detector bounds are uniform over that class.

Proof. Lorentz covariance of the Klein–Gordon equation makes u^Λ a solution with the same c, m. The preceding detector estimate applies at each time with $b = f(t, \cdot)$; its time integral is $|\mathcal{D}_h(f; q_h^w) - \mathcal{D}(f; w)| \le \mathcal{E}_h(f, w)$. Since $\det \Lambda = 1$ and both detector supports lie in the comparison window, change of variables gives $\mathcal{D}(f^\Lambda; u^\Lambda) = \mathcal{D}(f; u)$. The two approximation bounds and the triangle inequality prove the claim. □

The transformed preparations are supplied through the continuum reference on the transformed Cauchy surface; this comparison does not construct a boost map from one finite history or its initial data. The same estimate applies to smooth local preparation differences by linearity. Spacetime smearing is essential to the displayed comparison: a fixed-time detector instead transforms to a tilted hypersurface with its transformed measure. The result controls detector values for one common declared action, not exact finite-graph support, native source selection, or boosts of arbitrary finite records. Time integration and preparation errors, if present in an execution, require their own bounds.

B.13 Action measure, record production and geometric density

A supplied scalar response operator contains more information about its action measure than an unweighted event count. It does not identify that measure with physical volume. The distinction can be tested on the same source-addressed family, with its preparation, protected records, read law and model clock kept explicit.

Proposition B.9 (Identification of normalized action measure)**.** *Let A be a finite real operator whose nonzero off-diagonal support is connected. Suppose a positive diagonal matrix $M = \mathrm{diag}(m_i)$ makes MA symmetric. The equations*

$$m_i A_{ij} = m_j A_{ji}, \qquad \sum_i m_i = 1$$

determine m uniquely. If the two directed coefficients on every support edge are negative, a spanning tree constructs the candidate through $m_j/m_i = A_{ij}/A_{ji}$; the remaining edges test consistency.

Proof. For two positive solutions m, n, multiplying their edge equations and cancelling a nonzero coefficient gives $m_i n_j = m_j n_i$. Thus m_i/n_i is constant along every path. Connectivity makes the constant global, and normalization sets it to one. Tree products produce the candidate; positivity and every remaining edge equation certify its extension to the whole graph. Multiplying both M and the stiffness MA by one positive constant leaves A unchanged, so reciprocity cannot identify an absolute action scale. □

This reconstruction also has a finite conditioning bound. If each directed edge coefficient has relative error at most $\epsilon < 1$, each inferred edge ratio changes by a factor between r^{-1} and r, where $r = (1 + \epsilon)/(1 - \epsilon)$. For a tree of depth d, the root-relative weights change by factors in $[r^{-d}, r^d]$. Normalizing their sum gives factors in $[r^{-2d}, r^{2d}]$. This bounds the tree estimate; noisy cycle disagreement remains a model-consistency test.

On the authenticated $q = 5$ scalar family, the independent certificate recovers all 64 normalized weights from the directed operator alone, checks its 144 undirected edges, and compares the result with the separately reconstructed golden dual-cell quadrature. The tree has depth 9 and 81 remaining cycle edges. The parent mass vector is used for this comparison, not by the reconstruction algorithm. Operator and action selection remain supplied inputs.

A production control on the same finite family. Two preparations, with amplitude multipliers 1/2 and 1 relative to the retained source, give 2,944 independently replayed field writes. They have the same response operator and field ancestry; their initial energies differ by a factor of four. Local observer records read the actual writer, immutable version and value at the same site. Their extra read/write operations are charged separately and have no identified physical durations. The union of these local certificates is not a construction of remote record transport to one observer.

For an interior field-ancestry interval I, let $v_e = \tau m_{i(e)}$, where $\tau = \sqrt{5}/35$ is the retained action tick. Declare a production multiplier $r > 0$ and an integer exposure budget K. The finite local production rule

$$n_e = \lfloor K r v_e \rfloor, \qquad \widehat{V}_r(I) = \frac{1}{Kr} \sum_{e \in I} n_e$$

has the exact deterministic enclosure

$$0 \le \sum_{e \in I} v_e - \widehat{V}_r(I) < \frac{|I|}{Kr}. \tag{22}$$

The proof sums $0 \le K r v_e - n_e < 1$. Four controls cross the two source strengths with $r = 1, 2$ at $K = 4096$. All 324 local observer records are replayed. Changing r changes raw density while the field order and action measure stay fixed; dividing by the declared production law recovers the same finite measure within (22). Neither that production law nor the identification of τm_i with physical four-volume is derived by this test. No Poisson reference or old causal count-clock theorem is substituted for the actual field ancestry.

What a specified conformal action adds. Consider the separate control metric

$$g = u(x)\,(-dt^2 + dx^2 + dy^2 + dz^2), \qquad u > 0,$$

and the canonical massive scalar action in these coordinates. Its kinetic weight scales as u, whereas its metric four-volume scales as u^2. In the finite control, $M_i(u) = u_i M_i(1)$, each undirected gradient coefficient is multiplied by $(u_i + u_j)/2$, the Dirichlet boundary coefficient by its interior u_i, and the unit-mass potential coefficient by u_i^2. These terms define one symmetric quadratic action and reproduce the original operator at $u = 1$.

Reciprocity determines the relative profile u_i/u_j after division by the reference weights. The full operator supplies more: at an interior site with all six neighbours and the specified unit mass potential,

$$\sum_j A(u)_{ij} = u_i.$$

Gradient terms cancel in this row sum. It calibrates the common profile factor; all eight interior sites provide independent consistency checks. Thus no additional value of u is imported into this reconstruction. The unit scalar mass and the conformal action class are essential inputs. The controls $u = 1$ and $u = 1 + x/2$ recover both profiles exactly while retaining the same nonzero support and hence the same layered support order. Their normalized kinetic and four-volume measures differ for the nonconstant profile.

For an identified affine control $u = a + bx$, its specified continuum metric has scalar curvature

$$R = -6\Omega^{-3}\Box\Omega = \frac{3b^2}{2u^3}, \qquad \Omega = \sqrt{u},$$

with signature $(- + ++)$ and the displayed curvature convention. The nonzero control curvature is obtained through the supplied action and metric dictionary, not from an unweighted ordering statistic. No law makes either profile respond to the simulated source strength.

Scalar readouts do not determine tensor response. The same finite action also permits a direct comparison of directional responses. Write its potential as $V = G_x + G_y + G_z + V_m$, including every interior and Dirichlet edge, and consider the retained neighbouring configurations q_-, q. Their discrete action contribution is

$$L_d = T_d - \tau V, \qquad T_d = \frac{(q - q_-)^T M (q - q_-)}{2\tau}.$$

The potential uses the right endpoint q. Summing these contributions gives the recorded centred update through the discrete Euler–Lagrange equation. At fixed configurations, a stretch of coordinate axis k by s changes the contribution to

$$L_d(s) = sT_d - \tau\{s(V - G_k) + G_k/s\}, \qquad \left.\frac{dL_d}{d\log s}\right|_{s=1} = T_d - \tau(V - 2G_k).$$

An x/y permutation commutes with the complete cubic source operator. It preserves T_d, V, the action measure and the isotropic sum of these derivatives, but interchanges unequal directional responses. The certificate uses the two retained configurations and independently reassembles the stretched-grid action at $s = 2$ and $s = 1/2$; two thirds of their difference equals the displayed derivative exactly. Kinetic variation is included. These are finite action responses, with no identification of a physical stress source or gravitational coupling.

The connectedness and normalization theorem, common-scale ambiguities, finite production-error bound and conformal calibration algebra are checked in `Lean/Geometry/SourceActionMeasure.lean`. The original source histories, exact controls and independent verifier are at `evidence/source_density_geometry/` and `code/source_density_geometry/`. They identify a relative action measure and a profile within a specified conformal action class. They do not select the population, action, clock, production grammar, storage, source-to-metric update or physical tensor coupling. Those inputs are needed before the recovered quantity can be used as a sourced physical volume or an Einstein response.

B.14 A common source-population scalar and quantum detector packet

The prepared golden source addresses admit a local massive scalar action whose field edges fit inside the causal construction's existing read radius. The action below uses adjacent sorted source coordinates, rather than the radial kernel. It therefore has its own coefficients and boundary condition. The source addresses are protected preparation data, distinct from the mutable field registers. No invariance of an address under arbitrary live port repair is assumed.

Put $L = 2/\sqrt{\phi + 2}$ and use coordinates $y = x/L$, model time $s = ct/L$, $\hbar = 1$ and dimensionless mass one. For $q = F_n$, sort $\xi_b = b\phi - \lfloor b\phi \rfloor$, $0 \le b < q$, as $0 = x_0 < x_1 < \cdots < x_{q-1} < 1$, and append $x_q = 1$ as a fixed Dirichlet boundary marker. This marker is also a source-coded point. The mutable interior population $\{x_1, \ldots, x_{q-1}\}^3$ is contained in the original golden product population. Each coordinate is the normalized readback of the integer control $a = -\lfloor b\phi \rfloor, b$; the nodes are not replaced by their nearby uniform-grid labels.

Write $h_i = x_{i+1} - x_i$ and $w_i = (h_{i-1} + h_i)/2$ for interior nodes. Let $W = \mathrm{diag}(w_i)$ and let K be the Dirichlet stiffness matrix with diagonal $h_{i-1}^{-1} + h_i^{-1}$ and off-diagonal $K_{i,i+1} = -h_i^{-1}$. On the tensor population use product masses and the action

$$\mathcal{S}_q = \frac{1}{2}\int \left[\|\dot{u}\|^2_{W^{\otimes 3}} - \langle u, (I + \mathsf{L}_1 + \mathsf{L}_2 + \mathsf{L}_3)u\rangle_{W^{\otimes 3}}\right] ds, \qquad \mathsf{L} = W^{-1}K. \tag{23}$$

Here L_d acts on coordinate d. Each varying field read is between adjacent source sites; fixed boundary values contribute on-site terms. The golden gap is at most $2/q$, so $h_i < q^{-1/2}$ for $q > 4$.

The four finite certificates also check the stronger exact gap inequality $qh_i^2 < 1$ directly. Thus these field edges are permitted by the radius $a_q = L/\sqrt{q}$. This inclusion is not a construction of authenticated time-step events. Subcycling and the custody of a count-derived clock require an additional execution comparison.

Mass whitening gives the strictly positive real matrix

$$A_q = I + B \otimes I \otimes I + I \otimes B \otimes I + I \otimes I \otimes B, \qquad B = W^{-1/2} K W^{-1/2}.$$

Canonical quantization is the oscillator Fock space with vacuum-subtracted Hamiltonian $d\Gamma(\omega_q)$, $\omega_q = A_q^{1/2} \geq I$. The comparison theory is the massive Dirichlet free scalar on the same unit cube, with $\omega = (I - \Delta_D)^{1/2}$. Its nine comparison modes are

$$e_k(y) = \sqrt{8}\sin(k\pi y_1)\sin(\pi y_2)\sin(\pi y_3), \qquad \omega_k = \sqrt{1 + \pi^2(k^2+2)}, \quad 1 \leq k \leq 9.$$

The remaining continuum and lattice modes belong to their respective full Fock spaces; they are not discarded from an evolution residual.

A full frequency residual from small tridiagonal solves. Let S sample these modes with square roots of the tensor masses, $G = S^*S$, and suppose $\|G - I\| \leq d < 1$. Then $j = SG^{-1/2}$ is an isometry. Put

$$c_d = (1-d)^{-1/2} - 1, \qquad s_d = \sqrt{1+d}\, c_d, \qquad \Omega = \operatorname{diag}(\omega_k).$$

For $f_k(x) = \sqrt{2}\sin(k\pi x)$ define $s_k = W^{1/2} f_k|_S$ and $r_k = W^{1/2}(\mathsf{L} f_k - (k\pi)^2 f_k)|_S$. For each k compute the two nonnegative quadratic forms

$$Q_{kj} = r_j^*[B + (1+\omega_k^2)I]^{-1} r_j, \qquad j \in \{k, 1\},$$

and set

$$b_k = \sqrt{Q_{kk}}\|s_1\|^2 + 2\sqrt{Q_{k1}}\|s_k\|\|s_1\|, \qquad r = \Big(\sum_k b_k^2\Big)^{1/2}.$$

The quadratic forms require only the positive tridiagonal solve $(K + (1+\omega_k^2)W)y = W(\mathsf{L} f_j - (j\pi)^2 f_j)$.

Proposition B.10 (Full lattice leakage and coefficient-sensitive control)**.** *With the definitions above,*

$$\begin{aligned} \|\omega_q j - j\Omega\| &\leq r(1-d)^{-1/2} + 2\omega_9 s_d, \\ \|(\omega_q j - j\Omega)v\| &\leq \kappa(v) := \sum_k b_k|v_k| + (rc_d + 2\omega_9 s_d)\|v\|. \end{aligned} \tag{24}$$

The bound satisfies $\kappa(Dv) \leq \kappa(v)$ whenever D is diagonal and $|D_{kk}| \leq 1$. For $m = 1$,

$$\|\omega_q^{-1/2} j v - j\Omega^{-1/2} v\| \leq \tfrac{1}{2}\kappa(v). \tag{25}$$

Proof. For a positive scalar λ, $(\sqrt{\lambda} - \omega_k)^2 \leq (\lambda - \omega_k^2)^2/(\lambda + \omega_k^2)$. Functional calculus applies this inequality to the full matrix A_q. The stiffness residual on $Se_k = s_k \otimes s_1 \otimes s_1$ is the sum of its three tensor residuals. Minkowski's inequality in the $(A_q + \omega_k^2)^{-1}$ norm, followed by $A_q + \omega_k^2 \geq B_d + (1+\omega_k^2)I$, bounds the full frequency residual column by b_k. This step includes its component perpendicular to the sampled nine-mode space. Hence $\|\omega_q S - S\Omega\| \leq r$ and its action on v is bounded by $\sum_k b_k|v_k|$.

Write $G^{-1/2} = I + C$, with $\|C\| \le c_d$ and $\|SC\| \le s_d$. Expanding

$$\omega_q j - j\Omega = (\omega_q S - S\Omega)(I + C) + S[\Omega, C]$$

gives both bounds. In the commutator estimate use $\|S\|\|C\| \le s_d$; no commutation of G and Ω is assumed. Diagonal contraction decreases every term in κ. Finally, the resolvent formula for the inverse square root gives

$$-\frac{1}{\pi}\int_0^\infty t^{-1/2}(\omega_q + t)^{-1}(\omega_q j - j\Omega)(\Omega + t)^{-1} v\, dt.$$

Its norm is at most $\kappa(v)\pi^{-1}\int_0^\infty t^{-1/2}(1+t)^{-2}dt = \kappa(v)/2$. □

The isometry on the nine-mode comparison Fock space is $\Gamma(j)$. On its sector with at most N particles, the full generator residual is at most $N[r(1-d)^{-1/2} + 2\omega_9 s_d]$, by the sum of the N one-particle residuals on symmetric tensors. This is an actual generator estimate, not a bound on a compressed matrix. For the particular coherent states below, using their coefficients directly gives a substantially smaller comparison without an occupation cutoff.

Proposition B.11 (The same action has a free-field detector limit)**.** *Along the golden refinements, the sampled isometries on every fixed finite Dirichlet sine band have vanishing full frequency residual. For any fixed finite time interval, the classical smeared responses to bounded, piecewise-continuous L^2 momentum preparations, and the coherent-state expectations of finitely many bounded smeared Weyl effects, converge to those of the massive Dirichlet scalar on the cube. Smearings and their products must be Riemann integrable; a finite collection of slab-boundary jumps satisfies this condition. Initial position is zero for the momentum preparations in this statement.*

Proof. Let $\ell = \max_i h_i \to 0$. Taylor's formula at an interior node gives

$$|\mathsf{L}f(x_i) + f''(x_i)| \le \frac{h_{i-1}^2 + h_i^2}{3(h_{i-1} + h_i)}\|f'''\|_\infty \le \frac{\ell}{3}\|f'''\|_\infty.$$

This includes the generally nonzero error from unequal adjacent gaps. The product trapezoidal rule has error at most $\ell^2 \sum_d \|\partial_d^2 F\|_\infty/12$ on the unit cube. Boundary masses may be included: the Dirichlet sine products vanish there, so their contributions to the Gram matrix are zero. Consequently $G - I = O(\ell^2)$ on every fixed band and the full stiffness residual is $O(\ell)$. The square-root resolvent bound, or directly $(\sqrt{\lambda} + \omega_k)^{-1} \le 1/2$, makes the frequency residual $O(\ell)$ as well. These are actual residual estimates for this action; no ratio $H/a_q^2 \to 0$ is required.

Finite-band Duhamel and the negative-power estimate give uniform finite-time convergence of $\omega_q^{-1}\sin(s\omega_q)$ between sampled finite sine combinations, as well as the corresponding classical velocity readouts. The graph kinetic energy norm is the same product-mass norm. For general stated smearings, first approximate them in L^2 by finite sine combinations. Their squared sampling errors converge to the squared L^2 errors by product Riemann quadrature. The uniform bounds $\|\omega_q^{-1/2}\|, \|\omega^{-1/2}\| \le 1$ and unitarity control the tails independently of q and of time in the fixed interval. Take $q \to \infty$ for a fixed sine approximation, then let its error tend to zero. This also applies to the compact slab functions below.

The same argument controls coherent amplitudes, Weyl parameters, their inner products and their squared norms. The Gaussian characteristic function therefore gives the asserted bounded quantum readout convergence; finite Weyl products use the Weyl relations and the same controlled inner products. The limiting Fock theory is built on the full continuum L^2 one-particle space. Each finite-band isometry has its own domain; there is no isometry of that entire infinite one-particle space into a single finite lattice space. This observable convergence does not assert an interacting continuum or an operator-norm limit of all local effects. □

Compact preparation and a separated bounded detector. Normalize f to unit L^2 norm, where

$$f(y) = C\sin(\pi y_1)(1+\cos(\pi y_1))^8\sin(\pi y_2)\sin(\pi y_3), \qquad g(y) = f(1-y_1,y_2,y_3).$$

Both have exactly the nine modes above; denote their coefficient vectors by a and $\widetilde{a}_k = (-1)^{k+1}a_k$. The actual smearings are

$$f_c = f\,\mathbf{1}_{y_1\le 9/20}, \qquad g_c = g\,\mathbf{1}_{y_1\ge 11/20}.$$

Their supports are separated by $1/10$ in the normalized source coordinate, or $L/10$ in the source metric. Prepare $\exp(4i\Phi(f_c))|0\rangle$ and measure $E_c = (I+\sin\Phi(g_c))/2$, with $0\le E_c\le I$. Use the same sampled compact smearings and the same definitions in the finite graph oscillator theory. These are field instruments with spatial support, not a rank-one projector onto a named mode. The unperturbed vacuum probability and the prepared probability at time zero are both $1/2$.

For the smooth comparison, put $b = \|\Omega^{-1/2}a\|/\sqrt{2}$ and $\alpha = 4b$. Its exact induced probability is

$$P(s) - \tfrac{1}{2} = \tfrac{1}{2}e^{-b^2/2}\sin\left(4\sum_{k=1}^{9}(-1)^{k+1}a_k^2\frac{\sin(\omega_k s)}{\omega_k}\right). \tag{26}$$

This follows from the coherent-state characteristic function. For two coherent amplitudes differing by at most d_α and two Weyl parameters differing by at most d_b, the corresponding effects obey the explicit probability bound

$$\mathcal{E}(b,\alpha,d_b,d_\alpha) = \tfrac{1}{4}d_b(2b+d_b) + d_b(\alpha+d_\alpha) + bd_\alpha. \tag{27}$$

Indeed, $|e^{-x}-e^{-y}|\le|x-y|$ for $x,y\ge 0$ and $|\sin x-\sin y|\le|x-y|$ control the vacuum factor and the coherent phase. All norms refer to the full one-particle Hilbert spaces after the isometric comparison.

For the graph versus continuum smooth comparison, take

$$d_b = \frac{\kappa(a)/2+s_d}{\sqrt{2}}, \qquad d_\alpha = 4d_b + \frac{4|s|}{\sqrt{2}}\kappa(\Omega^{-1/2}a).$$

Equation (25) controls the initial smearing maps. Duhamel's formula on the finite spectral comparison space controls evolution; $\kappa(e^{-is\Omega}v) = \kappa(v)$ makes the bound uniform in time. The graph unitary acts on every lattice mode.

Each compact-continuum smearing differs from its smooth version by squared L^2 norm $\theta = \int_{9/20}^{1}|\sum_k a_k\sqrt{2}\sin(k\pi x)|^2dx$. Thus its extra d_b is at most $\sqrt{\theta/2}$ and its extra d_α at most $4\sqrt{\theta/2}$, using $\omega\ge 1$. The graph tails are computed separately from their actual lumped masses. Their two values need not equal the continuum tail or each other. Apply (27) to both localization steps and add their errors to the smooth comparison error. The continuum compact preparation has finite expected energy at most 8 above the vacuum. The graph energy is $8\|f_c|_S\|^2_{W^{\otimes 3}}$, which the certificate computes separately. A discontinuity at a slab boundary does not imply membership in the Hamiltonian's operator domain. Only its form domain and unitary norm preservation are used here; the Duhamel calculation uses the smooth finite-mode vector. For any $E>0$, the continuum energy tail is at most $8/E$ by the spectral Markov inequality.

Theorem B.12 (Resolved compact scalar quantum response)**.** *For the declared action, instruments and units above, the $q = 233$ golden population satisfies, uniformly for $19/20\le s\le 1$,*

$$|P_{233,c}(s) - P_c(s)| < 0.050055, \qquad |P_{233,c}(s) - \tfrac{1}{2}| > 0.137754.$$

The comparison error is smaller than the induced response throughout a nonzero time interval. The response is to a spatially nonuniform compact preparation at a spatially separated compact detector.

Proof. The executable certificate uses outward rational arithmetic, the exact golden coordinates, positive tridiagonal elimination and the bounds above. It encloses the continuum tail by 0.000222611 and the full nine-mode frequency residual by 0.053232. The coefficient-sensitive preparation residual is below 0.005360. One hundred closed time cells cover the entire interval; Taylor remainders enclose all trigonometric and exponential values. The final inequalities include both sampled localization tails and the continuum localization error. An independent verifier assembles edge fluxes, solves the positive systems in reverse order, and recomputes the probability bounds. The smaller populations $q = 13, 34, 89$ are reported without requiring them to satisfy the signal criterion. □

The tensor calculation uses one-dimensional arrays of length 232, while the mathematical finite model has 232^3 oscillators and an infinite-dimensional Fock space. No full read/write trace at that population is claimed. The theorem is a free massive scalar comparison with prescribed axes, boundary, action, protected addresses and model time. It does not select these inputs from native repair, identify physical clock units, prove an interacting or charged continuum, or identify Hamiltonian substeps with the retained causal-count history.

B.15 Time refinement of the same compact scalar detector

The golden tensor action admits a convergent local time discretization as well as the spatial comparison above. This statement concerns the mathematical trajectories of that action at each refinement. It does not assert that their complete observer event histories have been executed. Use the dimensionless coordinates, time and $\hbar = 1$ of (23). All norms below are in its mass inner product, or equivalently in the mass-whitened coordinates.

Proposition B.13 (Full-spectrum displacement and detector estimate)**.** *Let A be a finite positive self-adjoint operator, $A \geq I$, and use the canonical kick–drift–kick evolution with step $\tau > 0$, initial mean position zero and initial mean momentum v. Let $r = \tau^2\|A\| \leq r_0 < 4$, $s_0 = \sqrt{1 - r_0/4}$ and $d_0 = s_0(1 + s_0)$. At every integer $n \geq 0$ with $n\tau \leq T$,*

$$\|u_n - A^{-1/2}\sin(n\tau\sqrt{A})v\| \leq \tau^2\left[\frac{\|A^{1/2}v\|}{4d_0} + \frac{T\|Av\|}{12d_0}\right]. \tag{28}$$

Prepare the original action vacuum with coherent momentum displacement v, and consider $E_g = (I + \sin\Phi(g))/2$. The probability difference between this split evolution and continuous evolution of the same finite action is at most

$$\frac{\|g\|}{2}\tau^2\left[\frac{\|A^{1/2}v\|}{4d_0} + \frac{T\|Av\|}{12d_0}\right] + \frac{\tau^2\sqrt{G_0G_1}}{32(1 - r_0/4)}, \quad G_0 = \|g\|^2, \quad G_1 = \langle g, Ag\rangle. \tag{29}$$

Every mode is retained, including the split evolution of the original vacuum covariance.

Proof. For an eigenfrequency $\omega > 0$, put $z = \tau\omega/2$, $s = \sqrt{1 - z^2}$ and $\theta = 2\arcsin z$. The local canonical gates give the position recurrence $u_{n+1} = 2u_n - u_{n-1} - \tau^2 Au_n$, with $u_0 = 0$, $u_1 = \tau v$. In this mode its solution is $u_n = \sin(n\theta)v/(\omega s)$. Split its difference from the continuous solution into amplitude and phase errors. Since

$$\frac{1/s - 1}{\omega} = \frac{\tau^2\omega}{4s(1 + s)}, \qquad \theta - \tau\omega = 2\int_0^z \frac{a^2\,da}{\sqrt{1 - a^2}(1 + \sqrt{1 - a^2})} \leq \frac{\tau^3\omega^3}{12d_0},$$

the scalar error is at most $\tau^2[\omega/(4d_0) + T\omega^2/(12d_0)]|v|$. Functional calculus and the norm triangle inequality prove (28).

In the original vacuum the mode variances are $1/(2\omega)$ and $\omega/2$. The n-step canonical matrix has position row $(\cos(n\theta), \sin(n\theta)/(\omega s))$. Its extra position variance is

$$\delta V_n(\omega) = \frac{\tau^2 \omega \sin^2(n\theta)}{8(1 - \tau^2\omega^2/4)} \geq 0.$$

Therefore the smeared variance increase is at most $\tau^2\langle g, A^{1/2} g\rangle/[8(1 - r_0/4)]$ and hence at most $\tau^2\sqrt{G_0 G_1}/[8(1-r_0/4)]$. For mean m and variance V, the effect probability is $[1+e^{-V/2}\sin m]/2$. Its change is bounded by $|\delta m|/2+|\delta V|/4$; Cauchy–Schwarz bounds the mean change by $\|g\|$ times (28). This uses the original vacuum, not the invariant vacuum of a modified discrete Hamiltonian. □

A same-action continuum limit. Let $h = \min_i(x_{i+1} - x_i)$ be the *minimum* adjacent golden coordinate gap, including the fixed Dirichlet endpoint. The one-dimensional edge form satisfies

$$z^* K z = \sum_i \frac{|z_{i+1} - z_i|^2}{h_i} \leq \frac{4}{h} \sum_{i=1}^{q-1} |z_i|^2, \qquad z^* W z \geq h \sum_{i=1}^{q-1} |z_i|^2.$$

Consequently $\|A_q\| \leq \Lambda_q := 1 + 12/h^2$. For bounded mass norms of v_q, g_q, the displacement and probability bounds vanish if $\tau_q = o(h)$. Indeed, $\|A_q^{1/2} v_q\| \leq \sqrt{\Lambda_q}\|v_q\|$, $\|A_q v_q\| \leq \Lambda_q \|v_q\|$ and $\sqrt{G_0 G_1} \leq \sqrt{\Lambda_q}\|g_q\|^2$. This requires no uniform derivative norm of a discontinuous compact smearing. A concrete choice is the largest dyadic $\tau_q \leq h^2$. For $h \leq 1/4$, it gives $\tau_q^2 \Lambda_q \leq h^4 + 12h^2 < 1$.

Use exactly the sampled f_c, g_c of Theorem B.12, the same original vacuum, and $v_q = 4 f_c|_S$. The parent Gram bound $\|G - I\| \leq d_q$ and positive mass weights imply

$$\|g_c|_S\|^2 \leq 1 + d_q, \qquad \|v_q\|^2 \leq 16(1 + d_q).$$

Multiplication by either slab indicator cannot increase these norms. For $r_0 = 1$, $d_0 \geq 3/2$. On $0 \leq n\tau_q \leq 1$, the temporal probability bound is therefore

$$\mathcal{E}_{q,\mathrm{time}} \leq \tau_q^2 (1 + d_q) \left[\frac{3}{8}\sqrt{\Lambda_q} + \frac{1}{9}\Lambda_q\right]. \tag{30}$$

Combined with Proposition B.11, this proves convergence of the same compact displacement response and bounded smeared field effect for the chosen space–time refinement. Bounded, piecewise-continuous smearings whose products are Riemann integrable satisfy the spatial hypothesis. No velocity-norm convergence, full-Fock state-norm convergence across the growing lattice, or interacting field limit is inferred from this estimate.

Proposition B.14 (A resolved fully time-discrete comparison)**.** *For the same $q = 233$ source population, action, preparation and detector, the exact minimum gap is $233 - 144\phi$ and the declared dyadic step is $\tau = 2^{-17}$. At every grid time $19/20 \leq n\tau \leq 1$, the mathematical split evolution satisfies*

$$|P^{\mathrm{split}}_{233,c}(n\tau) - P_c(n\tau)| < 0.050063, \qquad |P^{\mathrm{split}}_{233,c}(n\tau) - \tfrac{1}{2}| > 0.137746.$$

For every reference time $t \in [19/20, 1]$, define the step readout by

$$k_0 = \lceil (19/20)/\tau \rceil, \quad k_1 = 1/\tau, \quad k(t) = \min\{k_1, \max\{k_0, \lfloor t/\tau + 1/2 \rfloor\}\},$$

$$P^{\mathrm{step}}_{233,c}(t) = P^{\mathrm{split}}_{233,c}(k(t)\tau).$$

This is the nearest update inside the window, with upward ties. Then $k(t)\tau \in [19/20, 1]$, $|k(t)\tau - t| \le \tau$, *and*

$$|P^{\mathrm{step}}_{233,c}(t) - P_c(t)| < 0.050078, \qquad |P^{\mathrm{step}}_{233,c}(t) - \tfrac{1}{2}| > 0.137746.$$

This readout selects a defined discrete state; it introduces no state at an intermediate time.

Proof. The separate exact certificate reconstructs the ordered golden coordinates and minimum gap, verifies the dyadic step and stability inequality, and replays the immutable spatial comparison. It bounds (30) by 0.000008071512. Adding this to the spatial error and subtracting it from the spatial response gives the grid-time assertions. In the continuum the vacuum variance is stationary and $|P_c'(t)| \le \|g_c\|\|4f_c\|/2 \le 2$. Rounding selects a nearest grid point before imposing the window, with distance at most $\tau/2$. The upper endpoint 1 is itself a grid point, so upper clipping cannot occur for these t. If rounding falls below k_0, clipping chooses $k_0\tau$, and $(k_0 - 1)\tau < 19/20 \le t \le k_0\tau$ gives distance less than τ. Thus the selected point stays inside the window, also at its endpoints. The extra comparison error is at most 2τ, and the discrete response lower bound is unchanged. □

Proposition B.15 (Pre-arrival detector limit)**.** *Use the same massive Dirichlet action on the cube, zero initial mean position, original vacuum, and coherent momentum* af, *with fixed real* a. *Let* f, g *be bounded, compactly supported* L^2 *functions satisfying the sampling hypotheses of Proposition B.11. Suppose their support distance is* $d > 0$. *For every* $T < d$, *the corresponding local split trajectories with* $\tau_q = o(h_q)$ *satisfy*

$$\sup_{0 \le n\tau_q \le T} \left| P^{\mathrm{split}}_{q,g}(n\tau_q) - \tfrac{1}{2} \right| \longrightarrow 0.$$

The same conclusion holds for the step readout $P^{\mathrm{split}}_{q,g}(\lceil t/\tau_q \rceil \tau_q)$, *uniformly for* $0 \le t \le T$.

Proof. The continuum mean solves $u_{tt} - \Delta u + u = 0$, with $u(0) = 0$, $u_t(0) = af$. Its finite propagation speed is one. Indeed the local energy $e = (|u_t|^2 + |\nabla u|^2 + |u|^2)/2$ satisfies $\partial_t e = \nabla \cdot (u_t \nabla u)$ and $|u_t \nabla u \cdot n| \le e$ on each unit normal n. Integrating over a shrinking backward light cone gives zero energy when its initial base misses the preparation. The fixed Dirichlet boundary has zero energy flux; smooth approximation in the energy norm extends the argument to these L^2 momentum data. Thus $\langle g, u(t) \rangle = 0$ for $t < d$. The stationary continuum vacuum covariance then gives the sine-effect probability exactly $1/2$ there.

The parent spatial comparison is uniform on every fixed finite time interval. Positive Riemann mass quadrature bounds the sampled norms, so (29) makes the additional split error vanish uniformly on that interval. This proves the grid assertion. For the ceiling readout choose $T < T' < d$; eventually $\lceil t/\tau_q \rceil \tau_q \le T + \tau_q < T'$ for all $t \le T$, and apply the same uniform estimate on $[0, T']$. □

This is a limit of operational detector responses for the declared preparation class. It does not identify microscopic read ancestry with continuum causal order or establish a point-event count law. The pre-arrival assertion is asymptotic; the finite error certificate above concerns only its stated comparison window and preparation.

The finite local canonical gates define these trajectories and full-Fock states for each fixed q and finite number of steps. The certificate uses source gaps, scalar norm bounds and the spatial comparison, not a stored 232^3-oscillator history. It does not import the separate $q = 5$ preparation, clock-error envelope or observer event receipt. The time refinement, protected population, routing, action, boundary and quantization remain declared. Native acceptance, a physical clock, outcome attestation and a count-volume interpretation of substeps are separate requirements.

B.16 An authenticated execution of the golden scalar action

A small execution retains the field reads that realize the tensor action. It uses the $q = 5$ golden source coordinates

$$0, \quad 2\phi - 3, \quad 4\phi - 6, \quad \phi - 1, \quad 3\phi - 4, \quad 1.$$

The first five are the existing golden population; the final point is the fixed Dirichlet boundary marker. The $4^3 = 64$ interior patches carry mutable field registers. Their source addresses and action coefficients are protected preparation data. Axes, boundary conditions, the action and model time remain declared inputs.

Use the product masses M, tensor operator A and normalized coordinates of (23). Thus A is positive and self-adjoint in the M norm, with $\mathsf{A} \geq I$. Put

$$\Delta = 1/\sqrt{5}, \qquad \tau = \Delta/7 = (2\phi - 1)/35, \qquad \tau^2 = 1/245.$$

The discrete action and its exact Euler–Lagrange update are

$$\begin{aligned} \mathcal{S}_\tau &= \sum_j \left[\frac{\|u_{j+1} - u_j\|_M^2}{2\tau} - \frac{\tau}{2} \langle u_j, \mathsf{A} u_j \rangle_M \right], \\ u_{j+1} &= 2u_j - u_{j-1} - \tau^2 \mathsf{A} u_j. \end{aligned} \tag{31}$$

The step reads the previous local value and the current local value and current nearest neighbours. Fixed boundary values contribute on-site terms. Two alternating registers per patch retain immutable versions of the two layers. A serialized event names every consumed resource, version, actual writer and exact value, and commits its new value. The serial hash parent is also retained; it is not inserted into the semantic field-read order.

The preparation has $u_0 = 0$ and

$$v_i = 256 \prod_{a=1}^{3} x_{ia}(1 - x_{ia})\, \mathbf{1}_{x_{i1} < 9/20}, \qquad g_i = 64 \prod_{a=1}^{3} x_{ia}(1 - x_{ia})\, \mathbf{1}_{x_{i1} > 11/20}.$$

The detector is $D(u) = \langle g, u \rangle_M$. The initial memory $u_{-1} = -\tau v$ gives the centered initial velocity v. It is integrator preparation, not a recorded exact solution at negative time. This algebraic preparation differs from the nine-sine preparation in Theorem B.12; its numerical accuracy is not transferred to this coarser population.

Proposition B.16 (Stable exact execution and preserved energy)**.** *For this action,* $\tau^2 \|\mathsf{A}\|_M < 3 < 4$. *Every step of* (31) *preserves*

$$\begin{aligned} \mathcal{E}_\tau(u_{j-1}, u_j) &= \frac{\|u_j - u_{j-1}\|_M^2}{2\tau^2} + \frac{1}{2} \langle u_j, \mathsf{A} u_{j-1} \rangle_M \\ &= \frac{1}{2} \|\pi_j\|_M^2 + \frac{1}{2} \langle u_j, \mathsf{A}(I - \tau^2 \mathsf{A}/4) u_j \rangle_M, \end{aligned} \tag{32}$$

where $\pi_j = (u_j - u_{j-1})/\tau - \tau \mathsf{A} u_j/2$. *In particular the energy is positive, and* $\mathcal{E}_\tau \geq \|\pi_j\|_M^2/2 + \|u_j\|_M^2/8$.

Proof. The smallest coordinate gap is $5 - 3\phi > 7/50$. The row sums of the absolute one-dimensional operator are bounded by $4/h_{\min}^2$. Similarity to its symmetric mass-whitened matrix therefore gives

$$\|\mathsf{A}\|_M \leq 1 + 12/h_{\min}^2 < 30049/49 < 614, \qquad \tau^2 \|\mathsf{A}\|_M < 30049/12005 < 3.$$

Subtract successive first-line energies and use the recurrence and self-adjointness. The difference is zero. Completing the square gives the second line, and $I - \tau^2 \mathsf{A}/4 > I/4$ proves coercivity. The executable verifier checks this equality exactly at all twenty-two retained states. It is a modified discrete energy; conservation of the original continuous Hamiltonian is not asserted at nonzero step size. □

Full continuous-action comparison. The reference here is the continuous-time *same finite* 64-mode operator, with the same $u_0 = 0, v_0 = v$:

$$U(t) = \mathsf{A}^{-1/2} \sin(t\sqrt{\mathsf{A}})v.$$

Its comparison to the execution includes every spectral mode. Let

$$P_{80}(t) = \sum_{k=0}^{80} \frac{(-1)^k t^{2k+1}}{(2k+1)!} \mathsf{A}^k v.$$

The field data obey $0 \le v_i \le 4$, $0 \le g_i \le 1$ and $\sum_i M_i \le 1$, hence $\|v\|_M \le 4$, $\|g\|_M \le 1$. Taylor's theorem for sine through degree 162, followed by spectral calculus and the preceding operator bound, gives for $0 \le t \le 3/\sqrt{5} < 3/2$

$$\|U(t) - P_{80}(t)\|_M \le R := \frac{4(3/2)^{163} 614^{81}}{163!} < 10^{-30}. \tag{33}$$

The polynomial, the executed values and their squared M-norm difference are exact elements of $\mathbb{Q}(\phi)$. Exact sign bisection encloses their square roots. Adding R gives a full state-error enclosure at every executed time $t_j = j\tau$, $1 \le j \le 21$, and detector error at most the same bound. The reference is not a truncated eigenvalue simulation.

The exact energy is approximately 0.276440884. At $t = 3/\sqrt{5}$, the full state discretization error is enclosed between 0.009539009052 and 0.009539009055. These are finite-operator comparisons at executed times, not spatial continuum errors or errors for intermediate unstored times. The independent verifier uses the $1, \sqrt{5}$ basis, assembles undirected edge energies and evaluates the same full Taylor polynomial in reverse order. It validates the scientific values as well as the commitments.

Canonical quantum compatibility. In mass-whitened canonical coordinates, the update is the Heisenberg map of the symmetric unitary product

$$e^{-i\tau V/(2\hbar)} e^{-i\tau T/\hbar} e^{-i\tau V/(2\hbar)}, \qquad T = \tfrac{1}{2} P^T P, \quad V = \tfrac{1}{2} X^T A_q X.$$

Thus the exact classical impulse response also determines the central commutator of the corresponding smeared canonical fields:

$$\frac{[\Phi(g)_j, \Phi(v/4)_0]}{i\hbar} = -\frac{1}{4} D(u_j)\, I.$$

This identity is state independent, initially on the common Schwartz domain and then as its constant quadratic-form extension. It uses all finite oscillator modes. No covariance or measurement probability is computed. In particular no modified-Hamiltonian vacuum replaces the original source-action vacuum. A probability comparison under these gates would also have to propagate that original covariance.

Actual reads and the clock boundary. One trace stores 128 preparation writes and $21 \times 64 = 1344$ update writes. There are $8 \cdot 4^3 - 6 \cdot 4^2 = 416$ dynamic field reads per update layer, or 8736 in total. These counts include every previous-local and current-layer field read; the immutable source and coefficient commitments are supplied once as preparation inputs. Baseline and intervention traces under ascending and descending site schedules have identical corresponding field layers. Their full audit chains are retained and differ with schedule.

Inclusive intervals in the actual field-read order, from the current-layer seed at site $(1, 1, 1)$ to the same site after $7, 14, 21$ steps, contain $144, 576, 1024$ events. They are not the intervals of the earlier all-neighbour source-count law. For example the prepared source plane is at $x = 2\phi - 3$ and the recorded target at $x = 3\phi - 4$ has a nonzero response after seven substeps. Their point displacement squared is $2 - \phi > 1/5$, exceeding the old radius squared at that coarse layer. This exposes a finite point-event ancestry mismatch. It is not a continuum superluminality claim: interpolation supports differ from source-point locations, and no spatial continuum error bound is claimed at this population.

All substeps remain visible. Counting a chosen subset of their events does not by itself preserve the causal intervals required for a count clock. Relating this execution to that clock requires an explicit common event or operational readout map with its intervention, order and volume errors. The present trace supplies the inputs for that comparison; it does not identify native time, primitive word costs or laboratory seconds.

B.17 Volume work and finite field equations of state

An equation of state needs more than an update at fixed geometry. Energy, volume work and state preparation must refer to the same dynamical system. The distinction can be made exact for scalar pair repair, the golden scalar action, and the finite Maxwell action. These statements concern bounded self-reading patches with local state, ports, retained readback and repair or feedback, together with their declared field encodings. Thermodynamic preparation and metric deformation are additional inputs.

Fixed-volume repair does not determine pressure. For $Q(x) = \frac{1}{2}\sum_i x_i^2$, pair averaging at (i, j) preserves $\sum_i x_i$ and decreases Q by $(x_i - x_j)^2/4$. An added cumulative loss record B therefore preserves $Q + B$. This identity does not identify the loss as heat or Q as physical energy.

Proposition B.17 (Nonuniqueness of an unprovided volume response)**.** *Fix $V_0 > 0$. For every real α, the positive extension*

$$H_\alpha(x, V) = Q(x)(V/V_0)^{-\alpha}, \qquad Q(x) > 0, \quad V > 0,$$

agrees with Q on every state at $V = V_0$. Under the supplied convention $p = -\partial_V H|_x$ and $\rho = H/V$, it has $p/\rho = \alpha$. Consequently complete fixed-volume repair histories and the function Q do not select a work ratio within this extension class.

Proof. At V_0 the multiplicative factor is one, independently of x. Differentiation gives $-\partial_V H_\alpha = \alpha H_\alpha/V$. In particular $\alpha = 1/3$ and $\alpha = 1/2$ give different positive pressures despite identical scalar histories at the reference volume. □

Adding a volume-independent constant to H also changes ρ while leaving its work derivative unchanged. The energy reference is therefore part of the required specification. This is neither an equilibrium construction nor an obstruction to supplying an independently justified volume response. Exact twelve-port traces retain both quadratic loss and the added bookkeeping record; the latter is an external recorder, not a derived thermal bath.

Stress from the golden scalar action. Let $s > 0$ dilate the reference cube, with $V = s^3$ in fixed reference units. The dual mass and gradient forms scale as $M_s = s^3 M_0$ and $K_s^{\mathrm{grad}} = sK_0^{\mathrm{grad}}$. Keep the mass parameter m fixed in those units. For original field coordinates q and canonical momenta p, define at $s = 1$

$$K = \tfrac{1}{2} p^T M_0^{-1} p, \qquad G = \tfrac{1}{2} q^T K_0^{\mathrm{grad}} q, \qquad U = \tfrac{1}{2} m^2 q^T M_0 q.$$

The full box volume is distinct from the sum of interior dual-cell masses.

Proposition B.18 (Canonical scalar work and thermal specialization)**.** *At fixed original* (q, p)*,*

$$H_s = K/s^3 + sG + s^3 U, \qquad p_{\mathrm{sc}} V = K - G/3 - U, \qquad \rho_{\mathrm{sc}} = (K + G + U)/V,$$

where the last two expressions use energies evaluated at the chosen scale. For nonnegative K, G, U *and positive total energy, the instantaneous ratio lies in* $[-1, 1]$*. If* $K = G + U$*, its value lies in* $[0, 1/3]$*; for* $U = 0$ *it equals* $1/3$*.*

Proof. Differentiate H_s and divide by $dV/ds = 3s^2$. Positivity gives the first bounds. Substitution of $K = G + U$ gives $p_{\mathrm{sc}} V = 2G/3$ and $H = 2(G + U)$, proving the remaining assertions. □

Holding mass-normalized canonical coordinates fixed during dilation would instead give $pV = 2G/3$. Its difference from the original-coordinate stress is $K - G - U = \frac{1}{2} d(q^T p)/dt$ under the continuous action. Stationarity or a controlled virial average removes this difference; transient records do not satisfy that condition automatically.

For the authenticated $q = 5$ execution, the centered velocity is $v_n = (q_n - q_{n-1})/\tau - \tau \mathsf{A} q_n/2$ and $p_n = M_0 v_n$. Exact arithmetic evaluates K, G, U and stress on all twenty-two retained states of all four histories. The conserved split quantity is $H - \tau^2 \|\mathsf{A} q\|_{M_0}^2/8$, not H. Full-operator sine and cosine Taylor polynomials, with outward rational remainders, separately enclose the continuous-action pressure at all twenty-two times. In detail, $\mathrm{spec}_{M_0}(\mathsf{A}) \subset [1, 614]$, $\|v_0\|_{M_0} \le 4$ and $t \le 3/2$ give the displacement and velocity remainder bounds

$$\delta_q = \frac{4(3/2)^{163} 614^{81}}{163!}, \qquad \delta_v = \frac{4(3/2)^{162} 614^{81}}{162!}.$$

These follow from the real sine and cosine Taylor remainders by spectral calculus, retaining every mode. The corresponding polynomial norms are bounded by 7 and 5. Expanding the stress quadratic form therefore gives

$$|\delta p| \le 5\delta_v + \delta_v^2/2 + \frac{616}{3}(7\delta_q + \delta_q^2/2) < 1.385 \cdot 10^{-33}.$$

Outward rational rounding on the 10^{-12} reporting grid is separate from this analytic tail. At $t = 3/\sqrt{5}$ and $V = 1$,

$$\begin{aligned} p_{\mathrm{continuous}} &\in [0.129161530903, 0.129161530906], \\ (p/\rho)_{\mathrm{continuous}} &\in [0.467230205001, 0.467230205004], \\ p_{\mathrm{split}} - p_{\mathrm{continuous}} &\in [-0.017284266362, -0.017284266359]. \end{aligned}$$

These are certified nonequilibrium stress calculations in the supplied model. The ratio exceeding $1/3$ does not contradict its thermal specialization.

For a separately supplied canonical Bose state, let $\lambda_j > 0$ be the complete spatial generalized spectrum of the same Dirichlet action. In fixed units $\hbar = k_B = c = 1$, with zero chemical potential and zero-point energy subtracted at each s, put

$$\omega_j^2 = m^2 + \lambda_j/s^2, \quad n_j = (e^{\omega_j/T} - 1)^{-1}, \quad E_j = \omega_j n_j, \qquad F = T\sum_j \log(1 - e^{-\omega_j/T}).$$

Differentiating this free energy at fixed T, m and mode count gives

$$p = -\partial_V F = \frac{1}{3V}\sum_j \frac{E_j \lambda_j}{s^2 \omega_j^2}, \qquad \rho = \frac{1}{V}\sum_j E_j.$$

The vacuum-subtracted thermal covariances obey $K = G + U$, so this pressure agrees with the canonical mechanical stress. The massive ratio depends on temperature and scale; the massless ratio is $1/3$ from the declared three-dimensional scaling. All 64 modes are retained in 36 numerical cases with $s \in \{1, 2, 8\}$, $m \in \{0, 1, 4\}$ and $T \in \{1/2, 2, 10, 50\}$. Their ratios range from approximately 0.0409561 to $1/3$. An independent high-precision spectrum and partition derivative check these numerical values; this thermal calculation is not an interval certificate. At fixed mode count the ultraviolet scale changes with s, so neither a bulk thermodynamic limit nor cutoff removal follows. The Bose occupation support is unbounded despite the finite mode count. Native preparation, thermalization, finite observer-memory realization and physical calibration are not supplied. Subtracting zero-point energy at every scale also subtracts its pressure; no vacuum-pressure prediction is asserted.

Maxwell field pressure on the supplied cone. For the declared three-dimensional Whitney action, let a be the edge-integrated potential and π its conjugate momentum. Under uniform metric dilation, the one-form mass is $M_s = sM_1$ and the two-form mass is $N_s = s^{-1}N_1$. With curvature $b = Ca$,

$$H_s = \tfrac{1}{2}\pi^T M_s^{-1}\pi + \tfrac{1}{2} b^T N_s b = H_1/s.$$

At fixed (a, π), $V_s = s^3 V_1$ therefore gives $p_{\mathrm{mean}} = H_s/(3V_s)$ for positive field energy. Independently the integrated Maxwell momentum-flux tensor

$$\Pi = \frac{1}{V}\int \left[\tfrac{1}{2}(|E|^2 + |B|^2)I - EE^T - BB^T\right] d^3x$$

has $\operatorname{tr}\Pi/3 = H/(3V)$. Directional stresses can include tension; the trace identity does not require an isotropic individual field. Zero field energy leaves the ratio undefined. This is field energy only, excluding matter, sources, clocks, walls and observer records.

On the complete cone, removing its twelve gradient gauge directions leaves thirty positive oscillator modes. A separately supplied classical Gibbs ensemble on a fixed canonical gauge quotient, with a geometry-independent Liouville cell normalization, has

$$\log Z = 30\log T - \sum_{j=1}^{30}\log\omega_j + \text{constant}, \qquad \langle H\rangle = 30T, \qquad pV = 10T.$$

The formulas follow by the finite Gaussian integral and $\omega_j(s) = \omega_j(1)/s$. Independent gauge quotients and spatial quadrature check nine scale/temperature cases. A separate source-free numerical episode retains eighty nonzero displacement-current updates and eighty-one full canonical states. It checks energy balance, Ampère, Faraday and Gauss with zero conduction current. The pressure

extension also consumes the decoded fields of the self-reading parent instrument; the new source-free trajectory is not itself another observer readout instrument. The Gibbs preparation is separate from that trajectory, and supplies no thermalization, quantum blackbody law or continuum limit.

The work nonuniqueness and finite scalar/Maxwell scaling identities are kernel-checked in `Lean/Thermodynamics/EquationOfState.lean`. The state, stress and independent verification artifacts are in `code/equation_of_state/`. Gaussian ensemble evaluation and Maxwell trajectory checks use numerical tolerances; only the scalar continuous comparison has the stated outward-rational pressure enclosures. These are consequences of supplied action and state laws, not a derivation of a thermodynamic equation of state from scalar pair averaging.

B.18 A clock reconstructed from the scalar records

The finite execution also determines an action-time readout without using the recorded step labels to assign its configuration layers. This construction retains the supplied positive mass inner product, operator A, resource-to-site map and action normalization. It concerns one duration shared by an entire configuration layer, not independent lapses at each site.

Recovering the layers. Authenticate each immutable write and every consumed resource, version, writer and value. Draw directed edges from consumed writers to their readers. The serial audit parent remains authenticated but is not a field input in this action. Define roots to have height zero and every other vertex to have height one plus its greatest parent height. An update reads its own field two layers earlier and the nonempty current stencil, including itself. Induction therefore gives height j to every update of u_j, $1 \le j \le 21$. No event identifier component or grouping of serialized positions enters this definition.

Both seed configurations have height zero. Their roles are distinguished in this particular graph by their numbers of height-one children: every older-memory root has one, whereas each current-field root has four, five, six or seven. The respective counts are $64, 8, 24, 24, 8$. With the declared resource-to-site map, each recovered layer contains exactly one write per site; the verifier checks alternating buffers, consecutive versions and the full adjacent-layer action stencil. Thus all 23 layers are recovered in each trace. The two root sets remain causally incomparable; the older seed is integrator memory, not a negative-time physical interval. The complete retained read graph is required. For two symmetric roots feeding a single update, graph symmetry prevents this seed-role distinction.

Proposition B.19 (Unique positive action durations)**.** *Let A be positive and self-adjoint in the supplied M inner product. For fixed endpoints consider the configuration-stationarity equations of*

$$\mathcal{S}_h[u] = \sum_{j=0}^{20} \left[\frac{\|u_{j+1} - u_j\|_M^2}{2h_j} - \frac{h_j}{4}\big(E(u_j) + E(u_{j+1})\big) \right], \qquad E(u) = \langle u, \mathsf{A}u \rangle_M, \tag{34}$$

where the $h_j > 0$ are fixed but unknown parameters. At an interior center put $x = u_j - u_{j-1}$, $y = \mathsf{A}u_j$, $z = u_{j+1} - u_j$ and suppose

$$D = \langle x, x \rangle_M \langle y, y \rangle_M - \langle x, y \rangle_M^2 > 0.$$

If the full vector equation $z = \alpha x + \beta y$ holds, positive adjacent durations exist exactly when $\alpha > 0$ and $\beta < 0$. They are unique:

$$h_{j-1}^2 = \frac{-2\beta}{\alpha(\alpha+1)}, \qquad h_j = \alpha h_{j-1}. \tag{35}$$

If every actual center $j = 1, \dots, 20$ has $\alpha = 1$, $\beta = -s$ with $s > 0$, all 21 positive durations equal $\sqrt{s}$.

Proof. The discrete variational framework is supplied [38]. Positive Gram determinant implies independence of x, y. Configuration variation in (34) gives

$$z/h_j - x/h_{j-1} + (h_j + h_{j-1})y/2 = 0.$$

Its independent coefficients imply $h_j/h_{j-1} = \alpha$ and $\beta = -h_j(h_j + h_{j-1})/2$, proving necessity and uniqueness. Substitution proves sufficiency. Shared intervals must agree between adjacent centers; for the stated uniform coefficients they do, including the endpoint intervals determined by centers 1 and 20. □

The coefficients are obtained by the inverse Gram matrix, and their full vector residual is separately checked; projection alone would miss a component orthogonal to x, y. Exact replay of both intervention schedules gives 40 positive determinants, all exceeding $2083218601/10^{12}$, and $(\alpha, \beta) = (1, -1/245)$. Hence the positive tick is $\sqrt{5}/35$ and the elapsed time from an assigned origin is $3/\sqrt{5}$. The zero baseline has no independent pair and defines no such clock. Six Lean lemmas formalize coefficient extraction, duration uniqueness and the Gram bridge; an additional theorem composes these implications. The variation and the actual finite-record values are established analytically and by exact replay.

At equal $h_j = \tau$, (34) differs from the left-endpoint action yielding (31) by the endpoint term $\tau(E(u_0) - E(u_{21}))/4$, leaving fixed-endpoint stationarity unchanged. Its variable-duration extension is a supplied quadrature choice. Durations are not varied in deriving the configuration equation: unconstrained variation in h_j would give the additional, generally nonzero derivative $-\|u_{j+1} - u_j\|_M^2/(2h_j^2) - (E(u_j) + E(u_{j+1}))/4$.

Proposition B.20 (Stable duration identification)**.** *In the same real Hilbert space let $F_0 = [x\ y]$, $z = F_0 c_0$, $c_0 = (\alpha_0, \beta_0)$, and $D > 0$. Choose positive bounds*

$$\ell_\alpha \geq \sqrt{\langle y, y\rangle_M / D}, \qquad \ell_\beta \geq \sqrt{\langle x, x\rangle_M / D}.$$

For perturbed data suppose the full equation $z + e_z = (F_0 + [e_x\ e_y])c + r$ holds, where $\|e_x\|_M \leq \epsilon_x$, $\|e_y\|_M \leq \epsilon_y$, $\|e_z\|_M \leq \epsilon_z$, $\|r\|_M \leq \rho$ and all budgets are nonnegative. If $q = \epsilon_x \ell_\alpha + \epsilon_y \ell_\beta < 1$, put

$$B = \frac{\epsilon_z + \rho + \epsilon_x|\alpha_0| + \epsilon_y|\beta_0|}{1 - q}.$$

Then $|\alpha - \alpha_0| \leq \ell_\alpha B$ and $|\beta - \beta_0| \leq \ell_\beta B$, and the perturbed columns remain independent. If these bounds give $0 < a_L \leq \alpha \leq a_U$ and $0 < b_L \leq -\beta \leq b_U$, define the associated positive durations by (35). They obey

$$\begin{aligned} \frac{2b_L}{a_U(a_U + 1)} \leq h_{j-1}^2 \leq \frac{2b_U}{a_L(a_L + 1)}, \\ \frac{2b_L a_L}{a_L + 1} \leq h_j^2 \leq \frac{2b_U a_U}{a_U + 1}. \end{aligned} \tag{36}$$

Proof. The two rows of $(F_0^* F_0)^{-1} F_0^*$ have the stated norm bounds. Applying them to the perturbed equation bounds each coefficient error by its ℓ times $b = \epsilon_z + \rho + \epsilon_x|\alpha| + \epsilon_y|\beta|$. Substitution gives $b \leq (1 - q)B + qb$, hence $b \leq B$. In the coefficient norm $\max(|c_1|/\ell_\alpha, |c_2|/\ell_\beta)$, the left-inverse times $[e_x\ e_y]$ has norm at most $q < 1$; a vector in the perturbed kernel must vanish. Formula (36) follows by monotonicity from (35). □

For the actual record, $\|\mathsf{A}\|_M < 614$. A hypothetical error at most $\epsilon = 10^{-8}$ in each complete configuration therefore permits $\epsilon_x = \epsilon_z = 2\epsilon$, $\epsilon_y = 614\epsilon$ and $\rho = 0$ for a hidden exactly configuration-stationary history. Every $q < 7.64\,10^{-6}$ and the coefficient sign margins stay positive. Intersecting bounds on shared intervals and summing all 21 durations gives, in either schedule, the exact rational enclosure

$$1.341635903357 \le t_{21} - t_0 \le 1.341645669638.$$

These are conditional stability bounds, not measured noise. Nonempty interval intersections do not by themselves establish existence of an arbitrary noisy stationary history. A nonzero residual of the configuration Euler equation must be multiplied by h_j before it is bounded by ρ; the residual units cannot be identified without that factor.

The code independently authenticates all 5,888 events and 34,944 reads, reconstructs the graph layers, and checks the Gram and stability calculations with exact rational and quadratic-field arithmetic. The same reconstructed values agree across the two schedules. This finite action clock selects neither physical seconds nor native dynamics, regional time-slice generation, a count-volume interpretation or a quantum clock. Resource pairing, the action, its quadrature, orientation and relative normalization remain explicit inputs.

B.19 Calibrated clocks and an undetermined light-deflection coefficient

The source action measure and a readable scalar clock do not by themselves select the spatial gravitational response. A constructive comparison uses the grounded massless stiffness $K = M(\mathsf{A} - I) > 0$ on the same prepared golden addresses, with two real fields u, v and an identical source j. The constitutive law, source preparation, update program and physical probe dictionary are declared. They are not consequences of A1–A3.

Proposition B.21 (Two-field clock nonidentifiability)**.** *For every $\kappa > 0$ and $\gamma \in \mathbb{R}$, let*

$$E_{\gamma,\kappa}(u, v; j) = \frac{1}{2} u^{\mathsf{T}} K u + \frac{\kappa}{2}(v - \gamma u)^{\mathsf{T}} K (v - \gamma u) - j^{\mathsf{T}} u. \tag{37}$$

Every positive symmetric two-field edge matrix whose clock Schur complement equals one has this form. The unique stationary fields satisfy $u = K^{-1} j$ and $v = \gamma u$. The entire clock susceptibility, Gaussian clock marginal at any fixed positive inverse temperature, and clock dynamics under the kinetic form $[\dot{u}^{\mathsf{T}} M \dot{u} + \kappa \dot{w}^{\mathsf{T}} M \dot{w}]/2$, where $w = v - \gamma u$, agree throughout this family for matching clock initial data.

Proof. For an edge matrix $S > 0$, set $\kappa = S_{22}$ and $\gamma = -S_{12}/S_{22}$. Its normalized Schur complement gives $S_{11} = 1 + \kappa\gamma^2$. The change to u, w has determinant one and separates two positive quadratic forms. Completing the square gives the common minimum $-j^{\mathsf{T}} K^{-1} j/2$ and a nonnegative excess energy. Stationarity, Gaussian integration and the separated kinetic equations then give the stated identities. □

The same separation works for any common C^2 clock energy $H(u; j)$ with a finite Gibbs partition. Coercivity alone does not imply that partition condition. In particular, the resting-source energy $H = u^{\mathsf{T}} K u/2 + \sum_i j_i(e^{-u_i} - 1)$, $j_i \ge 0$, has positive Hessian, a unique minimum and a Gaussian lower bound ensuring integrability. Its stationary equation $Ku = je^{-u}$ therefore preserves the spatial ambiguity with the same proper-time coupling used for probe clocks.

Use one metric and one universal matter/Maxwell dictionary for every member:

$$ds^2 = -e^{-2u} c^2 dt^2 + e^{2v} d\mathbf{x}^2, \qquad S_{\text{clock}} = -mc^2 \int d\tau, \qquad S_{\text{photon}} = S_{\text{Maxwell}}[g]. \tag{38}$$

An identical local gap ΔE has coordinate phase rate $e^{-u}\Delta E/\hbar$, with $\Delta E = c^2\Delta m$ for a rest-mass gap. Static exchange gives $\nu_r/\nu_e = e^{u_r-u_e}$, independent of γ. The optical index is $n = e^{u+v}$. For a calibrated exterior field $u = \mu/r$, $\mu = GM/c^2$, the infinity-to-infinity first-order deflection is $2(1+\gamma)\mu/b$ for fixed finite γ in the weak-field limit $|u|, |\gamma u| \ll 1$. The member $\gamma = -1$ is conformal and has exactly zero bending, while $\gamma = 1$ has the Einstein weak-field coefficient. The linearized exterior tensor is

$$G_{ij} = (\partial_i\partial_j - \delta_{ij}\Delta)(u - v).$$

For a nonzero compact source, $G_{ij} = 0$ selects $\gamma = 1$ in this class. All members have zero linearized exterior scalar curvature and a divergence-free linearized Einstein tensor. Scalar vacuum and Ward balance therefore do not supply this selector. The exponential $\gamma = 1$ metric is not asserted to solve the full nonlinear vacuum Einstein equation.

Sixteen independently replayed local histories cover both source laws, strengths, schedules and transverse seeds. Actual final writers feed clock and optical-index reads. Exact residuals and a maximum-principle barrier enclose the stationary fields, and the conformal and Einstein optical intervals are disjoint in every history. A compact C^1 density with a C^3 potential gives a controlled continuum sequence. For rays whose actual closest approach lies outside the source and $0 \le p = (1+\gamma)\mu/r_{\min} \le 10^{-3}$, direct differentiation of the exact optical integral gives $|\theta(p) - 2p| \le 11p^2$. Numerical quadrature supplies a separate check, not a certified quadrature enclosure or a finite-observer radio error budget.

The retrospective measurement comparison retains the Pound–Snider static redshift result and all four overlapping Fomalont solar-radio reductions, including their measurement conventions and statistical errors. The bounded redshift systematic is kept separate. These compressed published results are not independent repetitions or a likelihood fit of raw observations. The same-input general-relativistic reference is evaluated without a fitted per-channel normalization. The construction identifies a missing physical tensor condition; it supplies no selected constitutive law, source-to-stress identification, physical calibration or autonomous apparatus history (`code/paired_gravity/README.md` and `docs/research/PAIRED_SOURCE_GRAVITY.md`).

B.20 A common scalar history with readout and restoration

The finite golden scalar action admits an operational composition in which readout changes the field and retained records restore it before subsequent evolution. This connects the local instrument to the same history whose action and separated response are checked. It strengthens a free-field execution by accounting for the later consequences of readout error. The construction uses the declared 64-site operator A, masses M, left-supported velocity v, right-supported detector g, and $\tau^2 = 1/245$. It retains $\|v\|_M \le 3/4$, $\|g\|_M \le 1/4$, $\sum_i M_i \le 1$, and $0 < \tau^2 A < 3I$. The source and detector supports have coordinate separation greater than $3/8$. These are finite supplied coordinates and operations.

At each detector site a zero pointer and the updated field value x undergo the classical averaging probe $(x, 0) \mapsto (x/2, x/2)$. A retained response $r = x/2 + \eta$ is routed to the readout and used to restore the field to $2r$ and reset the pointer. Consequently the actual subsequent action consumes $x + 2\eta$, rather than an independently reloaded reference field. Every probe, retained response, feedback, route, accumulator reset and clock update names its actual consumed writers and values.

Proposition B.22 (Accumulated readout error on one action history)**.** *Let $\tau > 0$, let A be positive and self-adjoint in the M inner product, with $0 \le \tau^2 A \le 4I$, and let $\sum_i M_i \le 1$. For nonnegative error budgets ϵ_p, ϵ_r, suppose the initial velocity differs by at most ϵ_p in M norm, and every retained*

local response error obeys $|\eta_{j,i}| \le \epsilon_r$. *The restored history satisfies*

$$\widetilde{u}_j = (2I - \tau^2 A)\widetilde{u}_{j-1} - \widetilde{u}_{j-2} + 2\eta_j, \qquad \|\widetilde{u}_j - u_j\|_M \le j\tau\epsilon_p + j(j+1)\epsilon_r,$$

where u_j *is the ideal action history, both histories start at zero position, and* η_j *vanishes outside the detector support.*

Proof. Restoration has exact error $2\eta_j$, with norm at most $2\epsilon_r$. In a spectral mode put $c = 1 - \tau^2\lambda/2 \in [-1, 1]$. The response to an additive impulse after n further steps is the Chebyshev polynomial $U_n(c)$. Its representation $U_n(\cos\theta) = \sum_{k=0}^{n} e^{i(n-2k)\theta}$ gives $|U_n(c)| \le n + 1$, including the endpoints by continuity. Spectral calculus and the triangle inequality therefore give $j\tau\epsilon_p + 2\epsilon_r \sum_{k=1}^{j} k$. □

For two independently perturbed baseline and intervention histories, the corresponding detector error at step j is bounded by

$$2\|g\|_M\{j\tau\epsilon_p + j(j+1)\epsilon_r\} + 2\epsilon_d,$$

where ϵ_d bounds each final scalar readout error. No independence or probabilistic cancellation of the errors is assumed. Routing, reset, averaging, restoration arithmetic and accumulation are exact in this contract. Errors at those additional operation locations would require their own propagation budget. Take $\epsilon_p = \epsilon_r = 10^{-6}$, $\epsilon_d = 10^{-7}$ and model-clock uncertainty 10^{-6}. The nominal action-step clock selects the last completed record over $[15\tau, 17\tau]$; it consumes no future observation. The clock uncertainty bounds displacement of the comparison time while holding this nominal record selection fixed. For the continuous-time reference on the same finite action, $U(t) = A^{-1/2}\sin(t\sqrt{A})v$,

$$|D'(U(t))| \le \|g\|_M\|v\|_M \le 3/16.$$

Thus holding and clock uncertainty contribute at most $(3/16)(\tau + 10^{-6})$. The full finite-action discretization error is the independently reconstructed all-mode bound at the retained steps, multiplied by $\|g\|_M$. Preparation, propagated feedback, final readout, discretization, hold reconstruction and timing errors are added before comparison with the response. Exact algebraic arithmetic and exact finite supports contribute zero numerical and localization error on this declared model; this is not a spatial continuum error assertion.

The independently replayed packet resolves a continuous model-time window of length $2\tau > 0.127$. Its conservative total comparison error is below 0.01364, while every admitted paired held readout has magnitude above 0.01576. The strict positive margin also bounds the continuous same-action signal away from zero across that window. The explicit noisy histories are examples inside the uniform bound. A doubled-input history checks linear response. Disabling restoration produces a different field history and breaks the identification of the decoded records with the evolved state. The consumer rejects resealed substitutions of the action, population, clock, readout, record value or consumed writer.

This is a classical software realization with bounded registers at the declared cutoff, protected coefficients, supplied record channels and exact retained algebraic values. All auxiliary events and coefficient precision are counted separately from the model-action clock; their physical durations and costs are not inferred from that clock. Quantum instrument semantics, physical source selection and laboratory calibration are separate obligations. The replay is in `code/common_history_packet/verify.py`; its complete histories and comparison certificate are in `evidence/common_history_packet_20260925/`. The four declarations in `Lean/Screen/CommonHistoryFeedback.lean` verify exact restoration and its finite accumulation factor. The operator stability and spectral error argument above are analytical.

B.21 A bounded quantum probability of the same finite action

The canonical commutator of the preceding execution is state independent. A separate construction propagates the original action's vacuum covariance and a coherent preparation, giving an actual bounded-effect probability. It uses all 64 modes of the same $q = 5$ operator, without transferring the spatial continuum error of another population.

Set $\hbar = 1$, $Q = M^{1/2}u$, $B = M^{1/2}\mathsf{A}M^{-1/2}$ and $[Q_i, P_j] = i\delta_{ij}$. On $L^2(\mathbb{R}^{64})$, let

$$H = \tfrac{1}{2}(P^TP + Q^TBQ), \qquad \Phi(g) = (M^{1/2}g)^TQ, \qquad E_g = \tfrac{1}{2}(I + \sin\Phi(g)).$$

The effect satisfies $0 \le E_g \le I$ by spectral functional calculus. Since $B \ge I$, H has a unique normalized ground state Ω. The baseline is Ω; the intervention is $\exp(i\Phi(v))\Omega$, with initial canonical means $Q = 0$, $P = M^{1/2}v$ and the same initial covariance. Both states are propagated by the exact symmetric gates

$$S = e^{-i\tau Q^TBQ/4}e^{-i\tau P^TP/2}e^{-i\tau Q^TBQ/4}.$$

At operator level the integrator memory is $Q_{-1} = (I - \tau^2B/2)Q_0 - \tau P_0$; the recorded preparation fixes its mean, not a sharp quantum coordinate. The reference uses e^{-itH} on the same preparations. The reference baseline is stationary; the split baseline is generally squeezed. No modified-Hamiltonian vacuum is substituted.

Proposition B.23 (Original-vacuum probability comparison)**.** *Write $r_* = 30049/12005 < 3$, $\tau^2 = 1/245$, $G_0 = \langle g, g\rangle_M$ and $G_1 = \langle g, \mathsf{A}g\rangle_M$. Let V_0 denote the vacuum variance of $\Phi(g)$ and V_j its variance after j split steps, with or without the coherent displacement. Then*

$$0 \le V_j - V_0 \le D := \frac{\tau^2\sqrt{G_0G_1}}{8(1 - r_*/4)}, \qquad 0 \le V_0 \le G_0/2. \tag{39}$$

$$|p_j - p(t_j)| \le \tfrac{1}{2}\sqrt{G_0}\,e_j + \tfrac{1}{4}D, \qquad t_j = j\tau, \tag{40}$$

where p_j and $p(t_j)$ are the intervention probabilities of E_g, and e_j is any valid enclosure of $\|u_j - \mathsf{A}^{-1/2}\sin(t_j\sqrt{\mathsf{A}})v\|_M$. Both baseline probabilities are exactly $1/2$, so the same estimate bounds the difference of intervention-minus-baseline responses.

Proof. For a mode of frequency $\omega = \sqrt{\lambda}$, set $x = \tau^2\lambda/4$, $\theta = 2\arcsin(\tau\omega/2)$ and $\widetilde{\omega} = \omega\sqrt{1 - x}$. The split Heisenberg matrix is

$$\begin{pmatrix} 1 - 2x & \tau \\ -\tau\omega^2(1 - x) & 1 - 2x \end{pmatrix}, \qquad Q_j = \cos(j\theta)Q_0 + \frac{\sin(j\theta)}{\widetilde{\omega}}P_0.$$

The original vacuum has variances $1/(2\omega)$ and $\omega/2$, with zero symmetrized cross covariance. Hence, exactly,

$$\mathrm{Var}(Q_j) - \mathrm{Var}(Q_0) = \frac{\tau^2\omega\sin^2(j\theta)}{8(1 - \tau^2\lambda/4)} \ge 0.$$

Summing with the squared spectral components of $M^{1/2}g$ gives the first bound in (39), using $\langle g, \sqrt{\mathsf{A}}g\rangle_M \le \sqrt{G_0G_1}$. The inequality $\mathsf{A} \ge I$ gives $V_0 \le G_0/2$. For a Gaussian state with field mean m and variance V,

$$\langle E_g\rangle = \tfrac{1}{2}(1 + e^{-V/2}\sin m).$$

The displacement changes only the mean; its split mean is $\langle g, u_j\rangle_M$. Cauchy–Schwarz bounds the mean error by $\sqrt{G_0}e_j$. The sine is 1-Lipschitz and $e^{-V/2}$ is $1/2$-Lipschitz for $V \ge 0$, proving (40). Centered Gaussian states have zero expectation of the odd sine, including the squeezed split baseline. The mode calculations hold on the preserved Schwartz domain; the final effect is bounded on the full Hilbert space. □

Exact finite probability enclosures. The mass and energy moments are exact elements of $\mathbb{Q}(\phi)$. An independent rational verifier consumes the authenticated execution and its full-mode mean-error bounds at all 21 stored positive times. It obtains the uniform outward bound $V_j - V_0 \le 0.000646532908$. No covariance diagonalization is needed: rational bounds on the moments, $e^{-z} \ge 1 - z$, and $\sin y \ge y - y^3/6$ for $0 \le y \le 1$ give signed probability intervals. All times are retained; at 15 of them the certified response exceeds the total comparison error. For example, at $t = 16\sqrt{5}/35$,

$$p_{16} \in [0.490342191369,\ 0.490493790733], \qquad |p_{16} - p(t)| \le 0.000810888870.$$

Thus the split response to the baseline $1/2$ has magnitude at least 0.009506209267, and the continuous finite-action response has the same negative sign with magnitude at least 0.008695320397.

The action, boundary, canonical quantization, original vacuum, coherent preparation, detector normalization and model tick are explicit inputs. These are bounded quantum effect predictions at executed finite-action times. They do not attest observed outcomes, a quantum state encoded by the classical registers, a spatial continuum error, intermediate-time accuracy, an interacting quantum field, native time or laboratory units.

B.22 The same quantum detector with reconstructed action-time uncertainty

The original-vacuum probability comparison extends to the elapsed-time intervals obtained from Proposition B.20 on the same authenticated scalar records. Fix the positive 64-mode action, mass metric, original vacuum, coherent momentum preparation v, and detector g of Proposition B.23. The quantum state and operator are kept fixed throughout this comparison. Uncertain record-based timing is the additional input. The extension covers reference probabilities inside each time interval; it does not define a continuous trajectory for the split evolution.

Proposition B.24 (Probability comparison on a recovered time interval)**.** *Let p_j be the intervention probability of $E_g = (I + \sin\Phi(g))/2$ after j executed symmetric gates, and let $p(t)$ be its continuous finite-action probability on the same preparation. Suppose $|p_j - p(t_j)| \le \epsilon_j$, where $t_j = j\tau$ is the nominal action time. For any closed interval $I_j = [a_j, b_j]$, set*

$$L_g = \tfrac{1}{2}\|g\|_M\|v\|_M, \qquad \delta_j = \max\{|a_j - t_j|, |b_j - t_j|\}.$$

Then, simultaneously for every $t \in I_j$,

$$|p_j - p(t)| \le \epsilon_j + L_g\delta_j. \tag{41}$$

Both baseline probabilities are $1/2$, so this also bounds the difference of the corresponding intervention responses. If a certified signed split response has magnitude at least $s_j > \epsilon_j + L_g\delta_j$, the continuous response has that same sign throughout I_j, with magnitude at least $s_j - \epsilon_j - L_g\delta_j$.

Proof. The original vacuum is stationary under the continuous Hamiltonian. Its detector variance V_0 is constant and the coherent field mean is

$$m(t) = \langle g, \mathsf{A}^{-1/2}\sin(t\sqrt{\mathsf{A}})v\rangle_M, \qquad m'(t) = \langle g, \cos(t\sqrt{\mathsf{A}})v\rangle_M.$$

Self-adjoint spectral calculus gives $\|\cos(t\sqrt{\mathsf{A}})\|_M \le 1$ for every real t. The finite-dimensional mean is differentiable, and $p(t) = (1 + e^{-V_0/2}\sin m(t))/2$ therefore obeys $|p'(t)| \le e^{-V_0/2}\|g\|_M\|v\|_M/2 \le L_g$. The mean-value bound and triangle inequality prove (41). The split probability belongs only to its executed layer; no split evolution at unexecuted times is introduced by this argument. □

Exact comparison on all retained intervals. The two execution schedules have identical reconstructed field layers and elapsed-time enclosures. The conditional clock budget is 10^{-8} per configuration in the M norm, with a common hidden history stationary for the same supplied trapezoidal action and one positive duration per complete configuration interval. Exact source moments give $L_g \le 0.092011181733$. Outward rational arithmetic composes all 21 original probability rows with all 21 positive elapsed-time intervals, retaining every unresolved row. The additional timing error is at most 0.000000449304. The same 15 responses are resolved uniformly over their respective nonzero intervals.

For the sixteenth completed update, the recovered interval is

$$I_{16} = [1.022198489136,\ 1.022206518861].$$

The split probability remains in $[0.490342191369, 0.490493790733]$, and for every $t \in I_{16}$,

$$|p_{16} - p(t)| \le 0.000811258283, \qquad p(t) - \tfrac{1}{2} \le -0.008694950984.$$

These are interval-wide predictions of the same finite quantum action. The source, state, covariance and detector are unchanged.

The record-error budget conditions the clock inverse problem; it is not an error bound for a perturbed quantum preparation, field trajectory, Hamiltonian or detector. Existence of one common hidden stationary history is a hypothesis and does not follow from intersecting duration intervals. The graph reconstruction supplies complete configurations for this finite record, not a regional quantum time-slice theorem. The action normalization, variable-duration quadrature, orientation, vacuum and canonical quantization remain declared. No count-volume identification, physical seconds, quantum clock measurement, observed outcomes or spatial-continuum error is inferred from the timing comparison.

B.23 Repeated local quantum readout with controlled disturbance

The same finite scalar action admits an explicit instrument for its bounded sine detector. This gives a joint classical record algebra and a controlled comparison of sequential measurement probabilities. The construction assumes quantum pointer preparation, controlled field operations and Born readout; these operations are additional inputs. It does not identify the classical field registers with sampled quantum outcomes.

Use the Hamiltonian, original vacuum, coherent intervention and symmetric step S of the finite scalar probability construction. Write $A_g = \Phi(g) = \sum_i m_i g_i q_i$ and $W(g) = e^{iA_g}$, with $\hbar = 1$. A fresh pointer starts in $|+X\rangle$. Apply $U_g = \exp(-iZ \otimes A_g/2)$ and read the standard Pauli Y, labeling its positive eigenvalue by zero. The two outcome operators are

$$K_0 = \frac{e^{-iA_g/2} - ie^{iA_g/2}}{2}, \qquad K_1 = \frac{e^{-iA_g/2} + ie^{iA_g/2}}{2}.$$

In particular $K_0^* K_0 = (I + \sin A_g)/2$ and $\sum_s K_s^* K_s = I$. The sign convention matters: reversing the sign of U_g with the same pointer labels reverses the sine response.

Theorem B.25 (Centered sequential instrument)**.** *Apply this instrument after every step S. For a record $h = (h_1, \dots, h_n)$ let*

$$L_h = K_{h_n} S \cdots K_{h_1} S, \qquad \mathcal{J}_n(\rho) = \sum_h L_h \rho L_h^* \otimes |h\rangle\langle h|.$$

This is a completely positive trace-preserving map into a field algebra and a classical record algebra. Its finite prefixes are consistent when only the classical records are retained. The probability of a record is $\mathrm{Tr}(L_h \rho L_h^*)$*; these records need not be independent.*

Let p_j *be the original unmeasured probability after* j *steps and define* $[A_g(0), A_g(d\tau)] = ic_d$ *for the split evolution. Then the unconditional probability of outcome zero at the* j*th readout is exactly*

$$p_j^{\mathrm{seq}} - \tfrac{1}{2} = D_j(p_j - \tfrac{1}{2}), \qquad D_j = \prod_{d=1}^{j-1} \cos(c_d/2). \tag{*}$$

The nonselective readout preserves all canonical first moments. It increases the expected original Hamiltonian by $G_0/8$ *at each readout, where* $G_0 = \langle g, g \rangle_M$*. Both the split and continuous sequential vacuum baselines remain exactly* $1/2$*.*

Proof. The bounded outcome operators give the Kraus completeness identity above; iterating it proves normalization of every record distribution and its prefix marginals. Summing the two outcomes cancels the interference terms:

$$\mathcal{M}_g(\rho) = \tfrac{1}{2}\left(W(-g/2)\rho W(g/2) + W(g/2)\rho W(-g/2)\right).$$

In the interaction picture, conjugation of the future Weyl observable by either half kick multiplies it by $e^{\pm ic_d/2}$. Thus each earlier nonselective readout contributes the real factor $\cos(c_d/2)$. Taking the imaginary part of the Weyl expectation proves (*), without assuming that the measured state remains Gaussian. A centered vacuum has real Weyl expectation; hence its sine marginal stays zero after these real multipliers. The half kicks shift P by $\pm M^{1/2} g/2$, leave Q unchanged, and average to zero first-moment shift. Expanding $P^T P/2$ gives the energy increment $G_0/8$. These energy identities apply on the finite-energy domain; the instrument and its probability identities are bounded and apply to every density state. □

The detector occupies the 32-site half-volume of the original finite action. It can be compiled using qubits at those sites. Prepare their GHZ state along a 31-edge spanning tree of the *scalar action graph*, then apply the local operations $\exp(-iZ_i \otimes m_i g_i q_i/2)$. Read Y at the root and X at the other sites, retaining every local bit. For a fine record b with parity s, its field outcome operator is $2^{-(32-1)/2} K_s$. There are 2^{31} fine records of each parity, so coarse-graining to parity gives exactly the instrument above. An independent stabilizer calculation verifies the supplied tree against the actual detector support and action edges. The instrument lies in the original equal-time regional field algebra; it needs no exterior collar to measure this position-only detector.

This circuit has 32 zero preparations, one Hadamard, 31 CNOTs, 32 local controlled field kicks, 32 pointer reads and one parity decode per readout: 129 ideal operations, or 2,709 over 21 readouts. The initial coherent field intervention uses 16 further local Weyl factors; original-vacuum preparation is supplied separately. These counts exclude field evolution, hardware time, finite precision and preparation of the field vacuum. The scalar action graph is not identified with a native seam-routing implementation. The construction supplies the controlled field couplings and entangling links, and treats the readout as an instantaneous ideal instrument.

Proposition B.26 (Sequential comparison over recovered time intervals)**.** *Let* $[a_j, b_j]$ *be the ordered recovered time intervals of the same reference mean-field history. Suppose the earlier probability certificate gives* $|p_j - p(t_j)| \le \epsilon_j$ *for every* $t_j \in [a_j, b_j]$*, and* $|p_j - 1/2| \le u_j$*. Let* $d_j \le D_j \le 1$ *be a positive bound and set*

$$L_j = \frac{G_0^2}{8} \sum_{k<j} (b_j - a_k)^2 < 1, \qquad \Gamma_j = \max(1 - d_j, L_j).$$

For every ordered choice $t_k \in [a_k, b_k]$, the same continuous Hamiltonian with the same centered instrument at those times satisfies

$$|p_j^{\mathrm{seq}} - p_{\mathrm{cont}}^{\mathrm{seq}}(t_1, \dots, t_j)| \le \epsilon_j + \Gamma_j u_j.$$

If the unmeasured split response has a fixed sign and magnitude at least s_j, its measured split magnitude is at least $d_j s_j$.

Proof. The continuous cross-time commutator is

$$c(t) = \langle g, \mathsf{A}^{-1/2} \sin(t\sqrt{\mathsf{A}})g\rangle_M, \qquad |c(t)| \le G_0|t|.$$

The same interaction-picture argument gives a continuous attenuation product with factors $\cos(c(t_j - t_k)/2)$. The bound $\cos x \ge 1 - x^2/2$ and $L_j < 1$ show that all these factors are positive and their product lies in $[1 - L_j, 1]$. Therefore the two attenuation factors differ by at most Γ_j. Subtract their products with the unmeasured responses, multiplying the original probability error by the continuous attenuation (at most one), to obtain the stated bound. □

For the complete 64-mode action, the discrete commutators are computed exactly in $\mathbb{Q}(\phi)$ from

$$z_0 = 0, \quad z_1 = \tau g, \quad z_{d+1} = (2I - \tau^2 \mathsf{A})z_d - z_{d-1}, \qquad c_d = \langle g, z_d\rangle_M.$$

Independent verification instead uses first-order kick–drift–kick phase-space evolution and the independently assembled full stiffness. Rational bounds on c_d^2 give $d_j = \prod_{d<j}(1 - \overline{c_d^2}/8)$, with outward rounding retained. All 21 times and all 210 prior-readout pairs enter the comparison. At $j = 16$, the split sequential response magnitude exceeds 0.00950581279, its comparison error is below 0.00083418122, and the continuous sequential response has the same negative sign with magnitude above 0.00867163157. The same 15 time slots remain resolved. Through the final readout, the discrete attenuation loss is below 0.000044641.

The classical clock certificate is reused as a conditional reference-time enclosure. Centering preserves ensemble first moments; it does not make each measurement branch obey the original deterministic configuration history or derive a clock from the binary records. The certificate reports unconditional marginals of an explicit joint instrument, not sampled histories or independent trials. Original-vacuum preparation, the quantum control and Born readout laws, the physical meaning of the clock, and finite-duration hardware implementation remain supplied. No spatial continuum, interacting quantum limit or observed physical outcome follows.

B.24 Finite-duration quantum readout on the same scalar action

The centered instrument admits a finite-duration realization while the field continues to evolve. The control Hamiltonian and its precision remain supplied. The result below bounds the resulting probabilities; it does not identify a classical execution with a quantum outcome history.

Keep the positive scalar operator A, mass inner product, original vacuum, coherent velocity v and detector g used above. In canonical coordinates $Q = M^{1/2}q$, $P = M^{-1/2}p$, write

$$H_0 = \tfrac{1}{2}(P^T P + Q^T \Omega^2 Q), \qquad \Omega^2 = M^{1/2}\mathsf{A}M^{-1/2}.$$

The pointer has zero free Hamiltonian. During a rectangular pulse of duration $\delta > 0$, apply the simultaneous local coupling

$$H_{\mathrm{int}} = \frac{1}{2\delta} \sum_{i \in \operatorname{supp} g} Z_i \otimes m_i g_i q_i.$$

The coupling strength scales as $1/\delta$; pulse duration alone does not bound implementation cost or required control strength.

Proposition B.27 (Exact finite pulse on the prepared pointer code)**.** *On the GHZ code spanned by* $|0\rangle^{\otimes 32}$ *and* $|1\rangle^{\otimes 32}$*, every* Z_i *acts as the same logical* Z*. Define* $S_\delta = \operatorname{sinc}(\delta\sqrt{\mathsf{A}}/2)$ *and* $g_\delta = S_\delta g$*, with* $\operatorname{sinc}(0) = 1$*. The full pulse propagator on that code is*

$$e^{-i\delta H_0/2}\, e^{i\theta_\delta} e^{-iZ\otimes\Phi(g_\delta)/2}\, e^{-i\delta H_0/2},$$

where

$$\theta_\delta = \frac{\langle g, [\delta\mathsf{A}^{-1} - \mathsf{A}^{-3/2}\sin(\delta\sqrt{\mathsf{A}})]g\rangle_M}{8\delta^2}, \qquad 0 \le \theta_\delta \le \frac{\delta\langle g, g\rangle_M}{48}.$$

Pointer preparation and parity readout therefore realize the centered instrument with smear g_δ *at the pulse center, together with the surrounding free field evolution.*

Proof. In each logical pointer sector the field is a finite collection of harmonic oscillators with a constant linear force of opposite sign. Its exact Weyl solution has the averaged interaction-picture linear observable and a central second-order phase. Averaging symmetrically about the pulse center cancels the momentum term and multiplies each position mode by $\operatorname{sinc}(\delta\omega/2)$. The central phase is quadratic in the force, so both logical sectors have the displayed same phase. These identities can be computed on the oscillator Schwartz core and extended through the bounded unitary operators. The inequalities $0 \le x - \sin x \le x^3/6$ for $x \ge 0$ give the phase bound. The earlier parity calculation then applies with g_δ. □

The phase is scalar on the ideal GHZ code only. On the full pointer Hilbert space the second-order terms contain Z_iZ_k; they are not generally scalar. Noise comparisons below use the full pulse channel and do not assume that a noisy pointer stays in the code. Also, the local coupling support and the effective equal-time smear are different: $S_\delta g$ generally extends outside the original detector region. The instantaneous regional-algebra statement for g cannot be transferred to this averaged smear.

For centers $t_1 < \cdots < t_{21}$, the interaction-picture observable is $B_j = \Phi(g_\delta)(t_j)$. The same two Kraus formulas define a joint field/record instrument whose classical record prefixes are consistent. Its unconditional sine response is

$$R_j^\delta = \frac{1}{2}e^{-\nu_\delta/2}\sin\mu_\delta(t_j)\prod_{k<j}\cos\frac{c_\delta(t_j - t_k)}{2},$$

where

$$\begin{aligned}\mu_\delta(t) &= \langle g, S_\delta\mathsf{A}^{-1/2}\sin(t\sqrt{\mathsf{A}})v\rangle_M,\\ \nu_\delta &= \tfrac{1}{2}\langle g, S_\delta^2\mathsf{A}^{-1/2}g\rangle_M,\\ c_\delta(t) &= \langle g, S_\delta^2\mathsf{A}^{-1/2}\sin(t\sqrt{\mathsf{A}})g\rangle_M.\end{aligned}$$

The centered ideal baseline remains $1/2$. The ideal nonselective pulse preserves canonical first moments relative to free evolution and adds $\|g_\delta\|_M^2/8$ to the expected field energy. This is not a bound on pointer, controller or noisy implementation energy.

Proposition B.28 (Duration and channel-error comparison)**.** *Put* $G_0 = \langle g, g\rangle_M$*,* $G_1 = \langle g, \mathsf{A}g\rangle_M$ *and* $V_0 = \langle v, v\rangle_M$*. The finite pulse and the instantaneous instrument on the same continuously evolving field satisfy*

$$\begin{aligned}|\mu_\delta - \mu_0| &\le \delta^2\sqrt{G_1V_0}/24,\\ |\nu_\delta - \nu_0| &\le \delta^2\sqrt{G_0G_1}/24,\\ |c_\delta(d) - c_0(d)| &\le \delta^2 G_1|d|/12,\\ |\cos(c_\delta(d)/2) - \cos(c_0(d)/2)| &\le \delta^2 G_0G_1d^2/48.\end{aligned}$$

Suppose $t_k \in [a_k, b_k]$ are the inherited ordered center intervals and u_j bounds the unmeasured continuous response magnitude there. Set

$$B_\delta = \delta^2 \left(\frac{\sqrt{G_1 V_0}}{48} + \frac{\sqrt{G_0 G_1}}{96} \right), \qquad A_j = \frac{\delta^2 G_0 G_1}{48} \sum_{k<j} (b_j - a_k)^2.$$

Then $|R_j^\delta - R_j^0| \le B_\delta + u_j A_j$.

Assume each initial state has trace distance at most ϵ_0 from its specified preparation and each of the 129 declared fault locations per readout has channel distance at most ϵ in half the diamond norm. The error of one outcome probability through readout j is at most $\epsilon_0 + 129j\epsilon$. For the difference between noisy intervention and noisy baseline probabilities, add $2\epsilon_0 + 258j\epsilon$.

Proof. Use $|\operatorname{sinc} x| \le 1$ and $|1 - \operatorname{sinc} x| \le x^2/6$ in the finite spectral calculus. Cauchy–Schwarz gives the mean and variance bounds; in particular $\langle g, \sqrt{\mathsf{A}}g\rangle_M \le \sqrt{G_0 G_1}$. The inequality $|\sin(d\omega)| \le |d\omega|$ gives the commutator bound. Both commutators have absolute value at most $G_0|d|$, and $|\cos x - \cos y| \le |x - y|(|x| + |y|)/2$ proves the last line. The unmeasured probability is half a damped sine. Its mean contribution is bounded by half the mean error and its variance contribution by one quarter of the variance error, giving B_δ. Telescope the attenuation products, whose factors have magnitude at most one, to obtain A_j. Finally telescope the implemented channels and use trace-distance contractivity, retaining the complete field, pointers and classical records. The two preparations may have different errors; hence both probability errors enter their difference. A noisy baseline is not fixed at $1/2$. □

The channel-error assumptions do not follow from a bound on coefficients of an unbounded field Hamiltonian. The declared model places the 32 local pulse fault channels at its common endpoint. Pointer-only operations commute with the free field flow, and later operations preserve retained records. These are explicit control and record assumptions. Half-diamond closeness controls bounded probabilities, not energy or noisy first moments.

A nonzero duration and noise window. The independently replayed certificate retains all 21 centers and all 210 earlier-readout pairs. It uses a common pulse duration $1/1000 \le \delta \le 1/100$, pointer-only operation duration $\gamma = 1/10000$, and bounds $\epsilon \le 10^{-7}$, $\epsilon_0 \le 10^{-5}$. Each readout has 64 sequential preparation operations, 32 simultaneous local field couplings and 33 read/decode operations. Its occupied window is $\delta + 97\gamma \le 0.0197$, with all 2,709 operation locations retained. Every choice of centers in the inherited intervals leaves a gap greater than 0.04417 between these occupied windows.

Combining the pulse and noise bounds with the earlier same-instrument clock comparison retains all fifteen resolved responses. The smallest retained magnitude exceeds 0.0003659; at the sixteenth readout the noisy intervention-minus-baseline response is negative with magnitude exceeding 0.0082353. Its total comparison error is below 0.0012705. The final parity record becomes available in $[1.349935903357, 1.349945669638]$ at the maximum pulse duration, after its center-time interval. The ideal field-energy increase per pulse is enclosed by $[0.007652475342308, 0.007656300183578]$.

The package `code/source_scalar_finite_instrument/` contains the producer, independent verifier and mutation controls. It checks the complete parent action and instrument, moments, chronology, fault counts and outward error bounds. The observer-like detector has local pointers, coupling ports, retained readback records, fresh-pointer resets and declared control. The certificate is an operator and error calculation, not a sampled run.

Initial vacuum/coherent preparations are supplied at model time zero; the sixteen earlier ideal preparation factors are not given a finite-duration implementation by this result. The inherited

clock is a conditional reference-mean enclosure, not a noisy or branch-conditioned clock read from pointer bits. Quantum preparation, entangling operations, Born readout, physical regions and time units remain supplied or unattached. No native seam implementation, spatial continuum or interacting-field limit follows.

B.25 Finite local preparation of the scalar intervention

The coherent intervention can be prepared by a finite local force while the original field action continues to evolve. This removes the instantaneous displacement from the conditional instrument construction. The original vacuum, quantization and admissibility of the force control remain hypotheses; neither the vacuum nor quantum outcome statistics are selected by this result.

Use the same convention $\hbar = 1$. Let $\mathsf{A} \geq I$ be self-adjoint in the positive mass inner product $\langle \cdot, \cdot \rangle_M$, and let v be the compact velocity profile. Starting with the original vacuum at time $-\delta/2$, apply

$$H(t) = H_0 - \frac{1}{\delta} \sum_{i \in \operatorname{supp} v} m_i v_i q_i, \qquad -\delta/2 \leq t \leq \delta/2.$$

All force ports act in parallel. Their shared duration is δ, and their strength increases as $1/\delta$. Thereafter evolve with H_0 and apply the declared field–pointer instrument.

Proposition B.29 (Preparation with an active free field)**.** *Up to a scalar phase, the pulse prepares at its endpoint the free evolution for time $\delta/2$ of the coherent momentum displacement with velocity*

$$v_\delta = \operatorname{sinc}(\delta\sqrt{\mathsf{A}}/2)v.$$

Write $\rho_v(t)$ for free evolution from the ideal displaced vacuum at the virtual reference time zero, and $\rho_\delta(t)$ for the pulse-prepared state followed by free evolution. For every $t \geq \delta/2$,

$$D(\rho_\delta(t), \rho_v(t)) \leq \frac{\delta^2}{24\sqrt{2}} \|\mathsf{A}v\|_M, \qquad D(\rho, \sigma) = \tfrac{1}{2}\|\rho - \sigma\|_1.$$

If the initial state has trace distance at most ϵ_{vac} from that vacuum, and an additional real acceleration force $e(t)$ in the same controlled ports satisfies $\int \|e(t)\|_M \, dt \leq \epsilon_F$, then

$$D(\widetilde{\rho}_\delta(t), \rho_v(t)) \leq \epsilon_{\mathrm{vac}} + \frac{\epsilon_F}{\sqrt{2}} + \frac{\delta^2}{24\sqrt{2}} \|\mathsf{A}v\|_M.$$

This bounds every subsequent bounded-effect probability, and remains valid after a common sequential instrument including its classical record. Here $e(t)$ enters $\ddot{q} + \mathsf{A}q = v/\delta + e(t)$, equivalently through the additional Hamiltonian term $-\sum_i m_i e_i(t) q_i$.

Proof. Diagonalize the positive oscillator matrix in mass-normalized coordinates. A linear force changes only the coherent displacement, with a central phase. The exact forced mean at the pulse endpoint in each mode is

$$q(\delta/2) = \frac{1 - \cos(\delta\omega)}{\delta\omega^2} v, \qquad \dot{q}(\delta/2) = \frac{\sin(\delta\omega)}{\delta\omega} v.$$

These are the free evolution for time $\delta/2$ of position zero and velocity $\operatorname{sinc}(\delta\omega/2)v$. Two such coherent states with velocity difference w at their common reference time have squared overlap $\exp[-\|\mathsf{A}^{-1/4}w\|_M^2/2]$. Consequently their trace distance is at most $\|\mathsf{A}^{-1/4}w\|_M/\sqrt{2}$. The scalar inequality $|1 - \operatorname{sinc} x| \leq x^2/6$, together with $\mathsf{A} \geq I$, gives the stated ideal error. For the additional force, the displacement amplitude is a unitary mode rotation of $\mathsf{A}^{-1/4}e(t)/\sqrt{2}$ integrated over the pulse. The triangle inequality bounds its norm by $\epsilon_F/\sqrt{2}$. Unitary invariance propagates the initial error, and contractivity under the common channel gives the record-probability statement. □

The virtual reference state is not asserted to be the actual state halfway through the force pulse. The comparison starts at the pulse endpoint. The ideal energy added above the original vacuum is $\|v_\delta\|_M^2/2 \le \|v\|_M^2/2$. A trace-distance preparation bound alone supplies no noisy-energy or apparatus-power bound.

On the same 64-site action, all sixteen nonzero entries of v lie in the original preparation region. Exact quadratic-field arithmetic gives $\|v\|_M^2 \in [0.552881768009, 0.552881768010]$ and $\|\mathsf{A}v\|_M^2 \in [1965.046309667688, 1965.046309667689]$. For $10^{-4} \le \delta \le 2 \times 10^{-3}$ and $\epsilon_{\mathrm{vac}}, \epsilon_F \le 10^{-6}$, the conservative bound obtained by dropping each factor $1/\sqrt{2}$ is less than 9.389×10^{-6}. The baseline, with zero intended force, has error at most 2×10^{-6}. Both fit the separate 10^{-5} preparation budgets of the finite-duration instrument. Preparation ends more than 0.05148 model-time units before its first pointer operation. All 21 correlated readouts and all fifteen resolved response signs therefore retain their stated bounds.

The operation contract includes 16 force ports, 48 coefficient reads and 32 on/off writes, together with the integrated force-error limit. It is a specified quantum-control channel on the supplied action, not an authenticated quantum execution or a derivation of its control law. The independent verifier reconstructs positivity and mass self-adjointness of all 64 action rows, the preparation norm and the inherited timing/error interfaces. Altered pulse centers, omitted resets, incorrect coefficients, missing branch errors and promoted outcome claims are rejected.

B.26 Regional canonical reconstruction from two field layers

The scalar action permits a concrete regional algebra construction. Its access requirement is stronger than a stored classical configuration: one must admit the full real-parameter Weyl families of the specified linear field observables. No quantum outcomes or physical region assignment follow from a classical trace.

Let $M = \mathrm{diag}(m_i) > 0$, let $K = K^T$ be the complete stiffness matrix, and put $A = M^{-1}K$. Take $\hbar > 0$. In the finite Schrödinger representation, $[q_i, p_j] = i\hbar\delta_{ij}$, and the position component of one canonical kick–drift–kick step of duration $\tau \neq 0$ is

$$q^+ = q + \tau M^{-1}p - \frac{\tau^2}{2}Aq.$$

This is the full coupled action, including the entries connecting a region to its complement. For a site set R, write $E = R^c$, $r = |R|$, $B = A_{R,E}$, and let Cq_E be the admitted exterior linear fields. For a real space S of linear canonical observables, write $\mathcal{W}(S) = \{\exp(i\ell/\hbar) : \ell \in S\}''$.

Theorem B.30 (Sharp regional collar criterion)**.** *Let* $\mathcal{A}_R = \mathcal{W}(\mathrm{span}_{\mathbb{R}}\{q_i, p_i : i \in R\})$ *and*

$$\mathcal{D}_R(C) = \mathcal{W}(\mathrm{span}_{\mathbb{R}}\{q_R, q_R^+, Cq_E\}).$$

Then

$$\mathcal{A}_R \subseteq \mathcal{D}_R(C) \quad\Longleftrightarrow\quad \ker C \subseteq \ker B \quad\Longleftrightarrow\quad \mathrm{row}\, B \subseteq \mathrm{row}\, C.$$

Consequently the minimum number of independent exterior linear fields is $\mathrm{rank}\, B$*. Choosing a row basis* C *of* B *attains this minimum. With no exterior access the original regional algebra is recovered exactly if and only if* $B = 0$*.*

Proof. Using the local q_R directions and the invertibility of M_R, the real linear space defining $\mathcal{D}_R(C)$ is equivalently generated by

$$q_R, \qquad p_R - \frac{\tau}{2}M_R B q_E, \qquad Cq_E.$$

Every p_R direction is therefore present exactly when every row of B is in the row space of C. The finite-dimensional kernel criterion and the sharp rank bound follow by annihilators. Sufficiency also gives an explicit reconstruction: if $B = DC$, then

$$p_R = M_R \left[\frac{q_R^+ - q_R}{\tau} + \frac{\tau}{2}(A_{R,R} q_R + DC q_E) \right].$$

For necessity, if $Ce = 0$ but $Be \neq 0$, the perturbations $\delta q_R = 0$, $\delta q_E = e$ and $\delta p_R = (\tau/2) M_R B e$ leave all admitted linear data unchanged but change a local momentum.

The passage from linear spans to von Neumann algebras does not assume that observable products are linear. If a linear canonical direction v lies outside a real subspace S, nondegeneracy of the ambient symplectic form gives $z \in S^\sigma$ with $\sigma(z, v) \neq 0$. Every Weyl operator in direction z commutes with $\mathcal{W}(S)$, whereas a suitable real multiple does not commute with the Weyl family in direction v. That family cannot lie in $\mathcal{W}(S)$. Conversely the Weyl relations produce every direction in the real span. This proves the algebraic equivalence, including degenerate S. □

The two field layers themselves are noncommutative:

$$[q_i, q_j^+] = i\hbar \frac{\tau}{m_i} \delta_{ij}, \qquad [q_i^+, q_j^+] = 0.$$

Their $2r$-dimensional regional space is nondegenerate, even when it fails to contain the original local momenta. In fact

$$\frac{m_i}{\tau} q_i^+ = p_i + (Gq)_i, \qquad G = M/\tau - \tau K/2 = G^T.$$

Multiplication by $U = \exp[-i q^T G q/(2\hbar)]$ implements $UpU^* = p + Gq$. Hence $\mathcal{D}_R(0) = U \mathcal{A}_R U^*$: the two-layer family is an isotonic finite noncommutative net, and its disjoint site regions commute. The same unitary works for every region. This sheared net is not identified with the original site net when exterior couplings are present. For a minimal collar, the enlarged real space instead has dimension $2r + \operatorname{rank} B$ and a central exterior subspace of dimension $\operatorname{rank} B$.

For the same 64-site golden action, exact arithmetic gives the following minimal access counts. The coordinate column counts every exterior site with a nonzero incident coupling; a linear combination can require reads from several such sites.

region	r	exterior coordinate fields	$\operatorname{rank} B$
interior singleton	1	6	1
central eight-site cube	8	24	8
half-volume	32	16	16
interior four-site line	4	16	4
checkerboard	32	32	32
whole volume	64	0	0

The complete certificate also retains the empty region and corner singleton. It verifies both exact rank factorizations, an identity minor, the full cross-time canonical brackets, and 62 explicit perturbations showing that deletion of any selected collar row loses a local momentum. These are mathematical access counts, not implemented routing costs.

The kernel criterion and explicit velocity reconstruction have Lean proofs; the Weyl and von Neumann interpretation above is analytic. Native availability of the continuous Weyl families, preparation and outcome maps, physical clock and region calibration, and compatibility with a continuum regional net remain additional requirements.

C Local clocks, record gluing and information capacity

The proposed record-gluing law RG (Section B.8) connects response directions to classical record transport. It has distinct information, geometry and implementation premises. A simpler sufficient transport principle replaces exact dilation covariance and continuous rotation symmetry by a regular local clock and time-faithful refinement. This is a proposed physical extension, not a fourth adopted core axiom or a property certified for the simulator. Its target is the operational causal/count limit; it does not uniquely select a finite Fibonacci population or a microscopic read radius.

The local-clock interface. Let E be the three-dimensional simple response ideal with its positive trace metric. Its attachment to operational positions is supplied. Assume:

1. Independent classical bits, clean auxiliary registers, retained versions, preparation, discard, waiting and NOT/Toffoli gates admit arbitrary finite composition and parallel realization. All auxiliary work is counted. Local processing has zero ideal duration or arbitrarily small positive overhead for each complete finite program.

2. Every sufficiently small displacement v has a retained-record primitive of duration $\ell(v)$, independent of origin and covariant under carrier rotations, with $\ell(0) = 0$ and

$$\ell(v)^2 = Q(v) + o(\|v\|^2), \qquad Q > 0.$$

   The expansion is uniform in direction. Local directional coverage is a premise; finitely many rotated directions do not imply it.

3. Every native dependency, including control, timing and metadata, admits refinement into small primitives and waits with the same endpoints and nonnegative elapsed time converging to the original time. Local operations and waits have zero displacement. A local primitive costs at least $\ell(v)$. Finite programs have exact value-preserving realizations with arbitrarily small whole-program overhead. Local interventions at every designated intermediate output before publication are preserved while all other existing records and caches are fixed.

These are LC1–LC3. They constrain all admitted influences, including those outside a chosen observation program.

Finite symmetry and the limiting cone. The quadratic form inherits the latency symmetry by dividing $\ell(sRv)^2 = \ell(sv)^2$ by s^2 and taking $s \to 0$. In an orthonormal icosahedral frame, invariance of a symmetric quadratic form under $\mathrm{diag}(1, -1, -1)$ and $\mathrm{diag}(-1, 1, -1)$ makes its off-diagonal entries zero. The cyclic coordinate permutation equates its diagonal entries. These proper rotations preserve the twelve ports and generate a tetrahedral subgroup. Thus $Q(v) = \tau^2\|v\|^2$ for $\tau > 0$, and uniformly $\ell(v) = \tau\|v\| + o(\|v\|)$.

For any native path of elapsed time T and displacement v, refine until every step satisfies the local lower bound. For any $0 < \epsilon < \tau$,

$$T_n \geq \sum_j \ell(v_j) \geq (\tau - \epsilon) \sum_j \|v_j\| \geq (\tau - \epsilon)\|v\|.$$

Time-faithful refinement and then $\epsilon \to 0$ give $T \geq \tau\|v\|$. Conversely, N equal steps have total time $N\ell(v/N) \to \tau\|v\|$. Finite realization with a sufficiently small overhead meets every strict deadline $T > \tau\|v\|$; waiting fills the remaining time. The resulting access cone has speed $c = 1/\tau$. Exact null access is not asserted.

The example $\ell(v) = \tau r + \kappa r^2$, $r = \|v\|$, $\kappa > 0$, has subdivision time $\tau r + \kappa r^2/N$. Subdivision followed by waiting preserves an old primitive's duration, yet no nonzero finite path attains the null boundary. It violates exact dilation covariance while satisfying the local-clock principle. This separates LC from exact RG geometry; an approximate implementation of RG can have the same limiting behavior. The icosahedrally invariant polyhedral norm $\max_{u \in V} |u \cdot v|$ shows why finite symmetry without quadratic regularity is insufficient. A fast coarse shortcut preserves local Taylor data but violates time-faithful refinement. Both premises bear physical content.

For q^3 distinguished records in a window of side L, spacing L/q, and layers $\delta_q \to 0$ with $q\delta_q \to \infty$, finite independent programming realizes inner read menus $a_q = (1 - \eta_q) c\delta_q$, $\eta_q \to 0$, with sufficient deadline slack. The source-net covering theorem gives inner and outer radii $k(a_q - 2h_q)$ and ka_q, where $h_q = \sqrt{3}L/q$. Localized intermediate interventions place consumed-writer paths in complete native ancestry; the clock theorem excludes native spacelike shortcuts. The orders converge away from the null boundary. Cell assignment supplies count-volume weight $(L/q)^3 \delta_q$ for the distinguished records. The M1 choice $\delta_q = L/(c\sqrt{q})$ is one regulator; auxiliary work has a separate bill.

Classical compilation and finite algebra coherence. Every Boolean map has a clean reversible implementation: toggle an output for each enabled minterm, compute its multi-input conjunction using a Toffoli ladder, and reverse the ladder. NOT gates implement negative controls. CNOT is a derived Toffoli gate with a constant control; its clean control register, preparation and reset are counted. The input is retained, the desired value is XORed into the output, and all auxiliary bits return to zero. This construction is exponential in general. It proves finite programmability, not an efficiency or energy bound.

On a finite graph take private factors $\mathbb{C}_i^{12} \otimes Q_i$ and shared edge buffers $B_e = M_{12}$. The carrier algebra consisting of its private factors and incident buffers has center exactly $\mathbb{C}_i^{12}$. The channel with Kraus operators

$$K_{p,b} = e_p \otimes |p\rangle\langle b|, \qquad \sum_{p,b} K_{p,b}^\dagger K_{p,b} = I,$$

retains central port p and copies it into a quantum buffer's fixed basis. Reading that basis and preparing a target port completes a classical copy; unknown quantum states are not cloned. Off-diagonal buffer effects remain usable measurements. Private response actions commute with these copying maps. Buffer access, controller transport and readback receive their actual local-clock grades.

An explicitly included joint observer and its marginal constraints provide a nonempty finite state family. Choose the local references as restrictions of the same faithful joint tracial state. These compatible references and an injective positive-weight entropy cover give the joint tracial state as the unique unconstrained minimum. Independent preparations are interventions or constraints, not that unconstrained minimum's selected records. Adding blank factors and lifting old channels gives value-level refinement. This is a finite consistency witness. It does not furnish the all-level source-bound support, degree-one bridge, operational deletion tests or proof that the private response exhausts all admitted reversible source responses. A dormant support added to a circuit would not meet those obligations. Compatibility matters: on a shared classical bit with two equal-weight references $(1/2, 1/2)$ and $(3/4, 1/4)$, both faithful and tracial, the summed entropy derivative at probability $1/2$ is $-\log 3$, so the joint reference does not minimize that incompatible-reference problem.

There is also a finite process-to-state connection. For g gates on R bits, n of them independent input bits, let $D = R(g + 1)$ and admit all 2^D ambient histories. Initial blanks and actual gate

equations define the feasible family by linear indicator constraints. Each input x has exactly one feasible history $h(x)$. A faithful input prior $p(x)$ with fair references on all other history coordinates gives $Q(h(x)) = p(x)2^{-(D-n)}$. Hence every feasible P satisfies

$$D(P\|Q) = D(P_{\text{input}}\|p) + (D-n)\log 2.$$

With the full joint history observer as the injective scored cover, the unique minimum is $P_*(h(x)) = p(x)$, exactly the executed circuit ensemble. Feasibility uses actual transitions, not success at the target truth table: a defective program retains its defective histories with positive weight. This is an explicitly classical finite score problem. Additional scored marginals, the program and its physical clock are not selected by this identity. The native all-level source obligations remain separate.

Independent motivation and its boundary. Reversible classical computation motivates cleaning auxiliary work after copying an answer. Its relation to a transport clock is an extra premise. For a tracial matrix reference $\rho_0 = I/d$ and injective traceless Hermitian linear perturbations $X(v)$, direct expansion gives

$$D(I/d + X(v)\|I/d) = \frac{d}{2}\operatorname{Tr} X(v)^2 + O(\|v\|^3).$$

An equivariant perturbation makes this positive susceptibility isotropic by the same finite-symmetry argument. Identifying $\ell(v)^2$ with a positive multiple of this entropy cost to second order would supply a regular clock, but A3 supplies neither that calibration nor its coefficient. Equal-radius latency ratios and coarse/fine timing comparisons test this proposal without an M1 fit. Entropy curvature alone is not evidence for a physical clock law.

Information and resource consequences. If a receiver decodes m independently intervenable d-ary records from a finite transcript alphabet $\mathcal{T}$, with all caches fixed, the transcript map is injective. Therefore $|\mathcal{T}| \geq d^m$. Payload, timing, control and metadata all count. Additive accumulation cannot decode arbitrary tuples; preexisting correlations do not evade fresh independent interventions.

For $q = n^2$ and $n \geq 2$, put $m = \lfloor n/2 \rfloor$. In the M1 grid, all offsets in $[-m, m]^3$ lie strictly within the radius $L/\sqrt{q}$. At least $(n^2 - 2m)^3$ interior receivers each consume $(2m+1)^3$ records. Thus there are at least $n^9/8 = q^{9/2}/8$ read incidences per layer and $n^{10}/8 = q^5/8$ in a horizon L/c. Each interior receiver needs at least $(2m+1)^3 - 1 \geq n^3/2$ incoming binary transcript bits per layer under fresh independent interventions, hence bandwidth at least $cq^2/(2L)$. This excludes its own locally available record from the communication cut. Multicast can share transport paths; the aggregate statement counts output incidences rather than unicast packets. For inner menus, the same cube is included when $\eta_q < 1 - \sqrt{3}/2$, giving the asymptotic bounds; the all-n constants above concern the full-radius experiment. The rate bound presumes independent new interventions with prior caches fixed. It does not infer a fresh entropy rate from an unperturbed correlated history.

Cofinal workspace is consequently substantive. A fixed per-receiver bandwidth cannot support literal dense lossless observation at all refinements under these conditions. A single central $\mathbb{C}^{12}$ record has only twelve alternatives; an unbounded lossless tuple therefore requires a growing composite observer/register bank. The logical receiver cannot be identified with one unchanged primitive carrier. The same transport law also permits identity, parity and other local updates, so it selects no matter equation, energy law or Lorentz covariance of dynamics. Its consequences beyond M1 are communication, refinement timing and information-capacity constraints.

The complete analytic argument and controls are in `code/rg_principle/DERIVATION.md`. The compact receipt independently verifies the full named catalog and replays 263 finite circuits, including 7,880 localized interventions and 40 paired intermediate-writer comparisons, with every reference gate and work register counted. Coordinated circuit/table substitutions and cached-writer substitutions fail verification. Exact algebra controls cover all 1,728 basis units of the copying channel. Twelve kernel-checked results and a transitive standard-axiom audit are in `Lean/Geometry/RecordGluingPrincipleAxiomAudit.lean`. Finite controls do not replace the analytic all-size arguments or certify a native source realization.

D M1 necessity: sparse reads and observable dependence

The complete metric-neighbor menu is sufficient for the preceding causal/count limit. Explicit sparse menus give the same named limits and exhibit a downstream observable that changes. No new transport axiom or assumption of successful convergence enters the construction. The finite graphs and actions below are mathematical regulators; source selection, physical clocks and a native operation grammar remain separate requirements.

Three explicit families. For $n \geq 1$ put $R = 2^n$, $q = R^2$, $h = L/q$ and $\delta = hR/c = L/(cR)$. Place q^3 sites at hz for $z \in \{0, \ldots, q-1\}^3$. Event (j, z) has time $j\delta$ and weight δh^3. A symmetric integer stencil V containing zero specifies all preceding-layer reads, clipped at the box boundary. Let B contain every integer vector of Euclidean norm at most R. Construct S from the vectors

$$v(w)_i = \operatorname{sgn}(w_i) \left\lfloor \frac{R|w_i|}{\sqrt{\sum_j w_j^2}} \right\rfloor, \qquad w \in \mathbb{Z}^3, \quad \|w\|_\infty = n,$$

all $\pm 2^b e_i$ with $0 \leq b \leq n$, and zero. Remove duplicates. Finally let $A = S \cup \{\pm j e_i : 1 \leq j \leq R\}$. Then

$$S \subseteq A \subseteq B, \qquad |S| \leq D_n := 24n^2 + 6n + 9, \qquad |A| \leq D_n + 6R.$$

All three menus have coordinate-permutation and sign-change symmetry, and all use the same events, times and weights.

Constructive propagation bound. For a nonzero displacement $d \in \mathbb{Z}^3$, write $r = \|d\|_2$. Round $nd/\|d\|_\infty$ to its nearest integer vector w, with half ties away from zero. Normalizing and rounding towards zero gives

$$\|v(w) - Rd/r\|_1 \leq 3R/n + 3.$$

Take $k = \lfloor r/R \rfloor$ copies of $v(w)$. The residual $e = d - kv(w)$ satisfies $\|e\|_1 \leq 3r/n + 3r/R + 2R$. Correct each coordinate by repeated signed R-steps and the binary digits of its remainder. This takes at most $\|e\|_1/R + 3n$ additional steps and constructs an exact route with

$$\frac{r}{R} \leq N \leq \left(1 + \frac{3}{n} + \frac{3}{R}\right) \frac{r}{R} + 3n + 2. \tag{42}$$

The route remains within $3r/n + 3r/R + 2R$ of the straight segment: its first part follows the chosen flight, and the coordinate corrections approach the endpoint monotonically. Thus it fits in the box whenever this tube does. Zero steps fill any extra integer duration. In physical units, the excess route time is at most

$$\left(\frac{3}{n} + \frac{3}{R}\right) \frac{\|hd\|_2}{c} + \frac{(3n+2)L}{cR},$$

and the tube thickness tends uniformly to zero on bounded displacements.

Preserved spacetime and count conclusions. Every fixed interior strictly timelike pair is eventually connected by the constructed route. Every spacelike pair is eventually excluded because no step travels farther than R. The same holds for A and B by inclusion. Half-open space-time cells have weight δh^3 and vanishing diameter. Their indicators give continuity-set counts by quadrature. For generated intervals with converging interior tips, the path bound gives indicator convergence off the limiting null boundaries; dominated convergence gives the actual interval volume. Applying

the argument to product cells gives the strict-pair limit, since pair-null and diagonal sets have zero measure. The flat-diamond integrals therefore give volume $\pi c^3 T^4/24$, ordering fraction $1/10$ and the fourth-root count-clock ratio. A bounded continuous positive profile $\sigma(t)$ similarly gives the weighted limit $\sigma(t)^4\,dt\,d^3x$ and the same finite FLRW mass identities. The profile and its physical interpretation are inputs, not results of the comparison.

Over a horizon L/c, S requires at most $q^3 R D_n$ logical read incidences, of order $q^{7/2}\log^2 q$, while B has order q^5. These are incidences, not bit traffic, energy or native operation counts. Exact finite M1 outputs are not preserved: a deleted adjacent-layer edge has no replacement path within one layer, and an omitted independent input changes a lossless tuple read. The proved replacement concerns the named limiting observables.

Crossing-read entropy is not fixed by those limits. Use the coordinate cut $x = (q-1)/2$ in index coordinates. A vector with $v_x > 0$ contributes exactly $v_x(q-|v_y|)(q-|v_z|)$ unordered crossing pairs. Consequently

$$X_S \le D_n R q^2, \qquad X_B \ge R^4 q^2/32, \qquad \frac{X_S}{X_B} \le \frac{32 D_n}{R^3} \longrightarrow 0.$$

For the lower bound retain $1 \le v_x \le R/2$ and $|v_y|, |v_z| \le R/2$, all strictly inside the ball. Thus the existing one-nat-per-crossing-read dictionary changes by an unbounded factor despite the common causal/count limit.

The A family supplies a stronger counterexample. Its full axial subset has coordinate-cut count X_0 and count Y_0 across $x + y = q - 1/2$ given by

$$X_0 = \frac{q^2 R(R+1)}{2}, \qquad Y_0 = q^2 R(R+1) - \frac{qR(R+1)(2R+1)}{3}.$$

Indeed a positive step j along the x or y axis has $qj(q-j)$ allowed diagonal-crossing pairs, including the finite box correction. The remaining S edges add at most $D_n R q^2$ and $2D_n R q^2$ respectively. Since $D_n/R \to 0$, the full ratio $Y_A/(2X_A)$ tends to one. The section areas are L^2 and $\sqrt{2}(1 - 1/(2q))L^2$, hence

$$\frac{Y_A/[\sqrt{2}(1-1/(2q))L^2]}{X_A/L^2} \longrightarrow \sqrt{2}. \tag{43}$$

The complete-neighbor cap integral instead gives limit one. A common scalar entropy conversion cancels in this ratio, so a single Planck-area calibration cannot remove the orientation dependence. Cubic symmetry and a round limiting causal cone therefore do not select an isotropic crossing-read entropy law.

A positive sparse scalar action. For $V = S, A, B$ omit zero and put $M_2 = \sum_{v\in V} \|v\|^2$ and $\alpha = 6/(h^2 M_2)$. Exact sign and permutation symmetries imply $\sum v_i v_j = (M_2/3)\delta_{ij}$. Define

$$(A_V f)(x) = \alpha \sum_{v\in V} \big(f(x) - f(x+hv)\big),$$

with zero extension at the boundary. The operator is symmetric positive semidefinite, with the corresponding squared-difference energy including the fixed-zero boundary terms. Taylor expansion cancels odd terms and yields

$$|A_V f + \Delta f| \le \frac{a^2 M_4}{4}, \qquad a = hR,$$

where M_4 bounds unit-direction fourth derivatives and the reference is smoothly zero near the boundary. This exact moment normalization removes the $h/a^2 \to 0$ requirement of the earlier generic quadrature estimate for this explicit action, at the unchanged square-root radius scaling.

For a smooth continuum Klein–Gordon reference supported away from the boundary through time T, with M_4 uniform in time, the exact finite evolution $F_{tt} + (c^2 A_V + m^2 I)F = 0$ with sampled initial data satisfies

$$\|F(t) - f(t)\|_h \le \frac{c^2 t^2 a^2 L^{3/2} M_4}{8}, \qquad \|u\|_h^2 = h^3 \sum_x |u_x|^2.$$

This follows from Duhamel and $\|\sin(t\sqrt{B})/\sqrt{B}\| \le t$ for $B \ge 0$, including zero modes. Bounded-norm smeared detectors inherit the bound. This is a free-field limit of specified actions, not an exact finite causal-support statement or a native quantum/matter implementation.

A concrete time-discretization obstruction. For nonnegative symmetric weights of range at most a, whose displacements are nonzero modulo the periodic box, isotropic normalization gives $\sum w_v |hv|^2 = 6$, so $\sum w_v \ge 6/a^2$. On a periodic box the largest Laplacian eigenvalue is at least its mean diagonal, hence at least $6/a^2$. Centered leapfrog requires $c^2 dt^2 \lambda_{\max} \le 4$. At the M1 tick $dt = a/c$ the left side is at least six, so that direct scheme is unstable, for the dense family as well. The exact finite evolution above remains defined; alternative integrators and substeps need their own physical operation and clock accounting. The obstruction concerns this specified scheme, not all dynamics or M1 itself.

The radius exponent is not fixed by these limits. For any fixed integer $k \ge 2$, keep the same stencils and $R = 2^n$ but set $q = R^k$, $h = L/q$ and $\delta = hR/c$. The route bound's additive time cost is $(3n+2)LR/(cq) \to 0$, and the boundary tube thickness is $(3/n+3/R)\,\text{distance} + 2LR/q \to 0$. The same cell and strict-margin proofs give all the stated causal/count limits. The action error is $O((LR/q)^2) \to 0$; the cut anisotropy remains $\sqrt{2}$ because $R/q \to 0$. Thus the explicitly constructed radii $a = Lq^{-(k-1)/k}$, including $L/\sqrt{q}$ and $Lq^{-2/3}$, work for these named limits. This does not select a native radius or preserve finite counts between different schedules; it proves that the named limits do not uniquely require the square-root radius, Fibonacci population, or complete-neighbor law.

A sparse family preserving the crossing observable. The entropy separation has a constructive complement. For $t \ge 1$ set $q = 2^{8t}$, $m = 2^{4t}$, $K = 2^{3t}$ and $R = mK$. Let T contain every mu with $u \in \mathbb{Z}^3$, $\|u\| \le K$, together with the signed axis steps $2^b e_i$ for $0 \le b < 4t$. Use the same grid, tick hR/c and equal event cells. This has $\Theta(q^{9/8})$ parents, compared with $\Theta(q^{3/2})$ for the original dense radius $R_0 = \sqrt{q}$.

Round d/m to z, connect zero to z by rounded linear interpolation with $\lceil \|z\|/(K-2) \rceil$ coarse steps, and correct the residual by binary axis steps. Every step is legal and charged; the route satisfies

$$\frac{\|d\|}{mK} \le N \le \frac{\|d\|/m + 2}{K-2} + 1 + 12t$$

and stays in a tube of radius $2m$. Its physical excess time and tube width vanish. The same generated-interval, pair, count-clock and scalar proofs therefore apply, with radius $a = Lq^{-1/8}$ and scalar error $O(q^{-1/4})$.

Put $C_K = \sum_{\|u\| \le K,\, u_x > 0} u_x$. Exact finite enumeration gives

$$X_T = \sum_{\|u\| \le K,\, u_x > 0} m u_x (q - m|u_y|)(q - m|u_z|) + q^2(m-1).$$

Unions of unit cubes about the integer ball give $\pi(K-1)^4/4-(2K+1)^2/8 \le C_K \le \pi(K+1)^4/4$. Since $mK^4=q^2$, it follows that $X_T/q^4 \to \pi/4$, exactly the dense leading coefficient. The receiver and displacement meshes are h and mh; their cap-count error is $O(K^{-1}+mK/q)$ for each fixed positive-area regular plane section cutting the cube interior into two positive-volume parts; supporting boundary faces are excluded. Hence, for every such section Σ,

$$X_T(\Sigma)=\frac{\pi q^4}{4L^2}|\Sigma|\,(1+o(1)).$$

The added dyadic steps are lower order. Thus the original leading planar crossing entropy and Planck-area dictionary are preserved with one nat per unordered pair, without directional weights or scalar recalibration. Over horizon L/c the logical read count is $\Theta(q^{17/4})$ rather than $\Theta(q^5)$. The larger radius converges more slowly: this comparison fixes q, not accuracy, native cost or crossing traffic per unit model time.

The radius change has a proved role. For any symmetric translation-invariant sub-stencil $V \subseteq B_{R_0}$ at the original radius, let $E=B_{R_0}\setminus V$. The sum of the three coordinate-cut deficits is at least $(q-R_0)^2\sum_{v\in E}\|v\|_1/2$. Retaining their leading counts implies $\sum_E \|v\|_1=o(R_0^4)$. For every $\eta>0$,

$$\frac{|E|}{R_0^3} \le (2\eta+1/R_0)^3+\frac{\sum_E\|v\|_1}{\eta R_0^4},$$

so $|E|=o(R_0^3)$ and $|V|/|B_{R_0}|\to 1$. Substantial thinning at the original radius cannot retain these counts within this stencil class. The explicitly constructed T family avoids that obstruction.

Completing the field comparison beyond smooth probes. The uniform-weight T action has a concrete defect. On the periodic box, all m^3 Fourier indices $(q/m)\ell$ make its coarse displacement phases equal one. Only the $24t$ dyadic offsets act on them, giving eigenvalues at most $48t\alpha$. A sub-box of the coarse ball gives $M_2 \ge m^2K^5/64$, so this upper bound is $(18432/L^2)t2^{-7t}\to 0$. For positive mass these modes approach the mass frequency, making the fixed-temperature thermal energy diverge. Smooth-mode agreement alone therefore misses a real failure.

The completed replacement action fixes it using existing edges:

$$\widehat{A}_V=\tfrac12(A_V+A_{\mathrm{nn}}),\qquad (A_{\mathrm{nn}}f)(x)=h^{-2}\sum_{i=1}^{3}[2f(x)-f(x+he_i)-f(x-he_i)].$$

Every V edge has positive weight and nearest neighbors receive the additional $1/(2h^2)$. This specified one-half combination preserves exact second moments, positivity and the spatial error bound, since its remainder is at most $(a^2+h^2)M_4/8 \le a^2M_4/4$. It changes no read, crossing entropy, radius or degree. This is the scalar action used for the completed T replacement.

For every principal periodic wavevector, elementary sine bounds give

$$\frac{2}{\pi^2}|p|^2 \le \widehat{\lambda}_V(p) \le |p|^2.$$

The nonzero alias modes have eigenvalue at least $8/(h^2m^2)\to\infty$. Modewise convergence and this uniform whole-spectrum bound give convergence of the full positive-mass thermal partition function, normal-ordered energy and entropy, by summable exponential domination. No fitted vacuum-energy subtraction is used.

The result also holds on the clipped, zero-exterior box of the read graph. The form dominates half the nearest-neighbor energy, giving discrete-gradient compactness in L^2 and limits in H_0^1. For a smooth compact test f, positivity gives

$$\langle u_n,A_Vu_n\rangle_h \ge 2\langle u_n,A_Vf_n\rangle_h-\langle f_n,A_Vf_n\rangle_h.$$

Consistency and the supremum over such tests give the full continuum Dirichlet energy as the lower limit. Smooth recovery supplies the upper limit. Min-max with compactness proves convergence of each ordered eigenvalue with multiplicity. Moreover the nearest-neighbor sine spectrum gives

$$\widehat{\lambda}_{n,j} \geq j^{2/3}/(2L^2).$$

The resulting uniform thermal tails are summable, proving the full Dirichlet partition, energy and entropy limits. The clipped-box result allows mass $\mu \geq 0$, since its frequency gap is at least $\sqrt{\mu^2 + c^2/(2L^2)}$; the periodic result requires $\mu > 0$.

Canonical quantization also gives convergence of joint Weyl characteristic functions for every fixed finite collection of smooth compact position and momentum smearings, with sampled coherent means and bounded observation times. The position covariance is bounded by the inverse gap; the smooth-test bound on $\|\widehat{A}_V f_n\|_h$ controls the unbounded momentum covariance tail. The field commutator has coefficient

$$i\hbar \left\langle g_n, \frac{\sin((s-t)\sqrt{B_n})}{\sqrt{B_n}} f_n \right\rangle_h, \qquad B_n = c^2 \widehat{A}_V + \mu^2 I.$$

It converges to the Dirichlet Klein–Gordon commutator and vanishes in the limit before the supports can meet at speed c. Rounding each observation time to a declared layer tick changes this coefficient by at most $\hbar\delta_n \|g_n\|_h \|f_n\|_h$, which tends to zero. Thus the free quantum response and read/count cones agree in the limit. Exact finite support or native execution of the evolution operator is not asserted. Thermal Gibbs entropy and the crossing statistic remain distinct observables; preserving both does not derive their physical identification or a horizon.

The full constructions, boundary proofs and consumer inventory are in `code/m1_necessity/DERIVATION.md` and `code/m1_necessity/BALANCED_REPLACEMENT.md` and `code/m1_necessity/SPECTRAL_COMPLETION.md`, with the inventory in `code/m1_necessity/CONSUMERS.md`. Independent integer replay checks actual graph states, intermediate interventions, routes and crossing counts; seventeen kernel-checked reductions have a transitive standard-axiom audit. The universal stencil and continuum arguments are analytic. Historical golden-family, routing and resource theorems retain their own inputs, and this comparison establishes no physical source-selection premise.

E Variational crossing area and a uniform read action

Agreement on fixed smooth sections does not ensure that a crossing statistic selects macroscopic interfaces. A sparse read family can retain the leading planar area coefficient while admitting microscopic partitions whose entropy lies below the continuum isoperimetric value. An explicit addition of shorter reads restores compactness, the full perimeter limit and the limiting minimum shape. The same scale threshold removes an extra sector of the uniform read-action spectrum. These are statements about declared mathematical read graphs, coordinate volumes and observables; no physical horizon or source identification is an antecedent or a conclusion.

Finite definition. Set $L = 1$ initially. Use q^3 equal cells of side $h = 1/q$, either on the unit torus or in the unit cube. In the latter case all occupied cells must lie in a fixed interior cube $C = [\eta, 1-\eta]^3$, $0 < \eta < 1/2$; reads are clipped to the actual site population. For sufficiently small read radius, the cut count then equals its zero-extended count without any outside record. A binary site partition u has unordered crossing count and normalized value

$$X_V(u) = \frac{1}{2}\sum_x \sum_{v\in V} |u(x+v) - u(x)|, \qquad \mathcal{E}_V(u) = \frac{4X_V(u)}{\pi q^4}.$$

Every unordered pair has one nat, and its two layer directions are counted only once here. The physical area conversion is $L^2\mathcal{E}_V$.

For $n \geq 2$ put

$$q = 2^{8n}, \quad m = 2^{4n}, \quad K = 2^{3n}, \quad R = mK, \qquad \epsilon = m/q, \quad a = R/q.$$

The stencil T consists of every mv, $v \in \mathbb{Z}^3$, $\|v\| \leq K$, and all signed dyadic axis steps below m, with zero included once. Its previous-layer tick is a/c. The identities $q = m^2$, $mK^4 = q^2$ and $a^4 = q\epsilon^3$ fix its coarse entropy as

$$\mathcal{E}_{\mathrm{coarse}}(u_h) = \frac{2}{\pi q} \sum_{\|v\|\leq K} \|u_h(\cdot + \epsilon v) - u_h\|_{L^1},$$

where u_h is the piecewise-constant cell indicator. The ball flux $\int_{\|z\|\leq a} |z \cdot \nu|\, dz = \pi a^4/2$ gives coefficient one for every fixed finite-perimeter interface.

A microscopic partition with the wrong entropy. On the torus occupy precisely the sites with $x_1 \bmod m < m/2$. The volume is exactly one half. Coarse reads preserve the residue label. A dyadic x-step $s \leq m/2$ changes a fraction $2s/m$ of labels, so

$$X_T = \frac{2q^3(m-1)}{m}, \qquad \mathcal{E}_T = \frac{8(m-1)}{\pi m q} \longrightarrow 0.$$

These indicators converge weakly to $1/2$ and have no strongly L^1 convergent subsequence. Thus bounded crossing entropy does not imply a macroscopic region, and the finite half-volume minima tend to zero.

There is also a clipped-box violation of the expected minimum. Let a ball $F \Subset C$ have volume $2v$, and retain only the active residue half in F. Coarse displacements preserve that residue. Averaging it over its ϵ-period changes each crossing slab by at most its boundary area times $O(\epsilon)$; the normalized error is $O(\epsilon/a)$. The cell error is $O(h/a)$ and the dyadic contribution is $O(\log q/q)$. Consequently

$$|u_h| \to v, \qquad \mathcal{E}_T(u_h) \to \tfrac{1}{2}\operatorname{Per}(F) = 2^{-1/3}(36\pi)^{1/3}v^{2/3}.$$

This is strictly below the Euclidean isoperimetric value at volume v. The count error is $O(mq^2)$ cells. Correcting it to round(vq^3) changes the normalized cut by at most $O(mD_T/q^2) = O(K^{-1}) + o(1)$, so the exact finite volume constraint retains the counterexample.

Connecting the occupied slabs by a one-cell-wide axial path changes only $O(q)$ cells and adds entropy $O(D_T/q^3) \to 0$. This connected sequence has weak limit $\frac{1}{2}\mathbf{1}_F$ and no strong subsequence. Connectivity alone therefore does not restore compactness. This connected control uses volume convergence; the exact-volume minimum above imposes no separate connectivity constraint. Fixed-section area agreement is consistent with all these failures, because it did not quantify over microscopically varying cuts.

An explicit sparse repair. Set $U = T \cup B_r$, where $B_r = \{w \in \mathbb{Z}^3 : \|w\| \le r\}$ and $r \le m/2$. All read and entropy weights remain equal. The relevant window is

$$\frac{r^4}{mq} \longrightarrow \infty, \qquad \frac{r^4}{q^2} \longrightarrow 0.$$

The first condition will be proved sufficient for compactness; the second makes the added smooth-interface contribution vanish. The final construction chooses, for every $t \ge 1$,

$$q = 2^{16t}, \quad m = 2^{8t}, \quad K = 2^{6t}, \quad R = 2^{14t}, \quad r = 2^{7t}.$$

Both mq/r^4 and r^4/q^2 equal 2^{-4t}. The radius remains $a = q^{-1/8}$ and the degree is exactly

$$D_U = |B_K \cap \mathbb{Z}^3| + |B_r \cap \mathbb{Z}^3| - 1 + 6(t-1) = \Theta(q^{21/16}).$$

The last term counts the remaining dyadic lengths between r and m. Over a horizon $1/c$, logical read incidences are $\Theta(q^{71/16})$, compared with $\Theta(q^5)$ for the original dense radius-$\sqrt{q}$ family. This fixes population, with slower radius convergence; it does not establish equal-accuracy computational or native resource savings.

The threshold is necessary within this short-ball repair class. For the periodic residue partition the added ball contributes exactly

$$X_{B_r} = \frac{2q^3}{m} C_r, \qquad C_r = \sum_{\|w\| \le r,\, w_1 > 0} w_1 \sim \frac{\pi r^4}{4}.$$

Its normalized entropy is asymptotic to $2r^4/(mq)$. If this ratio has a bounded subsequence, the residue partitions have bounded repaired entropy and no strong subsequence. The same conclusion follows after localization inside F.

Compactness from the reads. Write $d(w) = \|u_h(\cdot + hw) - u_h\|_1$. For $\|s\| \le r/2$, average $d(s) \le d(w) + d(s+w)$ over integer $\|w\| \le r/2$. This gives

$$d(s) \le Cr^{-3} \sum_{\|w\| \le r} d(w) \le Cqr^{-3}\mathcal{E}_{\text{fine}}.$$

Split translations of length $O(m)$ into $O(m/r)$ such steps. Fractional cell translations are convex combinations of adjacent integer differences. If P_ϵ is averaging over an ϵ-cube, then

$$\|u_h - P_\epsilon u_h\|_1 \le C\frac{mq}{r^4}\mathcal{E}_{\text{fine}}.$$

Choose a fixed smooth nonnegative bump supported in the half unit ball, sample it on v/K, and normalize the weights w_v. They obey $w_v \le CK^{-3}$ and $|w_v - w_{v-e_i}| \le CK^{-4}$. Put $Qu = \sum_v w_v u(\cdot + \epsilon v)$ and $f_h = P_\epsilon Q u_h$. Move the derivative of the box average onto the weight differences and subtract the unshifted u_h, since those differences sum to zero. The identity $mK^4 = q^2$ yields

$$\|\nabla f_h\|_1 \le C\mathcal{E}_{\text{coarse}}, \qquad \|u_h - f_h\|_1 \le C\left(a + \frac{mq}{r^4}\right)\mathcal{E}_U.$$

Each component energy is bounded by $\mathcal{E}_U$ separately; overlaps are not charged twice. BV compactness therefore gives strongly convergent subsequences for every bounded-entropy family, with binary finite-perimeter limits. The explicit parameter choice proves this conclusion without assuming compactness or filtering for successful interfaces.

Perimeter, exact volume and minimum shapes. For $u_h \to \mathbf{1}_E$ strongly, assign each $y \in B_1$ to $v = \text{round}(Ky)$. The distributional difference quotient and lower semicontinuity give

$$\liminf a^{-1}\|u_h(\cdot + \epsilon v) - u_h\|_1 \ge |D_y \mathbf{1}_E|.$$

No convergence rate relative to a is required: the difference quotient acts on a fixed smooth test function after changing variables. Fatou and the polar decomposition of the BV derivative imply

$$\liminf \mathcal{E}_U(u_h) \ge \frac{2}{\pi}\int_{B_1} |D_y \mathbf{1}_E|\, dy = \text{Per}(E).$$

For recovery, apply the directional BV translation formula to any set of finite perimeter. It gives the coarse energy limit. Displacement-cell errors are $O(K^{-1})\,\text{Per}(E)$; the short ball adds at most $C(r^4/q^2)\,\text{Per}(E)$. Majority occupancy of each h-cell has L^1 error $O(h)$ by the cube BV Poincare inequality, including the fixed interior boundary restriction. Its energy error is $O(D_U/q^2) \to 0$. This gives binary recovery and the full perimeter Γ-limit in strong L^1.

For prescribed $0 < v < |C|$, the recovery needs only $O(q^2)$ label changes to have exactly $\text{round}(vq^3)$ occupied sites. A changed label affects at most D_U crossing pairs, so the additional normalized cost is again $O(D_U/q^2) \to 0$. Every finite problem has a minimum. Compactness, the lower bound and recovery prove convergence of those minimum values and of subsequences of all near-minimizers with $o(1)$ normalized excess. If a ball of volume v fits strictly inside C, the Euclidean isoperimetric equality case gives

$$\min \mathcal{E}_U \longrightarrow (36\pi)^{1/3} v^{2/3},$$

and every limiting minimizer is a ball, up to translation and null sets. Restoring units gives $(36\pi)^{1/3}V^{2/3}$ for physical volume V. This identifies the limiting shape of the declared cut problem, not a physical event horizon.

A uniform action and its hidden modes. The repair also permits equal positive action coefficients on every nonzero read. Set $M_2 = \sum_{v \in U} \|v\|^2$ and

$$A_U f = \alpha \sum_{v \in U \setminus \{0\}} [f(x) - f(x + hv)], \qquad \alpha = \frac{6}{h^2 M_2}.$$

Cubic symmetry gives exact continuum second moments and spatial error at most $a^2 B_4/4$, where B_4 bounds fourth directional derivatives of the test function. No specially weighted nearest-neighbor

term is added. The uniform coefficient here normalizes an action; the crossing entropy weights remain one. On the clipped cube the zero-exterior convention includes a boundary potential: if b_x counts missing nonzero stencil offsets, then

$$\langle f, A_U f\rangle_h = \alpha h^3 \left[\sum_{\{x,y\}\,\text{internal}} (f_x - f_y)^2 + \sum_x b_x f_x^2 \right].$$

Its diagonal is $\alpha(|U|-1)$ at every site. The boundary term uses prescribed zero exterior values and requires no outside record. Omitting it gives a clipped graph Laplacian with a constant zero mode, rather than the Dirichlet operator analyzed here. The finite evidence checks both the constant and an affine field against an independent edge-energy formula.

For an integer ball of radius $\rho \geq 8$ and principal phase θ, an elementary Dirichlet-sum estimate gives

$$\sum_{\|w\|\leq\rho} [1 - \cos(\theta \cdot w)] \geq \frac{\rho^3}{49152} \min(\rho^2\|\theta\|^2, 1).$$

To see uniformity away from zero, put an integer cube of side $N \geq \rho$ inside the ball. Its one-dimensional sum $d_N(x)$ satisfies $1 - |d_N(x)|/N \geq \min(N^2x^2, 1)/16384$: for $|x| \leq 4/N$, expand $N^2 - |d_N|^2 = 2\sum_{d=1}^{N-1}(N-d)(1-\cos dx)$; beyond that interval use $|d_N|/N \leq \pi/(N|x|)$. Summing the other two coordinates proves the displayed bound. Thus high-frequency control is not inferred from selected numerical probes.

Write $s = r/K$ and decompose each Fourier index uniquely as $k = m\ell + j$, $-m/2 \leq j_i < m/2$. The coarse phase depends only on j. The moment bound $M_2 \leq 56m^2K^5$ and the two ball-symbol bounds imply, for $s \geq 1$ and a fixed $C_0 > 0$,

$$\lambda_U(k) \geq C_0 \min(\|j\|^2 + s^5\|\ell\|^2, a^{-2}).$$

The same conclusion holds when s stays above any fixed positive constant, with an adjusted bound. Since $K^4 = m^3$, the entropy compactness parameter is exactly $r^4/(mq) = s^4$.

If $s \to \gamma \in (0,\infty)$, second-moment expansion for every fixed pair $(j,\ell) \in \mathbb{Z}^3 \times \mathbb{Z}^3$ gives

$$\lambda_U(m\ell + j) \longrightarrow 4\pi^2(\|j\|^2 + \gamma^5\|\ell\|^2).$$

The coarse integer phases cancel, the short-ball remainder vanishes because $r/m \to 0$, and the dyadic contribution is $O(n\alpha) \to 0$. This is the oscillator spectrum of a product of two three-tori with side lengths 1 and $\gamma^{-5/2}$. It describes additional spectral labels of the regulator, not extra physical spacetime dimensions. Smooth fixed spatial probes see only $\ell = 0$.

For positive mass, the complete thermal limit follows as well. Below the saturation scale the summands are bounded by a summable multiple of $\exp[-c_0(\|j\| + \|\ell\|)]$; above it their total is at most $Cq^3\exp(-c_0/a) \to 0$. Thus the thermal partition function, normal-ordered energy and Gibbs entropy converge to the product-torus values at critical scale. If $s \to 0$, arbitrarily many fixed aliases approach the mass frequency and thermal energy diverges. If $s \to \infty$, all $\ell \neq 0$ contributions vanish under the same summable bound, leaving the ordinary three-dimensional thermal limit. This last condition is precisely the one that restores cut compactness.

For the explicit repair, $s^5 = 2^{5t} \geq a^{-2} = 2^{4t}$, giving, for principal physical wave vectors $p = 2\pi k$, the stronger whole-spectrum estimate

$$C_1 \min(\|p\|^2, a^{-2}) \leq \lambda_U(p) \leq \|p\|^2.$$

On the clipped cube, zero extension to a doubled periodic cube and this bound give L^2 compactness and an H_0^1 limit. Positive-form duality, consistency and smooth recovery give the full Dirichlet form limit; min-max gives ordered eigenvalue convergence with multiplicities. The zero extension has at most one eighth of its norm in the constant mode, so there is a uniform Dirichlet gap, and compression yields $\lambda_{n,j} \geq C_2 \min(j^{2/3}, a^{-2})$. Summable low-frequency tails and the vanishing high-frequency total prove the full Dirichlet thermal limits, including zero mass. Smooth-test action norms control the momentum-covariance tail, giving the compact Gaussian Weyl and Klein–Gordon commutator limits too.

All T paths remain present. Charged coarse interpolation and binary corrections give the local round causal limit. Use diamonds compactly inside the clipped spatial cube and observation time slab; on the torus, use non-wrapping diamonds, for example with time height T satisfying $cT < 1/2$. The shrinking route tubes then stay inside the relevant patch. Their generated-diamond volumes tend to $\pi c^3 \tau^4/24$, where τ is the endpoint proper-time separation. The oriented strict-pair measure is $V^2/20$. For P causally ordered pairs among N interval events, $P/N^2 \to 1/20$ and the ordering fraction $2P/[N(N-1)] \to 1/10$. Fourth-root interval-count ratios give proper-time ratios. Truncated and wrapping intervals do not obey these unbounded flat-diamond formulas. The free response cone agrees in the limit, including observations rounded to the declared tick. Exact ODE evolution is not a native layer update: the positive normalized trace forbids stable centered leapfrog at a/c. Thermal Gibbs entropy and the crossing statistic remain distinct observables.

The complete proofs and explicit finite evidence are in `code/m1_interfaces/DERIVATION.md` and `code/m1_interfaces/receipt.json`. The independent checker reconstructs all labels and unordered edges for 96 cut cases on 12 small graphs, covering 3,340,024 logical read incidences. Larger moment certificates use exact column sums; their full populations are not claimed executed. Eight finite graph and scale reductions have transitive standard-axiom audits. The universal BV, isoperimetric and spectral arguments are analytic. The continuum nonlocal-perimeter framework is standard; see A. C. Ponce, *A new approach to Sobolev spaces and connections to* Γ*-convergence*, Calculus of Variations and Partial Differential Equations 19 (2004), 229–255. The atomic residue obstruction, its sharp sparse repair and the associated critical spectral sector require the separate analysis given here.

F Global vacuum fidelity of local scalar read programs

The finite original-vacuum comparison in Section B.21 distinguishes a stationary reference from a squeezed split-step baseline. On a growing population this distinction has a global consequence: stable evolution and agreement on smooth probes do not suffice to control the total energy produced above the original vacuum. This section proves sharp full-spectrum thresholds and an explicit repair. Every comparison retains the original Hamiltonian and its original vacuum.

F.1 Exact mode error and a constructive higher-order step

For a real symmetric positive matrix B on N sites, use the finite Schrödinger representation with

$$H = \tfrac{1}{2}(P^T P + Q^T B Q), \qquad [Q_i, P_j] = i\hbar\delta_{ij}.$$

Let Ω be its normalized ground state and let $\omega_k = \sqrt{\lambda_k(B)}$. A potential kick of signed duration s sends P to $P - sBQ$; an onsite kinetic drift sends Q to $Q + sP$. The kick–drift–kick step of

duration τ has, in the mode coordinates $X = \sqrt{\omega}Q$, $Y = P/\sqrt{\omega}$, matrix

$$S_2(z) = \begin{pmatrix} 1 - z^2/2 & z \\ -z + z^3/4 & 1 - z^2/2 \end{pmatrix}, \qquad z = \tau\omega.$$

Put $\rho = \sqrt[3]{2}$, $w = (2-\rho)^{-1}$ and $v = -\rho(2-\rho)^{-1}$. Use the standard symmetric fourth-order composition

$$S_4(z) = S_2(wz)S_2(vz)S_2(wz).$$

This composition is due to Yoshida, *Physics Letters A* 150 (1990), 262–268, DOI 10.1016/0375-9601(90)90092-3. Its middle inverse step is retained; inverse Hamiltonian gates are operations, not negative transport time.

Both matrices have the form $S = \left(\begin{smallmatrix} a & b \\ c & a \end{smallmatrix}\right)$ with $a^2 - bc = 1$. Exact multiplication gives

$$\begin{aligned} a_4 &= 1 - z^2/2 + z^4/24 + (1/48 + 5\rho/288 + \rho^2/72)z^6, \\ b_4 &= z - z^3/6 - (1/36 + \rho/36 + \rho^2/48)z^5, \\ c_4 &= -z + z^3/6 + (\rho + \rho^2)z^5/144 - (25/1728 + 5\rho/432 + \rho^2/108)z^7. \end{aligned}$$

For $0 < z \le 1/10$, these formulas imply $b > 0$, $c < 0$, $0 < a < 1$ and $0.9z^2 \le -bc \le 1.1z^2$. Indeed $1259/1000 < \rho < 126/100$ bounds every coefficient rationally, and the remaining powers obey $z^2 \le 1/100$. Moreover $b_2 + c_2 = z^3/4$ and

$$b_4 + c_4 = d_4 z^5 - Dz^7, \quad d_4 = -1/36 - \rho/48 - \rho^2/72 \neq 0, \quad D = 25/1728 + 5\rho/432 + \rho^2/108 > 0,$$

with $z^5/16 \le |b_4 + c_4| \le z^5/8$ on that domain.

Proposition F.1 (Exact original-vacuum excitation)**.** *Set* $\cos\theta = a$, $\sin\theta = \sqrt{-bc} > 0$ *and* $\chi = \sqrt{-c/b}$. *After* j *steps the original particle number in one mode and the full-population quantities are*

$$n_j(z) = \frac{(b+c)^2}{-4bc}\sin^2(j\theta), \qquad N_j = \sum_k n_j(\tau\omega_k), \tag{44}$$

$$\Delta E_j = \hbar\sum_k \omega_k n_j(\tau\omega_k), \qquad F_j = |\langle\Omega, U^j\Omega\rangle|^2 = \prod_k (1 + n_j(\tau\omega_k))^{-1/2}. \tag{45}$$

In particular the second-order step produces exactly $z^6/64$ *particles in one mode after one step.*

Proof. The exact power is

$$S^j = \begin{pmatrix} \cos(j\theta) & \sin(j\theta)/\chi \\ -\chi\sin(j\theta) & \cos(j\theta) \end{pmatrix}.$$

The initial covariance is $\hbar I/2$. Applying this matrix gives $n_j = (\chi - \chi^{-1})^2\sin^2(j\theta)/4$, which is (44). At $j = 1$, $n_1 = (b+c)^2/4$. The overlap of the normalized squeezed Gaussian with the original vacuum is $(1 + n_j)^{-1/2}$ in squared modulus. Orthogonal diagonalization gives N independent real modes, so their energies add and their overlaps multiply. Fourier multiplicities are counted in full. The quadratic identities hold on Schwartz space; the metaplectic unitaries and bounded effects act on the full finite-population Hilbert space. □

The invariant squeezed vacuum is a different reference. It carries original-mode excess energy $\hbar\omega(\chi + \chi^{-1} - 2)/4$. Substituting it does not remove an error relative to H and Ω.

F.2 Sharp full-population thresholds

Theorem F.2 (Global vacuum and energy fidelity)**.** *Let B range over positive population matrices satisfying*

$$\sigma^2 = N^{-1}\operatorname{Tr} B \longrightarrow \infty, \qquad \lambda_{\max}(B) \le 2\sigma^2.$$

For $p = 2$ or 4, take the respective step above, $\tau \to 0$ and $\tau\sqrt{\lambda_{\max}} \le 1/10$. For any fixed $T > 0$, average over every $j = 1, \dots, \lfloor T/\tau \rfloor$. Then

$$\overline{N_j} = \Theta(N\tau^{2p}\sigma^{2p}), \qquad \overline{-\log F_j} = \Theta(N\tau^{2p}\sigma^{2p}),$$
$$\overline{\Delta E_j} = \Theta(\hbar N\tau^{2p}\sigma^{2p+1}).$$

The corresponding upper bounds hold uniformly for every integer j. Consequently uniform vacuum fidelity on a fixed nonzero time interval is equivalent to $N\tau^{2p}\sigma^{2p} \to 0$, and uniform total energy fidelity is equivalent to $N\tau^{2p}\sigma^{2p+1} \to 0$.

Proof. At least $N/3$ eigenvalues exceed $\sigma^2/2$; otherwise the mean is less than σ^2 even if all the remaining eigenvalues are $2\sigma^2$. Thus for every fixed $s > 0$,

$$\frac{N}{3}(\sigma/\sqrt{2})^s \le \sum_k \omega_k^s \le N(\sqrt{2}\sigma)^s.$$

The explicit nonzero defects above give $c_p z^{2p} \le (b+c)^2/(-4bc) \le C_p z^{2p}$. The finite geometric sum gives

$$\left| \frac{1}{J}\sum_{j=1}^{J} \sin^2(j\theta) - \frac{1}{2} \right| \le \frac{1}{2J\sin\theta}.$$

For the bulk modes the right side is $O((T\sigma)^{-1})$. Using these modes for the lower bound and all modes for the upper bound proves the number and energy statements. The inequalities $x/(1+x) \le \log(1+x) \le x$ and the uniform bound on each n_j give the fidelity statement. Uniform convergence implies convergence of the time average, proving necessity. Sufficiency uses the all-time upper bound. □

If either scale diverges, the corresponding error is large on a positive fraction of the times, since its mean and maximum have matching orders. The theorem permits isolated revivals; it does not select favorable times. When $\tau\sigma \to 0$, the sharper energy average is

$$\overline{\Delta E_j} = \frac{\hbar d_p^2}{8}\tau^{2p}\sum_k \omega_k^{2p+1}(1 + o(1)), \qquad d_2 = 1/4.$$

The phase-average remainder has relative size $O((T\sigma)^{-1})$ and the coefficient remainder has size $O((\tau\sigma)^2)$. In particular state-norm convergence does not imply convergence of the unbounded energy: at a constant nonzero energy scale the number scale tends to zero, leaving rarer excitations of increasing frequency.

For the vacuum, every bounded effect has probability error at most $\sqrt{1 - F_j}$. More generally, take coherent inputs of original excess energy at most E_0, supported on $\omega \le \Omega_0$ with a fixed gap $\omega \ge \mu > 0$. In dimensionless mode coordinates their displacement norm is at most $\sqrt{2E_0/(\hbar\mu)}$. On the stated stability domain, $|\chi - 1|, |\chi^{-1} - 1| \le z^p$ and $|\theta - z| \le z^{p+1}$. For the last bound, $|a - \cos z| \le z^4/24$ for S_2 and $|a - \cos z| < 0.102z^6$ for S_4, whereas both angle sines exceed $0.94z$.

Thus comparison to the exact oscillator rotation gives, when $\tau\Omega_0 \le 1/10$ and for every bounded effect,

$$|\Delta \Pr| \le \sqrt{1-F_j} + \tau^p \Omega_0^p (1+T\Omega_0)\sqrt{E_0/(\hbar\mu)}.$$

This follows from Weyl covariance, the coherent overlap and the triangle inequality for trace distance. It is a global observable comparison for these preparations, not an unrestricted input-channel norm claim.

F.3 The sparse ultraviolet scale and its repair

On the unit periodic cube, take $N = q^3$ sites, $c = \mu = \hbar = 1$ and a symmetric nonzero integer stencil V without wrapped duplicate edges. Define

$$d = |V|, \quad M_2 = \sum_{v\in V} |v|^2, \quad Af(x) = \frac{6q^2}{M_2} \sum_{v\in V} (f(x) - f(x+v)), \quad B = I + A.$$

Writing $W = 6q^2 d/M_2$, one has $\sigma^2 = 1 + W$ and $\lambda_{\max}(B) \le 1 + 2W \le 2\sigma^2$. The theorem therefore applies to these actual graphs without a spectral success filter.

For integers $t \ge 1$, set

$$q = 2^{16t}, \quad m = 2^{8t}, \quad K = 2^{6t}, \quad r = 2^{7t}, \quad R = mK.$$

Let U be the union of m times the closed integer K-ball, the closed integer r-ball and the signed axial lengths 2^j, $0 \le j < 8t$, with zero removed. The two balls meet only at zero. Ball Riemann sums give

$$d \sim \frac{4\pi}{3} r^3 = \Theta(q^{21/16}), \qquad M_2 \sim \frac{4\pi}{5} m^2 K^5 = \Theta(q^{23/8}), \qquad W \sim 10 q^{7/16}.$$

The short ball dominates degree; the spaced ball dominates second moment. Thus $\sigma = \Theta(q^{7/32})$, whereas the longest physical read is $a = R/q = q^{-1/8}$. Its inverse is not the bulk frequency scale.

Corollary F.3 (An explicit globally faithful sparse program)**.** *Use S_4 on U with $\tau = q^{-5/8}/64$. At every level it is in the proved stability domain, and uniformly in the step index,*

$$\Delta E_j = O(q^{-1/32}), \qquad 1 - F_j = O(q^{-1/4}).$$

The bounded-effect comparison above follows for the stated coherent inputs. The same tick with S_2 has time-averaged excess energy $\Theta(q^{51/32})$ on each fixed nonzero time interval.

Proof. Insert $N = q^3$ and $\sigma = \Theta(q^{7/32})$ in the theorem. For all-level stability, integer cube bounds give $d \le 60r^3$ and $M_2 \ge m^2K^5/256$. The latter follows by retaining the box $K/4 \le v_1 \le K/2$, $|v_2|, |v_3| \le K/4$ in the integer K-ball. Thus $W \le 92160 q^{7/16}$ and $\tau\sqrt{1+2W} \le \sqrt{184321}\, q^{-13/32}/64 < 1/10$ for $q \ge 2^{16}$. □

F.4 Read cost, clock scope and an exact-preservation obstruction

Merging adjacent potential kicks, J second-order steps use $J+1$ kicks and J drifts; J fourth-order steps use $3J+1$ kicks and $3J$ drifts. Every reference kick reads each site's d current neighbors. The ordered nonlocal counts are exactly $(J+1)Nd$ and $(3J+1)Nd$. The quantum potential unitary factors into $Nd/2$ commuting two-mode phases and N onsite mass phases per kick; a drift

is N onsite kinetic unitaries. The classical writer ledger records reference calculations, not cloned quantum fields.

At fixed horizon and small energy tolerance ϵ, the matching bounds give the within-method cost

$$\text{reads} = \Theta\Big(TdN^{1+1/(2p)}\sigma^{1+1/(2p)}\epsilon^{-1/(2p)}\Big).$$

The resulting powers of q are 683/128 and 1263/256 for sparse S_2 and S_4. For the dense radius-$q^{-1/2}$ ball they are 47/8 and 87/16, respectively, at the same q and energy tolerance. The explicit repair uses $\Theta(q^{79/16})$ incidences and its stated vanishing energy tolerance. Equal q is not equal total continuum accuracy. These are declared arithmetic and gate counts, not native bit, energy or execution-time lower bounds.

Each kick expands canonical support by at most one stencil edge; each drift is onsite. The support bound after J macros is $(J+1)a$ or $(3J+1)a$. Here a/τ diverges. The simulation duration is therefore not identified with a physical transport clock, and inverse gates do not make it negative.

Proposition F.4 (Attained signal front and simulation-clock obstruction)**.** *Before periodic wrapping, the canonical support radii $(j+1)a$ for S_2 and $(3j+1)a$ for S_4 are attained by nonzero signals. A faithful implementation with complete influence speed at most c must assign at least these distances divided by c as elapsed time. The simulation tick of the explicit quantum repair cannot be this native clock at any fixed finite c.*

Proof. In each stated stencil, Re_1 is the unique displacement with first coordinate R. Its coefficient in B is $-\alpha$, where $\alpha > 0$. For S_4, require $(3j+1)R < q/2$. The coefficient from initial Q_y to final P_x, $x - y = (3j+1)Re_1$, is exactly

$$-D(2A)^{j-1}\tau^{6j+1}(-\alpha)^{3j+1} \neq 0,$$

where $A = 1/48 + 5\rho/288 + \rho^2/72 > 0$ and $D > 0$ is given above. Cayley–Hamilton gives the lower-left entry of S^j as $U_{j-1}(a)$ times that of S, and U_{j-1} has leading coefficient 2^{j-1}. The highest terms in unscaled (Q, P) coordinates are $A\tau^6B^3$ on the diagonal and $-D\tau^7B^4$ below it. Only the path using Re_1 at every factor of B^{3j+1} reaches that extreme site. Lower powers cannot reach it, and the non-wrapping condition excludes alternative periodic images. For S_2 the same argument gives $(-1)^{j-1}\tau^{2j+1}(-\alpha)^{j+1}/4 \neq 0$ at $(j+1)Re_1$.

A local Q displacement changes the final P_x mean by this nonzero coefficient. A bounded sine effect on a finite Gaussian baseline has a nonzero response for suitable finite displacement and probe strength. These are operational influences in the declared oscillator family, so a complete speed bound requires the stated elapsed times. On the repair, choose $j = \lfloor 1/(8a) \rfloor$. Its support distance tends to 3/8 without wrapping, while $j\tau = O(q^{-1/2}) \to 0$. No fixed finite c can supply this timing identification. □

The LC reduction bounds every admitted native influence, including small ones. The proposition excludes this clock attachment; it does not assert that the quantum approximation fails. Other operation grammars and implementations are outside this obstruction. Local ultraviolet interventions are not the fixed-band preparations of the global comparison. Six full 32-site ray controls verify the extreme coefficients and zeros outside support for both methods at $j = 1, 2, 3$.

There is also an exact obstruction. For a scalar polynomial symplectic update $S(B)$ preserving the original vacuum, its mode matrix preserves $\mathrm{diag}(\lambda, 1)$. Comparing the energy and symplectic inverses gives $d = a$, $c = -\lambda b$ and

$$a(\lambda)^2 + \lambda b(\lambda)^2 = 1.$$

If there are more distinct eigenvalues than this polynomial's degree, it is an identity. The two nonconstant terms have opposite degree parity, so their leading coefficients cannot cancel: $b = 0$ and $a = \pm 1$. A consistent near-identity step is the identity. The stated families have unboundedly many distinct eigenvalues: along $(k, 0, 0)$, for $0 \le k \le \lfloor q/(4R) \rfloor$, each nonzero axial projection contributes a strictly increasing $1 - \cos(2\pi k v_1/q)$, with angle at most $\pi/2$. Thus no fixed number of scalar kick/drift stages nontrivially preserves the original vacuum exactly on all refinements. This does not exclude nonpolynomial flows, growing-degree interpolation or other operation grammars.

Evidence and scope. The independent packet replays four complete periodic spectra, 512 population observations, 24 full spatial covariance comparisons, 702 forward and 702 inverse scalar writes, and 2,268 nonlocal reference reads. Exact cubic arithmetic checks the signed composition and its stability domain. The $q = 65536$ moment certificate uses integer columns; it does not execute that full population. Ten Lean reductions have transitive standard-axiom audits. The universal spectral and asymptotic arguments are analytic. This supplies a global quantum approximation and its explicit cost, not an all-level A1–A3 source, finite twelve-port quantum implementation, interacting matter law or physical clock. Existing physical premises are unchanged.

G Native time and coherent quantum transport

The local-clock proposal gives classical records a complete influence cone. A coherent extension must also charge the physical time of intermediate motion. We characterize a finite register-flight and onsite-scattering class, construct its optimal isotropic Weyl limit, retain every low-energy species, and identify the stronger coherent structure needed for a shared matter and record light speed. This is a theorem about a declared operation class, not an admission certificate for a complete A1–A3 source.

The native flight class. Let a flight of duration t have orthogonal internal projectors P_s summing to the identity and displacements b_s with $\|b_s\| \le ct$. Its Fourier action is $F(k) = \sum_s e^{-ik\cdot b_s} P_s$. The modes move along their straight segments during the flight. Onsite unitaries and waits may be inserted; all sequential stage durations add. Internal dimension is arbitrary and finite, and every intermediate event is retained. Flights move coherent modes without copying unknown states. Their exterior-power lifts act on fermionic Fock space. Arbitrary admitted record-flight directions attain the round cone of speed c; a fixed direction menu has its own polyhedral reachability cone.

Sharp velocity theorem. For a word $U(k)$ of total duration $T > 0$, let the isometry V select a two-dimensional degenerate eigenband at $k = 0$. Its first-order velocity matrices are

$$A_i = \frac{i}{T} V^* U(0)^* (\partial_i U)(0) V = \sum_a v_{a,i} E_a, \qquad E_a \ge 0, \quad \sum_a E_a = I_2, \quad \|v_a\| \le c.$$

Indeed differentiating the word produces positive compressions of each flight projector, weighted by that stage's fraction of the actual duration. Waits and onsite time contribute zero velocities. Ancillary states and quantum interference remain in the compressed effects. Write $E_a = w_a(I + r_a \cdot \sigma)$, where $w_a \ge 0$, $\|r_a\| \le 1$ and $\sum_a w_a = 1$. For $A_i = u_i I + \sum_j M_{ij}\sigma_j$,

$$M = \sum_a w_a v_a r_a^{\mathsf{T}}, \qquad \|M\|_* \le \sum_a w_a \|v_a\| \|r_a\| \le c. \tag{QT1}$$

Here $\|\cdot\|_*$ is the sum of singular values. Conversely, any real matrix $M = \sum_j s_j u_j r_j^{\mathsf{T}}$ with $\sum_j s_j \le c$ is attained without tilt: perform conditional flights with velocities $\pm c u_j$ and projectors $(I \pm r_j \cdot \sigma)/2$ for fractions s_j/c of the period, then wait. Thus (QT1) is the complete untilted first-order attainable set. General axes give finite addressed experiments; periodic lattice realizations require commensurate directions or controlled approximations. No characterization of the joint tilted pair (u, M) is asserted.

In particular an isotropic matrix $M = vR$, R orthogonal, requires $3|v| \le c$. A normalized trace witness gives the same bound for a three-component Clifford band: $(n \cdot \alpha)^2 = \|n\|^2 I$ bounds each positive-effect contribution. More coin states, directions, local gates or grouped periods cannot remove this factor within the stated grammar and regular linear band limit. This excludes an isotropic Weyl or Dirac sector with the complete record-cone speed, rather than merely failing to find one. It is not a theorem about every quantum cellular automaton.

Saturation derives the minimal geometry. For $M = (c/3)I$, equality in the trace bound requires every nonzero effect to have $\|v_a\| = c$, $\|r_a\| = 1$ and $v_a = c r_a$. Consequently

$$\sum_a w_a n_a = 0, \qquad \sum_a w_a n_a n_a^{\mathsf{T}} = I/3, \qquad \|n_a\| = 1. \tag{QT1a}$$

Conversely these weighted spherical two-design identities attain the bound. They are derived from optimality. At least four nonzero outcomes are needed: the three independent coordinate columns are orthogonal to the weight column. With exactly four, the matrix with rows $\sqrt{w_a}(1, \sqrt{3} n_a)$ has orthonormal columns and therefore rows. The row norms force $w_a = 1/4$ and orthogonality forces $n_a \cdot n_b = -1/3$ for $a \ne b$. Thus the regular equally weighted tetrahedron is the unique minimal optimizer, up to rotation, reflection and relabeling. This minimum counts distinct velocity outcomes across the whole word. A single flight therefore needs at least four internal channels, since its distinct nonzero projectors are orthogonal. A multi-stage word can reuse fewer channels: the three-axis example below uses two, with six velocity outcomes. None of these counts is an OPH observer population or a choice of central source ports.

An explicit minimal four-channel process. Use displacements as with

$$s \in \{(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)\}$$

and the onsite coin

$$C = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & i & -i \\ 1 & -i & 0 & i \\ 1 & i & -i & 0 \end{pmatrix}, \qquad C^* = C, \quad C^2 = I.$$

Apply C, then one simultaneous flight of the four channels at speed c. The duration is $T_a = \sqrt{3} a/c$ and $U_{\mathrm{tet}}(\theta) = \mathrm{diag}(e^{-i\theta \cdot s}) C$. The projectors $P_\pm = (I \pm C)/2$ have rank two. For $B_i = \mathrm{diag}(s_i)$, the matrices $X_i^\pm = \sqrt{3} P_\pm B_i P_\pm$ obey Pauli multiplication on their bands, with opposite orientations. The compressed flight effects are $(P_\pm + \sum_i s_i X_i^\pm/\sqrt{3})/4$. Hence the physical band speed is $(a/\sqrt{3})/T_a = c/3$. The native clock, coherent mixing and minimal direction geometry are all specified in the same process. It also transmits an exact native record at speed c: in an empty-register background, leave the source empty for zero and prepare the local one-particle state $C|s\rangle$ for one. The coin gives $C^2|s\rangle = |s\rangle$, and the flight reaches as in time $\sqrt{3} a/c$. The receiver occupation effect distinguishes them with probabilities zero and one. This attained response belongs to the

same walk. These inputs do not replace the fixed filled reference used below. The experiment uses the complete finite channel family, without a uniform continuum quasienergy bound on the preparation. Restricting interventions to a low-energy sector would change the complete-clock requirement.

Its entire four-channel characteristic polynomial is

$$\det(\lambda I - U_{\mathrm{tet}}) = \lambda^4 - \frac{2}{3}(\cos 2x + \cos 2y + \cos 2z)\lambda^2 + 1.$$

Writing $\cos 2\epsilon = (\cos 2x + \cos 2y + \cos 2z)/3$ with $0 \le \epsilon \le \pi/2$, the eigenvalues are $\pm e^{\pm i\epsilon}$. Exactly eight zero-quasienergy cones occur at $\{0,\pi\}^3$, with parity selecting four cones of each chirality. The same momenta contain a π band; the eight points $\{\pi/2, 3\pi/2\}^3$ are degeneracies at $\pm i$, not extra zeros. With the fixed logarithm convention below, both energies ϵ/T_a and $(\pi-\epsilon)/T_a$ occur twice at every momentum. The identity $\sin^2\epsilon = \sum_i \sin^2\theta_i/3$ gives $\epsilon \ge 2d(\theta, \{0,\pi\}^3)/(\pi\sqrt{3})$. It proves the same complete thermal limit (QT5); the second band is bounded below by $\pi/(2T_a)$ and its full thermal contribution vanishes in the refinement limit. No finite channel is discarded. Independent checks include all four-channel momentum matrices, both bands, the full sixteen-dimensional onsite Fock coin and all charged trajectories.

There is also a full four-channel dynamical bound, without selecting a momentum-dependent band. Put $B = \mathrm{diag}(s\cdot k)$ and $X_i = X_i^+ + X_i^- = \sqrt{3}(B_i + CB_iC)/2$. Exactly $U_{\mathrm{tet}}(ak)^2 = e^{-iaB}e^{-iaCBC}$; the product error is at most $a^2\|[B, CBC]\|/2 \le 3a^2\|k\|^2$. For $j = \lfloor t/T_a \rfloor$, $\|k\| \le K$ and $0 \le t \le T$,

$$\|U_{\mathrm{tet}}(ak)^j - C^j e^{-it(c/3)k\cdot X}\| \le \frac{\sqrt{3}cT}{2}aK^2 + \frac{4}{\sqrt{3}}aK.$$

The final unpaired tick costs at most $\sqrt{3}aK$ and time rounding at most $aK/\sqrt{3}$. At other nodes replace C by $(-1)^{\mathrm{parity}}C$. The actual zero band has unit carrier, while the other band keeps its alternating phase. Omitting this carrier makes an order-one error at odd ticks even at $k = 0$. Blocking two ticks cannot turn the retained π quasienergy sector into another physical zero-energy species.

A bound for arbitrary finite stencils. The flight factorization is not needed for a second obstruction. Let $U(k) = \sum_{b\in S} e^{-ik\cdot b}K_b$ be any translation-invariant finite-range unitary with finite internal dimension and period T. For $G(n) = iU(k)^*\partial_n U(k)$, $l = \min_b n\cdot b$ and $h = \max_b n\cdot b$,

$$lI \le G(n) \le hI. \tag{QT1b}$$

To prove this, take a lattice-rational direction and rescale to integer exponents. The polynomial $V(z) = \sum_b z^{n\cdot b - l}e^{-ik\cdot b}K_b$ is unitary on the circle and contractive in the disk by the maximum principle. The radial derivative of $\|V(r)\psi\|^2$ at $r = 1$ is nonnegative, giving the first inequality. The reversed polynomial gives the second. Rational-direction density gives the result for every n. Neither positivity nor orthogonality of the coefficient matrices is assumed.

Compression to a degenerate two-state band implies $u + M\overline{B}_1 \subseteq \mathrm{conv}(S/T)$. An untilted isotropic speed v therefore requires a ball of radius $|v|T$ inside that polytope. Exact native speed c requires $\|b\| \le cT$. A finite polytope inside the cT ball cannot contain the same round ball, so finite stencils cannot attain an exactly isotropic speed c. The caps $\{n \in S^2 : n\cdot b \ge |v|T\}$ must cover the unit sphere when $|v| > 0$. Each has area at most $2\pi(1 - |v|/c)$. If N counts nonzero displacement terms, sphere area and subadditivity give

$$N(1 - |v|/c) \ge 2. \tag{QT1c}$$

Thus fixed N excludes even a refinement limit with $|v|/c \to 1$; accuracy $|v|/c \geq 1-\eta$ requires $N \geq 2/\eta$. Increasing internal dimension alone cannot evade it. The bound also applies to approximately isotropic refinements: use $s_{\min}(M) - \|u\|$ in place of $|v|$ when it is strictly positive, since that centered ball lies in the velocity ellipsoid. A nonpositive value supplies no positive-radius cap bound. For example, $U(k) = e^{-ik_x} I_2$ with $T = c = 1$ has $N = 1$, $u = (1,0,0)$ and $M = 0$; its velocity set does not contain the origin. Clamping the radius to zero would falsely demand $1 \geq 2$. The stationary identity has $N = 0$ and is also outside this positive-radius argument. Exact scope controls retain both processes. For $M \to cR$, $u \to 0$, the radius is eventually positive and tends to c, so $N \to \infty$ follows without exact isotropy at any cutoff. The tetrahedral process saturates its support-polytope bound as well as the flight bound. The cube has inradius a and circumradius $\sqrt{3}a$, so even an endpoint-only clock for the next example gives at most $c/\sqrt{3}$; its actual sequential flights give $c/3$. A growing direction set is necessary, but does not prove existence of a luminal unitary refinement. The stronger flight bound excludes such a limit for every positively timed register-flight lowering, even with growing direction sets.

An optimal native process. On the unit three-torus take $q \in 4\mathbb{N}$, $a = 1/q$, and two fermionic modes at each site. Apply conditional axis flights of length a, with Pauli projectors along x, y, z in that order. Each costs a/c, so

$$T_a = 3a/c, \qquad U(\theta) = e^{-i\theta_z \sigma_z} e^{-i\theta_y \sigma_y} e^{-i\theta_x \sigma_x}. \tag{QT2}$$

Translations and onsite basis changes realize these exact unitaries. For physical momentum $k = \theta/a$, Taylor remainders and unitary telescoping give, for $\|k\| \leq K$ and $0 \leq t \leq T$,

$$\left\| U(ak)^{\lfloor t/T_a \rfloor} - e^{-it(c/3)k\cdot\sigma} \right\| \leq \frac{2cT}{3} aK^2 + aK. \tag{QT3}$$

The product remainder is at most $a^2(\sum_i |k_i|)^2/2$, the single exponential remainder at most $a^2\|k\|^2/2$, and the final incomplete period costs at most aK. The construction attains the speed bound. Assigning the whole word duration a/c would omit two actual flights. Onsite overhead can only lower the speed. An overhead fraction tending to zero preserves (QT3) in the limit; uniformly bounded control strength is not asserted to realize arbitrarily fast fixed-angle basis changes.

Complete spectrum and fixed reference. The eigenvalues are $e^{\pm i\epsilon}$, $0 \leq \epsilon \leq \pi$, with

$$\cos\epsilon = \cos x \cos y \cos z + \sin x \sin y \sin z. \tag{QT4}$$

There are exactly eight zeros: four points in $\{0,\pi\}^3$ with even π parity, and four in $\{\pi/2, 3\pi/2\}^3$ with even $3\pi/2$ parity. Cauchy–Schwarz gives the bound

$$|\cos\epsilon| \leq \sqrt{\cos^2 x \cos^2 y + \sin^2 x \sin^2 y} \leq 1.$$

Equality forces $\cos x \sin y = \sin x \cos y = 0$ and the stated z alignment. The opposite parities give eight π points. At the first four zeros the linear Pauli coefficient matrix is I; at the other four it is $\mathrm{diag}(1,-1,1)$. All eight Weyl cones propagate at $c/3$, four with each chirality. None is removed.

Define the finite reference using the principal Hermitian logarithm of the actual U, divided by T_a, with a fixed branch at $-I$. Fill strictly negative levels and leave zero levels empty. The Fock lift preserves this Slater vacuum exactly. Normal ordering gives two excitation choices per momentum of energy $E_a = \epsilon/T_a$. This is a declared Floquet quasienergy; the generally nonlocal logarithm is not a derived autonomous Hamiltonian or an OPH physical energy calibration.

For every $\beta > 0$, the complete finite partition function satisfies

$$\log Z_a = 2\sum_{\theta} \log(1 + e^{-\beta E_a(\theta)}) \longrightarrow 16 \sum_{n\in\mathbb{Z}^3} \log(1 + e^{-\beta(2\pi c/3)\|n\|}). \tag{QT5}$$

The normal-ordered energy and Gibbs entropy converge as well. An explicit global bound is $\epsilon(\theta) \geq d(\theta, Z)/(8\sqrt{3})$. For $\epsilon \geq \pi/8$, the torus diameter proves it. Otherwise $\cos(2x)\cos(2y) \geq \cos(2\epsilon) > 0$ puts x, y within ϵ of the same zero family. Writing $Re^{i\phi} = \cos x \cos y + i \sin x \sin y$, one obtains $|z-\phi| \leq \epsilon$ and an additional angular deviation at most $\arctan(\tan^2 \epsilon) \leq \epsilon$, hence $d(\theta, Z) \leq \sqrt{6}\epsilon$. Assigning each momentum to a nearest zero gives the uniform summable bound $E_a \geq \pi c\|n\|/(12\sqrt{3})$. Dominated summation and the eight linearizations prove (QT5), including all zero-level degeneracy. Fixed finite-excitation dynamics follow from exterior-product telescoping and (QT3). Finite smeared CAR covariances use projector convergence away from the explicitly treated zero levels. No global norm identification of infinite filled seas or interacting-field limit is claimed.

Why a finite Hamiltonian does not repair exact delay. Let H act on a finite tensor product of full matrix site factors at distinct fixed positions. If every local pair satisfies

$$[e^{itH} A_x e^{-itH}, B_y] = 0 \quad \text{for } 0 < t < \|x - y\|/c, \tag{QT6}$$

analyticity makes that commutator zero for all t. The evolving algebra at x therefore stays in that factor. Its derivative is an inner onsite derivation; subtracting all onsite implementers leaves a scalar. Thus H has no intersite interaction. Finite quadratic canonical propagators have the same analytic obstruction. The hypothesis is exact zero influence for all admitted local preparations and effects, not a detection threshold.

For two qubits and interaction coefficients h_{ab} in the Pauli basis, the complete derivative witness is

$$\sum_{i,j=1}^{3} \|[[H, \sigma_i \otimes I], I \otimes \sigma_j]\|_{\mathrm{HS}}^2 = 256 \sum_{a,b=1}^{3} h_{ab}^2.$$

For $H = \sigma_x \otimes \sigma_x$, first-qubit preparation $+y$ and remote preparations $\pm x$ give receiver-effect probabilities $(1 \pm \sin 2t)/2$. Thus an actual local preparation produces an arbitrarily early signal. Moving modes evade fixed algebra ownership; sampled automata lack the assumed continuous interpolation; continuum fields use infinite local algebras and singular generators.

The coherent alternative. If the response vector space acts on an operationally supplied fundamental SU(2) module, covariance forces a real-linear Hermitian velocity map to be $A(v) = \gamma v \cdot \sigma$: Hermitian matrices split into the scalar and adjoint representations. A covariant onsite term may add μI; covariance does not determine that energy offset. Choosing zero onsite offset, handedness and scale gives $i\partial_t \psi = -ic\sigma \cdot \nabla\psi$. Its generator squares to $-c^2\Delta$, and its conserved current obeys $\rho = \psi^*\psi$, $j = c\psi^*\sigma\psi$, $\|j\| \leq c\rho$. The exterior-ball flux identity proves exact propagation in radius ct, with dispersion $\pm c\|k\|$. A parity-paired Dirac module permits an anticommuting onsite mass without changing this cone. Standard CAR quantization supplies locality of even observables and positive excitation energies. These noncommuting luminal velocities cannot have the positive flight resolution above, which would imply $3c \leq c$.

This standard continuum witness identifies a precise structure to realize; it does not supply the fundamental module's source meaning, coherent spatial generator, physical clock or refining local instruments. Internal gauge SU(2) is not thereby identified with spin. Neither a new core axiom nor a finite twelve-port realization is asserted.

Evidence and relation to prior mathematics. The package at `code/m1_quantum_transport` independently reconstructs all six flight orders, exact momentum matrices, complete thermal catalogs, all sectors and intermediate stages of finite fermionic gate programs, positive velocity resolutions, Pauli delay probes and coherent currents. Hostile controls reject uncharged time, hidden species, changed phases and regenerated signless permutations. Thirteen Lean lemmas and a transitive axiom audit certify finite algebra reductions; the operator, SVD and asymptotic arguments above remain analytic proofs. Quantum-walk relativistic limits are standard; see Mlodinow and Brun, *Physical Review A* 97, 042131 (2018), arXiv:1802.03910, and Bisio, D'Ariano and Perinotti, arXiv:1608.02004. The tetrahedral qubit measurement is a standard SIC construction; see Renes, Blume-Kohout, Scott and Caves, Journal of Mathematical Physics 45, 2171 (2004). Tetrahedral Dirac walks are also studied by Nzongani and collaborators, arXiv:2404.09840. The strict-causal versus Hamiltonian distinction is discussed by Ranard, Walter and Witteveen, Annales Henri Poincaré (2022). The contribution here is the sharp native-time characterization, its optimal complete-spectrum construction and their implication for the proposed common record and matter clock.

H Massive operational clocks and resolved local reads

The tetrahedral flight process of Appendix G admits a massive extension in the same native time. The graph, coherent onsite gates, mass parameters and readout operations are supplied. Ideal onsite gates take zero duration in this grammar; a physical gate overhead must be charged separately. These assumptions define a mathematical construction, not a source-selected physical instrument. Throughout, $a, c, m > 0$ and $\hbar = 1$.

Complete massive dynamics. With the coin C and tetrahedral sign vectors s defined above, put $W_a(k) = \mathrm{diag}(e^{-iak\cdot s})C$, $\tau = \sqrt{3}a/c$, $\mathcal{R} = I_2 \otimes C$ and $\beta = \sigma_x \otimes I_4$. One parallel flight and onsite mixing give

$$U_m(k) = e^{-i\tau m\beta}\,\mathrm{diag}(W_a(k), W_a(k)^*).$$

For $0 < m\tau < \pi/2$, define $\sin^2\epsilon = \sum_i \sin^2(ak_i)/3$ and $\omega = \arccos(\cos(m\tau)\cos\epsilon)$, with $0 \le \epsilon \le \pi/2$. The eight eigenvalues are $e^{\pm i\omega}$ and $e^{\pm i(\pi-\omega)}$, each twice. All eight valleys are retained. Every smooth band has group speed at most $(c/3)\cos(m\tau)$, a sharp supremum. At fixed spacing, the ballistic distribution of any fixed normalized one-particle preparation has the same speed bound; this long-time statement is distinct from refinement.

Set $v = c/3$, $B(k) = \mathrm{diag}(k\cdot s)$, $X(k) = \sqrt{3}(B(k) + CB(k)C)/2$ and $H_m(k) = m\beta + v\sigma_z \otimes X(k)$. The Clifford identities give $H_m(k)^2 = (m^2 + v^2|k|^2)I_8$. For $|k| \le K$, $t = n\tau$, $b = \sqrt{3}aK$ and $\mu = m\tau$,

$$\|U_m(k)^n - \mathcal{R}^n e^{-itH_m(k)}\| \le D_n := \lfloor n/2\rfloor(b^2 + 4\mu b) + (n \bmod 2)(b + \mu b). \tag{MC1}$$

Pairing consecutive ticks and bounding the mass/flight commutators proves this full-channel estimate, including leakage and the odd-tick carrier. The carrier at valley $\pi z/a$ is $(-1)^{\sum z_i}\mathcal{R}$.

Actual spatially supported operations. For $f \in H^2(\mathbb{R}^3;\mathbb{C}^8)$ let $S_a f(x) = a^{3/2} f(ax)$, $M_2(f) = \||k|^2\widehat{f}\|_2$ and $A_a(f) = (a/\pi)^2\sqrt{4\pi^2 + \pi^4/45}\,M_2(f)$. Poisson summation bounds the sampling aliases by $A_a(f)$; (MC1) then gives, for $0 < K < \pi/(2a)$,

$$\|U_m^{\pm n} S_a f - \mathcal{R}^{\pm n} S_a e^{\mp itH_m} f\| \le Q_a(f) := 2A_a(f) + D_n\|f\|_2 + 2M_2(f)/K^2. \tag{MC2}$$

The full Dirac field has propagation speed v, including both energy signs. Let compactly supported $f, g \in H^2$ have separation greater than vt, and normalize their nonzero lattice samples to f_a, g_a. For the CAR number effects $N_h = a(h)^* a(h)$, encoding by $e^{i\theta N_{g_a}}$ changes the later N_{f_a} probability by at most

$$\min\{1,\, 2|\theta| Q_a(f)/\|S_a f\|_2\}. \tag{MC3}$$

This follows from the CAR commutator and Duhamel's formula, uniformly over Fock states and spectator ancillas. Finite products obey the sum bound. At fixed duration and envelopes, $K = a^{-1/4}$ in fixed units gives $Q_a = O(\sqrt{a})$. The specified operation class is essential: a single-site particle produces a perfect record at speed c in one flight.

A readable inertial clock. Take two coherently labelled masses $m_1 < m_2$ and a fixed $|u| < v$. For $\gamma = (1 - |u|^2/v^2)^{-1/2}$, prepare Gaussian envelopes of width σ around $k_j = \gamma m_j u/v^2$, with a common unit spinor ξ satisfying $H_{m_j}(k_j)\xi = E_j \xi$, $E_j = \gamma m_j$, and $\mathcal{R}\xi = \xi$. The normalized initial state is $(\Psi_1, \Psi_2)/\sqrt{2}$, where

$$\Psi_j(x) = (2\pi\sigma^2)^{-3/4} e^{-|x|^2/(4\sigma^2)} e^{ik_j \cdot x} \xi.$$

The actual walk starts from the normalized lattice samples of this state. These are common reference velocities; finite-walk drift is included in (MC1). Write $\Delta m = m_2 - m_1$, $\Delta k = k_2 - k_1$ and $\Delta E = E_2 - E_1$. A Gaussian spatial acceptance of radius R_d, centred at ut, and the mass-label effect $|{+_\theta}\rangle\langle{+_\theta}|$, with $|{+_\theta}\rangle = (|1\rangle + e^{i\theta}|2\rangle)/\sqrt{2}$, give the rigid-packet reference probability

$$\begin{aligned} p_\theta(t) &= \tfrac{1}{2}\{p_R + V_R \cos(\Delta m\, t/\gamma + \theta)\}, \qquad p_R = (1 + \sigma^2/R_d^2)^{-3/2}, \\ V_R &= p_R e^{-s^2|\Delta k|^2/2}, \qquad s^2 = \frac{\sigma^2 R_d^2}{\sigma^2 + R_d^2}. \end{aligned} \tag{MC4}$$

The beat follows from $\Delta E - u \cdot \Delta k = \Delta m/\gamma$, not an assumed proper-time accrual rule. The continuum norm error through time T is bounded by $e = \max_j\{\sqrt{3}v/(m_j\sigma) + \sqrt{15}Tv^2/(8m_j\sigma^2)\}$. Choose $\max_j |k_j| < K < \pi/(2a)$ and require $a|\Delta k| \le \pi$ and $b_\sigma \le 1/2$, where

$$B(z) = \sum_{\ell \in \mathbb{Z}^3 \setminus \{0\}} e^{-z|\ell|^2}, \qquad b_\sigma = B(2\pi^2\sigma^2/a^2), \qquad b_s = B(\pi^2 s^2/(2a^2)).$$

Writing $A = \max_j A_a(\Psi_j)$, $D = \max_j D_n(a, K, m_j)$ and h for the common Fourier tail norm beyond K, the actual lattice probability error at tick n with $n\tau \le T$ is at most

$$4(e + D + 4A + 2h) + 4(b_s + b_\sigma) + e^{-L_d^2/(2R_d^2)} + e_{\mathrm{prep}}. \tag{MC5}$$

The detector is cut off at radius L_d. For a product C^2 preparation cutoff taking values in $[0,1]$, equal to one on $[-P, P]^3$, and tapering to zero within $[-P - \sigma, P + \sigma]^3$, with $P > a$, one may use $e_{\mathrm{prep}} = 2\sqrt{3\,\mathrm{erfc}((P-a)/(\sqrt{2}\sigma))}$. The complete formulas and proof are in *Massive flights, resolved reads and observable clocks*. With $c = 3$, $(m_1, m_2) = (100, 100.1)$, $u = (0.6, 0, 0)$, $\sigma = 10$, $R_d = 100$, $L_d = 600$, $P = 80$ and $a = 10^{-9}$, outward interval arithmetic bounds the probability error below 0.035153 through a full moving cycle and the next tick. The reference fringe exceeds 0.7457 and the actual click-probability swing exceeds 0.6753 after all errors. This is a finite analytic existence witness; separate coarse periodic waveguide runs test evolution and readouts.

The positive Floquet accounting observable $|i \log U_m|/\tau$ has an empty Fock vacuum and bounded preparation cost under refinement for these packets, including their high-band leakage. It is not the signed generator or a derived physical energy. The construction supplies free inertial clock comparisons within its declared operations; physical source admission, mass selection, calibration, interacting or accelerated clocks, and distributed record-collection latency remain separate.

I Native quantum dynamics with live noisy records

The massive process of Appendix H has a realization in a declared coherent source, including its preparations, fermionic operations, feedback and public reads. Its apparatus consists of bounded patches with local processor state, ports, retained records and repair operations. Quantum protection and a maintained classical archive allow its complete named history to converge despite noisy local services. This construction retains the specified source, coherent transport, external schedule and clock parameters. It does not select them from A1–A3.

The source supplies and the constructed gates. The primitive processor is M_6, with twelve central record ports and the complete response algebra $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. The existing port generators satisfy $\sum_p D_p = \mathrm{diag}(iI_3, 0)$. The first two coordinates form a qubit code; their rotations are native $\mathfrak{u}(3)$ controls. In the four-coordinate pair code, the uniform port pulse restricts, up to a common phase, to $\mathrm{diag}(1, 1, 1, e^{i\theta})$. At $\theta = \pi$ it supplies controlled-Z. Packing and separation use fixed coordinate isometries, while every active basis change uses a source pulse. The pair code is a composite experiment, not an extra primitive fifteen-dimensional response.

Coherent code gluing requires agreement of complementary noncentral read experiments on the transported code. Classical record agreement alone allows complete phase loss and supplies no such channel. Pure blank preparation, destructive readback, resets and finite private inventories are declared interfaces; a pure blank is not inferred from an unconstrained entropy minimum. From these interfaces the circuits construct the massive walk on all occupation sectors, resolved CAR number instruments and a quartic interaction. The fermionic encoding uses the credited Bravyi–Kitaev edge construction; it does not derive fermionic statistics from classical records. Complete derivations are given in *Coherent source codes, entropy selection and charged clocks* and *Fermionic clocks and reads from the proper-code source*.

Complete leakage reduction and causal instruments. For $d = 2, 4$, let $V_d : \mathbb{C}^d \to \mathbb{C}^6$ be the fixed code embedding and $P_d = V_d V_d^*$. The admitted transfer distinguishes the code from its complement. Continue its failure output by the admitted reset, retaining the diagnostic flag. The complete Kraus family is

$$A_{\mathrm{ok}} = V_d^*, \qquad A_j = |0\rangle_d\langle j|_6 \quad (d \le j < 6), \qquad \sum_\alpha A_\alpha^* A_\alpha = I_6.$$

All A_j have one public failure label; j is a private Kraus index. Ignoring that flag only when defining the application marginal gives

$$\mathcal{R}_d(\rho) = V_d^* \rho V_d + \mathrm{Tr}((I - P_d)\rho)|0\rangle\langle 0|, \qquad \mathcal{R}_d(V_d x V_d^*) = x. \tag{46}$$

No experiment is conditioned on success. A noisy reduction, including a wrong flag or failed reset, belongs to its complete native gate box. Each input is reduced before a gate; the four-dimensional pair processor is internal to a two-operand box. Helpers never supply an uncounted third data operand. The noisy leakage-reduction theorem of Aliferis and Terhal, Lemma 2, then converts such boxes to ordinary local circuit faults with enlarged finite library constants. Their mathematical dilation grants no physical control of an environment or reversible access to all of M_6.

For a source instrument with public label y and private branches $K_{y\alpha}$, use the isometry $\psi \mapsto \sum_{y,\alpha} K_{y\alpha}\psi \otimes |y\rangle \otimes |y, \alpha\rangle$ and discard only the private factor. Reversible Boolean decisions and controlled operations preserve the resulting diagonal history blocks. Induction at successive

instrument cuts therefore gives the complete channel

$$\mathcal{I}(\rho) = \sum_h |h\rangle\langle h| \otimes \Phi_h(\rho),$$

including rejected application outcomes and arbitrary input references. Protection diagnostics are retained separately and never command a global abort. Private protected records drive feedback; the public archive consists of actual noisy central bits.

Noise hypotheses and the fixed-rate regime. Every gate, transfer, preparation, reset, flight and wait is a counted service, including its outcome register. For fresh-environment Markov noise with additional Hamiltonian B_t and Lindblad operators $L_{\alpha t}$, put $\Gamma(t) = 2\|B_t\| + 2\sum_\alpha \|L_{\alpha t}\|^2$. Duhamel's formula and complete-channel contractivity give $\delta \le \int \Gamma(t)\,dt$ in diamond norm, independently of the ideal drive amplitude. Any intrinsic event error must also enter δ. The Stinespring continuity theorem of Kretschmann, Schlingemann and Werner gives a local dilation-fault amplitude $\eta \le C\sqrt{\delta}$. This amplitude is not a failure probability; dissipative channels need not contain a positive multiple of the ideal channel.

For fixed bounded Γ, fixed finite local classical fault rates, $a = L/q$, complete-service exposure $O(a/c)$ and intrinsic event errors bounded by $O(q^{-1})$, one has $\delta = O(q^{-1})$ and $\eta = O(q^{-1/2})$. The measurement-free computation-code construction of Aharonov and Ben-Or, Sections 4 and 7–8, applies with the punctured RM$(5, 11)$ CSS code $[[2047, 1, 63]]$. Its 31-error capacity exceeds the conservative correction-and-gate spread 16. Disjoint rectangles tolerate one bad child, so a non-sparse six-level rectangle requires at least 64 native faults. The active volume is $O(q^5 \log^{16}(2q))$; its bad amplitude is consequently $O(q^{-27} \log^{16}(2q))$.

An all-to-all private-voter archive with $4r + 1$ bits, $r = \lceil \log_2(2q) \rceil$, keeps its live majority under r old errors and r fresh faulty locations. Its refresh size is $O(\log^4 q)$, and its marked-fault tail beats every inverse power of q in this shrinking-service-error regime. Summed rotation-synthesis error then gives

$$\epsilon_{\text{joint}} = O(q^{-20} \log^{16}(2q)), \qquad \epsilon_{\text{accounting}} = O(q^{-16} \log^{16}(2q)).$$

Six is a fixed concatenation depth here because complete-service errors shrink. A fixed nonzero error per service requires the separate construction below. The full-interface derivation is in *Full-interface leakage, dissipation and source refinement*.

Theorem I.1 (Fixed-strength computation and live public history)**.** *For the declared source and prepared experiments above, assume fresh-environment full-interface Markov noise and the specified local classical faults, with a joint local fault expansion: for every set S of scheduled locations, the sum $F(S)$ of paths faulty on S, with other locations unrestricted, satisfies $\|F(S)\| \le \eta^{|S|}$. Uniform proximity to the ideal complete services supplies this condition through the dilation bound above. There is a fixed $\eta_* > 0$ such that every fixed $0 < \eta \le \eta_*$ admits a cofinal apparatus family with $O(\log \log q)$ quantum protection levels and $O(\log q)$ bits per public record. For a fixed library constant κ,*

$$\epsilon_{\text{joint}} = O(q^{-20} \log^{16+2\kappa}(2q)),$$
$$\epsilon_{\text{accounting}} = O(q^{-16} \log^{16+2\kappa}(2q)).$$

The first estimate compares decoded quantum outputs and the complete majority-decoded named history, with every application abort retained. The public majority remains correct throughout each successful refresh. No elementary fault strength has to tend to zero in this regime.

Proof. Use the Margulis–Gabber–Galil graph on $\mathbb{Z}_m^2$, with the eight labelled neighbours $(x \pm 2y, y)$, $(x \pm (2y+1), y)$, $(x, y \pm 2x)$ and $(x, y \pm (2x+1))$, retaining loops and multiplicities. The formalized absolute spectral bound for its adjacency A is $5\sqrt{2}/8$ after normalization. For $N = m^2$, $D = 8^{12}$ and $P = A^{12}/D$, $\|P - J/N\|^2 \le (25/32)^{12} < 1/16$. If an error set E has $e \le N/8$ positions, voters with at least half their inputs in E satisfy $|T|(3/8)^2 \le \|P(1_E - e/N)\|_2^2 \le e/16$; thus $|T| \le 4e/9$.

Copy the D permutation matchings into private voter lanes, sort each lane using bounded Boolean services, and copy its lower middle bit into a separate output bank. Count every live-wire idle. A fault touches at most one shared input and one private voter. With at most $N/16$ old errors and $F \le N/64$ faulty services, the union of initially wrong and ever-damaged shared positions has size $e \le 5N/64$. New errors are at most $(4/9)(N/16 + F) + F \le 29N/576 < N/16$; the old bank keeps its majority throughout. Integer floors strengthen these inequalities. The complete cycle has at most $C_A N$ locations, with fixed finite C_A.

The sum of paths with at least M faults has the exact marked-set expansion

$$\sum_{|S| \ge M} (-1)^{|S|-M} \binom{|S|-1}{M-1} F(S).$$

A path with $k \ge M$ faults has coefficient one by the binomial identity; otherwise its coefficient is zero. Summing absolute coefficients bounds the norm by $\beta(V, M) \le e^{V\eta}(eV\eta/M)^M$. Taking $M = \lfloor N/64 \rfloor + 1$ and $\eta \le (1024 e C_A)^{-1}$ gives $\beta_{\text{archive}} \le e^{-N/32}$.

Export uses independently protected probes copied from a protected, dephased logical record. Decode each probe one concatenation level at a time, using lower-level protected operations. At each stage first correct its 2047 child blocks separately. The computation code's total-correction property also covers arbitrary damaged child inputs. After these entry corrections, at most 31 child positions carry arbitrary logical damage. The classical record decoder is exact on their entire support subspace: descending-degree cube parities decode $\mathrm{RM}(5, 11)$, since each coefficient has at least 64 disjoint votes and at most 31 are damaged. Try both puncture fills and retain only re-encodings within punctured distance 31; distance 63 makes the accepted word unique. An uncorrectable word returns a fixed fallback with its failure diagnostic, not a postselected experiment. The two correctable logical-label subspaces are orthogonal. Reversible evaluation is $|\psi\rangle \mapsto |a\rangle W_a |\psi\rangle$ on each, so discarding workspace preserves any conditional spectator state. The logical record is dephased before export; this is not an unknown-qubit coherent decoder.

For finite rectangle constants A_0, S_D, the sparse-path bound gives $\eta_j \le A_0^{-1}(A_0\eta)^{2^j}$. If $\theta = A_0\eta \le 1/2$, all decoder stages and the terminal read have summed bad strength $t \le S_D\eta/(1-\theta) + \eta$. Their union norm is at most $e^t - 1 \le C_E\eta$ for a fixed C_E and sufficiently small η. Distinct islands use disjoint native locations; their marked bounds multiply without conditioning on successful protected copying. Hence $\beta_{\text{export}} \le e^{-N/8}$ when $C_E\eta \le (256e)^{-1}$.

Enlarge A_0 to absorb the finite native constants and require $A_0 \ge \max(3S^2, 768C_A, 192C_E)$, where S bounds a complete correction/gate rectangle. The nonempty fixed interval $0 < \eta \le 1/(4A_0)$ meets all the preceding conditions. Put

$$\ell = \lceil \log_2(2q) \rceil, \quad m = \lceil \sqrt{2048\ell} \rceil, \quad N = m^2, \quad k = \lceil \log_2(16\ell) \rceil.$$

Then $\eta_k \le A_0^{-1}(2q)^{-32}$ and each archive/export contribution is at most $(2q)^{-64}$. Partition native locations among protected rectangles, export stages and refresh cycles, including every intervening wait. For their bad majorants β_i, disjoint marked sets give the joint bound $\|B_{\text{any}}\| \le \prod_i (1+\beta_i) - 1 \le \exp(\sum_i \beta_i) - 1$. This is a coherent norm estimate, not a probability union bound. The charged volume below and rotation synthesis to q^{-24} give the stated joint error after complete decoding. Pulling back a positive logical observable bounded by $E_{\max} = O(q^4)$ through that TP decoder multiplies the error bound by at most $E_{\max}$ and proves the accounting estimate. □

Causal schedule, cost and interpretation. If $s \geq 2$ bounds one protection level's width and depth overhead, set $\kappa = \lceil \log_2 s \rceil$. A conservative lifetime ledger is

active inventory	$O(q^4 \log^{8+\kappa}(2q))$,
scheduled depth	$O(q \log^{8+\kappa}(2q))$,
active native volume	$O(q^5 \log^{16+2\kappa}(2q))$,
retained diagnostic bits	$O(q^5 \log^{17+2\kappa}(2q))$,
complete storage-time volume	$O(q^6 \log^{25+3\kappa}(2q))$.

Passive diagnostics are never reused by control; their independent local Markov noise cannot change the active reduced state. This does not assert decoupling from an arbitrary common bath. A longest local depth $B_q = O(\log^{8+\kappa}(2q))$, service time $a/(1024cB_q)$ and local cluster radius at most $a/(131072B_q)$ fit all compiled local work into fixed padded source slots. Shared processors serialize; separate lanes operate in parallel. Encoded constituents cross each original link concurrently, with duration at least length$/c$ and correction at endpoints. A common q-independent wall-time dilation preserves the native clock and velocity comparison. Faster drives, parallel interfaces, resets and growing inventory are paid resources, not fixed-device predictions.

The fixed noise interval has a concrete inhabitant: apply the independent full-carrier channel $\mathcal{E}_p(\rho) = (1-p)\rho + p \operatorname{Tr}(\rho)|5\rangle\langle 5|$ and classical bit flips at every counted native component and storage location. Its dilation distance from the ideal is $\sqrt{2 - 2\sqrt{1-p}} \leq \sqrt{2p}$. A sufficiently small fixed $p > 0$ therefore meets the complete-box bound for every q. No physical selection of this noise law is inferred. The expander degree $8^{12} = 68{,}719{,}476{,}736$ and the library constants are large; no practical hardware threshold is numerically estimated.

The detailed proof is in *Fixed-strength quantum computation and live public records*. Independent verifiers in `code/m1_leakage/` and `code/m1_expander_archive/` check complete native channels, retained failure branches, decoder outputs, exact graph certificates and hostile controls. The 91,740 executed archive faults concern smaller unpowered circuits, including their failures; they are not executions of the enormous powered apparatus or proofs of the imported universal gadget theorem. Public majority is a declared statistic of available bits, not a noiseless physical controller. Neither unknown raw-input encoding, a perfect final bare bit, autonomous time, physical source selection nor laboratory energy calibration follows. Positive logical accounting is not drive work or an equality of raw native event counts.

J Filled source vacuum and positive excitation energy

The supplied massive walk and fermionic source compiler admit a filled vacuum experiment. Its bounded observer-like patches have local state, reflecting boundaries, resolved readback, repair operations and public records. The graph, masses, canonical anticommutation relations, coherent controls and external schedule are hypotheses. Particle/hole normal ordering and Slater preparation are standard; relevant precedents are Gupta and Short's discrete-time Dirac sea and Kivlichan et al.'s fermionic rotations.

Reflected gaps and the Fock generator. Let $\tau = \sqrt{3}a/c$, $\mu = m\tau \in (0, \pi/2)$ and $U = e^{-i\mu\beta}U_0$. A flight leaving a finite cube reflects into the opposite-chirality bank at its starting cell. The flight permutation S satisfies $\beta S \beta = S^*$; exchanging the precoin and postcoin gives

$$\beta U_0 \beta = U_0^*, \qquad \operatorname{Re} U = \cos\mu \operatorname{Re} U_0.$$

Hence every eigenphase lies in $[-\pi+\mu,-\mu]\cup[\mu,\pi-\mu]$, including at the boundary. Put $h = i\operatorname{Log} U/\tau$, $P = \mathbf{1}_{h<0}$ and $Q = I - P$. For the Slater state Ω_P filling precisely the negative modes,

$$\begin{aligned} E_{\mathrm{vac}} &= \operatorname{Tr}(hP), \qquad H_{\mathrm{sea}} = d\Gamma(h) - E_{\mathrm{vac}} I, \\ H_{\mathrm{sea}} &= \sum_{e_j>0} e_j c_j^* c_j + \sum_{e_j<0} |e_j| c_j c_j^* \geq 0, \\ e^{-in\tau H_{\mathrm{sea}}} &= e^{in\tau E_{\mathrm{vac}}} \Gamma(U)^n. \end{aligned}$$

The filled state is the unique ground vector on the active logical Fock space. Helpers have specified blank inputs and return conditions; no apparatus ground-state uniqueness is asserted. The identity holds in every parity sector, with external spectators, and gives the complete even-observable instrument. Its excitation spectrum comprises all subset sums of $|e_j|$. The empty-vacuum accounting observable $d\Gamma(|h|)$ has the same partition function and a different vacuum and negative-mode dynamics.

Local vacuum comparison. Write $K = (U^* - U)/(2i) = \sin(\tau h)$ and $\delta = \sin\mu$. Then $|K| \geq \delta$ and $P = (I - \operatorname{sign} K)/2$. For the boundary comparison take the infinite-lattice K, P, and embed the decoupled reflecting interior and exterior on that same lattice. For a bounded one-Lipschitz cell-distance function d, $\|e^{sd} K e^{-sd} - K\| \leq e^s - 1$. Taking $s = \log(1 + \delta/2)$, the resolvent identity and sign integral give

$$\|(P - \widetilde{P})\mathbf{1}_A\| \leq (2/\delta) e^{-sr}, \qquad r = \left\lceil \frac{\log(4/(\delta\epsilon))}{\log(1+\delta/2)} \right\rceil,$$

where $\widetilde{P}$ belongs to the decoupled reflecting interior and exterior, and A is r layers from their interface. Both pieces have the same gap. At $a = L/q$, choosing $\epsilon = q^{-12}$ costs physical margin $O(\log q)$ and $M = O(q^3 \log^3 q)$ modes around a fixed experiment. No comparison of global sea-vector fidelities is required.

In a low-momentum chart, set $\mathcal{R} = I_2 \otimes C$, $H_m(k) = m\beta + (c/3)\alpha \cdot k$ and $E(k) = \sqrt{m^2 + (c/3)^2|k|^2}$. A second-order one-step Taylor bound yields

$$\begin{aligned} z_a(k) &= \frac{(m\tau + \sqrt{3}a|k|)^2}{2\tau}, \qquad \|K/\tau - \mathcal{R}H_m(k)\| \leq z_a(k), \\ P_0(k) &= \tfrac{1}{2}(I - \mathcal{R}H_m(k)/E(k)), \qquad \|P_a(k) - P_0(k)\| \leq \min\{1, \pi z_a(k)/(4m)\}. \end{aligned}$$

Sampling and Fourier-tail bounds give $O(\sqrt{a})$ convergence for each fixed H^2 packet. The boundary estimate transfers this to finite cubes; Wick determinants give convergence for fixed finite polynomials in smeared fields. The $\mathcal{R} = +1$ sector has negative Dirac filling, whereas $\mathcal{R} = -1$ has positive Dirac filling. All eight components, both bands and all valleys are retained. The full limit is this quasi-free state, not one ordinary Dirac vacuum.

Finite-energy operations and complete readout. For normalized f, put $N_f = c(f)^* c(f)$, $p = \langle f, Pf\rangle$, $e_+ = \langle Qf, |h|Qf\rangle$ and $e_- = \langle Pf, |h|Pf\rangle$. Applying $e^{i\theta N_f}$ to the sea costs

$$\Delta E = 4\sin^2(\theta/2)[pe_+ + (1-p)e_-] \leq 4\langle f, |h|f\rangle.$$

The complete quantum-nondemolition read $\{N_f, I - N_f\}$ has channel $\mathcal{D}_f(\rho) = N_f \rho N_f + (I - N_f)\rho(I - N_f)$ and obeys

$$\|\mathcal{D}_f^*(H_{\mathrm{sea}}) - H_{\mathrm{sea}}\| \leq 2\|hf\|, \qquad \|hf\| \leq \frac{\pi}{2\tau}\|(I-U)f\|.$$

For fixed compact H^2 modes with $\mathcal{R}f = f$ away from the boundary, $\|(I-U)f\| = O(a)$. Thus the complete read has uniformly finite mean excitation cost, retaining both outcomes and the full compact CAR fields. This bound does not control rare conditional outcomes or intermediate mode-gathering drive work.

For two masses, use $v = c/3$, $\gamma = (1-|u|^2/v^2)^{-1/2}$ and $k_j = \gamma m_j u/v^2$. Translate the entire compact Gaussian mode, including carrier and cutoff: $g_j(t) = T_{ut}g_j(0)$, and set $f_\theta(t) = (g_1(t) + e^{i\theta}g_2(t))/\sqrt{2}$. The normalized sampled initial modes have carrier $e^{ik_j\cdot x}\xi$ with common unit spinor $\mathcal{R}\xi = \xi$ and $H_{m_j}(k_j)\xi = \gamma m_j \xi$. Prepare deterministically $P + |\chi\rangle\langle\chi|$, where $\chi = Qf_0(0)/\|Qf_0(0)\|$. The actual binary probability is

$$p_\theta(t) = \langle f_\theta(t), Pf_\theta(t)\rangle + |\langle f_\theta(t), U^n\chi\rangle|^2, \qquad t = n\tau.$$

The background is retained without subtraction or postselection. The rigid-packet reference is $[1+\cos((m_2-m_1)t/\gamma+\theta)]/2$ because $\Delta E - u\cdot\Delta k = \Delta m/\gamma$. Translating only the envelope defines a different read with frequency $\gamma\Delta m$. For $c = 3$, $(m_1, m_2) = (100, 100.1)$, $|u| = 0.6$, Gaussian width 10, cutoff half-width 80 and $a = 10^{-9}$, outward interval bounds give probability error below 0.038 and native-tick swing above 0.923 through $2\pi\gamma/\Delta m + \tau$. This three-dimensional reflecting apparatus is an analytic finite-size witness. Separate periodic packet controls and small reflecting operators are executed. The read has charged duration and exports its result to a live archive; its center is prescribed by the external schedule.

Preparation, protection and energy ambiguity. An occupied-orbital unitary is prepared by adjacent-row rotations routed on the connected source graph, with every syndrome repaired and retained. At most $O(M^3)$ native rotations and depth suffice, including fermionic swaps and restored helper banks. With angle synthesis and scheduling, a conservative request envelope is $G = O(q^9\log^{16}(2q))$. At fixed positive admitted local noise strength, the supplied protection and archive compiler uses $O(\log\log q)$ protection levels and $O(\log q)$-bit archive words. If s is its fixed per-level size factor and $\kappa = \lceil\log_2 s\rceil$, synthesis to q^{-64} gives complete joint decoded-instrument error $O(q^{-55}\log^{36+2\kappa}(2q))$. Since $0 \le H_{\mathrm{sea}} \le M\pi/\tau$, the corresponding mean excitation error is $O(q^{-51}\log^{39+2\kappa}(2q))$. Preparation, idles and retained outcomes are included. Offline orbital computation and physical drive work are not assigned native energy costs. With $N = O(\log q)$ archive bits per word and k protection levels, the complete diagnostic storage-time envelope is $O(G^3N^6s^{3k}\log q) = O(q^{27}\log^{55+3\kappa}(2q))$. These resources include the enlarged preparation prehistory; its duration is not held fixed as resolution increases.

Neither the sea nor these reads select a unique energy interpolation. With $\Pi_{\mathrm{hi}} = \mathbf{1}_{|h|>\pi/(2\tau)}$, the alternative

$$h_{\mathrm{UV}} = h + (2\pi/\tau)\,\mathrm{sign}(h)\Pi_{\mathrm{hi}}$$

has the same U and P, and $|h| \le |h_{\mathrm{UV}}| \le 5|h|$. It preserves the finite-energy read class; on every fixed smooth admitted mode, the energy difference is at most $8\tau\|hf\|^2/\pi \to 0$. Its ultraviolet excitation costs differ despite identical integer-time reads, vacuum and infrared limits. The positive stroboscopic generator therefore supplies neither an apparatus pulse Hamiltonian nor interacting vacuum stability or empirical energy calibration. Complete derivations and independent finite operator/circuit checks are supplied in `docs/research/SOURCE_SEA_ENERGY.md` and `code/m1_sea_energy`.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Affine Lorentz atlases, Poincare cocycles, celestial null directions, observer hyperboloids, and Lorentz-orthogonal rest spaces [180, 187, 184].
- Axiomatic reconstruction programs (AQFT, causal sets, holography) [145, 1, 106].
- Bisio-D Ariano-Perinotti QCA [97].
- Borchers-Wiesbrock theory, Araki-Zsido, Guido-Longo-Wiesbrock, Wiesbrock modular intersections, HJPW Markov structure [103, 215, 92, 144, 216, 151].
- Cauchy-Binet and total nonnegativity [157, 133].
- Causal-set faithful embeddings and order-plus-number reconstruction, causal-conformal reconstruction, and Lorentzian limit diagnostics [1, 14, 15, 2, 7, 5].
- Charge conservation and Gauss-law preservation for discrete Maxwell schemes on cell complexes [211, 105, 217].
- Classical store-and-forward channels and finite dataflow refinement [110, 95].
- Classical store-and-forward networks and dataflow refinement [110, 95].
- coercive finite differences [206, 131].
- Combinatorial surface classification, Rado theorem, Mobius cross-ratio geometry, and modular GNS theory [136, 189, 177, 93, 208].
- Consistent positive finite differences and CFL stability [179, 169, 119, 120].
- Deterministic floor discrepancy bounds and conformal scalar action identities [167, 212].
- Discrete Poynting theorems on cell complexes [207, 105].
- Disjoint projection monoids, finite observation kernels and quantum phase sensitivity [149, 100, 109].
- Doob conditioning and relative entropy [127, 168].
- Dual-rail arithmetic and finite-precision error propagation [201, 154].
- Dual-rail arithmetic, register lifetime allocation and finite-precision numerical composition [201, 188, 113, 154].
- Exact classical readback/restoration and interval arithmetic [178].
- Explicit models of axiom systems and satisfiability witnesses for typed premise bundles [155].
- Fibonacci Diophantine approximation and deterministic equal-cell quadrature [150, 182].
- finite conditioned KL and Dyck path combinatorics [168, 121, 202, 203].
- finite Fourier and graph spectral analysis [209, 115].
- Finite graph invariants and noninjective observation maps [140].
- Finite linear observability and canonical Weyl algebras [161, 135].
- finite linear observability and word counting [161, 173].
- Finite linear recurrence stability and Duhamel error bounds [128, 186].

- Finite null tomography and Ward/Bianchi Einstein-form tensor identities [212, 175].
- Finite operational channel theory [213, 152].
- finite propagation for Klein-Gordon fields [130, 205].
- Finite quantum information projections [37, 213].
- Finite symmetry [196].
- Finite-graph divergence theorems and discrete radial-shell bookkeeping [141, 172].
- free Gaussian fields [139, 124].
- Gaussian oscillator and polynomial preservation identities [135, 116].
- Gaussian oscillator covariance [214, 195].
- Geometric spanners and causal-set order/count limits [181, 1, 2].
- Graph subdivision diameter bounds and register lifetime allocation [125, 188].
- information capacity [197, 121].
- Inverse problems on hyperbolic space, finite frames, residual minimization [153, 114, 129].
- Jacobson 1995, Jacobson 2016, BW modular theory [159, 160, 98, 99].
- Jacobson 1995/2016 small-ball derivations [159, 160].
- Jacobson Clausius bookkeeping, entanglement first laws [159, 101].
- Kraus instruments, Weyl channels and GHZ parity measurements [166, 156, 142, 102].
- lattice quadrature [123, 199].
- Lattice surface tensions and ball-cap crossing integrals [126, 194].
- Linear observability under time-dependent averaging and triangular inversion [161, 200].
- Longest-chain bounds in layered orders, the sum-of-cubes identity and integration of causal-diamond volume [174, 4, 138].
- Lorentz boost identification of rest spaces and the inverse-square shell law [180, 158].
- Marsden and West, Discrete mechanics and variational integrators (2001) [38].
- Mlodinow-Brun quantum-walk Dirac limits [176].
- Modular Hamiltonians of Gibbs states and the Clausius form of the entanglement first law [146, 159, 101].
- Modular temperature and KMS parameter rescaling in finite quantum statistical mechanics [117, 146, 108].
- Nonlocal BV perimeter theory [122, 91].
- Nzongani et al. tetrahedral walks [183].
- Order embeddings of finite causal sets into Minkowski causal cones [219, 1].
- Parameter identifiability and thermodynamic work derivatives [193, 111].
- Partial-information decision rules and absorbing finite random walks [132, 164].
- Positive diagonal symmetrization and edge-ratio cycle consistency [165, 163].
- Projective null-cone geometry, PSL(2,C) Lorentz isomorphism, hyperboloid H3 model [187, 191].

- Quantum Gibbs-state mixing, conditional mutual information, approximate quantum Markov chains, Fawzi-Renner recovery [134, 162, 107, 171].
- Ranard-Walter-Witteveen strict causality [190].
- Renes et al. tetrahedral SIC [192].
- reversible classical computation [94, 137].
- Scalar field stress tensors, harmonic Bose ensembles and spectral numerical error bounds [170, 185, 148].
- Sparse geometric graphs [181].
- Spectral graph diffusion, incidence homology, root lattices, finite-group isometries, and metric completion of dense arithmetic modules [115, 96, 118].
- standard Gram inversion and perturbation estimates [157, 204].
- Stress tensor tomography, modular Hamiltonian densities in QFT [112, 99, 198].
- Symplectic Verlet spectral error analysis [210, 147].
- Textbook Gauss-law inverse-square arguments [158, 143].
- Watrous Section 5.2 [213].
- Whitney finite elements, Maxwell stress and classical canonical Gaussian ensembles [104, 158, 185].
- Yoshida higher-order symplectic composition [218].

References

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[2] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). https://doi.org/10.1007/s41114-019-0023-1.

[3] L. Bombelli and D. A. Meyer, *The origin of Lorentzian geometry*, Phys. Lett. A **141**, 226–228 (1989). https://doi.org/10.1016/0375-9601(89)90474-X.

[4] G. Brightwell and R. Gregory, *Structure of random discrete spacetime*, Phys. Rev. Lett. **66**, 260–263 (1991). https://doi.org/10.1103/PhysRevLett.66.260.

[5] S. Major, D. Rideout, and S. Surya, *On recovering continuum topology from a causal set*, J. Math. Phys. **48**, 032501 (2007). https://doi.org/10.1063/1.2435599.

[6] S. Major, D. Rideout, and S. Surya, *Stable homology as an indicator of manifoldlikeness in causal set theory*, Class. Quantum Grav. **26**, 175008 (2009). https://doi.org/10.1088/0264-9381/26/17/175008.

[7] L. Glaser and S. Surya, *Towards a definition of locality in a manifoldlike causal set*, Phys. Rev. D **88**, 124026 (2013). https://doi.org/10.1103/PhysRevD.88.124026.

[8] J. Myrheim, *Statistical geometry*, CERN preprint TH-2538 (1978).

[9] D. A. Meyer, *The Dimension of Causal Sets*, PhD thesis, Massachusetts Institute of Technology (1988).

[10] D. D. Reid, *Manifold dimension of a causal set: Tests in conformally flat spacetimes*, Phys. Rev. D **67**, 024034 (2003). https://doi.org/10.1103/PhysRevD.67.024034.

[11] D. M. T. Benincasa and F. Dowker, *The scalar curvature of a causal set*, Phys. Rev. Lett. **104**, 181301 (2010). https://doi.org/10.1103/PhysRevLett.104.181301.

[12] F. Dowker and L. Glaser, *Causal set d'Alembertians for various dimensions*, Class. Quantum Grav. **30**, 195016 (2013). https://doi.org/10.1088/0264-9381/30/19/195016.

[13] L. Bombelli, J. Henson, and R. D. Sorkin, *Discreteness without symmetry breaking: A theorem*, Mod. Phys. Lett. A **24**, 2579–2587 (2009). https://doi.org/10.1142/S0217732309031958.

[14] S. W. Hawking, A. R. King, and P. J. McCarthy, *A new topology for curved space–time which incorporates the causal, differential, and conformal structures*, J. Math. Phys. **17**, 174–181 (1976). https://doi.org/10.1063/1.522874.

[15] D. B. Malament, *The class of continuous timelike curves determines the topology of spacetime*, J. Math. Phys. **18**, 1399–1404 (1977). https://doi.org/10.1063/1.523436.

[16] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observable_normal_forms.pdf.

[17] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, 2026. Available at GitHub PDF.

[18] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, and D. Matscheko, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*, 2026. Available at GitHub PDF.

[19] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/reality_as_consensus_protocol.pdf.

[20] B. Müller, A. Osika, K. Xue, B. Cassie, and B. de La Fournière, *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf.

[21] B. Müller, M. Poneder, A. Osika, and K. Xue, *Deriving the Particle Zoo from Observer Consistency*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[22] B. Müller and D. Matscheko, *Observer-Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/cosmology/oph_dark_matter_paper.pdf.

[23] B. Müller and J. Hill, *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics: A Support-Visible OPH Route to the Clay Problem: A Conditional Identity*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/yang_mills_gap_clay_problem.pdf.

[24] B. Müller, *Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Witness*, 2026. Archived manuscript. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/archive/observer_patch_holography_as_string_vacuum_selector.pdf.

[25] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/flagship/from_observer_consensus_to_standard_physics.pdf.

[26] B. Müller and M. Poneder, *An Executable Observer for Observer Patch Holography: Specification, Certified Components and Acceptance Criteria*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/executable_observer_construction.pdf.

[27] B. Müller, *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf.

[28] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source available at https://github.com/muellerberndt/oph-physics-sim. Large-run source revision: https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/closure_loop. Active refinement-readout revision: https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8. Finite local-domain revision and data: https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506, https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506/data/local_domain. Exact repository-local archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/local_domain. Causal-poset evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/source_net_causal_poset. Closure-loop evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/closure_loop. Evidence index: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence.

[29] Pragma Research, *Observer-Patch Holography: machine-checked proofs and executable certificates*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography.

[30] Pragma Research, *Frozen Prediction Register and Primitive Twelve-Port Prediction Receipt*, 2026. Immutable source receipt: https://github.com/FloatingPragma/observer-patch

-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json. Append-only custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz11_2026-07-31.

[31] DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," 2025, arXiv:2503.14738. https://arxiv.org/abs/2503.14738. Official chains and checksum manifest: https://data.desi.lbl.gov/public/papers/y3/bao-cosmo-params/README.html.

[32] M. Filzinger et al., "Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons," *Physical Review Letters* **130**, 253001 (2023). https://doi.org/10.1103/PhysRevLett.130.253001.

[33] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astronomical Journal* **152**, 157 (2016), arXiv:1606.09251. https://arxiv.org/abs/1606.09251. CDS/VizieR catalogue: https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/.

[34] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Physical Review Letters* **117**, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917.

[35] L. Onsager, "Reciprocal Relations in Irreversible Processes. I," *Physical Review* **37**, 405–426 (1931), https://doi.org/10.1103/PhysRev.37.405.

[36] M. S. Green, "Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II," *Journal of Chemical Physics* **22**, 398–413 (1954), https://doi.org/10.1063/1.1740082; R. Kubo, "Statistical-Mechanical Theory of Irreversible Processes. I," *Journal of the Physical Society of Japan* **12**, 570–586 (1957), https://doi.org/10.1143/JPSJ.12.570.

[37] I. Csiszár, "I-Divergence Geometry of Probability Distributions and Minimization Problems," *The Annals of Probability* **3**, 146–158 (1975), https://doi.org/10.1214/aop/1176996454.

[38] J. E. Marsden and M. West, "Discrete Mechanics and Variational Integrators," *Acta Numerica* **10**, 357–514 (2001), https://doi.org/10.1017/S096249290100006X.

[39] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at https://www.nu-fit.org/?q=node/309.

[40] M. H. A. Newman, "On theories with a combinatorial definition of 'equivalence'," *Ann. of Math. (2)* **43** (1942), 223–243.

[41] E. H. Lieb and D. W. Robinson, "The finite group velocity of quantum spin systems," *Commun. Math. Phys.* **28** (1972), 251–257.

[42] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* **16** (1975), 985–1007.

[43] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17** (1976), 303–321.

[44] R. Brunetti, D. Guido, and R. Longo, "Modular Structure and Duality in Conformal Quantum Field Theory," *Commun. Math. Phys.* **156** (1993), 201–219, arXiv:funct-an/9302008.

[45] H.-W. Wiesbrock, "Half-Sided modular inclusions of von-Neumann-Algebras," *Commun. Math. Phys.* **157** (1993), 83–92.

[46] H. Araki and L. Zsidó, "Extension of the structure theorem of Borchers and its application to half-sided modular inclusions," *Rev. Math. Phys.* **17** (2005), 491–543, arXiv:math/0412061.

[47] D. Guido, R. Longo, and H.-W. Wiesbrock, "Extensions of Conformal Nets and Superselection Structures," *Commun. Math. Phys.* **192** (1998), 217–244, arXiv:hep-th/9703129.

[48] H.-W. Wiesbrock, "Modular Intersections of von-Neumann-Algebras in Quantum Field Theory," *Commun. Math. Phys.* **193** (1998), 269–285.

[49] W. G. Unruh, "Notes on black-hole evaporation," *Phys. Rev. D* **14** (1976), 870–892.

[50] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75** (1995), 1260–1263, doi:10.1103/PhysRevLett.75.1260, arXiv:gr-qc/9504004.

[51] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116** (2016), 201101, doi:10.1103/PhysRevLett.116.201101, arXiv:1505.04753.

[52] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15** (1977), 2738–2751, doi:10.1103/PhysRevD.15.2738.

[53] Y. Chen, D. Stanford, H. Tang, and Z. Yang, "Negative shocks versus static patch holography," arXiv:2607.14042 [hep-th] (2026). https://arxiv.org/abs/2607.14042

[54] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641** (2020), A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209.

[55] R. Bousso, Z. Fisher, J. Koeller, S. Leichenauer, and A. C. Wall, "Proof of the quantum null energy condition," *Phys. Rev. D* **93** (2016), 024017, arXiv:1509.02542.

[56] R. Bousso, Z. Fisher, S. Leichenauer, and A. C. Wall, "Quantum focusing conjecture," *Phys. Rev. D* **93** (2016), 064044, arXiv:1506.02669.

[57] S. Balakrishnan, T. Faulkner, Z. U. Khandker, and H. Wang, "A general proof of the quantum null energy condition," *J. High Energy Phys.* **2019** (2019), 20, arXiv:1706.09432.

[58] E. H. Lieb and M. B. Ruskai, "Proof of the strong subadditivity of quantum-mechanical entropy," *J. Math. Phys.* **14** (1973), 1938–1941.

[59] E. H. Wichmann, "Density matrices arising from incomplete measurements," *J. Math. Phys.* **4** (1963), 884–896.

[60] F. Hiai, M. Ohya, and M. Tsukada, "Sufficiency, KMS condition and relative entropy in von Neumann algebras," *Pacific J. Math.* **96** (1981), 99–109.

[61] M. Takesaki, "Conditional expectations in von Neumann algebras," *J. Funct. Anal.* **9** (1972), 306–321.

[62] D. Petz, "Sufficient subalgebras and the relative entropy of states of a von Neumann algebra," *Commun. Math. Phys.* **105** (1986), 123–131.

[63] D. Petz, "Sufficiency of channels over von Neumann algebras," *Quart. J. Math.* **39** (1988), 97–108.

[64] O. Fawzi and R. Renner, "Quantum conditional mutual information and approximate Markov chains," *Commun. Math. Phys.* **340** (2015), 575–611, arXiv:1410.0664.

[65] P. Hayden, R. Jozsa, D. Petz, and A. Winter, "Structure of states which satisfy strong subadditivity of quantum entropy with equality," *Commun. Math. Phys.* **246** (2004), 359–374.

[66] S. Doplicher and J. E. Roberts, "A new duality theory for compact groups," *Invent. Math.* **98** (1989), 157–218.

[67] S. Doplicher and J. E. Roberts, "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics," *Commun. Math. Phys.* **131** (1990), 51–107.

[68] T. Tannaka, "Über den Dualitätssatz der nichtkommutativen topologischen Gruppen," *Tohoku Math. J.* **45** (1938), 1–12.

[69] M. G. Krein, "A principle of duality for a bicompact group and a square block algebra," *Dokl. Akad. Nauk SSSR* **69** (1949), 725–728.

[70] H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32** (1974), 438–441.

[71] S. L. Glashow, J. Iliopoulos, and L. Maiani, "Weak interactions with lepton-hadron symmetry," *Phys. Rev. D* **2** (1970), 1285–1292.

[72] E. Witten, "An SU(2) anomaly," *Phys. Lett. B* **117** (1982), 324–328.

[73] S. Dimopoulos, S. Raby, and F. Wilczek, "Supersymmetry and the scale of unification," *Phys. Rev. D* **24** (1981), 1681–1683.

[74] U. Amaldi, W. de Boer, and H. Fürstenau, "Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP," *Phys. Lett. B* **260** (1991), 447–455.

[75] D. J. Gross and W. Taylor, "Two-dimensional QCD is a string theory," *Nucl. Phys. B* **400** (1993), 181–208, arXiv:hep-th/9301068.

[76] F. Peter and H. Weyl, "Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe," *Math. Ann.* **97** (1927), 737–755.

[77] A. W. Knapp, *Lie Groups Beyond an Introduction*, second edition, Progress in Mathematics **140**, Birkhäuser, 2002.

[78] A. Bullivant, M. Calçada, Z. Kádár, P. Martin, and J. Faria Martins, "Topological phases from higher gauge symmetry in 3+1D," *Phys. Rev. B* **95** (2017), 155118, arXiv:1606.06639.

[79] S. Chandrasekharan and U.-J. Wiese, "Quantum link models: A discrete approach to gauge theories," *Nucl. Phys. B* **492** (1997), 455–471, arXiv:hep-lat/9609042.

[80] W. Donnelly and A. C. Wall, "Entanglement entropy of electromagnetic edge modes," *Phys. Rev. Lett.* **114** (2015), 111603, arXiv:1412.1895.

[81] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, "Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence," *JHEP* **06** (2015), 149, arXiv:1503.06237.

[82] M. A. Levin and X.-G. Wen, "String-net condensation: A physical mechanism for topological phases," *Phys. Rev. B* **71** (2005), 045110, arXiv:cond-mat/0404617.

[83] D. Laghi, G. Carullo, J. Veitch, and W. Del Pozzo, "Quantum black hole spectroscopy: probing the quantum nature of the black hole area using LIGO-Virgo ringdown detections," *Class. Quantum Grav.* **38** (2021), 095005, arXiv:2011.03816.

[84] Clay Mathematics Institute, "Yang–Mills & the Mass Gap." Available at https://www.claymath.org/millennium/yang-mills-the-maths-gap/.

[85] A. Jaffe and E. Witten, "Quantum Yang–Mills Theory," official Clay Mathematics Institute problem description. Available at https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf.

[86] L. Wu, "Poincaré and transportation inequalities for Gibbs measures under the Dobrushin uniqueness condition," *Annals of Probability* **34** (2006), 1960–1989, arXiv:math/0611635.

[87] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions," *Communications in Mathematical Physics* **31** (1973), 83–112.

[88] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions II," *Communications in Mathematical Physics* **42** (1975), 281–305.

[89] Particle Data Group, "Review of Particle Physics," *Phys. Rev. D* **110** (2024), 030001. Available at https://pdg.lbl.gov/2024/download/db2024.pdf.

[90] Particle Data Group, "Electroweak Model and Constraints on New Physics," Available at https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf.

[91] Ambrosio, L., Fusco, N., and Pallara, D. (2000). *Functions of Bounded Variation and Free Discontinuity Problems*. Clarendon Press, Oxford. ISBN 0-19-850245-1.

[92] Araki, H., and Zsidó, L. (2005). "Extension of the structure theorem of Borchers and its application to half-sided modular inclusions." *Reviews in Mathematical Physics* **17**, 491–543. https://doi.org/10.1142/S0129055X05002388

[93] Beardon, A. F. (1983). *The Geometry of Discrete Groups*. Springer. https://doi.org/10.1007/978-1-4612-1146-4

[94] Bennett, C. H. (1973). "Logical Reversibility of Computation." *IBM Journal of Research and Development* **17**, 525–532. https://doi.org/10.1147/rd.176.0525

[95] Bertsekas, D. P., and Gallager, R. G. (1992). *Data Networks*. Prentice Hall. ISBN 0-13-200916-1.

[96] Biggs, N. (1993). *Algebraic Graph Theory*. Cambridge University Press. https://doi.org/10.1017/CBO9780511608704

[97] Bisio, A., D'Ariano, G. M., and Perinotti, P. (2016). "Special relativity in a discrete quantum universe." *Physical Review A* **94**, 042120. https://doi.org/10.1103/PhysRevA.94.042120

[98] Bisognano, J. J., and Wichmann, E. H. (1975). "On the duality condition for a Hermitian scalar field." *Journal of Mathematical Physics* **16**, 985–1007. https://doi.org/10.1063/1.522605

[99] Bisognano, J. J., and Wichmann, E. H. (1976). "On the duality condition for quantum fields." *Journal of Mathematical Physics* **17**, 303–321. https://doi.org/10.1063/1.522898

[100] Blackwell, D. (1953). "Equivalent comparisons of experiments." *Annals of Mathematical Statistics* **24**, 265–272. https://doi.org/10.1214/aoms/1177729032

[101] Blanco, D. D., Casini, H., Hung, L. Y., and Myers, R. C. (2013). "Relative entropy and holography." *Journal of High Energy Physics* **2013**, 60. https://doi.org/10.1007/JHEP08(2013)060

[102] Bollinger, J. J., Itano, W. M., Wineland, D. J., and Heinzen, D. J. (1996). "Optimal frequency measurements with maximally correlated states." *Physical Review A* **54**, R4649–R4652. https://doi.org/10.1103/PhysRevA.54.R4649

[103] Borchers, H. J. (1992). "The CPT-theorem in two-dimensional theories of local observables." *Communications in Mathematical Physics* **143**, 315–332. https://doi.org/10.1007/BF02099011

[104] Bossavit, A. (1988). "Whitney forms: a class of finite elements for three-dimensional computations in electromagnetism." *IEE Proceedings A: Physical Science, Measurement and Instrumentation, Management and Education, Reviews* **135**, 493–500. https://doi.org/10.1049/ip-a-1.1988.0077

[105] Bossavit, A. (1998). *Computational Electromagnetism*. Academic Press. https://doi.org/10.1016/B978-0-12-118710-1.X5000-4

[106] Bousso, R. (2002). "The holographic principle." *Reviews of Modern Physics* **74**, 825–874. https://doi.org/10.1103/RevModPhys.74.825

[107] Brandão, F. G. S. L., and Kastoryano, M. J. (2019). "Finite Correlation Length Implies Efficient Preparation of Quantum Thermal States." *Communications in Mathematical Physics* **365**, 1–16. https://doi.org/10.1007/s00220-018-3150-8

[108] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2*. Springer. https://doi.org/10.1007/978-3-662-03444-6

[109] Braunstein, S. L., and Caves, C. M. (1994). "Statistical distance and the geometry of quantum states." *Physical Review Letters* **72**, 3439–3443. https://doi.org/10.1103/PhysRevLett.72.3439

[110] Broy, M., and Stølen, K. (2001). *Specification and Development of Interactive Systems*. Springer. https://doi.org/10.1007/978-1-4613-0091-5

[111] Callen, H. B. (1985). *Thermodynamics and an Introduction to Thermostatistics*. Wiley. ISBN 0-471-86256-8.

[112] Casini, H., Huerta, M., and Myers, R. C. (2011). "Towards a derivation of holographic entanglement entropy." *Journal of High Energy Physics* **2011**, 36. https://doi.org/10.1007/JHEP05(2011)036

[113] Chaitin, G. J. (1982). "Register allocation and spilling via graph coloring." *ACM SIGPLAN Notices* **17**, 98–101. https://doi.org/10.1145/872726.806984

[114] Christensen, O. (2016). *An Introduction to Frames and Riesz Bases*. Birkhäuser/Springer (Applied and Numerical Harmonic Analysis). https://doi.org/10.1007/978-3-319-25613-9

[115] Chung, F. R. K. (1997). *Spectral Graph Theory*. American Mathematical Society. https://doi.org/10.1090/cbms/092

[116] Combescure, M., and Robert, D. (2012). *Coherent States and Applications in Mathematical Physics*. Springer. https://doi.org/10.1007/978-94-007-0196-0

[117] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[118] Conway, J. H., and Sloane, N. J. A. (1999). *Sphere Packings, Lattices and Groups*. Springer New York. https://doi.org/10.1007/978-1-4757-6568-7

[119] Courant, R., Friedrichs, K., and Lewy, H. (1928). "Über die partiellen Differenzengleichungen der mathematischen Physik." *Mathematische Annalen* **100**, 32–74. https://doi.org/10.1007/BF01448839

[120] Courant, R., Friedrichs, K., and Lewy, H. (1967). "On the partial difference equations of mathematical physics." *IBM Journal of Research and Development* **11**, 215–234. https://doi.org/10.1147/rd.112.0215

[121] Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory*. Wiley. https://doi.org/10.1002/047174882X

[122] Dávila, J. (2002). "On an open question about functions of bounded variation." *Calculus of Variations and Partial Differential Equations* **15**, 519–527. https://doi.org/10.1007/s005260100135

[123] Davis, P. J., and Rabinowitz, P. (1984). *Methods of Numerical Integration*. Academic Press. https://doi.org/10.1016/C2013-0-10566-1

[124] Dereziński, J., and Gérard, C. (2013). *Mathematics of Quantization and Quantum Fields*. Cambridge University Press. https://doi.org/10.1017/CBO9780511894541

[125] Diestel, R. (2017). *Graph Theory*. Springer, Berlin (Graduate Texts in Mathematics). https://doi.org/10.1007/978-3-662-53622-3

[126] Dobrushin, R., Kotecký, R., and Shlosman, S. (1992). *Wulff Construction: A Global Shape from Local Interaction*. American Mathematical Society (Translations of Mathematical Monographs 104). https://doi.org/10.1090/mmono/104

[127] Doob, J. L. (1957). "Conditional Brownian motion and the boundary limits of harmonic functions." *Bulletin de la Société Mathématique de France* **85**, 431–458. https://doi.org/10.24033/bsmf.1494

[128] Elaydi, S. (2005). *An Introduction to Difference Equations*. Springer. https://doi.org/10.1007/0-387-27602-5

[129] Engl, H. W., Hanke, M., and Neubauer, A. (1996). *Regularization of Inverse Problems*. Kluwer (Springer Netherlands), Dordrecht. https://doi.org/10.1007/978-94-009-1740-8

[130] Evans, L. C. (2010). *Partial Differential Equations*. American Mathematical Society. https://doi.org/10.1090/gsm/019

[131] Eymard, R., Gallouët, T., and Herbin, R. (2000). "Finite volume methods." In *Handbook of Numerical Analysis, Vol. VII*, 713–1018. Elsevier. https://doi.org/10.1016/S1570-8659(00)07005-8

[132] Fagin, R., Halpern, J. Y., Moses, Y., and Vardi, M. Y. (1995). *Reasoning About Knowledge*. The MIT Press. https://doi.org/10.7551/mitpress/5803.001.0001

[133] Fallat, S. M., and Johnson, C. R. (2011). *Totally Nonnegative Matrices*. Princeton University Press. https://doi.org/10.1515/9781400839018

[134] Fawzi, O., and Renner, R. (2015). "Quantum conditional mutual information and approximate Markov chains." *Communications in Mathematical Physics* **340**, 575–611. https://doi.org/10.1007/s00220-015-2466-x

[135] Folland, G. B. (1989). *Harmonic Analysis in Phase Space*. Princeton University Press. https://doi.org/10.1515/9781400882427

[136] Francis, G. K., and Weeks, J. R. (1999). "Conway's ZIP proof." *American Mathematical Monthly* **106**, 393–399. https://doi.org/10.1080/00029890.1999.12005061

[137] Fredkin, E., and Toffoli, T. (1982). "Conservative logic." *International Journal of Theoretical Physics* **21**, 219–253. https://doi.org/10.1007/BF01857727

[138] Gibbons, G. W., and Solodukhin, S. N. (2007). "The geometry of small causal diamonds." *Physics Letters B* **649**, 317–324. https://doi.org/10.1016/j.physletb.2007.03.068

[139] Glimm, J., and Jaffe, A. (1987). *Quantum Physics: A Functional Integral Point of View*. Springer, New York. https://doi.org/10.1007/978-1-4612-4728-9

[140] Godsil, C., and Royle, G. (2001). *Algebraic Graph Theory*. Springer. https://doi.org/10.1007/978-1-4613-0163-9

[141] Grady, L. J., and Polimeni, J. R. (2010). *Discrete Calculus*. Springer. https://doi.org/10.1007/978-1-84996-290-2

[142] Greenberger, D. M., Horne, M. A., and Zeilinger, A. (1989). "Going Beyond Bell's Theorem." In *Bell's Theorem, Quantum Theory and Conceptions of the Universe*, 69–72. Kluwer (Springer Netherlands), Dordrecht. https://doi.org/10.1007/978-94-017-0849-4_10

[143] Griffiths, D. J. (2017). *Introduction to Electrodynamics*. Cambridge University Press. https://doi.org/10.1017/9781108333511

[144] Guido, D., Longo, R., and Wiesbrock, H. W. (1998). "Extensions of conformal nets and superselection structures." *Communications in Mathematical Physics* **192**, 217–244. https://doi.org/10.1007/s002200050297

[145] Haag, R., and Kastler, D. (1964). "An algebraic approach to quantum field theory." *Journal of Mathematical Physics* **5**, 848–861. https://doi.org/10.1063/1.1704187

[146] Haag, R., Hugenholtz, N. M., and Winnink, M. (1967). "On the equilibrium states in quantum statistical mechanics." *Communications in Mathematical Physics* **5**, 215–236. https://doi.org/10.1007/BF01646342

[147] Hairer, E., Lubich, C., and Wanner, G. (2003). "Geometric numerical integration illustrated by the Störmer–Verlet method." *Acta Numerica* **12**, 399–450. https://doi.org/10.1017/S0962492902000144

[148] Hairer, E., Lubich, C., and Wanner, G. (2006). *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. Springer. https://doi.org/10.1007/3-540-30666-8

[149] Halmos, P. R. (1958). *Finite-Dimensional Vector Spaces*. Van Nostrand. https://doi.org/10.1007/978-1-4612-6387-6

[150] Hardy, G. H., and Wright, E. M. (2008). *An Introduction to the Theory of Numbers*. Oxford University Press. ISBN 978-0-19-921986-5.

[151] Hayden, P., Jozsa, R., Petz, D., and Winter, A. (2004). "Structure of states which satisfy strong subadditivity of quantum entropy with equality." *Communications in Mathematical Physics* **246**, 359–374. https://doi.org/10.1007/s00220-004-1049-z

[152] Heinosaari, T., and Ziman, M. (2011). *The Mathematical Language of Quantum Theory: From Uncertainty to Entanglement*. Cambridge University Press. https://doi.org/10.1017/CBO9781139031103

[153] Helgason, S. (1999). *The Radon Transform*. Birkhäuser, Boston (Progress in Mathematics). https://doi.org/10.1007/978-1-4757-1463-0

[154] Higham, N. J. (2002). *Accuracy and Stability of Numerical Algorithms*. SIAM. https://doi.org/10.1137/1.9780898718027

[155] Hodges, W. (1993). *Model Theory*. Cambridge University Press. https://doi.org/10.1017/CBO9780511551574

[156] Holevo, A. S. (2012). *Quantum Systems, Channels, Information: A Mathematical Introduction*. De Gruyter, Berlin. https://doi.org/10.1515/9783110273403

[157] Horn, R. A., and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press. https://doi.org/10.1017/CBO9781139020411

[158] Jackson, J. D. (1999). *Classical Electrodynamics*. Wiley, New York. ISBN 0-471-30932-X.

[159] Jacobson, T. (1995). "Thermodynamics of Spacetime: The Einstein Equation of State." *Physical Review Letters* **75**, 1260–1263. https://doi.org/10.1103/PhysRevLett.75.1260

[160] Jacobson, T. (2016). "Entanglement Equilibrium and the Einstein Equation." *Physical Review Letters* **116**, 201101. https://doi.org/10.1103/PhysRevLett.116.201101

[161] Kalman, R. E. (1963). "Mathematical description of linear dynamical systems." *Journal of the Society for Industrial and Applied Mathematics, Series A: Control* **1**, 152–192. https://doi.org/10.1137/0301010

[162] Kato, K., and Brandão, F. G. S. L. (2019). "Quantum Approximate Markov Chains are Thermal." *Communications in Mathematical Physics* **370**, 117–149. https://doi.org/10.1007/s00220-019-03485-6

[163] Kelly, F. P. (2011). *Reversibility and Stochastic Networks*. Cambridge University Press. ISBN 978-1-107-40115-0.

[164] Kemeny, J. G., and Snell, J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics). ISBN 0-387-90192-2.

[165] Kolmogoroff, A. (1936). "Zur Theorie der Markoffschen Ketten." *Mathematische Annalen* **112**, 155–160. https://doi.org/10.1007/BF01565412

[166] Kraus, K. (1983). *States, Effects, and Operations: Fundamental Notions of Quantum Theory*. Springer. https://doi.org/10.1007/3-540-12732-1

[167] Kuipers, L., and Niederreiter, H. (1974). *Uniform Distribution of Sequences*. Wiley. ISBN 0-486-45019-8.

[168] Kullback, S., and Leibler, R. A. (1951). "On information and sufficiency." *Annals of Mathematical Statistics* **22**, 79–86. https://doi.org/10.1214/aoms/1177729694

[169] Kuo, H. J., and Trudinger, N. S. (1992). "Discrete methods for fully nonlinear elliptic equations." *SIAM Journal on Numerical Analysis* **29**, 123–135. https://doi.org/10.1137/0729008

[170] Landau, L. D., and Lifshitz, E. M. (1975). *The Classical Theory of Fields*. Pergamon Press. ISBN 978-0-08-025072-4.

[171] Lieb, E. H., and Ruskai, M. B. (1973). "Proof of the strong subadditivity of quantum-mechanical entropy." *Journal of Mathematical Physics* **14**, 1938–1941. https://doi.org/10.1063/1.1666274

[172] Lim, L. H. (2020). "Hodge Laplacians on graphs." *SIAM Review* **62**, 685–715. https://doi.org/10.1137/18M1223101

[173] Lothaire, M. (1997). *Combinatorics on Words*. Cambridge University Press. https://doi.org/10.1017/CBO9780511566097

[174] Mirsky, L. (1971). "A Dual of Dilworth's Decomposition Theorem." *The American Mathematical Monthly* **78**, 876–877. https://doi.org/10.1080/00029890.1971.11992886

[175] Misner, C. W., Thorne, K. S., and Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman. ISBN 0-7167-0344-0.

[176] Mlodinow, L., and Brun, T. A. (2018). "Discrete spacetime, quantum walks, and relativistic wave equations." *Physical Review A* **97**, 042131. https://doi.org/10.1103/PhysRevA.97.042131

[177] Moise, E. E. (1977). *Geometric Topology in Dimensions 2 and 3*. Springer. https://doi.org/10.1007/978-1-4612-9906-6

[178] Moore, R. E., Kearfott, R. B., and Cloud, M. J. (2009). *Introduction to Interval Analysis*. SIAM. https://doi.org/10.1137/1.9780898717716

[179] Motzkin, T. S., and Wasow, W. (1952). "On the approximation of linear elliptic differential equations by difference equations with positive coefficients." *Journal of Mathematics and Physics* **31**, 253–259. https://doi.org/10.1002/sapm1952311253

[180] Naber, G. L. (2012). *The Geometry of Minkowski Spacetime: An Introduction to the Mathematics of the Special Theory of Relativity*. Springer. https://doi.org/10.1007/978-1-4419-7838-7

[181] Narasimhan, G., and Smid, M. (2007). *Geometric Spanner Networks*. Cambridge University Press. https://doi.org/10.1017/CBO9780511546884

[182] Niederreiter, H. (1992). *Random Number Generation and Quasi-Monte Carlo Methods*. SIAM. https://doi.org/10.1137/1.9781611970081

[183] Nzongani, U., Eon, N., Márquez-Martín, I., Pérez, A., Di Molfetta, G., and Arrighi, P. (2024). "Dirac quantum walk on tetrahedra." *Physical Review A* **110**, 042418. https://doi.org/10.1103/PhysRevA.110.042418

[184] O'Neill, B. (1983). *Semi-Riemannian Geometry with Applications to Relativity*. Academic Press, New York. ISBN 0-12-526740-1.

[185] Pathria, R. K., and Beale, P. D. (2011). *Statistical Mechanics*. Academic Press. ISBN 978-0-12-382188-1.

[186] Pazy, A. (1983). *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer. https://doi.org/10.1007/978-1-4612-5561-1

[187] Penrose, R., and Rindler, W. (1984). *Spinors and Space-Time, Volume 1*. Cambridge University Press. https://doi.org/10.1017/CBO9780511564048

[188] Poletto, M., and Sarkar, V. (1999). "Linear scan register allocation." *ACM Transactions on Programming Languages and Systems* **21**, 895–913. https://doi.org/10.1145/330249.330250

[189] Radó, T. (1925). "Über den Begriff der Riemannschen Fläche." *Acta Litterarum ac Scientiarum Regiae Universitatis Hungaricae Francisco-Josephinae, Sectio Scientiarum Mathematicarum* **2**, 101–121. http://acta.bibl.u-szeged.hu/13312/

[190] Ranard, D., Walter, M., and Witteveen, F. (2022). "A Converse to Lieb–Robinson Bounds in One Dimension Using Index Theory." *Annales Henri Poincaré* **23**, 3905–3979. https://doi.org/10.1007/s00023-022-01193-x

[191] Ratcliffe, J. G. (2019). *Foundations of Hyperbolic Manifolds*. Springer International Publishing. https://doi.org/10.1007/978-3-030-31597-9

[192] Renes, J. M., Blume-Kohout, R., Scott, A. J., and Caves, C. M. (2004). "Symmetric informationally complete quantum measurements." *Journal of Mathematical Physics* **45**, 2171–2180. https://doi.org/10.1063/1.1737053

[193] Rothenberg, T. J. (1971). "Identification in Parametric Models." *Econometrica* **39**, 577–591. https://doi.org/10.2307/1913267

[194] Santaló, L. A. (2004). *Integral Geometry and Geometric Probability*. Cambridge University Press. https://doi.org/10.1017/CBO9780511617331

[195] Serafini, A. (2017). *Quantum Continuous Variables: A Primer of Theoretical Methods*. CRC Press. https://doi.org/10.1201/9781315118727

[196] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[197] Shannon, C. E. (1948). "A Mathematical Theory of Communication." *Bell System Technical Journal* **27**, 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

[198] Sharafutdinov, V. A. (1994). *Integral Geometry of Tensor Fields*. De Gruyter. https://doi.org/10.1515/9783110900095

[199] Sloan, I. H., and Joe, S. (1994). *Lattice Methods for Multiple Integration*. Clarendon Press, Oxford. ISBN 0-19-853472-8.

[200] Sontag, E. D. (1998). *Mathematical Control Theory*. Springer. https://doi.org/10.1007/978-1-4612-0577-7

[201] Sparsø, J., and Furber, S. (2001). *Principles of Asynchronous Circuit Design*. Springer. https://doi.org/10.1007/978-1-4757-3385-3

[202] Stanley, R. P. (1999). *Enumerative Combinatorics, Volume 2*. Cambridge University Press. https://doi.org/10.1017/CBO9780511609589

[203] Stanley, R. P. (2015). *Catalan Numbers*. Cambridge University Press. https://doi.org/10.1017/CBO9781139871495

[204] Stewart, G. W., and Sun, J. g. (1990). *Matrix Perturbation Theory*. Academic Press. ISBN 0-12-670230-6.

[205] Strauss, W. A. (1989). *Nonlinear Wave Equations*. American Mathematical Society (CBMS Regional Conference Series in Mathematics 73). https://doi.org/10.1090/cbms/073

[206] Strikwerda, J. C. (2004). *Finite Difference Schemes and Partial Differential Equations*. SIAM. https://doi.org/10.1137/1.9780898717938

[207] Taflove, A., and Hagness, S. C. (2005). *Computational Electrodynamics: The Finite-Difference Time-Domain Method*. Artech House. ISBN 1-58053-832-0.

[208] Takesaki, M. (1970). *Tomita's Theory of Modular Hilbert Algebras and its Applications*. Springer. https://doi.org/10.1007/BFb0065832

[209] Terras, A. (1999). *Fourier Analysis on Finite Groups and Applications*. Cambridge University Press. https://doi.org/10.1017/CBO9780511626265

[210] Verlet, L. (1967). "Computer "Experiments" on Classical Fluids. I. Thermodynamical Properties of Lennard-Jones Molecules." *Physical Review* **159**, 98–103. https://doi.org/10.1103/PhysRev.159.98

[211] Villasenor, J., and Buneman, O. (1992). "Rigorous charge conservation for local electromagnetic field solvers." *Computer Physics Communications* **69**, 306–316. https://doi.org/10.1016/0010-4655(92)90169-Y

[212] Wald, R. M. (1984). *General Relativity*. University of Chicago Press. https://doi.org/10.7208/chicago/9780226870373.001.0001

[213] Watrous, J. (2018). *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142

[214] Weedbrook, C., Pirandola, S., García-Patrón, R., Cerf, N. J., Ralph, T. C., Shapiro, J. H., et al. (2012). "Gaussian quantum information." *Reviews of Modern Physics* **84**, 621–669. https://doi.org/10.1103/RevModPhys.84.621

[215] Wiesbrock, H. W. (1993). "Half-sided modular inclusions of von-Neumann-algebras." *Communications in Mathematical Physics* **157**, 83–92. https://doi.org/10.1007/BF02098019

[216] Wiesbrock, H. W. (1998). "Modular intersections of von Neumann algebras in quantum field theory." *Communications in Mathematical Physics* **193**, 269–285. https://doi.org/10.1007/s002200050329

[217] Yee, K. S. (1966). "Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media." *IEEE Transactions on Antennas and Propagation* **14**, 302–307. https://doi.org/10.1109/TAP.1966.1138693

[218] Yoshida, H. (1990). "Construction of higher order symplectic integrators." *Physics Letters A* **150**, 262–268. https://doi.org/10.1016/0375-9601(90)90092-3

[219] Zeeman, E. C. (1964). "Causality implies the Lorentz group." *Journal of Mathematical Physics* **5**, 490–493. https://doi.org/10.1063/1.1704140

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.